

Henryk Kotlarski

A Model—Theoretic Approach to Proof Theory

Edited by Zofia Adamowicz
Teresa Bigorajska
Konrad Zdanowski

Trends in Logic

Volume 51

TRENDS IN LOGIC

Studia Logica Library

VOLUME 51

Editor-in-Chief

Heinrich Wansing, *Department of Philosophy, Ruhr University Bochum,
Bochum, Germany*

Editorial Board

Arnon Avron, *Department of Computer Science, University of Tel Aviv, Tel Aviv, Israel*

Katalin Bimbó, *Department of Philosophy, University of Alberta, Edmonton, AB, Canada*

Giovanna Corsi, *Department of Philosophy, University of Bologna, Bologna, Italy*

Janusz Czelakowski, *Institute of Mathematics and Informatics, University of Opole,
Opole, Poland*

Roberto Giuntini, *Department of Philosophy, University of Cagliari, Cagliari, Italy*

Rajeev Goré, *Australian National University, Canberra, ACT, Australia*

Andreas Herzig, *IRIT, University of Toulouse, Toulouse, France*

Wesley Holliday, *UC Berkeley, Lafayette, CA, USA*

Andrzej Indrzejczak, *Department of Logic, University of Łódź, Łódź, Poland*

Daniele Mundici, *Mathematics and Computer Science, University of Florence, Firenze, Italy*

Sergei Odintsov, *Sobolev Institute of Mathematics, Novosibirsk, Russia*

Ewa Orłowska, *Institute of Telecommunications, Warsaw, Poland*

Peter Schroeder-Heister, *Wilhelm-Schickard-Institut, Universität Tübingen, Tübingen,
Baden-Württemberg, Germany*

Yde Venema, *ILLC, Universiteit van Amsterdam,*

Amsterdam, Noord-Holland, The Netherlands

Andreas Weiermann, *Vakgroep Zuivere Wiskunde en Computeralgebra, University of Ghent,
Ghent, Belgium*

Frank Wolter, *Department of Computing, University of Liverpool, Liverpool, UK*

Ming Xu, *Department of Philosophy, Wuhan University, Wuhan, China*

Jacek Malinowski, *Institute of Philosophy and Sociology, Polish Academy of Sciences,
Warszawa, Poland*

Assistant Editor

Daniel Skurt, *Ruhr-University Bochum, Bochum, Germany*

Founding Editor

Ryszard Wojcicki, *Institute of Philosophy and Sociology, Polish Academy of Sciences,
Warsaw, Poland*

The book series Trends in Logic covers essentially the same areas as the journal Studia Logica, that is, contemporary formal logic and its applications and relations to other disciplines. The series aims at publishing monographs and thematically coherent volumes dealing with important developments in logic and presenting significant contributions to logical research.

Volumes of Trends in Logic may range from highly focused studies to presentations that make a subject accessible to a broader scientific community or offer new perspectives for research. The series is open to contributions devoted to topics ranging from algebraic logic, model theory, proof theory, philosophical logic, non-classical logic, and logic in computer science to mathematical linguistics and formal epistemology. This thematic spectrum is also reflected in the editorial board of Trends in Logic. Volumes may be devoted to specific logical systems, particular methods and techniques, fundamental concepts, challenging open problems, different approaches to logical consequence, combinations of logics, classes of algebras or other structures, or interconnections between various logic-related domains. Authors interested in proposing a completed book or a manuscript in progress or in conception can contact either christi.lue@springer.com or one of the Editors of the Series.

Henryk Kotlarski
Author

A Model–Theoretic Approach to Proof Theory

Edited by: Zofia Adamowicz, Teresa Bigorajska
and Konrad Zdanowski

Author

Henryk Kotlarski (deceased)
Warsaw, Poland

Editors

Zofia Adamowicz
Polish Academy of Sciences
Institute of Mathematics
Warsaw, Poland

Teresa Bigorajska
Cardinal Stefan Wyszyński
University in Warsaw
Warsaw, Poland

Konrad Zdanowski
Cardinal Stefan Wyszyński
University in Warsaw
Warsaw, Poland

ISSN 1572-6126

Trends in Logic

ISBN 978-3-030-28920-1

ISSN 2212-7313 (electronic)

ISBN 978-3-030-28921-8 (eBook)

<https://doi.org/10.1007/978-3-030-28921-8>

© Springer Nature Switzerland AG 2019

This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland



Fig. 1 Henryk Kotlarski (1949–2008). A drawing by Andreas Weiermann based on a photograph by Nina Gierasimczuk



Fig. 2 Henryk with Zygmunt Ratajczyk. Their collaboration greatly influenced Henryk's book. Karpacz, Poland, 1979. Author unknown



Fig. 3 Henryk lecturing at Logic Workshop. Kazimierz Dolny, Poland, 2006. A photograph by Nina Gierasimczuk

A Note From the Editors

Henryk Kotlarski (1949–2008) was our friend or teacher or both. His scientific interests were concentrated around the model theory of arithmetic. The main topics of his research papers changed in time. He started with a study of well-ordered models in his Ph.D. Then he studied recursively saturated models of Peano arithmetic, especially their automorphism groups and satisfaction classes. With Zygmunt Ratajczyk he characterized the arithmetical strength of theories which are augmented by a satisfaction class and admit induction for Σ_n formulas mentioning the satisfaction class. He had a deep interest in incompleteness phenomena. He constructed his own proofs of the Gödel incompleteness theorems and was especially keen on eliminating the use of diagonal arguments in this context. In his last years, he devoted himself to the study of combinatorics of α -large sets in the style of Ketonen–Solovay. He liked the combinatorics of these results but he also treated them as a tool in the study of models of arithmetic. This book reflects this viewpoint. His publication record at MathSciNet contains 43 positions, many with co-authors.

During his last years, Henryk was occupied, among other things, with writing a book which would present his personal view on the model theory of arithmetic. The book was completed to a large extent, although many parts required careful proofreading and many corrections. We tried to do our best, but we also tried to respect Henryk’s own concepts and his style of presentation. We kept the working style of the book and we did not try to make it self-contained. On the other hand, we presented prerequisite notions and theorems which can be found in any standard textbook on formal arithmetic, e.g., Kaye [1]. We tried to make the reading comfortable, so we did not mark majority of our corrections. Occasionally, we marked additions to the text by a black vertical stroke at the side of the page. However, some proofs needed to be completely rewritten, especially the proofs of Theorems 5.2.3, 5.2.9, 5.2.11.

Moreover, we needed to remove substantial parts of the book. We removed the parts concerning the theory of an inductive full satisfaction class. The reason for this is a gap discovered by Heck and Visser in Henryk’s paper on arithmetic with bounded induction for a satisfaction class [2] (see an appendix of [3]). Recently, it has turned out that this gap may be filled (see Łęłyk’s Ph.D. [4] Theorem 141) but

we decided to skip the sections devoted to inductive full satisfaction classes as they would require major changes in the manuscript. We did so even though the results of Kotlarski in [2] and of Kotlarski and Ratajczyk in [5] and [6] are saved. The second part which we removed was based on Kotlarski's article [7] on quantifier rank of proofs of inconsistency. We found that it would take too much work to present these results with full rigor. Even proving them for Peano arithmetic would require carefully defining and inspecting the assumed logical formalism and style of axiomatization of Peano arithmetic.

During the last years, there were some new interesting developments in the field of arithmetics with satisfaction classes. We want to mention them as suggestions for further reading. Ali Enayat and Albert Visser found a new proof of the Kotlarski, Krajewski, and Lachlan theorem [8] stating that any countable, recursively saturated model of Peano arithmetic admits a satisfaction class. Their construction (see [9]) is considerably simpler and gives a satisfaction class in an elementary extension of a given model $\mathcal{M}$. Then, by resplendency, one concludes that $\mathcal{M}$ admits a satisfaction class, too. On the syntactic side, there is a cut-elimination style proof of conservativeness of PA with a satisfaction class over PA given by Leigh in [10]. Another line of research is related to satisfaction classes extended by some arithmetical principles. These are weak induction principles, like Δ_0 induction in the language with a satisfaction class, or reflection principles for, e.g., proofs in first-order logic, or the truth of internal induction for all formulas in the sense of a model. We mentioned some important findings of [4] in the previous paragraph. For more, a reader may consult [3, 11, 12] and a recent unpublished work by Fedor Pakhomov on satisfaction classes which are disjunctively correct. Cezary Cieřliński's recent book [13] contains a good presentation of a problem of conservativeness of various theories of truth. Finally, we have to recall a recent groundbreaking result by Patey and Yokoyama on the $\tilde{\Pi}_3^0$ conservativeness of Ramsey theorem for pairs and two colors, RT_2^2 over RCA_0 [14]. Among other things, they use substantially the combinatorics of α -large sets which is the topic of the first chapter of the book. Recently, Kołodziejczyk and Yokoyama extracted the combinatorial part of [14] giving a strengthening of their result with a more direct proof in the spirit of combinatorics of α -large sets, see [15].

During the long process of preparing Henryk's manuscript to be published, we had the support of many people, which was crucial for completing this task. First of all, we want to thank Andreas Weiermann for his strong encouragement of our work. Without him, we would not have the energy to finish this project. Then, our deep thanks go to Albert Visser. His ability to spot many subtleties in presented arguments saved the book from (some of) gaps or just inaccuracies. We also benefited from comments of Cezary Cieřliński, Ali Enayat, Leszek Kołodziejczyk, Mateusz Łećyk, and Bartosz Wcisćo with whom we discussed some parts of the book. We want to thank Nina Gierasimczuk who kindly agreed to use her photograph of Henryk during a lecture on a summer school. Last but not least, we thank the editors at Springer who were involved in the preparation of the book. Especially, our warm thanks go to Christi Lue who has helped us during all stages of the work.

The book concerns topics which are especially close to Henryk's interests. They are still at the heart of mathematical logic and philosophical inquiries. We hope that this book will present the unique approach to these problems which was developed by Henryk. We devote our work on this book to his memory.

Zofia Adamowicz
Teresa Bigorajska
Konrad Zdanowski

References

1. Kaye, R. (1991). *Models of peano arithmetic*. Oxford University Press. ISBN: 019853213X.
2. Kotlarski, H. (1986). Bounded induction and satisfaction classes. *Mathematical Logic Quarterly*, 32(31–34), 531–544. ISSN: 09425616. <https://doi.org/10.1002/malq.19860323107>.
3. Wcisło, B., & Łelyk, M. (2017). Notes on bounded induction for the compositional truth predicate. *The Review of Symbolic Logic*, 10(3), 455–480. ISSN: 1755-0203. <https://doi.org/10.1017/S1755020316000368>.
4. Łelyk, M. (2017). *Axiomatic theories of truth, bounded Induction and reflection principles*. Ph.D. Thesis. Warsaw University. <https://depotuw.ceon.pl/handle/item/2266>.
5. Kotlarski, H., & Ratajczyk, Z. (1990). Inductive full satisfaction classes. *Annals of Pure and Applied Logic*, 47(3), 199–223. ISSN: 01680072. [https://doi.org/10.1016/0168-0072\(90\)90035-Z](https://doi.org/10.1016/0168-0072(90)90035-Z).
6. Kotlarski, H., & Ratajczyk, Z. (1990). More on induction in the language with a satisfaction class. *Mathematical Logic Quarterly*, 36(5), 441–454. ISSN: 09425616. <https://doi.org/10.1002/malq.19900360509>.
7. Kotlarski, H. (1996). An addition to Rosser's theorem. *Journal of Symbolic Logic*, 61(1), 285–292. <https://projecteuclid.org:443/euclid.jsl/1183744940>.
8. Kotlarski, H., Krajewski, S., & Lachlan, A. H. (1981). Construction of satisfaction classes for nonstandard models. *Canadian Mathematical Bulletin*, 24(3), 283–293. ISSN: 0008-4395. <https://doi.org/10.4153/CMB-1981-045-3>.
9. Enayat, A. & Visser, A. (2015). *New constructions of satisfaction classes*. In T. Achourioti et al (Ed.). *Unifying the Philosophy of Truth*. Springer.
10. Leigh, G. E. (2015). Conservativity for theories of compositional truth via cut elimination. *Journal of Symbolic Logic*, 80(3), 845–865. <https://doi.org/10.1017/jsl.2015.27>.
11. Cieśliński, C., Łelyk, M., & Wcisło, B. (2017). Models of PT⁻ with internal induction for total formulae. *The Review of Symbolic Logic*, 10(1), 187–202. issn: 1755-0203. <https://doi.org/10.1017/S1755020316000356>.
12. Łelyk, M., & Wcisło, B. (2017). Models of weak theories of truth. *Archive for Mathematical Logic*, 56(5), 453–474. issn: 1432-0665. <https://doi.org/10.1007/s00153-017-0531-1>.
13. Cieśliński, C. (2017). *The epistemic lightness of truth*. Cambridge University Press. ISBN: 9781107197657. <https://doi.org/10.1017/9781108178600>.
14. Ratajczyk, Z. (1988). A combinatorial analysis of functions provably recursive in $\text{I}\Sigma_n$. *Fundamenta Mathematicae*, 130(3), 191–213. ISSN: 0016-2736. <https://doi.org/10.4064/fm-130-3-191-213>.
15. Kołodziejczyk, L. A., & Yokoyama K. (2019). *Some upper bounds on ordinal-valued Ramsey numbers for colourings of pairs*. preprint. <https://arxiv.org/abs/1807.00616>.

A Letter From Henryk Kotlarski

Let me write more about the book I have in mind.

1. I do not want to write a book on models of arithmetic. In my opinion, the subject of models of strong variants of arithmetic did not change enough to write a new book about it. My book will simply be very similar to the one written by Richard Kaye.
2. I want a book on selected topics on the border of combinatorics (partition properties), proof theory and models of arithmetic. The main part is supposed to be devoted to a construction due to Zygmunt Ratajczyk and Richard Sommer. Thus, I want to show how some purely combinatorial work allows one to give some local variant of a satisfaction class, so that, for example, consistency of PA may be derived with no cut elimination. I think simply that this is an alternative to cut elimination (despite the fact that this sort of idea is not developed enough to obtain results from proof theory with their huge recursive ordinals and weak variants of second-order arithmetic; the proof theorists have had more time to develop their ideas). I want to give an exposition only below ε_0 . I know of only three papers involving this sort of idea in which higher ordinals are considered. These are:
 - (i) A paper by Ratajczyk and me in *Zeitschr. Math. Log.* 1990,
 - (ii) Ratajczyk's paper in *APAL* 1994, and
 - (iii) The paper by Avigad and Sommer in *Bull. Symb. Log.* 1995.

But this seems not to be developed enough to be put to the book, it is too early. On the other hand, I would like to include the application of this idea to the construction of full inductive satisfaction classes.

3. As pointed out above, the main idea is to give an exposition of getting proof-theoretic results in model-theoretic fashion. This is also the idea of these proofs of the incompleteness theorems. In these proofs, one thinks what happens inside a model (despite the fact that the essence of my proof is that if the second incompleteness theorem failed then some function would grow quicker than it is

allowed to. But look at Kikuchi's argument). Of course, the Ratajczyk–Sommer construction also has this property: one looks at what happens inside a model.

The contents of the book are as follows.

1. Ordinals up to ε_0 . The main goal is to work out ordinals, the Hardy hierarchy and the main tool for the model–theoretic part, the so-called approximation lemma.
2. The incompleteness theorems in all directions. I personally understand the whole phenomenon much better after thinking in different terms than those of the diagonal lemma. Simply this gives additional insight into the essential reasons for the incompleteness theorems to hold. And in my opinion, this is essential.
3. Transfinite induction. Here I included the main part of Ratajczyk–Sommer. Thus, I show a logical indicator and how to obtain essentially the same indicator by an iterated use of the approximation lemma. Then I obtain results like Con_{PA} and reflection and the like using this idea.
4. Satisfaction classes. Here I include the main application of the Ratajczyk–Sommer idea to construct full inductive satisfaction classes. Of course, I prefer to write much more about satisfaction classes, not only the particular theorem. By the way, the characterization of theories $T \supseteq \text{PA}$ all of whose countable recursively saturated models admit a full inductive satisfaction class involves consistency of some version of ω -logic iterated to the transfinite (I have in mind the result of the paper by Ratajczyk and me in APAL 1990). A characterization in terms of transfinite induction provable in T is also known, but I only want to mention it. Indeed, it involves ordinals above ε_0 (the paper by Ratajczyk and me in Zeitschr. Math. Log. 1990 mentioned above).

Preface

This book is devoted to several themes on the border of combinatorics, proof theory, and model theory for Peano arithmetic. In a sense, it is a companion volume to two other books on foundations of arithmetic which appeared in the 90s [1, 2].

Let me tell a few words about the history of the subject. Gödel's celebrated incompleteness theorems were stated for Russell's *Principia Mathematica*. But it was soon noticed that they hold for first-order arithmetic. T. Skolem constructed nonstandard models of arithmetic by means of some restricted ultrapower construction, but his result followed from Gödel's completeness and incompleteness theorems. Gödel also introduced the class of primitive recursive functions; the class of general recursive functions was introduced several years later. Since then, recursion theory was being developed. In order to overcome Gödel's incompleteness theorem, Gentzen worked out his sequent calculus and cut-elimination procedure, thus obtaining a proof of consistency of arithmetic. He also isolated the principle beyond PA he had to use: this was transfinite induction up to ε_0 . Since then, proof theory was being developed. Model theory for arithmetic was developed very slowly, since the 30s up to the 70s the main discoveries were: non-finite axiomatizability (due to C. Ryll–Nardzewski), the standard systems (due to D. Scott), nonrecursiveness of nonstandard models (S. Tennenbaum), and the existence of end elementary extensions (MacDowell, Specker).

The breakthrough happened in the 70s. H. Friedman rediscovered Scott's result on the standard system. He also used some purely logical means to prove that every (countable nonstandard) model of arithmetic has nonstandard initial segments which are still models of this theory. Simultaneously, several people worked on applications of combinatorics (Ramsey theorem) to models of arithmetic. This sort of work was begun already by Ehrenfeucht and Mostowski in model theory, developed in set theory (Silver and Solovay showed that under the existence of some large cardinal the structure of Gödel's class of constructible sets trivializes), and then H. Gaifman developed his machinery of definable and minimal types in arithmetic. (By the way, he also gave an important corollary to a nontrivial result, due to Y. Matiasievitch.)

J. Paris and L. Kirby were working on initial segments in models of arithmetic. They applied many combinatorial and logical means. Paris introduced the notion of an indicator and found first independence results. L. Harrington gave a very elegant variant of finite Ramsey theorem which is independent of arithmetic. Paris and Harrington realized that the essence of their result is that some recursive function grows so quickly that its totality cannot be proved in arithmetic. J. Ketonen and R. Solovay worked out several hierarchies of rapidly growing functions; their work was preceded by some work by S. Wainer. Ketonen and Solovay gave very sharp information about the rate of growth of functions considered by Paris and Harrington and its variants. Later it was realized that hierarchies considered by Wainer and Ketonen and Solovay allow one to obtain several results from proof theory by model-theoretic means. Paris' 1980 paper seems to precede this idea, developed later independently and simultaneously by Z. Ratajczyk and R. Sommer (their ideas are related to a 1987 paper by P. Hájek and J. Paris), in Ratajczyk's exposition the so-called Pudlák's principle plays essential role.

Later, in the 80s and 90s, many people worked on models of weak systems of arithmetic, a subject which seems to be initiated by A. Wilkie. It is important because of its interconnections with complexity theory. The only topic in models of strong versions of arithmetic studied in the 90s seems to be that of automorphisms of countable recursively saturated models of arithmetic (R. Kaye, R. Kossak, the present author and J. Schmerl). The main results were obtained by D. Lascar (the so-called small index property of the automorphism group, provided that the model is arithmetically saturated), R. Kossak and J. Schmerl (encoding the standard system of the model by means of its automorphism group, also provided the model is arithmetically saturated), and by R. Kaye (description of closed normal subgroups in $\text{Aut}(\mathcal{M})$, provided $\mathcal{M}$ is recursively saturated).

Also in the 90s, the following problem was attacked. K. Gödel in the process of his proof of the incompleteness theorem worked out the so-called arithmetization of the metasystem. Then he constructed a sentence asserting its own unprovability (he mimicked the well-known liar paradox "the sentence I am just expressing is false", which he changed to "the sentence I am just expressing admits no proof"). Later R. Carnap distilled from Gödel's argument the so-called diagonal lemma. This part of Gödel's proof, being very intuitive in the natural language, is highly unnatural in formal languages like that of arithmetic. The question was how to eliminate this from proofs of Gödel's results. This was done in the 90s by three persons independently (T. Jech, M. Kikuchi, and the present author). Later two other arguments (C. Cieśliński and Z. Adamowicz) appeared.

Warsaw, Poland

Henryk Kotlarski

References

1. Hájek, P., & Pudlák, P. (1993). *Metamathematics of first-order arithmetic. Perspectives in mathematical logic*. Springer.
2. Kaye, R. (1991). *Models of Peano arithmetic*. Oxford University Press. ISBN: 019853213X.

Contents

1	Some Combinatorics	1
1.1	Infinite Ramsey Theorem	1
1.2	Finite Ramsey Theorem	4
1.3	Some Lower Bounds: Classic	5
1.4	Ordinals Below ε_0	7
1.5	Hardy Hierarchy	14
1.6	Approximating Functions	18
1.7	Hardy Largeness and Partitioning Elements	20
1.8	Iterations of Hardy Functions	26
1.9	An Upper Bound	27
1.10	Some Lower Bounds: Hardy	32
	References	36
2	Some Model Theory	39
2.1	Unions of Chains	39
2.2	The Recursive Saturation and Resplendency	40
2.3	The Theorem of Chang and Makkai	41
	References	41
3	Incompleteness	43
3.1	The Arithmetized Completeness Theorem	45
3.2	The Original Approach	48
3.3	Formalized Incompleteness Theorems	52
3.4	Satisfaction Classes	53
3.5	Tarski's Theorem	56
3.6	Scott and Kreisel's Proofs	56
3.7	Jech's Argument	58
3.8	Nonstandard Models and Incompleteness	59
3.9	Incompleteness Theorems via Berry's Paradox	62
3.10	Incompleteness via Grelling's Antinomy	65

3.11	Σ_1 -Closed Models	66
3.12	Some Extensions of PA	68
	References	70
4	Transfinite Induction	73
4.1	Indicators	73
4.2	Transfinite Induction in PA	74
4.3	Totality of Functions in Hardy Hierarchy	77
4.4	Hardy Largeness and Indicators	80
	References	87
5	Satisfaction Classes	89
5.1	Satisfaction Classes: Generalities	89
5.2	Noninductive Satisfaction Classes	91
5.3	Inductive Full Satisfaction Classes	104
	References	105
	Index	107

Chapter 1

Some Combinatorics



In this chapter we give some material about partitions. In Sects. 1.1 and 1.2 we prove classical Ramsey-type results. In later sections we introduce ordinals up to ε_0 and we work out the so-called Hardy hierarchy of functions. This hierarchy determines a notion of a finite set of natural numbers being α -large, $\alpha < \varepsilon_0$. We prove some partition properties for this notion of largeness. We point out that this type of results is just a technical refinement of results worked out by Ketonen and Solovay [1].

1.1 Infinite Ramsey Theorem

In this and next section we prove some Ramsey-type results. Our coverage of the material is far from complete, we have to refer the reader to special literature for more information in this direction, see, e.g. [2].

Let A be a set. By $P_m(A)$ we denote the set of all m -element subsets of A . If A is linearly ordered then we may identify $P_m(A)$ with the set of all increasing m -tuples of elements of A . It is customary to denote the set of increasing m -tuples by $[A]^m$. Observe that for $m = 1$ we may identify $P_m(A)$ and $[A]^m$ directly with A .

By a *partition* (or *coloring*) of $P_m(A)$ (or, equivalently, $[A]^m$) we mean a function $P : [A]^m \rightarrow I$. If I is finite, we may identify it with $I = \{0, \dots, r-1\}$, in this situation we say that P is a partition into r parts or into r colors.

Let $P : [A]^m \rightarrow I$ be a partition. Putting

$$A_i = \{(a_0, \dots, a_{m-1}) \in [A]^m : P(a_0, \dots, a_{m-1}) = i\}$$

we see that we may identify the partition P with a family of disjoint subsets $A_i : i \in I$ of $[A]^m$ whose union is $[A]^m$. Usually one requires also that each A_i is nonempty. (We do not need this, so we do not put it to the definition.)

Let $P : [A]^m \rightarrow I$ be a partition. Let $B \subseteq A$. We say that B is *homogeneous* for P (or *monochromatic*) if $[B]^m \subseteq A_i$ for some $i \in I$. We stress that we partition $[A]^m$, but the homogeneous set is supposed to be a subset of A (not of $[A]^m$).

It is customary to use the *arrow notation*: we write $\alpha \rightarrow (\beta)_\gamma^m$ (where α, β, γ are finite or infinite cardinals) as a shorthand for “whenever we have a partition of m -tuples from a set of size α into γ parts then there exists a homogeneous set of size β ”. The following result is known as the infinite version of Ramsey theorem, its proof will be a prototype for several arguments in later parts of the book.

Theorem 1.1.1 *If P is a partition of m -tuples from an infinite set into finitely many parts then there exists an infinite homogeneous set. In other words,*

$$\forall m, k \in \mathbb{N} \quad \aleph_0 \rightarrow (\aleph_0)_k^m.$$

In order to prove Theorem 1.1.1 we shall need several notions concerning trees. A partial ordering $\langle B, < \rangle$ is called a *tree* (in the set-theoretic sense, later we shall work with trees in a slightly another sense of the word) if there exists a smallest element in B and for every $b \in B$ the set $\{u \in B : u < b\}$ is well ordered. The smallest element of B is called the *root* of the tree $\langle B, < \rangle$. Suppose we are given a tree $\langle B, < \rangle$ and let $b \in B$. We say that $u \in B$ is a *predecessor* of b (or b is a *successor* of u) if $u < b$. We say that b is an *immediate successor* of u if it is a successor of u and there is nothing strictly between them, i.e., $\neg \exists c \ u < c < b$. We define *levels* by induction. We let U_0 be the set whose only element is the root of our tree. If U_n was defined then we let U_{n+1} be the set of all immediate successors of all elements from U_n . In set theory one considers levels defined by transfinite induction, then the above is the successor step of the definition; the limit step is as follows: U_λ is the set of all minimal elements of the set $B \setminus \bigcup_{\alpha < \lambda} U_\alpha$. But we shall develop a theory of ordinals below ε_0 only in Sect. 1.4, we will not need further levels in the proofs of the results of this section.

Let $\langle B, < \rangle$ be a tree. A set $G \subseteq B$ is a *branch* if G is linearly ordered by $<$ and maximal with respect to this property. The following fact is known as König’s infinity lemma.

Lemma 1.1.2 *Let $\langle B, < \rangle$ be an infinite tree such that every $b \in B$ has only finitely many immediate successors. Then there exists an infinite branch in the tree B .*

Proof By induction on n we construct an increasing sequence $b_n \in U_n$ of elements of B such that for every $n \in \mathbb{N}$ the set $\{z \in B : z > b_n\}$ of all, not necessarily immediate, successors of b_n is infinite. Obviously, we may take the root of our tree as b_0 . If b_n is constructed, we consider the set of all its successors and partition it in the following way. Let C be the set of all immediate successors of b_n . We write $\{z \in B : z > b_n\} = \bigcup_{c \in C} \{z \in B : z \geq c\}$. This is a partition of an infinite set into finitely many parts, so at least one part is infinite. Let $b_{n+1} = c$ for any $c \in C$ which has infinitely many successors. The sequence $b_n : n \in \mathbb{N}$ is thus linearly ordered. It need not be maximal, but by Kuratowski–Zorn Lemma it has a maximal linearly ordered superset. $\square$

Proof of Theorem 1.1.1 We proceed by induction on m . For $m = 1$ the result states that if an infinite set was partitioned into finitely many parts then at least one part is infinite, hence it is obvious. So assume the conclusion for m , we derive it for $m + 1$.

Observe that it suffices to obtain the conclusion in the case of a partition into two parts, indeed, granted this one can apply an obvious induction on the number of parts. So let $[\mathbb{N}]^{m+1} = A \cup B$ be a partition of $\mathbb{N}$ into two parts. We associate with this partition a relation $<$ so that $(\mathbb{N}, <)$ will become a tree. In order to do this we define a sequence $<_r$ of relations on $\{0, \dots, r\}$ by induction on r . We let $<_0$ be empty. If $<_r$ is defined we let $u <_{r+1} w$ be

$$\begin{aligned} & (u, w < r + 1 \ \& \ u <_r w) \vee \\ & \{u < r + 1 \ \& \ w = r + 1 \ \& \\ & \forall z_0 <_r \dots <_r z_{m-1} < u \ [\langle z_0, \dots, z_{m-1}, u \rangle \in A \equiv \langle z_0, \dots, z_{m-1}, w \rangle \in A] \}. \end{aligned}$$

It is easy to check that each $<_r$ is a tree on $\{0, \dots, r\}$ and the union of all these relations is a tree on $\mathbb{N}$.

We assert that $(\mathbb{N}, <)$ is finitely branching. Let a be at the k th level of this tree. We associate with every $b \succ a$ the set

$$\{ \langle z_0, \dots, z_{m-2} \rangle : z_0 < \dots < z_{m-2} < a \ \& \ \langle z_0, \dots, z_{m-2}, a, b \rangle \in A \}.$$

We obtained a function which associates with every $b \succ a$ a set of sequences as above. There are $\binom{k}{m-1}$ such sequences, ($\binom{k}{m-1}$ denotes the usual binomial coefficient) indeed, $a \in U_k$, hence there are at most $2^{\binom{k}{m-1}}$ such sets of sequences. It is also easy to see that if we are given such a set of sequences (and there exists b determining this set) then the smallest such b is the only immediate successor of a determining this set. It follows that every $a \in \mathbb{N}$ has at most $2^{\binom{k}{m-1}}$ immediate successors.

By Lemma 1.1.2 there exists an infinite branch G in the tree $(\mathbb{N}, <)$. We partition $[G]^m$ as follows. Observe that for an increasing sequence $z_0, \dots, z_{m-1}, u, w$ we have $\langle z_0, \dots, z_{m-1}, u \rangle \in A$ iff $\langle z_0, \dots, z_{m-1}, w \rangle \in A$. We let $\langle z_0, \dots, z_{m-1} \rangle \in A'$ iff there exists $u \succ z_{m-1}$ such that $\langle z_0, \dots, z_{m-1}, u \rangle \in A$ and $\langle z_0, \dots, z_{m-1} \rangle \in B'$ otherwise. This gives a partition of $[G]^m$ into two parts, so by the inductive assumption there exists an infinite homogeneous set D for this partition. We check that D is homogeneous for the original partition $[\mathbb{N}]^{m+1} = A \cup B$. So let $\langle u_0, \dots, u_m \rangle$ and $\langle w_0, \dots, w_m \rangle$ be in $[D]^{m+1}$. Pick $g \in D$ greater than u_m and w_m . Then if $\langle u_0, \dots, u_{m-1}, g \rangle \in A$ then both $\langle u_0, \dots, u_{m-1}, u_m \rangle$ and $\langle w_0, \dots, w_{m-1}, w_m \rangle$ are in A ; similarly if $\langle u_0, \dots, u_{m-1}, g \rangle \in B$. $\square$

We remark that in the exposition of the proof of Ramsey theorem above we used induction on the number of parts of the partition. This was inessential, indeed, one could use the same argument for a partition into c parts, but the appropriate tree would have a slightly weaker property. If $P : [\mathbb{N}]^m \rightarrow (< c)$ and we construct a tree as above then every $a \in U_k$ has at most $c^{\binom{k}{m-2}}$ immediate successors.

1.2 Finite Ramsey Theorem

In this section we prove the finite case of Ramsey theorem. The proof we give is just a variation of the argument from Sect. 1.1. Two other proofs are given in Graham, Rotschild, Spencer [2].

Theorem 1.2.1 *For every $b, c, m \in \mathbb{N}$ there exists $d \in \mathbb{N}$ such that $d \rightarrow (b)_c^m$.*

We use the arrow notation as introduced in Sect. 1.1.

Proof of Theorem 1.2.1 Exactly as in the proof of Theorem 1.1.1 we proceed by induction on m . If $m = 1$ then $d = c \cdot (b - 1) + 1$ satisfies our demand. Indeed, if $\text{Card}(X) = d$ and $X = \cup_{i < c} X_i$ is a partition of X into c parts and each X_i has at most $b - 1$ elements, then the union would have at most $c \cdot (b - 1) < d$ elements.

Let a set $X \subseteq \mathbb{N}$ and a partition $P : [X]^{m+1} \rightarrow (< c)$ be given. We construct a tree $<$ on X as in the proof of Theorem 1.1.1. Exactly as in that proof, every element w in the k th level has at most $c^{\binom{k}{m-1}}$ immediate successors. But $\binom{k}{m-1} \leq 2^k$ as is seen by substituting $x = y = 1$ in the expansion $(x + y)^k = \sum_{i=0}^k \binom{k}{i} x^i \cdot y^{k-i}$, hence every element of U_k has at most c^{2^k} immediate successors. By an easy induction we see that $\text{Card}(U_k) \leq c^{2^{k+1}-1}$.

Fix k . Then the union $U_0 \cup \dots \cup U_k$ has cardinality less than

$$c^{2^{2 \cdot 0 + 1} - 1} + c^{2^{2 \cdot 1 + 1} - 1} \dots c^{2^{2 \cdot k + 1} - 1}.$$

There are $k + 1$ summands, each smaller or equal $c^{2^{2 \cdot k + 1} - 1}$, hence the union is of size at most $(k + 1) \cdot c^{2^{2 \cdot k + 1} - 1}$. It follows that

if X has cardinality $\geq c^{2^{2 \cdot k + 1}}$ then
there exists a sequence $b_0 < \dots < b_k$ of length $k + 1$ in the tree $\langle X, < \rangle$.

Let G be a branch in the tree $\langle X, < \rangle$. Let $G' = G \setminus \{\max G\}$. Exactly as in the proof of Theorem 1.1.1 we consider the partition P' of $[G']^m$ into c parts by putting

$$P'(z_0, \dots, z_{m-1}) = i \quad \text{iff} \quad P(z_0, \dots, z_{m-1}, \max G) = i.$$

Now let k be large enough so that the partition P' has a homogeneous set Y of cardinality b . Then Y is homogeneous for the original partition P . $\square$

We remark that the estimate of d in terms of m, b, c which may be distilled from this proof are not optimal, see [2] for more information.

1.3 Some Lower Bounds: Classic

In this section we give some lower bounds for the finite Ramsey theorem. The bounds we give are very weak, but in Sect. 1.10 we shall modify the proofs given below. For far stronger results concerning the finite Ramsey theorem, see [2], in particular Sect. 4.2 in [2] for the nonconstructive method of proving the existence of partitions of pairs admitting no large homogeneous sets (by the way, one needs the so-called Stirling formula to make the proof in [2] complete) and Sect. 4.7 in [2] for the *stepping up lemma* which allows one to obtain lower bound for partitioning k -tuples, $k > 2$.

The starting point of our considerations is the following simple construction. Let A be any set of cardinality at most m^n . We may identify A with a subset of $\{0, \dots, m-1\}^{\{0, \dots, n-1\}}$. Thus, we work with the set of sequences of length n of numbers smaller than m . We regard A as a subset of the family of sequences as above. For $a, b \in A$ we put

$$P(a, b) = \min j : a_j \neq b_j \quad (1.1)$$

and treat P as a partition of $[A]^2$.

Lemma 1.3.1 *If $D \subseteq A$ is homogeneous for P then $\text{Card}(D) \leq m$.*

Proof Let D be monochromatic under P . Hence there exists $j < m$ such that $\forall(a, b) \in [D]^2, a_j \neq b_j$. It follows that D has at most m elements, indeed, there are at most m possible values for a_j or b_j 's. $\square$

If we partition triples then the number of parts may be reduced to 3 for a little price. We let

$$K(a, b, c) = \begin{cases} 0 & \text{if } P(a, b) > P(b, c) \\ 1 & \text{if } P(a, b) = P(b, c) \\ 2 & \text{if } P(a, b) < P(b, c). \end{cases} \quad (1.2)$$

We shall write P^A and K^A if needed.

Lemma 1.3.2 *Under the notations and assumption as above, if D is homogeneous for K then $\text{Card}(D) \leq \max(m, n+1)$.*

Proof If the color of $[D]^3$ is 1 then we apply the previous lemma. In other cases we let $D = \{d_0, \dots, d_{r-1}\}$ and put $j_i = P(d_i, d_{i+1})$. It follows that the sequence $j_i : i < r-2$ is strictly monotonic, hence one-to-one, hence it has at most n elements, so D has at most $n+1$ elements. $\square$

Before giving the general inductive step of the construction of partitions that we have in mind we give the step $k=4$ for getting better intuition of what will happen. Thus we shall construct a partition of $[A]^4$. The assumption on A is that its cardinality is at most m^{n^n} . Thus, we identify A with the set of sequences of length smaller than n^n of numbers smaller than m .

For $a \in [A]^4$ we let $G(a) = K^A(a_0, a_1, a_2)$. We defined a partition G of $[A]^4$ into 3 parts. Observe that if $D \subseteq A$ is of cardinality strictly greater than 4 and is homogeneous for G , then K^A is constant on $D \setminus \{\max(D)\}$.

We define an additional partition R of $[A]^4$ in the following way. For $a \in [A]^4$ we let $b_i = P(a_i, a_{i+1})$. Let B be the natural set of cardinality n^n , i.e., the set of all sequences of length n of numbers $0, \dots, n-1$. Then each b_i being smaller than the length of sequences from A is an element of B . Let $a \in [A]^4$. We let

$$R(a) = \begin{cases} K^B(b_0, b_1, b_2) & \text{if } b_0 < b_1 < b_2 \\ K^B(b_2, b_1, b_0) & \text{if } b_0 > b_1 > b_2 \\ 0 & \text{in other cases.} \end{cases} \quad (1.3)$$

Finally we put

$$L(a) = \langle G(a), R(a) \rangle. \quad (1.4)$$

Obviously, L is a partition of $[A]^4$ into 3^2 parts.

Lemma 1.3.3 *Under the above assumption (i.e., $\text{Card}(A) \leq m^{n^n}$) we have: whenever $D \subseteq A$ is homogeneous for L , then $\text{Card}(D) \leq \max(m+1, n+3)$.*

Proof Let $D \subseteq A$ be homogeneous for L . Then D is homogeneous for G , hence $D' = D \setminus \{\max(D)\}$ is homogeneous for K^A . If K^A colors D' by 1, then $\text{Card}(D) \leq m+1$ by Lemma 1.3.1. If D gets a color 0 or 2 under G then the sequence $b_i = P(d_i, d_{i+1})$ is either strictly increasing or strictly decreasing. In both cases, K^B is constant on the set of b_i 's. By Lemma 1.3.2, B has at most $n+1$ elements, hence D' has at most $n+2$ elements, so D has at most $n+3$ elements. $\square$

Let us describe further steps of construction of partitions. As we shall need the function which associates $n^{\cdot n}$ with n (the tower of k n 's), let us introduce a notation for it. Let $\text{tow}_1(n) = n$, and $\text{tow}_{k+1}(n) = n^{\text{tow}_k(n)}$.

Theorem 1.3.4 *For every $k > 2$ and every m, n and every set A of cardinality at most $m^{\text{tow}_{k-2}(n)}$ there exists a partition L of $[A]^k$ into 3^k parts such that every homogeneous set with at least $k+1$ elements is of cardinality at most $\max(m+k-3, n + \frac{(k-1)(k-2)}{2})$.*

Proof We proceed by induction on $k \geq 3$, for $k=3$ the result was proved above as Lemma 1.3.2. Assume the result for k . Let A have at most $m^{\text{tow}_{k-1}(n)}$ elements. We regard it as the set of sequences of length $\text{tow}_{k-1}(n)$ of numbers up to $m-1$. Let the partition P^A be given by (1.1) and let K^A be defined by (1.2). Let $B = \{0, \dots, \text{tow}_{k-1}(n)\}$ be the set of values of P . We define the partition G as before, i.e., for $a \in [A]^{k+1}$ such that $a = (a_0, \dots, a_k)$ we put $G(a) = K^A(a_0, a_1, a_2)$. Once again, every subset of A with more than k elements homogeneous for G is monochromatic with respect to K^A on all but its greatest $k-2$ elements. Let W be the partition of $[B]^k$ given by the inductive assumption (with $m=n$). We let $b_i = P(a_i, a_{i+1})$. We define the partition L exactly as above, but now we work with k -tuples rather than triples. That is, we put

$$R(a) = \begin{cases} W(b_0, \dots, b_{k-1}) & \text{if } b_0 < \dots < b_{k-1} \\ W(b_{k-1}, \dots, b_0) & \text{if } b_0 > \dots > b_{k-1} \\ 0 & \text{in other cases.} \end{cases} \quad (1.5)$$

We ask the reader to check that the partition L given by

$$L(a) = \langle G(a), R(a) \rangle$$

has the desired property. We merely point out that if D is homogeneous for L , then the set D' defined as D without its last $k - 2$ elements is homogeneous for G . It follows that if the value of G on $[A]^{k+1}$ is 1, then $\text{Card}(D') \leq m$, and if the appropriate value is 0 or 2, then the set $B = \{P(d_i, d_{i+1}) : i\}$ has the cardinality estimated by the cardinality of sets homogeneous for W . $\square$

1.4 Ordinals Below ε_0

In this section we give a treatment of ordinals below ε_0 . Moreover we work out some machinery, the so-called fundamental sequences, this will be needed in Sect. 1.5 to prove the main properties of Hardy hierarchy of quickly growing functions.

The idea is as in the high school, where one is taught that the set of all polynomials with integer coefficients is well ordered by the relation of domination between functions: $f < g \equiv \exists x \forall y > x \ f(y) < g(y)$. This is equivalent to the ordering obtained by comparing at first degrees of polynomials f and g , if they are equal, then one compares the coefficients at $x^{\deg(g)}$, etc. As a matter of fact we must work with a slightly more complicated family of functions. It is customary to denote the variable by ω in these considerations, thus we have in mind polynomials in variable ω .

We define two sequences Pol_n of families of functions and relations $<_n$ on Pol_n for natural n . We let Pol_0 be the family of all constant functions, where constants are non negative integers, and we let $<_0$ be the ordering of constant functions (the order between values). We let Pol_{n+1} be the set of all functions of the form

$$\omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_{r-1}} \cdot m_{r-1} + \beta,$$

where $\alpha_0, \dots, \alpha_{r-1}$ are elements of $\text{Pol}_n \setminus \text{Pol}_{n-1}$, $\beta \in \text{Pol}_n$, $\alpha_0 >_n \dots >_n \alpha_{r-1}$ and $m_0 \neq 0, \dots, m_{r-1} \neq 0$. We define the relation $<_{n+1}$ in the natural manner, that is $\alpha >_{n+1} \gamma$ if whenever α is written as above and

$$\gamma = \omega^{\gamma_0} \cdot b_0 + \dots + \omega^{\gamma_t} \cdot b_t + \delta$$

then either there exists $i \leq r$ such that

$$(\forall k < i \ \alpha_k = \gamma_k \ \& \ a_k = b_k) \ \& \ [\alpha_i >_n \gamma_i \vee (\alpha_i = \gamma_i \ \& \ a_i > b_i)]$$

$$\text{or } (r = t \ \& \ \forall k \leq r \ \alpha_i = \beta_i \ \& \ b_i = a_i) \ \& \ \beta \succ_n \delta.$$

Elements of Pol_n are called *polynomials of order n* . We shall treat elements of $\bigcup_{n \in \mathbb{N}} \text{Pol}_n$ as ordinals, or, to be more specific, notations for ordinals. It is immediate to check that the set of ordinals (in the above sense) is well ordered by $\bigcup_{n \in \mathbb{N}} \prec_n$, its order type is denoted in the literature as ε_0 . (The reader knowing some theory of well orderings will easily check that ε_0 is the least ordinal solution to the equation $\alpha = \omega^\alpha$.)

Of course, 0 is an ordinal (as a trivial polynomial). Every $0 < \alpha < \varepsilon_0$ may be written uniquely as

$$\alpha = \omega^{\alpha_0} \cdot a_0 + \cdots + \omega^{\alpha_s} \cdot a_s \quad (1.6)$$

for some $\alpha_0 > \alpha_1 > \cdots > \alpha_s$ with $\alpha > \alpha_0$ and $a_0, \dots, a_s \in \mathbb{N} \setminus \{0\}$. It is customary to refer to a_i 's as to *coefficient*, and to α_i 's as to *exponents* in the Cantor normal form of α . We shall refer to (1.6) as to the *Cantor normal form* of an ordinal below ε_0 . (In more advanced theories of ordinals one works with ordinals greater than ε_0 , then one has a similar Cantor normal form expansion, but one cannot require $\alpha > \alpha_0$.)

Observe that if α is written in its Cantor normal form (1.6) then all exponents may be written in their Cantor normal forms, etc. This process (when iterated sufficiently many times so that we only natural numbers occur in the expansion) yields *full* Cantor normal form of α .

The relation $<$ between ordinals written in the form (1.6) is given just by comparing exponents and coefficients. From now on we write the usual inequality sign $<$ rather than $\prec$ to denote inequality between ordinals below ε_0 .

If $0 < \alpha < \varepsilon_0$ we denote by $\text{LM}(\alpha)$ the *leftmost exponent* in its Cantor normal form, i.e., α_0 in (1.6). Similarly, by $\text{RM}(\alpha)$ we denote the *rightmost exponent*, that is α_s in (1.6).

We say that the Cantor normal form (1.6) is *trivial* if $s = 0$ and $a_0 = 1$ (i.e., α is of the form ω^δ) and *nontrivial* otherwise.

We shall need several notions concerning ordinals below ε_0 . The first two ones are those of a limit ordinal and successor. An ordinal $\alpha < \varepsilon_0$ is *successor* iff it is of the form $\beta + 1$ for some β . Observe that α is a successor iff in its normal form expansion (1.6), $\alpha_s = 0$. An ordinal α is *limit* iff it is not 0 and $\alpha_s > 0$. Observe that an ordinal is a successor (of β) iff $\alpha > \beta$ and there is no ordinal γ with $\beta < \gamma < \alpha$. Also, α is limit iff for every $\beta < \alpha$ there exists γ such that $\beta < \gamma < \alpha$.

We used above the addition $\beta + 1$. This is a general notion of *addition* of ordinals. It is defined by transfinite induction. We let $\alpha + 0 = \alpha$, $\alpha + 1$ is the successor of α (that is, if α is written in its normal form (1.6)), then the Cantor normal form of $\alpha + 1$ is the expansion of α followed by $1 = \omega^0$ (or if $\alpha_s = 0$ then the Cantor normal form of $\alpha + 1$ is the same as that of α but with $a_s + 1$ in place of a_s). If λ is limit then $\alpha + \lambda$ is the smallest ordinal greater than all $\alpha + \beta$ where $\beta < \lambda$. Observe that addition of ordinals is not commutative (e.g. $1 + \omega = \omega \neq \omega + 1$), but it is associative. Moreover if we are given two ordinals α, β then the Cantor normal form of $\alpha + \beta$ need not be the expression obtained from the expansion of α by writing the expansion of β as

following the expansion of α . In order to ensure that it is so we introduce the next notion. It is as follows. We write $\alpha \gg \beta$ if $\alpha = 0$ or $\beta = 0$ or all exponents in the Cantor normal form of α are $\geq$ all the exponents in the Cantor normal form of β . We remark that $\alpha \gg \beta$ does not imply $\alpha \geq \beta$, indeed, $\omega^\alpha \cdot m \gg \omega^\alpha \cdot k$ for all m, k .

In later parts of the book we shall also use the relation $\alpha \ggg \beta$, which is defined like $\alpha \gg \beta$, but with strict inequality.

One defines also *multiplication* of ordinals by induction. We let $\alpha \cdot 0 = 0$, $\alpha \cdot (\beta + 1) = \alpha \cdot \beta + \alpha$ and for limit λ , $\alpha \cdot \lambda$ = the smallest ordinal exceeding all $\alpha \cdot \beta$ for $\beta < \lambda$. We point out that addition and multiplication of ordinals are associative but not commutative.

One more remark should be added here. Every $0 < \alpha < \varepsilon_0$ may be written in the form

$$\alpha = \beta + \omega^\rho, \quad \text{where } \beta \gg \omega^\rho. \quad (1.7)$$

This form is obtained from (1.6) by putting

$$\beta = \omega^{\alpha_0} \cdot a_0 + \cdots + \omega^{\alpha_{s-1}} \cdot a_{s-1} + \omega^{\alpha_s} \cdot (a_s - 1) \quad \text{and} \quad \rho = \alpha_s.$$

We shall refer to (1.7) as to *short Cantor normal form* of α . Observe that the Cantor normal form of α is trivial iff $\beta = 0$ in (1.7), moreover if $\beta \neq 0$ then $\beta < \alpha$ and $\omega^\rho < \alpha$.

We remark also that if $\alpha = \beta + \omega^\rho$ and $\gamma = \delta + \omega^\xi$ are in their short Cantor normal forms, then $\gamma < \alpha$ exactly in the following cases:

1. $\beta > \delta \vee (\beta = \delta \ \& \ \rho > \xi)$
2. $\beta = 0 \ \& \ \delta \neq 0 \ \& \ \rho > \text{LM}(\gamma) = \text{LM}(\delta)$
3. $\beta \neq 0 \ \& \ \delta = 0 \ \& \ \text{LM}(\beta) > \xi$.

The next notion we need is that of a *fundamental sequence*. For every limit $\lambda < \varepsilon_0$ we choose a sequence $\{\lambda\}(n)$ converging to λ from below. We let $\{\omega\}(m) = m$, more generally $\{\omega^{\alpha+1}\}(m) = \omega^\alpha \cdot m$. For α limit we let $\{\omega^\alpha\}(m) = \omega^{\{\alpha\}(m)}$. Finally, if $\alpha = \varrho + \omega^\mu$ with $\varrho \gg \omega^\mu$ then we let $\{\alpha\}(m) = \varrho + \{\omega^\mu\}(m)$. It is easy to see that these conditions determine exactly one sequence $\{\lambda\}(m)$ for each limit $\lambda < \varepsilon_0$, the *fundamental sequence* for λ . We remark that Ketonen and Solovay [1] use slightly different fundamental sequences.

We shall need several other notions. We extend the notion of a fundamental sequence to non limit ordinals by putting $\{0\}(n) = 0$ and $\{\alpha + 1\}(n) = \alpha$.

Observe that $\{\alpha\}(b) < \alpha$ and whenever $\gamma \gg \alpha$ then $\gamma \gg \{\alpha\}(b)$ for all b . We shall use these observations without explicit mention.

For $\beta, \alpha < \varepsilon_0$ we write $\beta \rightarrow_n \alpha$ iff there exists a finite sequence $\alpha_0, \dots, \alpha_k$ of ordinals such that $\alpha_0 = \beta$, $\alpha_k = \alpha$ and for every $m < k$ there exists $j_m \leq n$ such that $\alpha_{m+1} = \{\alpha_m\}(j_m)$. We write $\beta \Rightarrow_n \alpha$ if there exists a sequence as above, but with each $j_m = n$. Observe that both relations $\rightarrow_n, \Rightarrow_n$ are transitive and imply $\beta \geq \alpha$.

In the following lemma we collect the main properties of the relation $\Rightarrow_b$. The lemma is known from Ketonen Solovay [1], they write that part 6 of it is related to some earlier work of Diana Schmidt.

Lemma 1.4.1 ([1])

1. For every α , b $\alpha \Rightarrow_b 0$.
2. If $\beta \gg \alpha$ and $\alpha \Rightarrow_b \gamma$ then $\beta + \alpha \Rightarrow_b \beta + \gamma$.
3. If $k < l$ and $b \geq 0$ then $\omega^\alpha \cdot l \Rightarrow_b \omega^\alpha \cdot k$.
4. If $\beta \Rightarrow_b \alpha$ and $b > 0$ then $\omega^\beta \Rightarrow_b \omega^\alpha$.
5. $\alpha \Rightarrow_b \{\alpha\}(j)$ and $\{\alpha\}(b) \Rightarrow_b \{\alpha\}(j)$ for $j \leq b$.
6. $\{\alpha\}(b) \Rightarrow_1 \{\alpha\}(j)$ for $0 < j \leq b$.
7. If $n \leq b$ and $\alpha \Rightarrow_n \beta$ then $\alpha \Rightarrow_b \beta$.
8. $\beta \Rightarrow_n \alpha$ iff $\beta \rightarrow_n \alpha$.
9. If $\alpha < \beta$ then there exists b such that $\beta \Rightarrow_b \alpha$.

Proof Part 1 is immediate by induction on α . Indeed, for $\alpha = 0$ there is nothing to prove, if $\alpha = \beta + 1$ and $\beta_0, \dots, \beta_{r-1}$ is a sequence witnessing the relation $\beta \Rightarrow_b 0$ then the sequence $\alpha, \beta_0, \dots, \beta_{r-1}$ witnesses $\alpha + 1 \Rightarrow_b 0$. If α is limit then $\alpha, \{\alpha\}(b)$, the sequence witnessing $\{\alpha\}(b) \Rightarrow_b 0$ does the job.

For part 2 let $\alpha_0, \alpha_1, \dots, \alpha_{r-1}$ be the sequence witnessing $\alpha \Rightarrow_b \gamma$. Then it is easy to see that the sequence $\beta + \alpha_0, \beta + \alpha_1, \dots, \beta + \alpha_{r-1}$ witnesses $\beta + \alpha \Rightarrow_b \beta + \gamma$.

For part 3 let α be fixed. Then $\{\omega^\alpha \cdot l\}(b) = \omega^\alpha \cdot (l - 1) + \{\omega^\alpha\}(b)$. By part 1 there exists a sequence $\gamma_0, \dots, \gamma_{r-1}$ witnessing the relation $\{\omega^\alpha\}(b) \Rightarrow_b 0$. By part 2, the sequence $\omega^\alpha \cdot (l - 1) + \gamma_0, \dots, \omega^\alpha \cdot (l - 1) + \gamma_{r-1}$ witnesses the relation $\omega^\alpha \cdot l \Rightarrow_b \omega^\alpha \cdot (l - 1)$. Thus, if $k = l - 1$, we are done. Otherwise we continue the same procedure, that is we go down to $\omega^\alpha \cdot (l - 2)$ etc.

Let us go to the next part, i.e., to 4. Let $\beta_0, \dots, \beta_{r-1}$ be the sequence witnessing the relation $\beta \Rightarrow_b \alpha$. Consider the sequence $\omega^{\beta_0}, \dots, \omega^{\beta_{r-1}}$. We shall put between items of this sequence many more items to obtain the right one. Consider two consecutive elements of this sequence. These elements look like $\omega^\gamma, \omega^{\{\gamma\}(b)}$ for some γ . If γ is limit then we do not put anything between these two items. Otherwise $\gamma = \delta + 1$ for some δ . Then $\{\omega^{\delta+1}\}(b) = \omega^\delta \cdot b$. We put between the two items all the sequence witnessing the relation $\omega^\delta \cdot b \Rightarrow_b \omega^\delta$, its existence follows from parts 1 and 3. Clearly, the sequence obtained in this manner has the required properties.

Let us go to the main part of the lemma, i.e., to 5. Obviously, it suffices to prove the second claim. We proceed by induction on α . If $\alpha = 0$ there is nothing to prove (one element sequence works). Also the non limit step is trivial. So assume that α is limit and the claim holds for all $\beta < \alpha$. If α admits a nontrivial Cantor normal form then we may write $\alpha = \delta + \beta$, where $\beta < \alpha$ and $\delta \gg \beta$. By the inductive assumption there exists a sequence $\beta_0, \dots, \beta_{r-1}$ witnessing the relation $\{\beta\}(b) \Rightarrow_b \{\beta\}(j)$. Then the sequence $\delta + \beta_0, \dots, \delta + \beta_{r-1}$ witnesses $\{\alpha\}(b) \Rightarrow_b \{\alpha\}(j)$ by part 2. So let us go to the case when the Cantor normal form of α is trivial, $\alpha = \omega^\delta$. If δ is non limit, $\delta = \rho + 1$ we must show $\{\omega^{\rho+1}\}(b) = \omega^\rho \cdot b \Rightarrow_b \{\omega^{\rho+1}\}(j) = \omega^\rho \cdot j$. This is a direct consequence of part 3. Finally, let δ be limit. Then by the inductive assumption $\{\delta\}(b) \Rightarrow_b \{\delta\}(j)$ and we may apply part 4.

The proof of part 6 is very similar to that of part 5, so is left to the reader.

Let us go to the next item of the lemma, i.e., to part 7. Let $T(\alpha)$ denote the property¹

$$\forall \beta, n, b [(n \leq b \ \& \ \alpha \Rightarrow_n \beta) \Rightarrow (\alpha \Rightarrow_b \beta)]$$

and we prove $\forall \alpha < \varepsilon_0 \ T(\alpha)$ by induction on α . If $\alpha = 0$ then $\beta = 0$ and the sequence with only one item equal 0 witnesses the relation $\alpha \Rightarrow_b \beta$. Also the implication $T(\alpha) \Rightarrow T(\alpha + 1)$ is easy. Indeed, if $\alpha + 1 \Rightarrow_n \beta$ then $\beta \leq \alpha + 1$. If $\beta = \alpha + 1$ then one element sequence with the only item $\alpha + 1$ works, otherwise $\beta \leq \alpha$ and $\alpha \Rightarrow_n \beta$. We apply the inductive assumption to α . So let us concentrate at the limit step. Thus, let α be limit and assume the conclusion for all ordinals strictly smaller than α . We write both of these ordinals in their Cantor normal forms:

$$\alpha = \omega^{\alpha_0} \cdot a_0 + \cdots + \omega^{\alpha_s} \cdot a_s$$

and

$$\beta = \omega^{\beta_0} \cdot b_0 + \cdots + \omega^{\beta_t} \cdot b_t.$$

By $\alpha \Rightarrow_n \beta$ we have $\alpha \geq \beta$. If they are equal then one element sequence works, so assume that $\beta < \alpha$ and consider two cases.

CASE 1. $s > t \ \& \ \forall i \leq t [a_i = b_i \ \& \ \alpha_i = \beta_i]$, that is $\alpha = \beta + \delta$ for some $\delta \ll \beta$, $\delta \neq 0$. By part 1 $\delta \Rightarrow_b 0$, hence $\alpha = \beta + \delta \Rightarrow_b \beta$ by part 2.

CASE 2. $\exists j \leq s \{(\forall k < j \ a_k = b_k \ \& \ \alpha_k = \beta_k) \ \& \ [\alpha_j > \beta_j \vee (\alpha_j = \beta_j \ \& \ a_j > b_j)]\}$. Fix such j . Let δ be the common initial part of Cantor normal forms of α and β . Thus we may write

$$\alpha = \delta + \omega^{\alpha_j} \cdot a_j + \cdots + \omega^{\alpha_s} \cdot a_s$$

and

$$\beta = \delta + \omega^{\beta_j} \cdot b_j + \cdots + \omega^{\beta_t} \cdot b_t.$$

We may assume that $\alpha_j > \beta_j$, otherwise we shall move the part $\omega^{\alpha_j} \cdot (a_j - b_j)$ to the left (i.e., we change δ). Consider the sequence $\gamma_0, \dots, \gamma_{r-1}$ witnessing the relation $\alpha \Rightarrow_n \beta$. It is determined uniquely by conditions $\gamma_0 = \alpha$, $\gamma_{i+1} = \{\gamma_i\}(n)$ for $i < r - 1$. It is strictly decreasing and $\gamma_{r-1} = \beta$. It follows that $\gamma_i = \delta + \rho_i$ for some ρ_i . The sequence $\rho_0, \dots, \rho_{r-1}$ witnesses the relation

$$\omega^{\alpha_j} \cdot a_j + \cdots + \omega^{\alpha_s} \cdot a_s \Rightarrow_n \omega^{\beta_j} \cdot b_j + \cdots + \omega^{\beta_t} \cdot b_t.$$

It follows that since these ordinals are strictly smaller than α and β respectively (i.e., if $\delta \neq 0$), we may apply the inductive assumption and get a sequence $\xi_0, \dots, \xi_{m-1}$ witnessing the relation

¹The $\Rightarrow$ sign without index denotes implication; the relation $\Rightarrow_b$ is as above. We are taking notation from different sources, I hope that this will not lead misunderstanding.

$$\omega^{\alpha_j} \cdot a_j + \dots + \omega^{\alpha_s} \cdot a_s \Rightarrow_b \omega^{\beta_j} \cdot b_j + \dots + \omega^{\beta_t} \cdot b_t.$$

Then the sequence $\delta + \xi_0, \dots, \delta + \xi_{m-1}$ witnesses the relation $\alpha \Rightarrow_b \beta$. Thus it remains to show what to do if $\delta = 0$, i.e., $\alpha_0 > \beta_0$. We may assume that $a_0 = 1$, for the sequence $\gamma_0, \dots, \gamma_{r-1}$ witnessing the relation $\alpha \Rightarrow_n \beta$ must go through ω^{α_0} , hence $\alpha \Rightarrow_n \omega^{\alpha_0} \Rightarrow_n \beta$ and we may apply the inductive assumption in both places to change to $\Rightarrow_b$ (if $b = 0$ then we apply part 3). So assume that $\alpha = \omega^{\alpha_0}$. If $\alpha_0 = \zeta + 1$ then $\alpha \Rightarrow_b \omega^\zeta \cdot b$, so by part 3 $\alpha \Rightarrow_b \omega^\zeta \cdot n$ and we may apply the inductive assumption $T(\omega^\zeta \cdot n)$. So assume that α_0 is limit. Consider the sequence $\gamma_0, \dots, \gamma_{r-1}$ witnessing the relation $\alpha \Rightarrow_n \beta$. Thus, $\gamma_1 = \omega^{\{\alpha_0\}(n)}$. Let $\delta_0 = \omega^{\alpha_0}$, $\delta_1 = \omega^{\{\alpha_0\}(b)}$. Further part of the required sequence is the whole sequence witnessing the relation $\omega^{\{\alpha_0\}(b)} \Rightarrow_b \omega^{\{\alpha_0\}(n)}$, whose existence is a consequence of parts 5 and 4. Further part of the required sequence δ_i is given by the inductive assumption, applied to obtain the sequence witnessing the relation $\omega^{\{\alpha_0\}(n)} \Rightarrow_b \beta$.

Part 8 is now immediate by part 7. Indeed, one direction is immediate (one may take each $j_m = n$), for the converse we observe that we may increase each j_m if necessary using part 7.

For part 9 we proceed by induction on β . Thus the induction thesis is $T(\beta)$:

$$\forall \alpha < \beta \exists b \beta \Rightarrow_b \alpha,$$

and we prove $\forall \beta < \varepsilon_0 T(\beta)$. If $\beta = 0$ then the conclusion holds vacuously (there is no α to consider). Also the non limit step is immediate. Let β be limit and assume the conclusion for all ordinals smaller than β . Pick $\alpha < \beta$, then $\alpha < \{\beta\}(n)$ for some n . By $T(\{\beta\}(n))$ there exists d such that $\{\beta\}(n) \Rightarrow_d \alpha$. By part 7 $b = \max(n, d)$ has the required property. $\square$

It turns out that part 9 of Lemma 1.4.1 may be strengthened considerably, namely the b to be found may be chosen to depend only on α (see Lemma 1.4.3). But in order to obtain this strengthening we need an auxiliary notion.

Let $\alpha < \varepsilon_0$. We define *pseudonorm* of α as the greatest natural number which occurs in its (full) Cantor normal form. Technically we define the function psn sending ordinals below ε_0 into $\mathbb{N}$ by putting $\text{psn}(n) = n$ for $n < \omega$ and

$$\text{psn}(\alpha) = \max(\text{psn}(\alpha_0), \dots, \text{psn}(\alpha_s), a_0, \dots, a_s)$$

where α is written in its Cantor normal form (1.6). (We write “pseudonorm” because Ketonen and Solovay [1] use a slightly another function, the *norm* of α , for similar purpose.)

The following properties of pseudonorm are easy:

1. If $\mu \gg \nu$ then $\text{psn}(\mu + \nu) \geq \text{psn}(\mu), \text{psn}(\nu)$,
2. $\text{psn}(\omega^\rho) = \text{psn}(\rho)$

We ask the reader to check these properties. We merely point out that the inequality in point 1 may be strict (and if this happens then $\text{RM}(\mu) = \text{LM}(\nu)$).

- Lemma 1.4.2** 1. If $\alpha < \beta < \varepsilon_0$, $m \geq 2$, $\text{psn}(\alpha) \leq m$ and β is limit then $\alpha \leq \{\beta\}(m)$.
2. For every α and every β , if $\text{RM}(\beta) > \alpha$ and $a = \text{psn}(\alpha)$ then $\{\beta\}(a) \gg \omega^\alpha$ and $\{\beta\}(a) + \omega^\alpha < \beta$.

Proof The first part is proved by induction on β . That is we denote by $T(\beta)$ the assertion

$$\forall \alpha < \beta \forall m \geq 2 [\text{psn}(\alpha) \leq m \ \& \ \text{Lim}(\beta) \Rightarrow \alpha \leq \{\beta\}(m)]$$

and prove it by transfinite induction.

The first limit ordinal is ω . But if $\beta = \omega$ and $\text{psn}(\alpha) \leq m$ then $\alpha \leq m = \{\beta\}(m)$ and we are done.

Let β be limit and assume the conclusion for each ordinal $< \beta$. Let $\alpha < \beta$. We select the (leftmost) common parts of Cantor normal forms of both of these ordinals, i.e., we write $\alpha = \delta + \omega^\gamma + \mu$ and $\beta = \delta + \omega^\xi + \nu$, where $\mu \ll \omega^\gamma \ll \delta$ and $\nu \ll \omega^\xi \ll \delta$. We have $\gamma < \xi$. If $\delta \neq 0$ then we may apply directly the inductive assumption to $\beta' = \omega^\xi + \nu$ (and $\alpha' = \omega^\gamma + \mu$). So assume that $\delta = 0$, that is $\beta = \omega^\xi + \nu$ and $\alpha = \omega^\gamma + \mu$. If $\nu \neq 0$ then $\{\beta\}(m) = \omega^\xi + \{\mu\}(m) > \alpha$ because $\xi > \gamma$. So assume that $\nu = 0$, i.e., $\beta = \omega^\xi$ and $\alpha = \omega^\gamma + \mu$. Then $\xi > \gamma$ because $\beta > \alpha$. If ξ is non limit, say $\xi = \zeta + 1$ then $\{\beta\}(m) = \omega^\zeta \cdot m > \omega^\gamma + \mu = \alpha$ (recall that $m \geq 2$), so assume that ξ is limit. But then $\gamma < \{\xi\}(m)$ by the inductive assumption $T(\xi)$. It follows that $\{\omega^\xi\}(m) = \omega^{\{\xi\}(m)} > \omega^\gamma + \mu = \alpha$ because $\omega^\gamma \gg \mu$ and we are done.

We prove the second part. We write $\beta = \delta + \omega^{\beta'}$ where $\delta \gg \omega^{\beta'}$. By the assumption, $\beta' > \alpha$. If $\beta' = \beta'' + 1$ then $\{\beta\}(a) = \delta + \omega^{\beta''} \cdot a$ with $\beta'' \geq \alpha$, so the first conclusion is immediate. The second one follows from the fact that in the decisive step the exponent β' was changed to the smaller one, i.e. β'' . If β' is limit then $\{\delta + \omega^{\beta'}\}(a) = \delta + \omega^{\{\beta'\}(a)}$. By part 1 $\{\beta'\}(a) \geq \alpha$, so the first conclusion holds. The second does as well because in the decisive step the exponent β' was lowered to $\{\beta'\}(a)$. $\square$

The following lemma is the promised strengthening of part 9 of Lemma 1.4.1.

Lemma 1.4.3 If $\alpha < \beta < \varepsilon_0$, $b \geq 1$, $\text{psn}(\alpha) \leq b$ then $\beta \Rightarrow_b \alpha$ and $\{\beta\}(b) \Rightarrow_b \alpha$.

Proof By induction on β . Thus the inductive thesis is $T(\beta)$:

$$\forall \alpha, b [(b \geq 1 \ \& \ \alpha < \beta \ \& \ \text{psn}(\alpha) \leq b) \Rightarrow (\beta \Rightarrow_b \alpha)].$$

For $\beta = 0$ there is nothing to prove. Assume the conclusion for β and let $\alpha < \beta + 1$. Then $\alpha = \beta$ or $\alpha < \beta$. If $\alpha = \beta$ then the one element sequence, whose only item is β , works, and if $\alpha < \beta$ then we take the sequence witnessing the relation $\{\beta\}(b) \Rightarrow_b \alpha$ and add to it β as the first item.

Let β be limit and assume the conclusion for all $\beta' < \beta$. Pick α, b such that $b > 1$, $\alpha < \beta$ and $\text{psn}(\alpha) \leq b$. Then $\alpha \leq \{\beta\}(b)$ by Lemma 1.4.2, so we may apply the assumption to $\beta' = \{\beta\}(b)$. We add to the sequence obtained in this manner the item $\{\beta\}(b)$ as the first item. $\square$

1.5 Hardy Hierarchy

In this section we describe Hardy hierarchy and the notion of largeness determined by it. The technical version of our considerations were distilled from the material of Ketonen–Solovay [1] by Zygmunt Ratajczyk [3], see also [4] and Ratajczyk’s final [5]. But the notion of Hardy hierarchy is much older, see, e.g. [6] or [7] for an exposition and history.

Let h be a finite increasing function (in the usual sense of the word, that is $\forall x, y \in \text{Dom}(h) [x < y \Rightarrow f(x) < f(y)]$). Assume moreover that $\forall x \ x < h(x)$. As an example, let $A \subseteq \mathbb{N}$ and let $h = h^A$ be the successor in the sense of A , i.e., the function defined on A if A is infinite and defined on $A \setminus \{\max A\}$ if A is finite, which associates with every a in its domain the next element of A .

For every $\alpha < \varepsilon_0$ we define a function h_α , by induction on α . We put $h_0(x) \simeq x$ and $h_{\alpha+1}(x) \simeq h_\alpha(h(x))$. We let $h_\lambda(x) \simeq h_{\{\lambda\}(x)}(x)$ for λ limit. We point out that the domain of h is a subset of $\mathbb{N}$, which may be finite or infinite. The family $h_\alpha : \alpha < \varepsilon_0$ is called *Hardy hierarchy* based on h . We remark that Odifreddi [8], vol. 2 p. 308 calls it *moderate growing hierarchy* (and uses another notation). As usual, $\simeq$ means “either both sides are undefined or both are defined and equal”, but we shall use directly the equality sign in this situation.

In order to see the exact mechanism of Hardy hierarchy, the reader is suggested to check the following. Let $h(x) = x + 1$ be the usual successor function. Then $h_{\omega \cdot n}(x) = 2^n \cdot x$, and, hence, $h_{\omega^2}(x) = 2^x \cdot x$. Moreover one should write the formula for $h_{\omega^3}(x)$ in order to see that this function grows quicker than $\mathfrak{J}_x(x)$, where the *iterated exponentiation* $\mathfrak{J}_i(x)$ is defined by $\mathfrak{J}_0(x) = x$, $\mathfrak{J}_{i+1}(x) = 2^{\mathfrak{J}_i(x)}$.

The notion of Hardy hierarchy allows us to define a set A of natural numbers to be α –large. That is A is α –large iff $(h^A)_\alpha(a)$ is defined, where h^A denotes the successor in the sense of A (i.e., the function with domain $A \setminus \{\max A\}$ which associates with every b in its domain the next element of A) and $a = \min A$. We shall write just h if the meaning of A is clear from the context.

A set A is α –small if it is not α –large. It will be also convenient to say that a set A is *exactly* α –large if it is α –large but $A \setminus \{\max A\}$ is α –small. We shall also say that A is *at most* α –large if it is either α –small or exactly α –large.

One can restate the definition of largeness in the following manner. A set A is 0–large iff it is nonempty. A is $\alpha + 1$ –large iff $A \setminus \{\min A\}$ is α –large. A is λ –large, λ limit, iff it is $\{\lambda\}(\min A)$ –large. We remark that Ketonen and Solovay [1] use a slightly different notion of largeness.

The following propositions show that this notion of largeness has the expected properties.

Proposition 1.5.1 *For every $\alpha < \varepsilon_0$ and every infinite $A \subseteq \mathbb{N}$ there exists $u \in A$ such that $\{a \in A : a \leq u\}$ is α –large.*

Proof By induction on α . For $\alpha = 0$ $u = a_0 = \min A$ satisfies our demand. Assume the lemma for α . Let A be infinite. Then $A \setminus \{\min A\}$ is infinite. Let $u \in A \setminus \{\min A\}$ be such that $\{a \in A \setminus \{\min A\} : a \leq u\}$ is α –large. Then u satisfies our demand for the set A . We leave the limit step to the reader. $\square$

Below we write $f(x) \downarrow$ for “ $f(x)$ is defined”, i.e., $x \in \text{Dom}(f)$.

Lemma 1.5.2 *Let h be a function as above. Then for every $\alpha < \varepsilon_0$*

1. h_α is increasing.
2. For every β, b if $\alpha \Rightarrow_b \beta$ then if $h_\alpha(b)$ exists then $h_\beta(b)$ exists and $h_\alpha(b) \geq h_\beta(b)$.

Proof By simultaneous induction on α . For $\alpha = 0$ both claims are evident. Assume both claims for α , we prove them for $\alpha + 1$. Then $h_{\alpha+1} = h_\alpha \circ h$ is increasing as a composition of two increasing functions. Further, if $\alpha + 1 \Rightarrow_b \beta$ and $h_{\alpha+1}(b) \downarrow$, then $\alpha + 1 \Rightarrow_{h(b)} \beta$ by Lemma 1.4.1 part 7, and $h_\alpha(h(b)) \downarrow$. By inductive assumption for α and $h(b)$ we infer that $h_\beta(h(b)) \downarrow$, so also $h_\beta(b) \downarrow$.

Assume both claims for all $\alpha' < \alpha$, where α is limit. We check firstly the second claim. Pick β, b such that $\alpha \Rightarrow_b \beta$ and $h_\alpha(b) \downarrow$. Then $h_{\{\alpha\}(b)}(b) \downarrow$ and $h_{\{\alpha\}(b)}(b) \geq h_\beta(b)$ by part 2 applied to $\alpha' = \{\alpha\}(b)$. Let us show part 1 for α . So let $x < y$. Then $\{\alpha\}(y) \Rightarrow_y \{\alpha\}(x)$ by Lemma 1.4.1, part 5, hence

$$h_\alpha(y) = h_{\{\alpha\}(y)}(y) \geq h_{\{\alpha\}(x)}(y)$$

by part 1, and this expression is $\geq h_{\{\alpha\}(x)}(x) = h_\alpha(x)$ because $h_{\{\alpha\}(x)}$ is increasing by the inductive assumption. $\square$

Below if we write $A = \{a_0, \dots, a_{\text{Card}A-1}\}$ we assume that this enumeration is the natural one, i.e., in increasing order.

- Lemma 1.5.3** 1. For every α if A, B are finite sets of the same cardinality and such that for every $i < \text{Card}A$ $b_i \leq a_i$ then for every $i < \text{Card}A$ if $(h^A)_\alpha(a_i)$ exists then $(h^B)_\alpha(b_i)$ exists and $(h^A)_\alpha(a_i) \geq (h^B)_\alpha(b_i)$.
2. If A, B are finite sets, A is α -large, $\text{Card}A = \text{Card}B$ and for every $i < \text{Card}A$ $b_i \leq a_i$ then B is α -large.
 3. If $A \subseteq B$ and A is α -large then B is α -large.

Proof The first part is immediate by induction on α , the second is a direct consequence of the first one. The third part follows from the observation that if $A \subseteq B$ then B has an initial segment of cardinality $\text{Card}A$. But obviously, if a set has an α -large initial segment then it is α -large itself, so the second part may be applied. $\square$

The following is a minor variant of Lemma 1.5.3 in which we speak of sets of different cardinalities.

Lemma 1.5.4 For every α and every D, E , if $D \subseteq E$, $x \in D$ and $(h^D)_\alpha(x) \downarrow$ then $(h^E)_\alpha(x) \downarrow$ and $(h^E)_\alpha(x) \leq (h^D)_\alpha(x)$.

Proof By induction on α . If $\alpha = 0$ the conclusion is obvious. Assume the conclusion for α , we derive it for $\alpha + 1$. So let D, E satisfy the assumption. Let $x \in D$ be such that $(h^D)_{\alpha+1}(x)$ exists. Then $(h^D)_{\alpha+1}(x) = (h^D)_\alpha((h^D)(x))$. Let $y = (h^D)(x)$. We apply the inductive assumption to y . Thus we infer $(h^E)_\alpha(y) \downarrow$ and $(h^E)_\alpha(y) \leq$

$(h^D)_\alpha(y)$. But $(h^E)(x) \leq (h^D)(x) = y$, hence $(h^E)_{\alpha+1}(x) = (h^E)_\alpha((h^E)(x)) \leq (h^E)_\alpha((h^D)(x)) \leq (h^D)_\alpha((h^D)(x)) = (h^D)_{\alpha+1}(x)$ because $(h^E)_\alpha$ is increasing by Lemma 1.5.2. We leave the limit step to the reader. $\square$

Lemma 1.5.5 *Let h be as above. Then for every α and every $\beta \gg \alpha$ $h_{\beta+\alpha} = h_\beta \circ h_\alpha$.*

Proof By induction on α . For $\alpha = 0$ there is nothing to prove. Assume the conclusion for α . Let $\beta \gg \alpha + 1$. Then $\beta \gg \alpha$, hence

$$h_{\beta+(\alpha+1)} = h_{(\beta+\alpha)+1} = h_{\beta+\alpha} \circ h = (h_\beta \circ h_\alpha) \circ h = h_\beta \circ (h_\alpha \circ h) = h_\beta \circ h_{\alpha+1}.$$

Assume the conclusion for all $\alpha' < \alpha$, where α is limit. Let $\beta \gg \alpha$. Then $\beta \gg \{\alpha\}(b)$ for all b . It follows that

$$h_{\beta+\alpha}(b) = h_{\beta+\{\alpha\}(b)}(b) = h_\beta \circ h_{\{\alpha\}(b)}(b) = h_\beta \circ h_\alpha(b)$$

as desired. $\square$

Let us restate this fact in the following manner.

Lemma 1.5.6 *Let A be a finite set and let $\beta \gg \alpha$. Then A is $\beta + \alpha$ -large iff there exists $u \in A$ such that $\{x \in A : x \leq u\}$ is α -large and $\{x \in A : u \leq x\}$ is β -large.*

Observe that it may happen that $h_\alpha(b) = h_\beta(b)$ also if $\alpha \neq \beta$, e.g., $h_\omega(b) = h_b(b)$. It will be convenient to be able to choose some ordinal uniquely in such situations. This is done as follows. Let a set A be given. Let $\mu < \varepsilon_0$. We define two sequences μ_j, b_j by the following induction. We let $\mu_0 = \mu$ and $b_0 = a_0 = \min A$. Assume that μ_j and b_j are constructed. If $\mu_j = 0$ then the construction terminates. If $\mu_j > 0$ and μ_j is limit we let $\mu_{j+1} = \{\mu_j\}(b_j)$ and $b_{j+1} = b_j$. If μ_j is non limit then the construction terminates if $b_j = a_{r-1} = \max A$, otherwise we let $\mu_{j+1} = \mu_j - 1$ and $b_{j+1} = h^A(b_j)$, the next element of A . This completes the definition of the sequences μ_j, b_j .

Observe that the sequence μ_j is decreasing. We remark also that the construction terminates in two cases: $\mu_j = 0$ or μ_j is nonlimit and $b_j = \max A$. It should be also noticed that in the process of this construction we passed from b_j to the next element of A only if μ_j is nonlimit.

The following proposition is (essentially) the original definition of a μ -large set, cf. Ketonen–Solovay [1].

Proposition 1.5.7 *Under the notation introduced above, A is μ -large iff there exists j such that $\mu_j = 0$.*

Proof We claim that the following are equivalent:

1. A is μ -large,
2. for every j the set $\{x \in A : b_j \leq x\}$ is μ_j -large,
3. for some j the set $\{x \in A : b_j \leq x\}$ is μ_j -large.

Both non obvious implications are proved by induction on j . They follow immediately from the definitions and are left to the reader.

Granted the claim we see that if A is μ -large and no $\mu_j = 0$ then the construction cannot terminate. Indeed, let μ_t, b_t be its last item. Consider the sequence $\gamma_0 = \mu_t$, $\gamma_{i+1} = \{\gamma_i\}(b_t)$ till we get a nonlimit γ_i . Then the set $\{x \in A : b_t \leq x\}$ is γ_i -large, where $\gamma_i > 0$, so it has an element $> b_t$ and we have b_{j+1} and μ_{j+1} . But this is impossible as the sequence μ_j would be then a decreasing sequence of ordinals.

If A is μ -small and $\mu_j = 0$ for some j then the set $\{x \in A : b_j \leq x\}$ being nonempty is 0-large. $\square$

This proposition allows one to associate with every $a \in A$ an ordinal. That is, given a fixed μ such that A is μ -small (or at least $A \setminus \{\max A\}$ is μ -small) we associate with every $a \in A$ the nonlimit μ_j such that $a = b_j$. But, of course, this assignment of ordinals to elements of A depends on μ . We shall write $\text{KS}(\mu; a)$ (or $\text{KS}^A(\mu; a)$ if necessary) for the last μ_j with $a = b_j$. Observe that for some μ , $\text{KS}(\mu; a)$ is not defined for all elements of A , but they are defined if A is at most μ -large.

We generalize slightly this idea. We define the relation the relation $\text{RKS}^A(\mu; \alpha, a)$. This is an abbreviation for: $\exists j [\alpha = \mu_j \ \& \ a = b_j]$. Thus, $\text{KS}(\mu; a)$ is the last α for which $\text{RKS}^A(\mu; \alpha, a)$ holds. Once again, we omit the superscript A unless it is really necessary.

As the reader has noticed, every $h_\alpha(a)$ is $h_m(a)$ for some finite m . Let us make this point precise. We define $\text{Nit}(h; \alpha, a)$, the *number of iteration function* by induction on α : $\text{Nit}(h; 0, a) = 0$, $\text{Nit}(h; \alpha + 1, a) = \text{Nit}(h; \alpha, h(a)) + 1$, and $\text{Nit}(h; \lambda, a) = \text{Nit}(h; \{\lambda\}(a), a)$ for limit λ . As usual, we shall write Nit^A if necessary.

Lemma 1.5.8 *Under the notation introduced above we have $h_\alpha(a) = h^{\text{Nit}(h; \alpha, a)}(a)$.*

Proof By induction on α , left to the reader. $\square$

Let us introduce an additional hierarchy of quickly growing functions. Let G be a function of which we assume, as usual, that it is increasing and $\forall x \ x < G(x)$. We let $G_0(x) = G(x)$, $G_{\alpha+1}(x) = (G_\alpha)^x(x)$ and $G_\lambda(x) = G_{\{\lambda\}(x)}(x)$ for limit λ . Here F^m denotes the m th iterate of F , i.e., $F^0(x) = x$, $F^{m+1}(x) = F^m(F(x))$. This hierarchy is called *Grzegorzcyk–Wainer hierarchy* or *Schwichtenberg–Wainer hierarchy* or *fast-growing hierarchy* in the literature.

Proposition 1.5.9 *Let G be as above, let h_α denote the Hardy hierarchy based on G . Then for every $\alpha < \varepsilon_0$ $G_\alpha = h_{\omega^\alpha}$.*

The proof of the above proposition is an easy induction on α which uses Lemma 1.5.5 in the case of α being a successor.

Of course, one can define a set $A \subseteq \mathbb{N}$ to be α -large in Grzegorzcyk–Wainer sense if $G_\alpha(\min A) \downarrow$, where G_α denotes the α th function in Grzegorzcyk–Wainer hierarchy based on the successor in the sense of A .

Yet another hierarchy of quickly growing functions is defined as follows. Let h satisfy the same assumptions as above, i.e., it is increasing and increases the argument.

We define a sequence sl_α of functions exactly like in the definition of Hardy hierarchy, but in nonlimit step we reverse the order of composition, that is $sl_{\alpha+1} = h \circ sl_\alpha$. The hierarchy gotten in this manner is called *slow growing hierarchy based on h* . We point out that this small change causes a drastic difference in the behavior of slow growing hierarchy and that of Hardy hierarchy. Once again, we get the notion of a set A being α -large in the sense of slow growing hierarchy.

From now on we write α -large for the notion given by Hardy hierarchy, otherwise we make explicit the hierarchy we are working with.

It will be also convenient to have the following notion. Say that $E \subseteq \mathbb{N}$ is f -scattered if for every $e \in E \setminus \{\max E\}$, $f(e) \leq h^E(e)$.

1.6 Approximating Functions

In this section we work out a combinatorial notion which we shall use in the proofs of independence results. Let $A \subseteq \mathbb{N}$ and let g be a function (defined on a subset of $\mathbb{N}$, which may be finite or infinite). We say that A *approximates g* if

$$\forall a \in A \forall x < a - 2 \{g(x) \downarrow \Rightarrow [g(x) < h^A(a) \vee g(x) \geq \max A]\}.$$

Of course, if A is infinite then the clause $g(x) \geq \max A$ should be omitted.

We define a sequence of families of finite sets of natural numbers. We say that $A \subseteq \mathbb{N}$ is *approximatively 0-large* if $\text{Card} A > \min A$. Say that A is *approximatively $(n+1)$ -large* if for every finite function g there exists an approximatively n -large $B \subseteq A$ which approximates g .

The goal of this section is the following result.

Theorem 1.6.1 *For every $a, n \in \mathbb{N}$ there exists an approximatively n -large set $A \subseteq \mathbb{N}$ with $\min A \geq a$.*

Soon after the famous paper by J. Paris and L. Harrington [9] appeared, a variant of the statement of Theorem 1.6.1 was shown to be unprovable in arithmetic by Pavel Pudlák (unpublished), who refers the idea to H. Friedman (unpublished). Since then, it is called *Pudlák's principle*, see, e.g. [10].

We shall see later that this is meaningful and unprovable in Peano arithmetic (though for each fixed n it is provable). In fact, our exposition of the Paris–Harrington result will be based on some variation of Theorem 1.6.1, cf. Theorem 4.4.4. The reformulation of Pudlák's principle in this manner and the proof of Theorem 1.6.2 and its application to construct initial segments in models of PA are due to Zygmunt Ratajczyk [3, 4]. Richard Sommer [11, 12] uses a similar tool (see [12], lemma 5.24).

Theorem 1.6.2 *If A is ω^α -large then for every finite function g there exists an α -large $B \subseteq A$ which is an approximation of g .*

Of course, Theorem 1.6.2 implies Theorem 1.6.1 immediately. Precisely let $\omega_1 = \omega$ and $\omega_{m+1} = \omega^{\omega_m}$.

Corollary 1.6.3 *Every ω_n -large set is $(n - 1)$ -approximatively large.*

In order to prove Theorem 1.6.2 we need a minor generalization of a set being an approximation for a function. Let (A, B) be a pair of finite sets. We say that the pair (A, B) is an *approximation* for a function g if $\max A = \min B$ and

$$\forall a \in A \setminus \{\max A\} \forall x < a - 2 \{g(x) \downarrow \Rightarrow [g(x) < h^A(a) \vee g(x) \geq \max B]\}.$$

For ordinals $\alpha, \beta, \gamma < \varepsilon_0$ we write $\alpha \rightarrow (\beta, \gamma)$ for “whenever A is an α -large set and g is a function there exist $B_1, B_2 \subseteq A$ such that

1. $\min B_1 = \min A$,
2. $\max B_1 = \min B_2$,
3. the pair (B_1, B_2) is an approximation for g ,
4. B_1 is β -large and B_2 is γ -large”.

The idea here is as follows. As a matter of fact we want to work with B_1, B_2 is just a reserve for further steps of induction needed for the following lemma.

Lemma 1.6.4 *If $\alpha \gg \beta$ and $\beta > 0$ then $\omega^{\alpha+\beta} \rightarrow (\beta, \omega^\alpha)$.*

Observe that Lemma 1.6.4 implies Theorem 1.6.2 immediately (just put $\alpha = 0$).

Proof of Lemma 1.6.4 We denote by $T(\beta)$ the formula

$$\beta > 0 \Rightarrow \forall \alpha \gg \beta [\omega^{\alpha+\beta} \rightarrow (\beta, \omega^\alpha)]$$

and prove it by induction on β . In each case the subcase $\alpha = 0$ is the same as the subcase when $\alpha \neq 0$, so we do not separate these subcases.

Case 1 $\beta = 1$. Let A be $\omega^{\alpha+1} = \omega^\alpha \cdot \omega$ -large. Let g be a function. Let $a_0 = \min A$ and $b_0 = \max A$. Hence $h_{\omega^{\alpha+1}}^A(a_0) = h_{\omega^\alpha, a_0}^A(a_0) \leq b_0$. Let $a_k = h_{\omega^\alpha, k}^A(a_0)$ for $k = 0, \dots, a_0$. Of course, $a_0 < \dots < a_{a_0}$. By the pigeon-hole principle there exists j_0 with $1 \leq j_0 \leq a_0 - 1$ and $[a_{j_0}, a_{j_0+1}) \cap g * [0, a_0 - 2) = \emptyset$ because there are at most $a_0 - 2$ images of $x < a_0 - 2$ and there are $a_0 - 1$ intervals $A \cap [a_1, a_2), \dots, A \cap [a_{a_0-1}, a_{a_0})$ of the set A . We put $B_1 = \{a_0, a_{j_0}\}$ and $B_2 = A \cap [a_{j_0}, a_{j_0+1}]$. Then B_1 is 1-large as $j_0 \neq 0$, so B_1 has exactly two elements. B_2 is ω^α -large since $h_{\omega^\alpha}^{B_2}(a_{j_0}) = a_{j_0+1}$, so the pair (B_1, B_2) has the desired properties.

Case 2 The successor step $T(\beta) \Rightarrow T(\beta + 1)$. Let $\alpha \gg \beta + 1$, so $\alpha \gg \beta$. Assume $\omega^{\alpha+\beta} \rightarrow (\beta, \omega^\alpha)$. Let A be $\omega^{\alpha+\beta+1}$ -large and let a function g be given. By the initial step there exists a pair (A_1, A_2) of subsets of A approximating g and such that $\min A_1 = \max A_2$, $\text{Card} A_1 = 2$ and A_2 is $\omega^{\alpha+\beta}$ -large. By the inductive assumption $\omega^{\alpha+\beta} \rightarrow (\beta, \omega^\alpha)$ there exists a pair (A_3, A_4) approximating g and such that $\max A_3 = \min A_4$, A_3 is β -large and A_4 is ω^α -large. Consider the pair $(A_1 \cup A_3, A_4)$. Clearly, it is an approximation for g . Thus it remains to show

that $A_1 \cup A_3$ is $\beta + 1$ -large. Let $a_0 = \min A = \min A_1$. Then $A_1 \cup A_3 = \{a_0\} \cup A_3$. Thus $h^{A_1 \cup A_3}(a_0) = \min A_3$, so $h_{\beta+1}^{A_1 \cup A_3}(a_0) \simeq h_{\beta}^{A_1 \cup A_3}(\min A_3) \simeq h_{\beta}^{A_3}(\min A_3) \downarrow$.

Case 3 β limit. The inductive assumption is

$$\forall \gamma < \beta \forall \alpha \gg \gamma [1 \leq \gamma \Rightarrow \omega^{\alpha+\gamma} \rightarrow (\gamma, \omega^{\alpha})].$$

Let $\alpha \gg \beta$. Then $\alpha \gg \{\beta\}(n)$ and $\{\beta\}(n) < \beta$ for all n . Let A be $\omega^{\alpha+\beta}$ -large and let a function g be given. Let $a_0 = \min A$. Then A is $\{\omega^{\alpha+\beta}\}(a_0)$ -large, i.e., $\omega^{\alpha+\{\beta\}(a_0)}$ -large. By the inductive assumption there exists an approximation (A_1, A_2) for g such that $\min A_1 = a_0$, A_1 is $\{\beta\}(a_0)$ -large and A_2 is ω^{α} -large. Thus, A_1 is β -large, so the same pair (A_1, A_2) has the required properties. $\square$

1.7 Hardy Largeness and Partitioning Elements

In this section we prove a partition theorem for elements (i.e., 1-tuples). The result is taken from [13], strengthening a result of Ketonen and Solovay ([1], theorem 4.7).

Let A be a finite subset of $\mathbb{N}$. We say that the partition $A = \cup_{0 \leq i \leq e} B_i$ of A is α -large if the set $E = \{\min B_0, \dots, \min B_e\}$ is α -large. A partition is α -small otherwise.

For ordinals $\alpha, \beta, \gamma < \varepsilon_0$ we write $\alpha \rightarrow (\beta)_{\gamma}^1$ if for every α -large set A with $\min A > 0$ and every partition $A = \cup_{0 \leq i \leq e} B_i$ of A which is not γ -large, there exists $i \leq e$ such that B_i is β -large. We keep the superscript 1 in the above notation just to follow the usual notation in Ramsey theory, cf. Sect. 1.1.

The main result of this section is the following theorem.

Theorem 1.7.1 *If $\alpha, \beta < \varepsilon_0$, $\alpha \geq 1$ and $\beta \gg \text{LM}(\alpha)$ then $\omega^{\beta} \cdot \alpha \rightarrow (\omega^{\beta})_{\alpha}^1$.*

In order to prove this we shall need several lemmas. The next one seems to be of independent interest. One might think of it as follows: the notion of largeness behaves like if it were just the counting measure.

Lemma 1.7.2 *For every α and every $\beta \gg \alpha$ and every A, B , if A is $\beta + \alpha$ -large, $B \subseteq A$ and B is α -small, then $A \setminus B$ is β -large.*

In order to derive Theorem 1.7.1 from Lemma 1.7.2 we need some auxiliary facts about fundamental sequences.

Lemma 1.7.3 1. $\forall \gamma > 0 \forall b > 0 \gamma \Rightarrow_b 1$.

2. for all limit α we have $\forall u > b > 1 [\{\alpha\}(u) \Rightarrow_b \{\alpha\}(b) + 1]$.

3. $\forall \alpha \gg \omega \forall \delta \gg \omega^{\alpha} \forall u > b > 1 \delta + \omega^{\{\alpha\}(u)} \Rightarrow_b \delta + \omega^{\{\alpha\}(b)} \cdot b$.

4. If a set D is $\delta + \omega^{\{\alpha\}(u)}$ -large, $b = \min D$ satisfies $u > b > 1$ then D is $\delta + \omega^{\{\alpha\}(b)} \cdot b$ -large.

Proof Part 1 is immediate by induction on γ . Part 2 is proved by induction on α . Cases $\alpha = \omega$ and $\alpha \rightarrow \alpha + \omega$ are immediate, so we show only the step $\alpha \gg \omega^2$. Write $\alpha = \delta + \omega^{\tau}$, where $\delta \gg \omega^{\tau}$. Thus, $\tau > 1$. If $\tau = \varrho + 1$ then

$$\{\alpha\}(u) = \{\delta + \omega^{\varrho+1}\}(u) = \delta + \omega^{\varrho} \cdot u = \delta + \omega^{\varrho} \cdot b + \omega^{\varrho} \cdot (u - b).$$

We use part 1 to infer $\{\alpha\}(u) \Rightarrow_b \{\alpha\}(b) + 1$ as required. So let τ be limit. Then $\{\tau\}(u) \Rightarrow_b \{\tau\}(b) + 1$ by the inductive assumption, so by Lemma 1.4.1, part 4,

$$\{\alpha\}(u) = \{\delta + \omega^{\tau}\}(u) = \delta + \omega^{\{\tau\}(u)} \Rightarrow_b \delta + \omega^{\{\tau\}(b)+1}.$$

Moreover, $\omega^{\{\tau\}(b)+1} \Rightarrow_b \omega^{\{\tau\}(b)} \cdot b = \omega^{\{\tau\}(b)} + \omega^{\{\tau\}(b)} \cdot (b - 1)$ and the same argument as above works.

Part 3 follows from part 2 and part 3 of Lemma 1.4.1.

In order to prove part 4, let D, u, b satisfy the assumption. That is, we have $h_{\delta+\{\omega^{\alpha}\}(u)}(\min D) \downarrow$. By part 3 and Lemma 1.5.2, $h_{\delta+\omega^{\{\alpha(b)\}} \cdot b}(\min D) \downarrow$ as required. $\square$

The lemma below is due to Teresa Bigorajska.

Lemma 1.7.4 *Let λ be a limit ordinal smaller than ε_0 . Then if $\beta \gg \text{LM}(\lambda)$ then for every $n \in \omega$, $\{\omega^{\beta} \cdot \lambda\}(n) = \omega^{\beta} \cdot \{\lambda\}(n)$.*

Proof Let λ be limit and let ϱ be the smallest exponent in the Cantor normal form expansion of λ . Then $\lambda = \delta + \omega^{\varrho}$ for some $\delta \gg \omega^{\varrho}$. Let $\beta \gg \text{LM}(\lambda)$ and $n \in \omega$. We have

$$\{\omega^{\beta} \cdot \lambda\}(n) = \{\omega^{\beta}(\delta + \omega^{\varrho})\}(n) = \{\omega^{\beta} \cdot \delta + \omega^{\beta+\varrho}\}(n) = \omega^{\beta} \cdot \delta + \{\omega^{\beta+\varrho}\}(n).$$

The last equality holds because $\omega^{\beta} \cdot \delta \gg \omega^{\beta+\varrho}$. Obviously $\beta \gg \varrho$, hence if $\varrho = \alpha + 1$ for some α then

$$\begin{aligned} \omega^{\beta} \cdot \delta + \{\omega^{\beta+\varrho}\}(n) &= \omega^{\beta} \cdot \delta + \{\omega^{\beta+\alpha+1}\}(n) = \omega^{\beta} \cdot \delta + \omega^{\beta+\alpha} \cdot n = \omega^{\beta} \cdot (\delta + \omega^{\alpha} \cdot n) = \\ &= \omega^{\beta} \cdot (\delta + \{\omega^{\alpha+1}\}(n)) = \omega^{\beta} \cdot \{\delta + \omega^{\varrho}\}(n) = \omega^{\beta} \cdot \{\lambda\}(n). \end{aligned}$$

Let ϱ be limit. By the assumption $\beta \gg \varrho$ we get

$$\begin{aligned} \omega^{\beta} \cdot \delta + \{\omega^{\beta+\varrho}\}(n) &= \omega^{\beta} \cdot \delta + \omega^{\{\beta+\varrho\}(n)} = \omega^{\beta} \cdot \delta + \omega^{\beta+\{\varrho\}(n)} = \\ \omega^{\beta} \cdot (\delta + \omega^{\{\varrho\}(n)}) &= \omega^{\beta} \cdot (\delta + \{\omega^{\varrho}\}(n)) = \omega^{\beta} \cdot \{\lambda\}(n). \end{aligned} \quad \square$$

Derivation of Theorem 1.7.1 from Lemma 1.7.2 By induction on α . Case $\alpha = 1$ is obvious, indeed, if a partition is 1-small then there is only one part.

Assume the conclusion for α , we derive it for $\alpha + 1$. Let A be an $\omega^{\beta} \cdot (\alpha + 1)$ -large subset of $\mathbb{N}$ and let $A = \cup_{0 \leq i \leq e} B_i$ be an $\alpha + 1$ -small partition of A . Let $E = \{\min B_0, \dots, \min B_e\}$, so E is $\alpha + 1$ -small. We may assume that $\min E = \min B_0$. We put $C = A \setminus B_0$. If B_0 is not ω^{β} -large then by Lemma 1.7.2, C is $\omega^{\beta} \cdot \alpha$ -large. Consider the partition $C = \cup_{1 \leq i \leq e} B_i$ of C . Let $E_1 = \{\min B_1, \dots, \min B_e\}$. But the partition of A is $\alpha + 1$ -small, hence $h_{\alpha+1}(\min B_0) \uparrow$ (where h denotes the successor in the sense of E). It follows that $h_{\alpha}(h(\min B_0)) \uparrow$. We have $h(\min B_0) = \min E_1$. Thus the above partition of C is α -small. We apply the inductive assumption to

the set C and the above-mentioned partition. Summing up, B_0 or at least one of B_i , $1 \leq i \leq e$, is ω^β -large.

Assume the conclusion for all ordinals smaller than λ , λ limit. Let A be an $\omega^\beta \cdot \lambda$ -large set, where $\beta \gg \text{LM}(\lambda)$. Let a partition $A = \bigcup_{0 \leq i \leq e} B_i$ be given and λ -small. Exactly as above, let $E = \{\min B_0, \dots, \min B_e\}$. Then A is $\{\omega^\beta \cdot \lambda\}(\min A)$ -large. Thus A is $\omega^\beta \cdot \{\lambda\}(\min A)$ -large by Lemma 1.7.4. Obviously, E is not $\{\lambda\}(\min A)$ -large and $\beta \gg \text{LM}(\{\lambda\}(\min A))$. By the inductive assumption, at least one of B_i , $i \leq e$ is ω^β -large. $\square$

Lemma 1.7.2 will be proved by induction on α and almost each step of it by induction on β . Our life will be slightly easier if we prove it for α with trivial Cantor normal form firstly and only then derive the general version. That is why we shall prove

Lemma 1.7.5 *For every α and every $\beta \gg \omega^\alpha$ and every A, B , if $\min A > 0$ and A is $\beta + \omega^\alpha$ -large, $B \subseteq A$ and B is not ω^α -large then $A \setminus B$ is β -large.*

Derivation of Lemma 1.7.2 from Lemma 1.7.5 Let A be $\beta + \alpha$ -large, where $\beta \gg \alpha$ and let B be its not α -large subset. Write $\alpha = \omega^{\alpha_s} + \dots + \omega^{\alpha_0}$, where $\alpha_s \geq \dots \geq \alpha_0$. Let $e = \max\{i \leq s : B \text{ is } \omega^{\alpha_i} + \dots + \omega^{\alpha_0}\text{-large}\}$. Let h denote the successor in the sense of B . Let $B_0 = \{x \in B : x < h_{\omega^{\alpha_0}}(\min B)\}$, $B_{i+1} = \{x \in B : h_{\omega^{\alpha_i}}(\min B_i) \leq x < h_{\omega^{\alpha_{i+1}}}(h_{\omega^{\alpha_i}}(\min B_i))\}$ for $i < e$. We let $B_{e+1} = B \setminus \bigcup_{0 \leq i \leq e} B_i$. Then $B = \bigcup_{0 \leq i \leq e+1} B_i$. Observe that no B_i , $i \leq e+1$, is ω^{α_i} -large. By Lemma 1.7.5, by induction on i , we infer that $A \setminus (B_0 \cup \dots \cup B_i)$ is $\beta + \omega^{\alpha_s} + \dots + \omega^{\alpha_{i+1}}$ -large. It follows that $A \setminus B$ is β -large. $\square$

Proof of Lemma 1.7.5 Let $T(\beta, \alpha)$ be the following property

for every A, B , if A is $\beta + \omega^\alpha$ -large and $B \subseteq A$ is not ω^α -large then $A \setminus B$ is β -large

and we shall prove the statement $\forall \alpha \forall \beta \gg \omega^\alpha T(\beta, \alpha)$ by induction on α .

CASE $\alpha = 0$. Then A is $\beta + \omega^0 = \beta + 1$ -large and B is not 1-large. If $\beta = 0$ then A is 1-large, i.e., has at least two elements, but B being not 1-large has at most one element, so $A \setminus B$ is nonempty, so 0-large. If $\beta > 0$ then A is $\beta + 1$ -large so $A \setminus \{a_0\}$ is β -large. Also, B being not 1-large has at most one element. It follows that $A \setminus \{a_0\}$ and $A \setminus B$ satisfy the assumption of Lemma 1.5.3, (these sets have the same cardinality and the i 'th element of $A \setminus B$ is $\leq i$ 'th element of $A \setminus \{a_0\}$), hence $A \setminus B$ is β -large.

CASE $\alpha = 1$. Exactly as above, the case $\beta = 0$ is obvious. For other cases we proceed by induction on β .

Let $\beta = \omega$. So let A be $\omega + \omega$ -large and let $B \subseteq A$ be not ω -large. Let $u = (h^A)_\omega(a_0)$. *Case 1.* $b_0 > u$. Then $\{x \in A : x \leq u\} \subseteq A \setminus B$. The first of these sets is ω -large, so the second is as well by Lemma 1.5.3. *2.* $b_0 = u$. Then there exists $z \in A \setminus B$ with $z > u$ (otherwise $\{x \in A : u \leq x\} \subseteq B$, so B is ω -large by Lemma 1.5.3). It follows that $\{x \in A : x < u\} \cup \{z\} \subseteq A \setminus B$, so this set is ω -large, again by Lemma 1.5.3. *Case 3.* $a_0 < b_0 < u$. In order to show that $A \setminus B$ is ω -large it suffices to show that it has more than a_0 elements, indeed, $\min(A \setminus B) = a_0$. But A has more than $a_0 + u$ elements and B , being not ω -large, has at most $b_0 < u$ of

them. *Case 4.* $a_0 = b_0$. Then B has at most a_0 elements, so $A \setminus B$ has more than u elements. If $\min(A \setminus B) \leq u$ then we are done. Otherwise $\{x \in A : x \leq u\} \subseteq B$, so this set is ω -large, contrary to the assumption.

Assume $T(\beta, 1)$, we prove $T(\beta + \omega, 1)$. So let A be $\beta + \omega + \omega$ -large and let $B \subseteq A$ be not ω -large. Let $u = (h^A)_\omega(a_0)$ and $w = (h^A)_\omega(u)$. *Case 1.* $b_0 > u$. Let $A' = A \setminus \{x \in A : x < u\}$. Thus $B \subseteq A'$. By $T(\beta, 1)$, $A' \setminus B$ is β -large, hence $\{x \in A : x < u\} \cup (A' \setminus B) = A \setminus B$ is $\beta + \omega$ -large by Lemma 1.5.6. *Case 2.* $b_0 = u$. Let A' be $\{x \in A : u \leq x\}$. By $T(\beta, 1)$, the set $C = A' \setminus B$ is β -large. Let c_0 be, as usual, the smallest element of C . Then $A \setminus B = \{x \in A : x < u\} \cup C = (\{x \in A : x < u\} \cup \{c_0\}) \cup C$ is $\beta + \omega$ -large by Lemma 1.5.6. *Case 3.* $a_0 < b_0 < u$. Obviously, B has at most b_0 elements (otherwise it is ω -large), so B has less than u elements. Let $k = \text{Card}(\{x \in A : w < x\})$. Thus, $A \setminus B$ has at least $a_0 + k + 1$ elements. Let $c_0 = a_0, c_1, \dots, c_{a_0}$ be list of the first $a_0 + 1$ elements of $A \setminus B$ in increasing order. We claim that $c_{a_0} \leq w$. For otherwise there are at least k elements of $A \setminus B$ which are $> c_{a_0} > w$. But this is impossible, there are at most $k - 1$ such elements of A . Let $E = \{e_0, \dots, e_k\}$ be the set of the $k + 1$ consecutive elements of $A \setminus B$, beginning with $e_0 = c_{a_0}$. Then E is β -large, indeed, its cardinality is $k + 1$ and its elements are $\leq$ of the corresponding elements of $\{x \in A : w \leq x\}$. It follows that $A \setminus B$ contains $\{c_0, \dots, c_{a_0}\} \cup E$, so is $\beta + \omega$ -large by Lemma 1.5.6. *Case 4.* $b_0 = a_0$. Then B , being not ω -large, has at most a_0 elements, hence $A \setminus B$ has more than $u + k$ elements. Let $E = \{e_0, \dots, e_{u+k}\}$ be the set of the first $u + k + 1$ of them. Then E is $\beta + \omega$ -large because its elements are $\leq$ corresponding elements of $\{x \in A : u \leq x\}$.

Assume $\forall \beta' < \beta \ T(\beta', 1)$ and $\text{RM}(\beta) > 1$, we check $T(\beta, 1)$. So let A be $\beta + \omega$ -large. Let $u = (h^A)_\omega(a_0)$ as usual, so $A = \{x \in A : x \leq u\} \cup \{x \in A : u \leq x\}$. The first of these sets is ω -large and the second one is β -large, i.e., $\{\beta\}(u)$ -large. By $\text{RM}(\beta) > 1$ we have (i) $\{\beta\}(u) \gg \omega$ and (ii) $\{\beta\}(u) + \omega < \beta$. Let B be a not ω -large subset of A . The set A is $\{\beta\}(u) + \omega$ -large and by $T(\{\beta\}(u), 1)$, $A \setminus B$ is $\{\beta\}(u)$ -large. Observe that $\min(A \setminus B) = c_0 \leq u$, for otherwise $\{x \in A : x \leq u\} \subseteq B$, so B is ω -large contrary to the assumption. If $c_0 = u$ then obviously $A \setminus B$ is β -large, so assume that $c_0 < u$. By Lemma 1.4.1, point 5, $\{\beta\}(u) \Rightarrow_u \{\beta\}(c_0)$, so $\{\beta\}(u) \Rightarrow_{c_0} \{\beta\}(c_0)$ by the same lemma, part 7. By Lemma 1.5.2, $(h^{A \setminus B})_{\{\beta\}(c_0)}(c_0)$ exists, so $A \setminus B$ is β -large.

We show non limit step in the proof of Lemma 1.7.5, i.e.,

$$\forall \alpha [(\forall \beta \gg \omega^\alpha T(\beta, \alpha)) \Rightarrow (\forall \beta \gg \omega^{\alpha+1} T(\beta, \alpha + 1))].$$

Once again, the case $\beta = 0$ is obvious. Indeed, if A is $\omega^{\alpha+1}$ -large and B is its not $\omega^{\alpha+1}$ -large subset, then $A \setminus B$ is nonempty, so 0-large.

CASE $\beta = \omega^{\alpha+1}$. Let A be $\omega^{\alpha+1} + \omega^{\alpha+1}$ -large. Let $u = (h^A)_{\omega^{\alpha+1}}(a_0)$. Then $A = \{x \in A : x \leq u\} \cup \{x \in A : u \leq x\}$ and both of these sets are $\omega^{\alpha+1}$ -large. Let B be a not $\omega^{\alpha+1}$ -large subset of A . *Case 1.* $b_0 > u$. Then $\{x \in A : x \leq u\}$ is contained in $A \setminus B$, so this set is $\omega^{\alpha+1}$ -large. *Case 2.* $b_0 = u$. Then there exists $z \in A \setminus B$ with $z > u$ (otherwise $\{x \in A : u \leq x\} \subseteq B$ and hence B is $\omega^{\alpha+1}$ -large what contradicts the assumption), so $\{x \in A : x < u\} \cup \{z\}$ is contained in $A \setminus B$, so this set is $\omega^{\alpha+1}$ -large. *Case 3.* $a_0 < b_0 < u$. We let $c_0 = b_0 = \min B$ and $c_{i+1} = (h^B)_{\omega^\alpha}(c_i)$. This

induction breaks after r steps, where $r \leq b_0$, otherwise B is $\omega^{\alpha+1}$ -large. That is, the last c_i is c_{r-1} . We let $A_0 = A$ and $A_{i+1} = A_i \setminus \{x \in B : c_i \leq x < c_{i+1}\}$ and $A_r = A_{r-1} \setminus \{x \in B : c_{r-1} \leq x\}$. Observe that A_0 is $\omega^\alpha \cdot (a_0 + u)$ -large, and (by the inductive assumption), A_i is $\omega^\alpha \cdot (a_0 + u - i)$ -large. In particular, A_r is $\omega^\alpha \cdot a_0$ -large, i.e., $\omega^{\alpha+1}$ -large, indeed $a_0 = \min(A \setminus B)$. *Case 4.* $b_0 = a_0$. Arguing like in case 3 we see that $A \setminus B$ is $\omega^\alpha \cdot u$ -large. Thus if $\min(A \setminus B) = u$ this set is $\omega^{\alpha+1}$ -large. If $d = \min(A \setminus B) < u$ then $\{\omega^{\alpha+1}\}(u) \Rightarrow_d \{\omega^{\alpha+1}\}(d)$ by Lemma 1.4.1, part 3, hence $\omega^\alpha \cdot u \Rightarrow_d \omega^\alpha \cdot d$ by the same lemma, part 7. By Lemma 1.5.2, $(h^{A \setminus B})_{\omega^\alpha \cdot d}(d)$ exists (because $(h^{A \setminus B})_{\omega^\alpha \cdot u}(d)$ exists).

We prove the implication $T(\beta, \alpha + 1) \Rightarrow T(\beta + \omega^{\alpha+1}, \alpha + 1)$ for $\beta \gg \omega^{\alpha+1}$. So let A be $\beta + \omega^{\alpha+1} \cdot 2$ -large and let B be its not $\omega^{\alpha+1}$ -large subset. Let $u = (h^A)_{\omega^{\alpha+1}}(a_0)$ and $w = (h^A)_{\beta + \omega^{\alpha+1}}(a_0)$. *Case 1.* $b_0 \geq u$. Let $A' = \{x \in A : x \geq u\}$. Then $B \subseteq A'$. By $T(\beta, \alpha + 1)$, $A' \setminus B$ is β -large, hence $A \setminus B = (\{x \in A : x < u\} \cup \{c_0\}) \cup (A' \setminus B)$, where $c_0 = \min(A' \setminus B)$, is $\beta + \omega^{\alpha+1}$ -large. *Case 2.* $a_0 < b_0 < u$. We put $d_0 = b_0 = \min B$ and $d_{i+1} = (h^B)_{\omega^\alpha}(d_i)$. Let r be the greatest i such that d_i exists. We must have $r < b_0$ for otherwise B would be $\omega^{\alpha+1}$ -large. Let $D_i = \{x \in B : d_i \leq x < d_{i+1}\}$ and $D_r = \{x \in B : d_r \leq x\}$. Observe that none of these sets is ω^α -large. On the other hand A is $\beta + \omega^\alpha(u + a_0)$ -large. It follows that $A \setminus D_0$ is $\beta + \omega^\alpha \cdot (u + a_0 - 1)$ -large, etc., $A \setminus B = A \setminus \bigcup_{i \leq r} D_i$ is $\beta + \omega^\alpha \cdot (u + a_0 - r)$ -large. But $r + 1 \leq u$, hence $A \setminus B$ is $\beta + \omega^\alpha \cdot a_0$ -large, so it is $\beta + \omega^{\alpha+1}$ -large because its minimum is a_0 . *Case 3.* $b_0 = a_0$. Exactly as above, by subtracting B from A in parts which are not ω^α -large we derive that $A \setminus B$ is $\beta + \omega^\alpha \cdot u$ -large. Indeed, there are only a_0 parts as above because $\min B = a_0$ and this set is not $\omega^{\alpha+1}$ -large. If $\min(A \setminus B) = u$ then we are done. Otherwise $e = \min(A \setminus B) < u$. But $\omega^\alpha \cdot u \Rightarrow_1 \omega^\alpha \cdot e$ by Lemma 1.4.1, part 5, and hence $\omega^\alpha \cdot u \Rightarrow_e \omega^\alpha \cdot e$ by part 7 of the same lemma. By Lemma 1.5.2, $(h^{A \setminus B})_{\omega^\alpha \cdot e}(e)$ exists because $(h^{A \setminus B})_{\omega^\alpha \cdot u}(e)$ exists.

Thus in order to prove the nonlimit step $\alpha + 1$ in the proof of Lemma 1.7.5 it remains to check the case $\text{RM}(\beta) > \alpha + 1$. So let $\text{RM}(\beta) > \alpha + 1$ and assume that for all $\beta' < \beta$, $T(\beta', \alpha + 1)$ holds. Let A be $\beta + \omega^{\alpha+1}$ -large and let B be its not $\omega^{\alpha+1}$ -large subset. As usual, we let $u = (h^A)_{\omega^{\alpha+1}}(a_0)$, so that $A = \{x \in A : x \leq u\} \cup \{x \in A : u < x\}$, the first of these sets is $\omega^{\alpha+1}$ -large, the second being β -large. It follows that A is $\{\beta\}(u) + \omega^{\alpha+1}$ -large. Observe that $u = (h^A)_{\omega^{\alpha+1}}(a_0) \geq \text{psn}(\alpha + 1)$. By Lemma 1.4.2, $\{\beta\}(u) \gg \omega^{\alpha+1}$ and $\{\beta\}(u) + \omega^{\alpha+1} < \beta$. By $T(\{\beta\}(u), \alpha + 1)$, $A \setminus B$ is $\{\beta\}(u)$ -large. Observe that $\min(A \setminus B) = c \leq u$, otherwise $\{x \in A : x \leq u\} \subseteq B$, so B is $\omega^{\alpha+1}$ -large, contrary to the assumption. If $c = u$ then we are done, $A \setminus B$ is $\{\beta\}(\min(A \setminus B))$ -large. So assume that $c < u$. Then $(h^{A \setminus B})_{\{\beta\}(u)}(c) \downarrow$. Also we have $\{\beta\}(u) \Rightarrow_1 \{\beta\}(c)$ by Lemma 1.4.1, part 5, hence, by the same lemma, part 7, $\{\beta\}(u) \Rightarrow_c \{\beta\}(c)$. By Lemma 1.5.2, $(h^{A \setminus B})_{\{\beta\}(c)}(c) \downarrow$ and $A \setminus B$ is β -large.

CASE α limit. So, by the assumption we have $\forall \alpha' < \alpha \forall \beta \gg \omega^{\alpha'} T(\beta, \alpha')$, we want to prove $\forall \beta \gg \omega^\alpha T(\beta, \alpha)$. As usual, case $\beta = 0$ is obvious.

Let $\beta = \omega^\alpha$. Let A be $\omega^\alpha + \omega^\alpha$ -large and let B be its not ω^α -large subset. As usual, let $u = (h^A)_{\omega^\alpha}(a_0)$. *Case 1.* $b_0 > u$. Then $\{x \in A : x \leq u\} \subseteq A \setminus B$, so this set is ω^α -large as required. *Case 2.* $b_0 = u$. Then there exists $z > u$ with $z \in A \setminus B$,

for otherwise $\{x \in A : u \leq x\} \subseteq B$, so B is ω^α -large contrary to the assumption. Thus $\{x \in A : x < u\} \cup \{z\} \subseteq A \setminus B$ and $A \setminus B$ is ω^α -large. *Case 3.* $a_0 < b_0 < u$ (the main case). Let $D = \{x \in A : b_0 \leq x\}$ and let $E = \{x \in A : u \leq x\}$. Then E is ω^α -large, i.e. it is $\omega^{\{\alpha\}(u)}$ -large. It follows that D is $\omega^{\{\alpha\}(u)}$ -large, indeed, it contains E . By part 4 of Lemma 1.7.3, D is $\{\omega^\alpha\}(b_0) \cdot b_0$ -large, in particular, it is $\{\omega^\alpha\}(b_0) + \{\omega^\alpha\}(b_0)$ -large (reason: $a_0 < b_0$, hence $b_0 > 1$). We apply the inductive assumption $T(\{\omega^\alpha\}(b_0), \{\alpha\}(b_0))$ and infer that $D \setminus B$ is $\{\omega^\alpha\}(b_0)$ -large. By part 3 of Lemma 1.5.3, $A \setminus B$ is $\{\omega^\alpha\}(b_0)$ -large. We also have $\{\omega^\alpha\}(b_0) \Rightarrow_{a_0} \{\omega^\alpha\}(a_0)$ by Lemma 1.4.1, part 7, hence $A \setminus B$ is $\{\omega^\alpha\}(a_0)$ -large, i.e., ω^α -large. *Case 4.* $b_0 = a_0$. In this case B is not $\{\omega^\alpha\}(a_0)$ -large. But A is $\{\omega^\alpha\}(u) + \{\omega^\alpha\}(a_0)$ -large. By the inductive assumption $T(\{\omega^\alpha\}(u), \{\alpha\}(a_0))$, $A \setminus B$ is $\{\omega^\alpha\}(u)$ -large. Let $s = \min(A \setminus B)$. If $s = u$ then we are done. If $s < u$ then $\{\omega^\alpha\}(u) \Rightarrow_s \{\omega^\alpha\}(s)$, hence $A \setminus B$ is $\{\omega^\alpha\}(s)$ -large, i.e., ω^α -large. The case $s > u$ cannot happen, for if it does then $\{x \in A : x \leq u\} \subseteq B$, so B is ω^α -large, contrary to the assumption.

Assume $T(\beta, \alpha)$, where $\beta \gg \omega^\alpha$, we prove $T(\beta + \omega^\alpha, \alpha)$. So let a set A be $\beta + \omega^\alpha + \omega^\alpha$ -large and let B be its not ω^α -large subset. Let u, w be as before, i.e., $u = (h^A)_{\omega^\alpha}(a_0)$ and $w = (h^A)_{\omega^\alpha}(u)$. *Case 1.* $b_0 \geq u$. Then $B \subseteq \{x \in A : u \leq x\}$ and by the inductive assumption $T(\beta, \alpha)$, $\{x \in A \setminus B : u \leq x\}$ is β -large. It follows that $A \setminus B = \{x \in A : x \leq u\} \cup \{x \in A \setminus B : u \leq x\}$ is $\beta + \omega^\alpha$ -large. *Case 2.* $a_0 < b_0 < u$. Let $E = \{x \in A : u \leq x\}$ and $D = \{x \in A : b_0 \leq x\}$. Then E is $\beta + \omega^\alpha$ -large, hence it is $\beta + \{\omega^\alpha\}(u)$ -large. It follows that D is $\beta + \{\omega^\alpha\}(u)$ -large as well. Exactly as above, it follows that D is $\beta + \{\omega^\alpha\}(b_0) \cdot b_0$ -large, hence it is $\beta + \{\omega^\alpha\}(b_0) \cdot 2$ -large. By the inductive assumption $T(\beta + \{\omega^\alpha\}(b_0), \{\alpha\}(b_0))$, $D \setminus B$ is $\beta + \{\omega^\alpha\}(b_0)$ -large, i.e. $\beta + \omega^\alpha$ -large. Hence $A \setminus B$ is $\beta + \omega^\alpha$ -large as a superset of $D \setminus B$. *Case 3.* $b_0 = a_0$. Then B is not $\{\omega^\alpha\}(a_0)$ -large. By the inductive assumption $T(\beta + \omega^\alpha + \{\omega^\alpha\}(a_0), \{\alpha\}(a_0))$, $A \setminus B$ is $\beta + \omega^\alpha$ -large.

Finally, let $\text{RM}(\beta) > \alpha$. Let, as usual, A be $\beta + \omega^\alpha$ -large and let B be its not ω^α -large subset. Let also $u = (h^A)_{\omega^\alpha}(a_0)$. Clearly $u \geq \text{psn}(\alpha + 1)$, hence A is $\{\beta\}(u) + \omega^\alpha$ -large. By Lemma 1.4.2, $\{\beta\}(u) \gg \omega^\alpha$ and $\{\beta\}(u) + \omega^\alpha < \beta$. By the inductive assumption $T(\{\beta\}(u), \alpha)$, $A \setminus B$ is $\{\beta\}(u)$ -large. Let $s = \min(A \setminus B)$. Exactly as above, $s \leq u$ for otherwise $\{x \in A : x \leq u\} \subseteq B$, so B is ω^α -large contrary to the assumption. If $s = u$ then we are done. Otherwise, $s < u$, hence $\{\beta\}(u) \Rightarrow_s \{\beta\}(s)$, so $A \setminus B$ is $\{\beta\}(s)$ -large, i.e. β -large. $\square$

Let us observe that the result of Theorem 1.7.1 is optimal. The reason is that if $\beta \gg \omega^\alpha$ and A is $\omega^\beta \cdot \alpha$ -small then by letting $b_0 = \min A$, $b_{j+1} = h_{\omega^\beta, j}(\min A)$ we obtain a sequence such that $B = \{b_j : j \text{ is such that } b_j \downarrow\}$ is α -small and the partition $A = \bigcup_j (A \cap [b_j, b_{j+1}))$ (the last of these sets should be $\{x \in A : x \geq \max B\}$) admits no ω^β -large monochromatic set.

It is also not clear whether one can introduce a sensible notion of a partition of m -tuples being α -large, $m \geq 2$.

1.8 Iterations of Hardy Functions

Let the function h satisfy the usual assumptions, thus we worked out in Sect. 1.5 its Hardy iterations h_β for $\beta < \varepsilon_0$. Let us give an information about Hardy iterations of the function h_β , or to be more specific, of h_{ω^β} . Thus we have $(h_\beta)_0 = \text{id}$, $(h_\beta)_{\alpha+1} = (h_\beta)_\alpha \circ h_\beta$ and $(h_\beta)_\lambda(b) = (h_\beta)_{\{\lambda\}(b)}(b)$ for any fixed $\beta < \varepsilon_0$.

We shall use the following observations only in Sect. 4.

Lemma 1.8.1 *If $\alpha, \beta < \varepsilon_0$ and $\beta \gg \text{LM}(\alpha)$ then $(h_{\omega^\beta})_\alpha = h_{\omega^\beta \cdot \alpha}$.*

Proof By induction on α , using Lemma 1.7.4 in limit steps. $\square$

In particular, $(h_{\omega^k})_{\omega^k} = h_{\omega^{2k}}$, that is, $(h_{\omega_1(k)})_{\omega_1(k)} = h_{\omega_1(2k)}$. For $m > 1$ the result is less exact.

Lemma 1.8.2 *For $m, b \geq 2$ we have $(h_{\omega_m(k)})_{\omega_m(k)}(b) \leq h_{\omega_m(k+1)}(b)$.*

Proof By Lemma 1.8.1

$$(h_{\omega_m(k)})_{\omega_m(k)}(b) = h_{\omega_m(k) \cdot 2}(b) = h_{\omega_{\omega_{m-1}(k) \cdot 2}}(b),$$

hence by Lemma 1.5.2 it suffices to show that

$$\omega^{\omega_{m-1}(k+1)} \Rightarrow_b \omega^{\omega_{m-1}(k) \cdot 2}. \quad (*)$$

We proceed by induction on m . Let $m = 2$. Thus we must show that $\omega^{\omega^{k+1}} \Rightarrow_b \omega^{\omega^k \cdot 2}$. Consider the sequence $\omega^{\omega^{k+1}}, \omega^{\omega^k \cdot b}$. If $b = 2$ then this sequence has the desired properties. Otherwise $\omega^k \cdot b \Rightarrow_b \omega^k \cdot 2$ by Lemma 1.4.1, part 3, hence by part 4 of the same lemma we have $\omega^{\omega^k \cdot b} \Rightarrow_b \omega^{\omega^k \cdot 2}$ and we may take concatenation of these two sequences.

Assume $(*)$ for m . By the inductive assumption and part 4 of Lemma 1.4.1 we have

$$\omega^{\omega^{\omega_{m-1}(k+1)}} \Rightarrow_b \omega^{\omega^{\omega_{m-1}(k) \cdot 2}},$$

hence it suffices to show that we may go down one step with this 2, i.e., to show that

$$\omega^{\omega^{\omega_{m-1}(k) \cdot 2}} \Rightarrow_b \omega^{\omega^{\omega_{m-1}(k)} \cdot 2}.$$

By Lemma 1.7.3 part 1 we have $\omega_{m-1}(k) \cdot 2 \Rightarrow_b \omega_{m-1}(k) + 1$, hence $\omega^{\omega_{m-1}(k) \cdot 2} \Rightarrow_b \omega^{\omega_{m-1}(k)+1}$, so $\omega^{\omega^{\omega_{m-1}(k) \cdot 2}} \Rightarrow_b \omega^{\omega^{\omega_{m-1}(k)+1}}$. Once again we go down from b to 2 as above. $\square$

Let us restate these facts in terms of large sets.

Lemma 1.8.3 *1. A is ω^{2k} -large iff it has an ω^k -large subset B such that for every $b \in B \setminus \{\max B\}$, the interval $A \cap [b, h^B(b)]$ of A is ω^k -large.*

2. If $m > 1$ and A is $\omega_m(k+1)$ -large then it has an $\omega_m(k)$ -large subset B such that for every $b \in B \setminus \{\max B\}$ the interval $A \cap [b, h^B(b)]$ is $\omega_m(k)$ -large.
3. If $m > 0$ and A is $\omega_m(k+1)$ -large and C is the set of all elements of A which have even indices (in the increasing enumeration of A), then C is $\omega_m(k)$ -large.

Proof The first two parts are just restatements of previous two lemmas, the third one is immediate by the second part. Indeed, let B be as in the second part. Let $b = \text{Card}(B)$ and let $C' = \{a_{2i} : i < b\}$. Then $a_{2i} \downarrow$ for $i < b$, indeed, $a_{2i} \leq b_i$, hence C' is $\omega_m(k)$ -large by Lemma 1.5.3, so C is $\omega_m(k)$ -large as well by the same lemma. $\square$

1.9 An Upper Bound

In this section we prove the following result (due to Ketonen and Solovay [1], the version for the notion of largeness determined by Hardy hierarchy is taken from [13]).

For every $\alpha < \varepsilon_0$ and every $c \in \mathbb{N} \setminus \{0\}$ we define $\omega_{(0)}(\alpha, c) = 1$, $\omega_{(1)}(\alpha, c) = \omega^\alpha \cdot c$, $\omega_{(2)}(\alpha, c) = \omega^{\omega_{(1)}(\alpha, c)}$, $\omega_{(n+1)}(\alpha, c) = \omega^{\omega_{(n)}(\alpha, c) \cdot 3}$.

Theorem 1.9.1 *Let $n \in \mathbb{N} \setminus \{0\}$, let A be a set $\omega_{(n)}(\alpha, c)$ -large, where $\alpha < \varepsilon_0$, $c < \min(A)$. If $P : [A]^n \rightarrow (< c)$ is a partition of the set $[A]^n$ into at most c parts then there exists an ω^α -large homogeneous set.*

We change the notion of tree. Say that $\langle A, < \rangle$ is a *tree (in the number-theoretic sense)* if A is a finite subset of $\mathbb{N}$, the relation $<$ is a tree (in the usual set-theoretic sense) on A and $x < y$ implies $x < y$ for all $x, y \in A$. From now on we use the notion of a tree in the sense introduced above. If we have in mind the notion from Sect. 1.1 then we will make it explicit. Let $\gamma < \varepsilon_0$. We say that the tree $\langle A, < \rangle$ is γ -large if its underlying set A is γ -large. We say that the tree $\langle A, < \rangle$ is γ -unbranching if for every $a \in A$, $\{a\} \cup \{b \in A : b \text{ is an immediate successor of } a\}$ is not γ -large. In particular, at most binary tree is 3-unbranching in this terminology. We remark that G. Mills [14] studied trees (in the above sense) in their own right, we prove below only some results on such trees which will be needed in Ramsey style results.

Theorem 1.9.2 *If the tree $\langle A, < \rangle$ is ω^α -large, ω -unbranching and $\min A > 1$, then it has a branch G such that $G \setminus \{\max G\}$ is α -large.*

In order to prove Theorem 1.9.2 we shall prove the following:

Lemma 1.9.3 *For every α we have: for every $\beta \gg \alpha$ for every tree $\langle A, < \rangle$, if A is $\omega^{\beta+\alpha}$ -large, the tree $\langle A, < \rangle$ is ω -unbranching, and $\min A > 1$, then there exists $c \in A$ such that $\{a \in A : a \leq c\}$ is α -large and $\{a \in A : c \leq a\}$ is ω^β -large.*

Lemma 1.9.3 implies Theorem 1.9.2 immediately (just substitute $\beta = 0$).

Proof of Lemma 1.9.3 By induction on α . If $\alpha = 0$ then $c = \min A$, i.e., the root of A satisfies our demand.

Assume the lemma for α , we prove it for $\alpha + 1$. So let $\beta \gg \alpha + 1$, so $\beta \gg \alpha$. Let the tree $\langle A, < \rangle$ be $\omega^{\beta+\alpha+1}$ -large. Let $a = \min A$ be its root. Let U_1 denote its

first level, i.e., the set of all immediate successors of a . By the assumption, $\{a\} \cup U_1$ is not ω -large, so has at most a elements. It follows that U_1 has strictly less than a elements. But the tree A itself is $\omega^{\beta+\alpha+1}$ -large, so is $\omega^{\beta+\alpha} \cdot a$ -large. It follows that $A \setminus \{a\}$ is $\omega^{\beta+\alpha} \cdot (a-1)$ -large. Moreover, we have a partition $A \setminus \{a\} = \cup_{u \in U_1} B_u$, where $B_u = \{x \in A : u \leq x\}$. By the result of Sect. 1.7, at least one of these parts, say B_{u_0} , is $\omega^{\beta+\alpha}$ -large. By the inductive assumption, there exists $c \in B_{u_0}$ such that $\{x \in B_{u_0} : x \leq c\}$ is α -large and $\{x \in B_{u_0} : c \leq x\}$ is ω^β -large. This c satisfies our demand.

Assume the lemma for all $\alpha < \lambda$, λ limit. Let $\beta \gg \lambda$. Then $\beta \gg \{\lambda\}(a)$ for all a , in particular for $a = \min A$. The tree $\langle A, < \rangle$ is $\omega^{\beta+\{\lambda\}(a)}$ -large, so by the inductive assumption there exists $c \in A$ such that $\{x \in A : x \leq c\}$ is $\{\lambda\}(a)$ -large and $\{x \in A : c \leq x\}$ is ω^β -large; this c satisfies our demand. $\square$

Now we repeat the construction from Sect. 1.2. Let $P : [A]^2 \rightarrow c$ be a partition of (increasing) pairs of elements of A , as usual we shall also use the notation $[A]^2 = \cup_{i < c} B_i$, where $B_i = \{(x, y) \in [A]^2 : P(x, y) = i\}$ for $i < c$. This partition determines an ordering $<$ on A so that $\langle A, < \rangle$ is a tree. We write $A = \{a_0, \dots, a_{s-1}\}$ in increasing order and define a sequence $<_m$ of relations on $\{a_0, \dots, a_m\}$. We let $<_0 = \emptyset$, $<_1 = \{(a_0, a_1)\}$. Further steps are inductive. We let $a_i <_{m+1} a_j$ iff

$$(i < j \leq m \text{ \& } a_i <_m a_j) \vee \\ (i \leq m \text{ \& } j = m+1 \text{ \& } \forall t [a_t <_m a_i \Rightarrow P(a_t, a_i) = P(a_t, a_j)]).$$

The relation $< \stackrel{\text{def}}{=} <_{s-1}$ is a tree on A . Its main property is:

$$\forall x, y \in A [x < y \equiv (x < y \text{ \& } \forall z < x P(z, x) = P(z, y))]. \quad (1.8)$$

We claim that every $b \in A$ has at most c immediate successors in the tree $\langle A, < \rangle$. So let $b = a_m \in A$. Put $f(i) = \min j > m : P(a_m, a_j) = i$. This function is defined on a subset of $(< c)$ and every $u > b$ is $<$ -greater than some $f(i)$. It follows that if $\min A > c$ then this tree is ω -unbranching. By Theorem 1.9.2 we obtain:

Lemma 1.9.4 *If $\min A > c$, A is ω^α -large and $[A]^2 = \cup_{i < c} B_i$ is a partition of $[A]^2$ into c parts, then the tree $\langle A, < \rangle$ has a branch G such that $G \setminus \{\max G\}$ is α -large.* $\square$

Let $G = \{a_{i_0}, \dots, a_{i_r}\}$ be a branch in the tree $\langle A, < \rangle$. Thus, by (1.8), we have

$$\forall x, y \in G [x < y \Rightarrow \forall z < x P(z, x) = P(z, y)].$$

It follows that for $z \in G \setminus \{\max G\}$ the following function, F , is well defined: $F(z) =$ the unique i such that for every $x \in G$, $x > z$ we have $P(z, x) = i$. This function F determines a partition of $G \setminus \{\max(G)\}$ into c parts. By the results of Sect. 1.7 we obtain:

Lemma 1.9.5 *Under the notation introduced above, if the branch G is such that $G \setminus \{\max G\}$ is $\omega^\alpha \cdot c$ -large then there exists an ω^α -large set which is homogeneous for F .*

Let us sum up.

Theorem 1.9.6 *Let A be an $\omega^{\omega^\alpha \cdot c}$ -large set and let $P : [A]^2 \rightarrow (< c)$ be a partition of $[A]^2$ into c parts as indicated. Assume also $\min A > c$. Then there exists an ω^α -large homogeneous set for this partition.*

Proof If A is $\omega^{\omega^\alpha \cdot c}$ -large and P is a partition of $[A]^2$ into at most c parts, then there exists a branch G in the tree $\langle A, < \rangle$ such that $G \setminus \{\max G\}$ is $\omega^\alpha \cdot c$ -large, by Lemma 1.9.4. By Lemma 1.9.5, the partition F of this branch (without its maximum) as described above, has an ω^α -large homogeneous set. It is easy to check that such a set is homogeneous for the original partition P , this follows from (1.8). $\square$

The following is an analogue of the arrow notation from Sect. 1.2.

$\alpha \rightarrow (\beta)_c^n$ iff for every α -large set A with $\min A > c$ and every partition $P : [A]^n \rightarrow (< c)$ there exists a β -large homogeneous set.

Theorem 1.9.6 may be stated as the following partition property:

$$\omega^{\omega^\alpha \cdot c} \rightarrow (\omega^\alpha)_c^2. \quad (1.9)$$

Let us go to a proof of a version of the Ramsey theorem for n -tuples. As usual at first we work out a lemma about trees. We need one more notation. Let $\alpha < \varepsilon_0$. Let

$$\alpha = \omega^{\alpha_0} \cdot a_0 + \cdots + \omega^{\alpha_s} \cdot a_s$$

be the Cantor normal form expansion of α , i.e., $\alpha > \alpha_0 > \cdots > \alpha_s$. For every $n \in \mathbb{N}$ we define $\alpha(\cdot)n = \omega^{\alpha_0} \cdot (a_0 \cdot n) + \cdots + \omega^{\alpha_s} \cdot (a_s \cdot n)$.

Theorem 1.9.7 *For every $\alpha < \varepsilon_0$, every $n \in \mathbb{N} \setminus \{0\}$ and every tree $\langle A, < \rangle$, which is ω^n -unbranching and A is $\omega^{\alpha(\cdot)n}$ -large and $\min(A) > 1$ there exists a branch B in $\langle A, < \rangle$ such that $B \setminus \{\max(B)\}$ is α -large.*

Theorem 1.9.7 is a corollary to the following lemma.

Lemma 1.9.8 *For every $\alpha < \varepsilon_0$ for every $n \in \mathbb{N} \setminus \{0\}$, every $\beta < \varepsilon_0$ and every tree $\langle A, < \rangle$ such that*

1. $\{\beta\}(\min(A)) \gg \alpha$,
2. $\min(A) > 1$,
3. The tree $\langle A, < \rangle$ is ω^n -unbranching,
4. The set A is $\omega^{\beta + \alpha(\cdot)n}$ -large,

there exists $c \in A$ such that $\{x \in A : x \preceq c\}$ is α -large and $\{x \in A : c \preceq x\}$ is ω^β -large.

The proof of Lemma 1.9.8 will be inductive on α . In the limit step we shall need the following lemma.

Lemma 1.9.9 *For every $\alpha < \varepsilon_0$ every $\beta < \varepsilon_0$ for every set A such that*

1. $\{\beta\}(\min(A)) \gg \text{LM}(\alpha)$
2. $\min(A) > 1$
3. A is $\omega^\beta \cdot \alpha$ -large

then A is $\omega^{\{\beta\}(\min(A))} \cdot \alpha$ -large.

Lemma 1.9.9 is a particular case of the following observation.

Lemma 1.9.10 *For every $\alpha, \beta < \varepsilon_0$ and every A such that $\{\beta\}(\min(A)) \gg \text{LM}(\alpha)$ and every $x \in A$, if $h_{\omega^\beta \cdot \alpha}(x) \downarrow$, then $h_{\omega^{\{\beta\}(\min(A))} \cdot \alpha}(x) \downarrow$.*

Proof of Lemma 1.9.10 (by induction on α). If $\alpha = 0$ then the conclusion is obvious. Assume the conclusion for α . Pick β and A such that $\{\beta\}(a) \gg \text{LM}(\alpha + 1) = \text{LM}(\alpha)$, where $a = \min(A)$. By Lemma 1.5.2, parts 4 and 7, for every $x > a$ $\omega^\beta \Rightarrow_x \omega^{\{\beta\}(a)}$. It follows that

$$\text{for every } x \in A, \text{ if } h_{\omega^\beta}(x) \downarrow, \text{ then } h_{\omega^{\{\beta\}(a)}}(x) \downarrow \text{ and } h_{\omega^{\{\beta\}(a)}}(x) \leq h_{\omega^\beta}(x) \quad (1.10)$$

(see part 1 of Lemma 1.5.3). Let $x \in A$ and $h_{\omega^\beta \cdot (\alpha+1)}(x) \downarrow$. Then we have $h_{\omega^\beta \cdot \alpha}(h_{\omega^\beta}(x)) \downarrow$. By (1.10) and the inductive assumption we get $h_{\omega^{\{\beta\}(a)} \cdot \alpha}(h_{\omega^{\{\beta\}(a)}}(x)) \downarrow$, so the conclusion for $\alpha + 1$ holds.

Let λ be limit and assume $\{\beta\}(a) \gg \text{LM}(\lambda)$. Assume the conclusion for all $\alpha < \lambda$. If $x \in A$ and $h_{\omega^\beta \cdot \lambda}(x) \downarrow$ then $h_{\omega^\beta \cdot \lambda}(x) \downarrow$. But for every $\gamma \gg \text{LM}(\lambda)$ and every x , $\{\omega^\gamma \cdot \lambda\}(x) = \omega^\gamma \{\lambda\}(x)$ (see Lemma 1.7.4). By the inductive assumption we get the conclusion for λ . $\square$

Proof of Lemma 1.9.8 (by induction on α). Let $\alpha = 0$. Let n, β and the tree $\langle A, < \rangle$ satisfy the hypothesis. Then $c = \min(A)$ has the desired property. Assume the conclusion for α . Let the set A be $\omega^{\beta+(\alpha+1)(\cdot)^n}$ -large. Let $a = \min(A) > 1$, $\{\beta\}(a) \gg \alpha + 1$ and let the tree $\langle A, < \rangle$ be ω^n -unbranching. Let $W_1 = \{a_0, \dots, a_k\}$ be the set consisting of a and all its immediate successors in the tree $\langle A, < \rangle$. Assume also that $a_0 = a$. Consider the partition $A = \bigcup_{i \leq k} B_i$, where $B_0 = \{a_0\}$, $B_i = \{x \in A : a_i \leq x\}$ for $i > 0$. This partition is ω^n -small (because $\langle A, < \rangle$ is ω^n -unbranching), so by the main result of Sect. 1.7 there exists $i_0 \leq k$ such that B_{i_0} is $\omega^{\beta+\alpha(\cdot)^n}$ -large. Obviously, $i_0 \neq 0$. By the inductive assumption applied to the tree $\langle B_{i_0}, < \rangle$ there exists c such that $\{x \in B_{i_0} : x \leq c\}$ is α -large and $\{x \in B_{i_0} : c \leq x\}$ is ω^β -large. This c has the desired property in the original tree $\langle A, < \rangle$.

Assume the conclusion for all $\alpha < \lambda$, where λ is limit. Let

$$\lambda = \omega^{\alpha_1} \cdot a_1 + \dots + \omega^{\alpha_s} a_s$$

be the Cantor expansion of λ , i.e. $\alpha_1 > \dots > \alpha_s$. Let us denote

$$\gamma = \omega^{\alpha_1} a_1 + \dots + \omega^{\alpha_s} \cdot (a_s - 1),$$

so that $\lambda = \gamma + \omega^{\alpha_s}$ and $\gamma \gg \omega^{\alpha_s}$. Let A be an $\omega^{\beta+\lambda(\cdot)n}$ -large, where $\beta \gg \lambda$. Hence A is $\omega^{\beta+\gamma(\cdot)n+\omega^{\alpha_s} \cdot n}$ -large. We apply Lemma 1.9.9 to the ordinals

$$\begin{array}{ll} \beta + \gamma(\cdot)n + \omega^{\alpha_s} \cdot (n-1) & \text{and } \omega^{\{\omega^{\alpha_s}\}(a)} \\ \beta + \gamma(\cdot)n + \omega^{\alpha_s} \cdot (n-2) & \text{and } \omega^{\{\omega^{\alpha_s}\}(a) \cdot 2} \\ \dots\dots\dots & \dots\dots\dots \\ \beta + \gamma(\cdot)n + \omega^{\alpha_s} & \text{and } \omega^{\{\omega^{\alpha_s}\}(a) \cdot (n-1)} \end{array}$$

and infer that the set A is $\omega^{\beta+\gamma(\cdot)n+\{\omega^{\alpha_s}\}(a) \cdot n}$ -large. Hence A is $\omega^{\beta+\{\lambda\}(a)(\cdot)n}$ -large. By the inductive assumption there exists c with the desired properties. $\square$

Once again we apply the same proof of Ramsey theorem.

Proof of Theorem 1.9.1 By induction on n . Case $n = 1$ is the main result of Sect. 1.7. The case $n = 2$ was proved above. Assume the conclusion for n , we derive it for $n + 1$. Let A be an $\omega_{(n+1)}(\alpha, c)$ -large set, where $\alpha < \varepsilon_0$ and $c < \min(A)$. Let $P : [A]^{n+1} \rightarrow (< c)$ be a partition of (increasing) $n + 1$ -tuples of elements of A , we shall also use the notation $[A]^{n+1} = \cup_{i < c} B_i$, where $B_i = \{(x_0, \dots, x_n) \in [A]^{n+1} : P(x, y) = i\}$ for $i < c$. This partition determines an ordering $<$ on A so that $\langle A, < \rangle$ is a tree. We write $A = \{a_0, \dots, a_{s-1}\}$ in increasing order and define a sequence $<_m$ of relations on $\{a_0, \dots, a_m\}$. We let $<_0 = \emptyset$, $<_1 = \{(a_0, a_1)\}$. Further steps are inductive. We let $a_i <_{m+1} a_j$ iff

$$\begin{aligned} & (i, j \leq m \text{ \& } a_i <_m a_j) \vee \\ & \{i \leq m \text{ \& } j = m+1 \text{ \& } \forall t_0, \dots, t_{n-1} [a_{t_0} <_m \dots <_m a_{t_{n-1}} <_m a_i \Rightarrow \\ & P(a_{t_0}, \dots, a_{t_{n-1}}, a_i) = P(a_{t_0}, \dots, a_{t_{n-1}}, a_j)]\}. \end{aligned}$$

Once again, it is well known (and easy to verify) that the relation $< \stackrel{\text{def}}{=} <_{s-1}$ is a tree on A . Its main property is:

$$\begin{aligned} \forall x, y \in A [x < y \equiv (x < y \text{ \& } \forall z_0 < \dots < z_{n-1} < x \\ P(z_0, \dots, z_{n-1}, x) = P(z_0, \dots, z_{n-1}, y))]. \end{aligned} \quad (1.11)$$

Let, for each $x \in A$, $\text{rank}(x) = \text{Card}(\{y \in A : y < x\})$, the *rank* of x in the tree $\langle A, < \rangle$ and W_x denote the set consisting of x and all its immediate successors. It is easy to see that for every $x \in A$ if $x < a_{n-1}$ then $\text{Card}(W_x) = 2$, if $x \geq a_{n-1}$ then $\text{Card}(W_x) \leq c^{\binom{\text{rank}(x)+1}{n}} + 1$. The binomial coefficient in the exponent is just the cardinality of the set of all n -tuples of elements of the set $\{z \in A : z \leq x\}$. Each set W_x is ω^3 -small because $c < a_0 \leq x$ and $\text{rank}(x) \leq x$. By Theorem 1.9.7 the tree $\langle A, < \rangle$ has a branch G such that $G \setminus \{\max(G)\}$ is $\omega_{(n)}(\alpha, c)$ -large. Let $G = \{a_{i_0}, \dots, a_{i_r}\}$ be such a branch in the tree $\langle A, < \rangle$. Thus, by (1.11), we have

$$\begin{aligned} & \forall x, y \in G [x < y \Rightarrow \\ & \forall z_0 < \dots < z_{n-1} < x P(z_0, \dots, z_{n-1}, x) = P(z_0, \dots, z_{n-1}, y)]. \end{aligned}$$

It follows that for an increasing n -tuple $z_0, \dots, z_{n-1} \in [G \setminus \{\max G\}]^n$ the function $F(z_0, \dots, z_{n-1}) = i$ such that for every $x \in G, x > z_{n-1}$ we have $P(z_0, \dots, z_{n-1}, x) = i$ is well defined. This function F determines a partition of $[G \setminus \{\max(G)\}]^n$ into c parts. Clearly, every set homogeneous for F is homogeneous for P . By the inductive assumption we infer the conclusion for $n + 1$. $\square$

Still using the notation taken from Ramsey theory, Theorem 1.9.1 may be stated as the following partition property:

$$\omega_{(n)}(\alpha, c) \rightarrow (\omega^\alpha)_c^n. \quad (1.12)$$

1.10 Some Lower Bounds: Hardy

In² this section we give some lower bounds for partition results for α -large sets. We suggest the reader to look through Sect. 1.3, indeed, below we shall mimic the arguments given there. The arguments given below are due to Ketonen and Solovay [1], but use has been made of some serious technical simplifications from section 6.3 in Graham, Rotschild and Spencer [2].

In Sect. 1.3 we used the structure of the set A under consideration given by sequences. As pointed out there, we could equally well work with expansions to the base m . The ordinals below ε_0 have a similar structure given by their Cantor normal form expansions, hence we shall work with ordinals and use the association KS as introduced in Sect. 1.5 to work with the set $A \subseteq \mathbb{N}$ under consideration.

It will be convenient to have some more notation concerning ordinals. We let $v_\delta(\alpha)$ be the coefficient of ω^δ in the Cantor normal expansion of α (and $v_\delta(\alpha) = 0$ if ω^δ does not occur in this expansion). Thus, if α is written in the form (1.6), then $v_{\alpha_i}(\alpha) = a_i$ for all $i \leq s$ and $v_\delta(\alpha) = 0$ for other δ . In particular, $v_\delta(0) = 0$ for every δ .

For ordinals $\alpha, \beta < \varepsilon_0$ we also define the leftmost difference in the expansions of ordinals α, β by

$$\text{LD}(\alpha, \beta) = \max \delta : v_\delta(\alpha) \neq v_\delta(\beta). \quad (1.13)$$

Let an ordinal μ and a set $A \subseteq \mathbb{N}$ be given and assume that A is μ -small. Thus, $\text{KS}(a) = \text{KS}^A(\mu; a)$ is defined for all $a \in A$ and is always greater than 0. We need some simple observations concerning the sequence μ_j used in the definition of the relation RKS.

Lemma 1.10.1 *For each μ, b we have $\text{psn}(\{\mu\}(b)) \leq \max(b, \text{psn}(\mu))$.*

Proof By induction on μ . If $\mu = 0$ the result is immediate. Also the successor step is obvious. If μ is of the form $\omega^{\tau+1}$ then $\{\mu\}(b) = \omega^\tau \cdot b$, hence $\text{psn}(\mu) = \max(b, \text{psn}(\tau))$, so the result holds. If μ is of the form ω^τ with limit τ , then $\{\mu(b)\} = \omega^{\{\tau\}(b)}$, so the result follows from the inductive assumption. Finally, if μ

²I would like to thank Teresa Bigorajska for discussions of the material of this section.

has nontrivial Cantor normal form expansion the result is immediate by the inductive assumption. $\square$

Corollary 1.10.2 *For every set A and every μ if μ_j and b_j are like in the definition of the relation RKS^A , then $\text{psn}(\mu_{j+1}) \leq \max(b_j, \text{psn}(\mu_j))$. In particular, if $\text{psn}(\mu) \leq \min A$, then for each j we have $\text{psn}(\mu_j) \leq b_j$. $\square$*

The analogue of the partition P , cf. (1.1), is defined by

$$\text{PO}(a, b) = \text{LD}(\text{KS}(a), \text{KS}(b)). \quad (1.14)$$

We shall also write $\text{PO}^A(\mu; a, b)$ if necessary. The partition PO has ordinals as values. This partition and the partition KE defined below (see formula (1.15)) depends on μ .

The next proposition will not be necessary in further considerations, but it may help the reader's intuition. (Recall the definition of $\ggg$ on Sect. 1.4).

Proposition 1.10.3 *Let $a \in A$ and let δ be an ordinal. The set $\{b \in A : a < b \text{ \& PO}(a, b) = \delta\}$ is empty iff $v_\delta(\text{KS}(a)) = 0$. If $v_\delta(\text{KS}(a)) = m > 0$ then we write $\text{KS}(a) = \varrho + \omega^\delta \cdot m + \xi$ with $\varrho \ggg \omega^\delta \ggg \xi$. Let $t \in A$ be such that $\text{RKS}(\varrho + \omega^\delta \cdot m, t)$ and let $s \in A$ be such that $\text{RKS}(\varrho, s)$. Then the above set is just $\{b \in A : t < b < s\}$.*

Proof Left to the reader. $\square$

The analogue of Lemma 1.3.1 is as follows.

Lemma 1.10.4 *Let $\mu = \omega^\eta$ and let A be a μ -small set such that $\text{psn}(\mu) \leq \min(A)$. Then every set $D \subseteq A$ homogeneous for PO is at most ω -large.*

Proof Let D be homogeneous for PO . Thus, there exists δ such that for every $(a, b) \in [D]^2$ we have $\text{PO}(a, b) = \delta$. We write $D = \{d_0, \dots, d_r\}$ in increasing order as usual. For each $i \leq r$ we write

$$\text{KS}(d_i) = \varrho_i + \omega^\delta \cdot m_i + \xi_i,$$

where $\varrho_i \ggg \omega^\delta \ggg \xi_i$. Observe that all ϱ_i 's are equal; for if $\varrho_i \neq \varrho_j$ then $\text{PO}(d_i, d_j) > \delta$. On the other hand all m_i are different, indeed, if $i \neq j$ and $m_i = m_j$ then $\text{PO}(d_i, d_j) < \delta$. The sequence $d_0, \dots, d_r$ is increasing, hence the sequence $\text{KS}(d_0), \dots, \text{KS}(d_r)$ is decreasing, so the sequence $m_0, \dots, m_r$ must be decreasing. It follows that the number of possible m_i 's does not exceed $1 + m_0 \leq 1 + \text{psn}(\text{KS}(d_0)) \leq 1 + d_0$ and the same applies to the cardinality of D . Thus, D has at most $d_0 + 1$ elements, i.e., it is at most ω -large. $\square$

Let

$$\text{KE}(a, b, c) = \begin{cases} 0 & \text{if } \text{PO}(a, b) > \text{PO}(b, c) \\ 1 & \text{if } \text{PO}(a, b) = \text{PO}(b, c) \\ 2 & \text{if } \text{PO}(a, b) < \text{PO}(b, c). \end{cases} \quad (1.15)$$

As usual, we shall write $\text{KE}^A(\mu; a, b, c)$ if needed.

The next lemma is variant of a result due to P. Erdős and G. Mills [15].

Lemma 1.10.5 *If A is ω^ω -small and $\min(A) \geq 1$ and D is homogeneous for KE then D has at most $2 + \min(D)$ elements.*

In the terminology introduced in Sect. 1.5, D is at most $\omega + 1$ -large in the sense of the slow growing hierarchy.

Proof Let $\mu = \omega^{a_0}$, where $a_0 = \min(A)$. If the color of $[D]^3$ is 1 then D is at most ω -large. Otherwise we let $D = \{d_0, \dots, d_r\}$ in increasing order and put $n_i = \text{PO}(d_i, d_{i+1})$ for $i < r$. These are natural numbers $\leq a_0$ and in both cases they form a one-to-one sequence, so there are at most $a_0 + 1$ of them. It follows that $r \leq 2 + a_0 \leq 2 + d_0$. $\square$

We begin working out the case $\mu > \omega^\omega$. Exactly like in Sect. 1.3 we shall now consider partitions of the family of possible values of PO. Thus we are obliged to work with partitions of sets of ordinals below ε_0 . After having some information about such partitions we shall transfer them to the set A under consideration by means of KS.

Let us make the convention that when working with partitions of sets of ordinals we write sets and sequences of them in *decreasing* order. This will be convenient because the mapping KS reverses the order, so this will fit the scheme of writing sets of integers in increasing order.

Let E denote the set of all ordinals (of course, as in all our considerations, ordinals below ε_0). The first partition to be considered is LD. This is a partition of $[E]^2$ into ε_0 parts. But if Z is the set of all ordinals below ω_k then the set of values of $\text{LD} \upharpoonright [Z]^2$ is the set of all ordinals below ω_{k-1} .

The next lemma is an analogue of Lemma 1.3.1.

Lemma 1.10.6 *Let $D \subseteq E$ be homogeneous for LD. Then $\text{Card}(D) \leq 1 + \text{psn}(\max(D))$.*

Proof Let D satisfy the assumption. Write $D = \{\delta_0, \dots, \delta_r\}$ in decreasing order. Thus $\max(D) = \delta_0$. By the assumption there exists γ such that $\text{LD}(\delta_i, \delta_j) = \gamma$ for all $i \neq j$. Write

$$\delta_i = \varrho_i + \omega^\gamma \cdot m_i + \xi_i,$$

where $\varrho_i \gg \omega^\gamma \gg \xi_i$. Then all ϱ_i 's are equal for if $\varrho_i \neq \varrho_j$ then $\text{LD}(\delta_i, \delta_j) > \gamma$. All m_i 's are different, indeed, if $i \neq j$ but $m_i = m_j$ then $\text{LD}(\delta_i, \delta_j) < \gamma$. Moreover, the sequence m_i is decreasing because the sequence δ_i is decreasing. Thus $r + 1$ (the number of elements of D) cannot exceed the number of possible m_i 's. But these cannot exceed $\text{psn}(\delta_0)$. $\square$

The analogue of the partition K (cf. (1.2))

$$\text{KO}(\alpha, \beta, \gamma) = \begin{cases} 0 & \text{if } \text{LD}(\alpha, \beta) < \text{LD}(\beta, \gamma) \\ 1 & \text{if } \text{LD}(\alpha, \beta) = \text{LD}(\beta, \gamma) \\ 2 & \text{if } \text{LD}(\alpha, \beta) > \text{LD}(\beta, \gamma). \end{cases} \quad (1.16)$$

Here is the basic property of this partition. It is analogous to Lemma 1.3.2.

Lemma 1.10.7 *Let Z denote the set of all ordinals smaller than ω^ω . Then every $D \subseteq Z$ monochromatic under KO satisfies $\text{Card}(D) \leq 2 + \text{psn}(\max(D))$.*

Proof Let KO be constant on $[D]^3$. If this constant is 1 then Lemma 1.10.6 does the job, so assume that this constant is either 0 or 2. Write D as in the proof of Lemma 1.10.6. Let $\gamma_i = \text{LD}(\delta_i, \delta_{i+1})$ for $i < r$. These are exponents in ordinals below ω^ω , hence they are natural numbers. By the assumption of the case, the sequence γ_i is either strictly increasing or strictly decreasing. Moreover, all of them must be $\leq$ the pseudonorm of δ_0 , indeed, if $\delta_0 = \omega^d \cdot m_d + \dots + \omega^0 \cdot m_0$ then every δ_i being smaller than δ_0 must have all exponents $\leq d$. Thus $r + 1 \leq \text{psn}(\delta_0) + 2$ as desired. $\square$

Let Z_k denote the set of all ordinals smaller than ω_k .

Lemma 1.10.8 *For every $k \geq 3$ there exists a partition $L_k : [Z_{k-1}]^k \rightarrow 3^{k-2}$ such that every set homogeneous $D \subseteq Z_{k-1}$ satisfies*

$$\text{Card}(D) \leq \text{psn}(\max(D)) + \frac{(k-1)(k-2)}{2} + 1.$$

Proof We proceed by induction on k . For $k = 3$ the result was proved above, that is we take $\text{KO}|[Z_2]^3$ as L_3 .

Assume the result for k . For $\alpha = \langle \alpha_0, \dots, \alpha_k \rangle \in [Z_k]^{k+1}$ we let $G(\alpha) = \text{KO}(\alpha_0, \alpha_1, \alpha_2)$. Exactly as in the proof of Lemma 1.3.2, every set D homogeneous for this partitions is homogeneous for KO. For α as above we let $\gamma_i = \text{LD}(\alpha_i, \alpha_{i+1})$. These are elements of Z_{k-1} . We let

$$W(\alpha) = \begin{cases} L_k(\gamma_0, \dots, \gamma_{k-1}) & \text{if } \gamma_0 > \dots > \gamma_{k-1} \\ L_k(\gamma_{k-1}, \dots, \gamma_0) & \text{if } \gamma_0 < \dots < \gamma_{k-1} \\ 0 & \text{otherwise} \end{cases}$$

and

$$L_{k+1}(\alpha) = \langle G(\alpha), W(\alpha) \rangle.$$

It is routine to verify that this partition has desired properties. $\square$

As pointed out above, we transfer these partitions of sets of ordinals to sets of natural numbers using the map KS.

Theorem 1.10.9 *Let $k \geq 2$. Then for every ω_k -small set $A \subseteq \mathbb{N}$ with $\min(A) > k$ there exists a partition $L : [A]^{k+1} \rightarrow 3^{k-1}$ such that every $D \subseteq A$ homogeneous for L satisfies $\text{Card}(D) \leq \min(D) + \frac{k(k-1)}{2} + 1$.*

In other words, D is at most $\omega + 2k + 3$ -large in the sense of the slow growing hierarchy.

Proof For $k = 2$ see Lemma 1.10.5. For arbitrary k we argue as follows. Let A satisfy the assumption. Let $\mu = \omega_k$, we write $\text{KS}(x)$ for $\text{KS}^A(\mu; x)$. Then for every $a \in A$

$\text{KS}(a)$ is defined, so we may apply the machinery worked out above. We take the partition $L_{k+1} : [Z_k]^{k+1} \rightarrow 3^{k-1}$ having the property described in Lemma 1.10.8. We let

$$L(a_0, \dots, a_k) = L_{k+1}(\text{KS}(a_0), \dots, \text{KS}(a_k))$$

and we assert that this partition has the desired property. So let D be monochromatic under L . Then the set $\{\text{KS}(a) : a \in D\}$ is monochromatic under L_{k+1} , hence both these sets have at most $\text{psn}(\text{KS}(a_0)) + \frac{k(k-1)}{2} + 1$ elements. On the other hand, $\text{psn}(\text{KS}(a_0)) \leq a_0$ by Corollary 1.10.2. $\square$

References

1. Ketonen, J., & Solovay, R. (1981). Rapidly growing Ramsey functions. *The Annals of Mathematics*, 113(2), 267 (1981). <https://doi.org/10.2307/2006985>.
2. Graham, R. L., Rothschild, B. L., & Spencer, J. H. (1990). *Ramsey theory*. Wiley-Interscience series in discrete mathematics and optimization (2nd ed.). New York: Wiley. ISBN: 0471500461.
3. Ratajczyk, Z. (1988). A combinatorial analysis of functions provably recursive in Σ_n . *Fundamenta Mathematicae*, 130(3), 191–213 (1988). ISSN: 0016-2736. <https://doi.org/10.4064/fm-130-3-191-213>.
4. Kotlarski, H., & Ratajczyk, Z. (1990). Inductive full satisfaction classes. *Annals of Pure and Applied Logic*, 47(3), 199–223. ISSN: 01680072. [https://doi.org/10.1016/0168-0072\(90\)90035-Z](https://doi.org/10.1016/0168-0072(90)90035-Z).
5. Ratajczyk, Z. (1993). Subsystems of true arithmetic and hierarchies of functions. *Annals of Pure and Applied Logic*, 64(2), 95–152. ISSN: 01680072. [https://doi.org/10.1016/0168-0072\(93\)90031-8](https://doi.org/10.1016/0168-0072(93)90031-8).
6. Fairtlough, M. V. H., & Wainer, S. S. (1992). Ordinal complexity of recursive definitions. *Information and Computation*, 99(2), 123–153. ISSN: 08905401. [https://doi.org/10.1016/0890-5401\(92\)90027-D](https://doi.org/10.1016/0890-5401(92)90027-D).
7. Fairtlough, M. V. H., & Wainer, S. S. (1998). Hierarchies of provably recursive functions. *Handbook of proof theory*. Studies in logic and the foundations of mathematics (Vol. 137, pp. 149–207). Amsterdam: Elsevier. ISBN: 9780444898401. [https://doi.org/10.1016/S0049-237X\(98\)80018-9](https://doi.org/10.1016/S0049-237X(98)80018-9).
8. Odifreddi, P. (1999). *Classical recursion theory, vol. I (1989), vol. II (1999)*. Studies in logic and the foundations of mathematics (Vol. 125, pp. 143). North Holland. ISBN: 9780444894830.
9. Paris, J., & Harrington, L. (1977). A mathematical incompleteness in Peano arithmetic. *Handbook of mathematical logic*. Studies in logic and the foundations of mathematics (Vol. 90, pp. 1133–1142). Amsterdam: Elsevier. ISBN: 9780444863881. [https://doi.org/10.1016/S0049-237X\(08\)71130-3](https://doi.org/10.1016/S0049-237X(08)71130-3).
10. Hájek, P., & Paris, J. (1987). Combinatorial principles concerning approximations of functions. *Archiv für mathematische Logik und Grundlagenforschung*, 26, 13–28. <http://eudml.org/doc/138050>.
11. Sommer, R. (1990). *Transfinite induction and hierarchies of functions generated by transfinite recursion within Peano arithmetic*. Ph.D. thesis. University of California, Berkeley.
12. Sommer, R. (1995). Transfinite induction within Peano arithmetic. *Annals of Pure and Applied Logic*, 76(3), pp. 231–289. ISSN: 01680072. [https://doi.org/10.1016/0168-0072\(95\)00029-G](https://doi.org/10.1016/0168-0072(95)00029-G).
13. Bigorajska, T., & Kotlarski, H. (1999). A partition theorem for α -large sets. *Fundamenta Mathematicae*, 160, 27–37.

14. Mills, G. (1980). A tree analysis of unprovable combinatorial statements. In L. Pacholski, J. Wierzejewski & A. J. Wilkie (Eds.), *Model theory of algebra and arithmetic* (pp. 248–311). Berlin: Springer. ISBN: 978-3-540-38393-2.
15. Erdős, P., & Mills, G. (1981). Some bounds for the Ramsey-Paris-Harrington numbers. *Journal of Combinatorial Theory, Series A*, 30(1), 53–70. ISSN: 00973165. [https://doi.org/10.1016/0097-3165\(81\)90040-6](https://doi.org/10.1016/0097-3165(81)90040-6).

Chapter 2

Some Model Theory



In this chapter we introduce some model-theoretic ideas. No completeness is assumed, the reader will find much more material in special monographs, e.g. Chang and Keisler [1] and Hodges [2]. We refer to these textbooks for a detailed account of, e.g., the completeness theorem.

2.1 Unions of Chains

In this section we present the construction of models by means of unions of chains, a variant of this construction will be used further, in Sect. 3.11.

Let L be a fixed language. Let $\mathcal{K}$ and $\mathcal{M}$ be structures for this language. We say that $\mathcal{K}$ is a *submodel* or *substructure* of $\mathcal{M}$ (or $\mathcal{M}$ is an *extension*) if the universe of $\mathcal{K}$ is a subset of the universe of $\mathcal{M}$ and

1. for every relation symbol R and every $a_0, \dots, a_{r-1} \in \mathcal{K}$, R is true of them in $\mathcal{K}$ iff it is true of them in $\mathcal{M}$,
2. for every function symbol F and every $a_0, \dots, a_{r-1} \in \mathcal{K}$, the value $F(a_0, \dots, a_{r-1})$ in $\mathcal{K}$ is the same as this value in $\mathcal{M}$ (in particular, this value is in $\mathcal{K}$),
3. for every constant symbol c , the value named by it in $\mathcal{K}$ and $\mathcal{M}$ is the same (once again, in particular, this value is in $\mathcal{K}$).

It is customary to denote this relation between structures exactly like inclusion, that is we write $\mathcal{K} \subseteq \mathcal{M}$. We point out that this relation does not imply that $\mathcal{K}$ and $\mathcal{M}$ satisfy the same first order sentences. Thus, for example, the ordered field of rationals is a substructure of the ordered field of the reals, but only the reals satisfy the sentence $\exists x \cdot x \cdot x = 1 + 1$. Say that the structures $\mathcal{K}, \mathcal{M}$ for the language L are *elementary equivalent* if they satisfy the same first order sentences of L . We write $\mathcal{K} \equiv \mathcal{M}$ if $\mathcal{K}$ and $\mathcal{M}$ are elementary equivalent. The relation $\equiv$ is difficult to deal

with. Though, in some cases, e.g., when we work with a theory which admits the so-called elimination of quantifiers, this is possible, consult [2] for more in this direction. In order to overcome the difficulties which occur here Tarski and Vaught [3] introduced the notion of an elementary substructure. Say that $\mathcal{K}$ is an *elementary substructure* of $\mathcal{M}$ (or $\mathcal{M}$ is an *elementary extension* of $\mathcal{K}$) iff $\mathcal{K}$ is a substructure of $\mathcal{M}$ and for every formula $\varphi \in L$ and every $a_0, \dots, a_{r-1} \in \mathcal{K}$, $\mathcal{K} \models \varphi[a_0, \dots, a_{r-1}]$ iff the same holds in $\mathcal{M}$. We write $\mathcal{K} \prec \mathcal{M}$ if $\mathcal{K}$ is an elementary submodel of $\mathcal{M}$. Obviously, if $\mathcal{K} \prec \mathcal{M}$ then $\mathcal{K} \equiv \mathcal{M}$.

Suppose that $\langle I, \leq \rangle$ is a linear ordering and we associated an L -structure $\mathcal{M}_i$ to each $i \in I$. We say that the family $\mathcal{M}_i : i \in I$ is a *chain* of structures if $i \leq j$ iff $\mathcal{M}_i \subseteq \mathcal{M}_j$ for every $i, j \in I$. We define a new structure $\mathcal{M} = \cup_{i \in I} \mathcal{M}_i$ called a *union of the chain* $\mathcal{M}_i$ in this situation. We let the universe of $\mathcal{M}$ be just the union of the universes of all $\mathcal{M}_i$. We let the interpretations of relation symbol R be the union of their interpretations in $\mathcal{M}_i$ and the same for function symbols. The constants of L have natural interpretations (they are the same in all $\mathcal{M}_i$).

The following fact (due to Tarski and Vaught [3]) is well known in model theory, later we shall use an analogue of it (cf. Lemma 3.11.1 in Sect. 3).

Theorem 2.1.1 *For every elementary chain $\{\mathcal{M}_i : i \in I\}$ of structures, the union $\cup_{i \in I} \mathcal{M}_i$ is an elementary extension of each $\mathcal{M}_i$.*

Proof Pick $i_0 \in I$. We remark that for every term $t(v_0, \dots, v_{r-1})$ and every $a_0, \dots, a_{r-1} \in \mathcal{M}_{i_0}$, the value of t on $a_0, \dots, a_{r-1}$ in $\mathcal{M}$ is the same as the value of t on the same valuation in $\mathcal{M}_{i_0}$, this observation is immediate (by induction on terms). Granted this we prove (by induction on φ) that truth of every formula φ on every valuation $a_0, \dots, a_{r-1} \in \mathcal{M}_{i_0}$ is the same in $\mathcal{M}$ as in $\mathcal{M}_{i_0}$. We leave the details to the reader. $\square$

2.2 The Recursive Saturation and Resplendency

Let $p(x) = \{\varphi_i(x)\}_{i \in \mathbb{N}}$ be a sequence of formulas in the language of $\mathcal{M}$, possibly with some finite set of parameters from $\mathcal{M}$. We say that a set of sentences is a *type over* $\mathcal{M}$ if for each $n \in \mathbb{N}$, $\mathcal{M}$ satisfies $\exists x (\bigwedge_{i \leq n} \varphi_i(x))$. A type $p(x)$ is Σ_n if all formulas φ_i are Σ_n . We apply the same conventions to other classes of formulas. The type $p(x)$ is recursive if $p(x)$ is a recursive sequence of formulas. A model $\mathcal{M}$ realizes $p(x)$ if there exists $a \in \mathcal{M}$ such that for each $\varphi_i(x)$, $\mathcal{M} \models \varphi_i[a]$. A models $\mathcal{M}$ is *saturated* if $\mathcal{M}$ realizes any type over $\mathcal{M}$. A models $\mathcal{M}$ is *recursively saturated* if $\mathcal{M}$ realizes any recursive type over $\mathcal{M}$. We have the following fundamental theorem.

Theorem 2.2.1 *For any countable model $\mathcal{M}$ in a countable language there exists a countable elementary extension $\mathcal{K} \succ \mathcal{M}$ such that $\mathcal{K}$ is saturated.*

It follows that if a theory T in a countable language has only infinite models and T does not prove a given sentence φ then this fact is witnessed in some countable and

saturated model of T and $\neg\varphi$. This fact is essential for conservativeness of a theory of Peano arithmetic with a satisfaction class over Peano arithmetic, see Corollary 5.2.2.

For a model $\mathcal{M}$ in a language $\mathcal{L}$ we define $\mathcal{L}(\mathcal{M})$ as the language $\mathcal{L}$ extended by constants denoting all elements from $\mathcal{M}$. Then, the *diagram* of $\mathcal{M}$, $\text{Diag}(\mathcal{M})$, is a set of atomic formulas in $\mathcal{L}(\mathcal{M})$ true in $\mathcal{M}$. The *elementary diagram* of $\mathcal{M}$, $\text{ElemDiag}(\mathcal{M})$, is a set of all sentences in $\mathcal{L}(\mathcal{M})$ true in $\mathcal{M}$. A model $\mathcal{M}$ in a recursive language $\mathcal{L}$ is *resplendent* if for any recursive theory T in a language $\mathcal{L}'$ extending $\mathcal{L}$, involving only finitely many parameters from $\mathcal{M}$ and consistent with $\text{ElemDiag}(\mathcal{M})$ there exists an expansion of $\mathcal{M}$ which satisfies T . The following theorem due to Barwise and, independently, Ressayre characterizes resplendent models.

Theorem 2.2.2 (Barwise [4], Ressayre [5]) *Let $\mathcal{M}$ be a countable model recursively saturated model in a recursive language. Then, $\mathcal{M}$ is resplendent.*

Let us observe that a converse implication also holds. Indeed, we can treat a recursive type $p(x)$ over $\mathcal{M}$ as a recursive theory $p(c)$ in a language with a new constant. Since $p(x)$ is a type over $\mathcal{M}$ it is consistent with $\text{ElemDiag}(\mathcal{M})$ and any expansion of $\mathcal{M}$ satisfying $p(c)$ would give an interpretation for the new constant c satisfying $p(x)$.

2.3 The Theorem of Chang and Makkai

The following result is one of the main results concerning subsets of a given structure which are not definable. Observe that if $\mathcal{M}$ is a recursively saturated structure and $A \subseteq \mathcal{M}$, then the structure $\langle \mathcal{M}, A \rangle$ need not be recursively saturated, indeed, there are more formulas (and, hence, more types) in the language for the structure $\langle \mathcal{M}, A \rangle$ than for the language of $\mathcal{M}$. The result below was proved firstly for saturated and special structures; the countable version of it is due to Schlipf [6].

Theorem 2.3.1 *Let $\mathcal{M}$ be a countable structure (for any language) and let $A \subseteq \mathcal{M}$ be such that $\langle \mathcal{M}, A \rangle$ is recursively saturated and A is not definable in $\mathcal{M}$ with parameters. Then A has $2^{\aleph_0}$ automorphic images, i.e. $\{f * A : f \in \text{Aut}(\mathcal{M})\}$ is of power continuum.*

References

1. Chang, C. C., & Keisler, H. J. (1977). *Model theory* (2nd ed.). North Holland.
2. Hodges, W. (1993). *Model theory*. Cambridge: Cambridge University Press. ISBN: 9780511551574. <https://doi.org/10.1017/CBO9780511551574>.
3. Tarski, A., & Vaught, R. L. (1957). Arithmetical extensions of relational systems. *Compositio Mathematicae*, 13, 81–102.
4. Barwise, J. (1975). *Admissible sets and structures: An approach to definability theory*. Perspectives in mathematical logic. Berlin: Springer. ISBN: 3-540-07451-1.

5. Ressayre, J. P. (1977). Models with compactness properties relative to an admissible language. *Annals of Mathematical Logic*, 11(1), 31–55. ISSN: 00034843. [https://doi.org/10.1016/0003-4843\(77\)90009-2](https://doi.org/10.1016/0003-4843(77)90009-2).
6. Schlipf, J. S. (1978). Toward model theory through recursive saturation. *Journal of Symbolic Logic*, 43(2), 183–206. ISSN: 0022-4812. <https://doi.org/10.2307/2272817>.

Chapter 3

Incompleteness



In this chapter we present several proofs of Gödel's incompleteness theorems and related results. Each of these proofs gives us a better understanding of the whole phenomenon of incompleteness of mathematics. The content of this chapter is rather advanced. We assume that the reader has basic knowledge in logic, proof theory, model theory, recursion theory and foundations of arithmetic. Most of the material required in this chapter can be found in [1] or [2] or [3].

As for logic we assume knowledge on the level of a graduate course, e.g. [1], in particular familiarity with the predicate calculus (the notion of a proof, of a rule of inference including the cut rule), the notion of a theory and a model and of an independent sentence. We assume that the reader is familiar with the notion of a primitive recursive function; and of a recursive set or function and with the representability theorem on representing recursive sets in PA .

In model theory we assume that the reader is familiar with the notion of completeness of a theory, completeness with respect to some set of sentences, completion of a theory, elementary equivalence, elementary embedding, elementary submodel and extension, simple elementary extension, Skolem functions, Skolemization, Henkin construction of a model, diagram of a model $\text{Diag}(M)$ and, more specifically, a Δ_0 diagram of a model, $\text{Diag}_{\Delta_0}(M)$, saturation, recursive saturation, a minimal model, a pointwise definable model.

In foundations of arithmetic we need familiarity with the notion of the arithmetical hierarchy of formulas $(\Sigma_n, \Pi_n, \Delta_n)$, the universal formula for Σ_n or Π_n formulas, the axiomatization of first order Peano Arithmetic PA , the notion of a fragment of PA of the form $I\Delta_0$ —induction for bounded formulas (bounded induction), $I\Delta_0 + \text{Exp}$ —induction for bounded formulas plus the totality of exponentiation, $I\Sigma_n$ —induction for Σ_n formulas (Σ_n induction), the notion of the standard model and of a non standard model, of an end-extension of a model, of an object in the sense of a model (for instance a syntactical object, e.g. a sentence), of coding sets and sequences, of the arithmetization of the language.

We also assume some familiarity with set theory and, in particular, the set theory of the continuum. For instance we use freely the notion of dominating one function

by another. We shall denote by $\ulcorner \phi \urcorner$ the Gödel number of a formula ϕ . Often we will identify a formula with its Gödel number and omit the corners.

Let us state here explicitly some of the very preliminaries of Peano Arithmetic—the language that we choose and the axioms of Peano Arithmetic which we choose.

The language of Peano arithmetic L_{PA} consists of one constant 0, one unary function symbol S (for the successor function) and two binary function symbols $+$ and $\cdot$. Its axioms are

- $Q1 \quad \forall x \, Sx \neq 0$
- $Q2 \quad \forall x, y \, [Sx = Sy \Rightarrow x = y]$
- $Q3 \quad \forall x \, [x \neq 0 \Rightarrow \exists y \, x = Sy]$
- $Q4 \quad \forall x \, x + 0 = x$
- $Q5 \quad \forall x, y \, [x + Sy = S(x + y)]$
- $Q6 \quad \forall x \, x \cdot 0 = 0$
- $Q7 \quad \forall x, y \, [x \cdot Sy = x \cdot y + x]$

together with the scheme of induction

$$\{\varphi(0) \ \& \ \forall x \, [\varphi(x) \Rightarrow \varphi(S(x))]\} \Rightarrow \forall x \, \varphi(x). \quad \text{IND}$$

The formula φ in the scheme IND may contain free variables other than x . The theory based on Q1–Q7 is known as *Robinson's Q* or PA^- in the literature. We extend it by introducing the inequality symbol and putting

$$x \leq y \equiv \exists z \, x + z = y$$

and

$$x < y \equiv x \leq y \ \& \ x \neq y.$$

A formula $\varphi \in L_{PA}$ is *open* or *quantifier-free* if it contains no quantifier. We abbreviate $\forall x \, [x < y \Rightarrow \varphi]$ as $\forall x < y \, \varphi$ and $\exists x \, [x < y \ \& \ \varphi]$ as $\exists x < y \, \varphi$. Such quantifiers are called *bounded*. A formula is called *bounded* if all quantifiers which occur in it are bounded (i.e., the formula either contains no quantifier or only bounded ones). We also say that our formula is of the class Δ_0 . We denote this class also as Σ_0 or Π_0 . Then, a formula is Σ_{n+1} if it is in the closure of Π_n under existential quantification and, similarly, a formula is Π_{n+1} if it is in the closure of Σ_n under universal quantification. For each $n \geq 0$, classes Σ_n and Π_n are closed under conjunction, disjunction and bounded quantification. However, Tarski's theorem on undefinability of truth they are not closed under negation. Indeed, for each $n \geq 1$ we have *universal formulas* Tr_{Σ_n} for formulas in Σ_n and Tr_{Π_n} for formulas in Π_n and these formulas belong to the corresponding class. That is Tr_{Σ_n} is Σ_n and Tr_{Π_n} is Π_n . For Δ_0 class the universal formula Tr_{Δ_0} may be written in both a Σ_1 form or a Π_1 form. For brevity we write often Tr_n for Tr_{Σ_n} and Tr_0 for Tr_{Δ_0} .

Occasionally, we use a pairing function. We denote it by $\langle x, y \rangle$ and we fix it as $\langle x, y \rangle = \frac{(x+y)(x+y+1)}{2} + x$. It can be easily checked that this function is a bijection between a set of ordered pairs from $\mathbb{N}$ and $\mathbb{N}$. Moreover, it has a property that each element of a pair is less or equal to the code of the pair. By $(z)_0$ and $(z)_1$ we denote

two projections, so $(\langle x, y \rangle)_0 = x$ and $(\langle x, y \rangle)_1 = y$. Observe, that $(z)_0$ and $(z)_1$ are well defined for any natural number z for $\langle x, y \rangle$ is a bijection.

Let T be a primitive recursive theory containing $I\Delta_0 + \text{Exp}$. Let $\text{Pr}_T(\ulcorner \varphi \urcorner)$ be a Σ_1 arithmetical formula expressing the fact that the formula φ is provable in T . We assume that $\text{Pr}_T(x)$ is of the form $\exists y \text{Prov}_T(y, x)$, for a Δ_0 formula $\text{Prov}_T(y, x)$. The formula $\text{Prov}_T(y, x)$ expresses the fact that y is a code of a proof of a formula with code x . We assume here some fixed coding of proofs. Moreover, since $\text{Prov}_T(y, x)$ is Δ_0 , T can decide all substitutions $\text{Prov}_T(S^n 0, S^k 0)$, for $n, k \in \mathbb{N}$. The main properties of arithmetization of the language are the so-called *derivability conditions*:

- D1 If $T \vdash \varphi$ then $T \vdash \text{Pr}_T(\ulcorner \varphi \urcorner)$ for every sentence φ ,
- D2 $T \vdash \forall \varphi [\text{Sent}(\varphi) \ \& \ \text{Pr}_T(\varphi) \Rightarrow \text{Pr}_T(\ulcorner \text{Pr}_T(\varphi) \urcorner)]$,
- D3 $T \vdash \forall \varphi, \psi [\text{Sent}(\varphi) \ \& \ \text{Sent}(\psi) \ \& \ \text{Pr}_T(\varphi) \ \& \ \text{Pr}_T(\varphi \Rightarrow \psi) \Rightarrow \text{Pr}_T(\psi)]$,
and the same for other rules of inference.

Yet another property of arithmetization is the so-called *demonstrable Σ_1 completeness*:

$$\text{for every } \Sigma_1 \text{ formula } \varphi(x), \quad T \vdash \forall x [\varphi(x) \Rightarrow \text{Pr}_T(\ulcorner \varphi(S^x 0) \urcorner)]. \quad (3.1)$$

It is easy to see that the condition D2 is a consequence of (3.1). Indeed, let $\mathcal{M} \models T$ and let $\varphi \in \mathcal{M}$ be such that $\mathcal{M} \models \text{Pr}_T(\varphi)$. Then $\mathcal{M} \models \text{Pr}_T(\ulcorner \text{Pr}_T(\varphi) \urcorner)$ by (3.1), for $\text{Pr}_T(\varphi)$ is a Σ_1 formula.

3.1 The Arithmetized Completeness Theorem

D. Hilbert and P. Bernays showed that the completeness theorem is formalizable in PA. We follow Smoryński's presentation [14] with some minor changes from [4]. In this section we assume that T is a primitive recursive theory in the language of PA and containing $I\Sigma_2$.

Let Tr_2 denote the universal formula for Σ_2 formulas. If $C(x)$ is a Σ_2 formula defining a set of formulas then, for convenience, we shall often write $\mathcal{M} \models \text{Tr}_2(C; \varphi)$ rather than $\mathcal{M} \models \text{Tr}_2(C(\varphi))$. Let $\text{Compl}_T(C)$ denote the formula which expresses “ C is the Gödel number of some Σ_2 -formula which describes a complete and consistent extension of T ”. Thus $\text{Compl}_T(C)$ is

$$C \in \Sigma_2 \ \& \ \forall x [\text{Tr}_2(C; x) \Rightarrow \text{Sent}(x)] \ \& \ \forall \varphi \{ \text{Sent}(\varphi) \Rightarrow [\text{Tr}_2(C; \varphi) \vee \text{Tr}_2(C; \neg \varphi)] \} \\ \& \ \forall (\varphi_0, \dots, \varphi_{r-1}) \{ [\forall i < r \text{Tr}_2(C; \varphi_i)] \Rightarrow \neg \text{Pr}_T(\neg \bigwedge_{i < r} \varphi_i) \}.$$

We read “ C is a completion of T (in $\mathcal{M}$)”. We remark that the restriction to Σ_2 formulas describing completions comes out from the proof of Theorem 3.1.1, where one obtains a Σ_2 -formula describing the completion. We could equally well work with Σ_n -completions for $n \geq 2$ or with Π_n -completions for $n \geq 2$. It is easy to see that a Σ_n -completion is also Π_n (and conversely), for if C is a completion then $\varphi \in C \Leftrightarrow \neg \varphi \notin C$.

We want to point out that if we are given a model $\mathcal{M}$ for T and $C \in \mathcal{M}$ satisfying Compl_T then C need not be a standard object. This is just an element of $\mathcal{M}$ of which $\mathcal{M}$ thinks that it is the Gödel number of a Σ_2 -formula, etc. Observe that $\text{Compl}_T(\cdot)$ is Π_3 .

The following fact is known as the Hilbert–Bernays arithmetized completeness theorem.

Theorem 3.1.1 *If T proves Σ_2 induction then T proves*

$$\forall \varphi \{ \text{Sent}(\varphi) \Rightarrow [\neg \text{Pr}_T(\neg \varphi) \equiv \exists C (\text{Compl}_T(C) \ \& \ \text{Tr}_2(C; \varphi))] \}.$$

Proof Let $\mathcal{M} \models T$ and let φ be a sentence in the sense of $\mathcal{M}$. Assume that there exists a completion $C \in \mathcal{M}$ making φ true, i.e., $\mathcal{M} \models \text{Tr}_2(C; \varphi)$. Assume $\mathcal{M}$ thinks that $\text{Pr}_T(\neg \varphi)$ and let $d \in \mathcal{M}$ be a proof, i.e., $\mathcal{M} \models \text{Prov}_T(d, \neg \varphi)$. Then, in $\mathcal{M}$, d is a sequence, $d = \langle d_0, \dots, d_{r-1} \rangle$ such that every d_i is either an axiom of T or is a consequence of previous items of d by means of inference rules. By induction on i one easily checks that every d_i is made true by C , i.e., $\mathcal{M} \models \forall i \leq r-1 \text{Tr}_2(C; d_i)$. In particular, this happens to $d_{r-1} = \varphi$, so d cannot be a proof of $\neg \varphi$. This proves the implication $\Leftarrow$. For the converse assume that $\mathcal{M} \models \neg \text{Pr}_T(\neg \varphi)$. We construct a completion by the standard procedure. That is, working inside $\mathcal{M}$, we take a primitive recursive enumeration $\{\xi_i\}$ of all sentences and add to C either ξ_i or $\neg \xi_i$ without destroying consistency. Let us check a single step at first. We claim that T proves the following

$$\forall \alpha, \beta \{ [\text{Sent}(\alpha) \ \& \ \text{Sent}(\beta) \ \& \ \neg \text{Pr}_T(\neg \alpha)] \Rightarrow [\neg \text{Pr}_T(\neg(\alpha \ \& \ \beta)) \vee \neg \text{Pr}_T(\neg(\alpha \ \& \ \neg \beta))] \}. \quad (3.2)$$

For, let α, β are given and $\text{Pr}_T(\neg(\alpha \ \& \ \beta))$ and $\text{Pr}_T(\neg(\alpha \ \& \ \neg \beta))$. Then by logic (i.e., derivability conditions) $\text{Pr}_T(\neg \alpha \vee \neg \beta)$ and $\text{Pr}_T(\neg \alpha \vee \neg \neg \beta)$. By the derivability condition for the cut rule $\text{Pr}_T(\neg \alpha)$, contrary to the assumption. This proves (3.2).

Granted (3.2) we construct C by induction. We let η_0 be ξ_0 if $\neg \text{Pr}_T(\neg(\varphi \ \& \ \xi_0))$ and $\eta_0 = \neg \xi_0$ otherwise. If $\eta_0, \dots, \eta_i$ are given and $\neg \text{Pr}_T(\neg(\varphi \ \& \ \bigwedge_{j < i} \eta_j))$ holds then we let η_{i+1} be ξ_{i+1} if $\neg \text{Pr}_T(\neg(\varphi \ \& \ \bigwedge_{j < i} \eta_j \ \& \ \xi_{i+1}))$ holds and $\eta_{i+1} = \neg \xi_{i+1}$ otherwise. We let $C(\gamma)$ be the formula $\exists i (\gamma = \eta_i)$. Thus, $C(\cdot)$ is a description of the set $\{\eta_i : i\}$. It is easy to see that C is a Σ_2 -formula. In order to write it explicitly it is convenient to write $\text{Con}_T(\gamma)$ as an abbreviation for a Π_1 formula $\neg \text{Pr}_T(\neg \gamma)$. Now we write the formula $C(\gamma)$ as

$$\begin{aligned} \exists \eta \{ \text{Seq}(\eta) \ \& \ \forall i < \text{lh}(\eta) [\text{Con}_T(\varphi \ \& \ \bigwedge_{j < i} \eta_j \ \& \ \xi_i) \Rightarrow \eta_i = \xi_i] \ \& \\ [\neg \text{Con}_T(\varphi \ \& \ \bigwedge_{j < i} \eta_j \ \& \ \xi_i) \Rightarrow \eta_i = \neg \xi_i] \ \& \ \exists i < \text{lh}(\eta) \ \gamma = \eta_i \} \end{aligned}$$

Clearly, $C(\gamma)$ is equivalent to a Σ_2 -formula. Also, writing “for every η as above, $\neg \gamma$ does not occur in the sequence η ”, one can write a formula equivalent to the one above, but of the form Π_2 . $\square$

Suppose we are given $\mathcal{M} \models T$ and let $\mathcal{M} \models \text{Compl}(C)$, i.e. C is a completion of T in $\mathcal{M}$. These data determine a new model. It is constructed as follows. Let $\text{Tm}^{\mathcal{M}}$

denote the set of all (Skolem) constant terms in the sense of $\mathcal{M}$. Divide it by the equivalence relation $t_1 \sim t_2 \equiv \mathcal{M} \models \text{Tr}_2(C; t_1 = t_2)$. Clearly this is an equivalence relation and the following definition of addition makes sense:

$$t_1^\sim + t_2^\sim = t_3^\sim \quad \equiv \quad \mathcal{M} \models \text{Tr}_2(C; t_1 + t_2 = t_3).$$

We treat other atomic symbols in the language of PA similarly. We denote the model constructed above by $\text{ACT}(\mathcal{M}; C)$. ACT stands for the Arithmetized Completeness Theorem. We call this model the *ACT-extension of $\mathcal{M}$* .

Lemma 3.1.2 *Let $\mathcal{M}$ be a model of T and let C be a completion in $\mathcal{M}$. Then for every formula $A(v_0, \dots, v_{r-1})$ and every r -tuple $t_0, \dots, t_{r-1}$, we have*

$$\text{ACT}(\mathcal{M}; C) \models A(t_0^\sim, \dots, t_{r-1}^\sim) \quad \text{iff} \quad \mathcal{M} \models \text{Tr}_2(C; A(t_0, \dots, t_{r-1})).$$

Proof This is a standard Henkin-like argument. Let us give one minor observation. Let T' be a theory for which we construct a completion. If T' contains the full induction then a function which associates to a formula its Skolem term given by the scheme of minimum is definable, so we can work with it freely inside $\mathcal{M}$. It follows that in this case all Skolem terms are definable in T' and C has in $\mathcal{M}$ the properties of a complete Skolemized theory, so the usual argument works smoothly. Otherwise, we should extend the language of T' by a recursive set of Skolem functions. $\square$

Let $\mathcal{M} \models T$ and let C be a completion of T in $\mathcal{M}$. It turns out that there exists a natural embedding of $\mathcal{M}$ onto an initial segment of $\text{ACT}(\mathcal{M}; C)$. It is defined as follows: we map $b \in \mathcal{M}$ to $S^b 0^\sim$: the equivalence class of the numeral $S^b 0$ in $\text{ACT}(\mathcal{M}; C)$. Let j denote this embedding. Thus we have

Lemma 3.1.3 *Let $\mathcal{M} \models T$ and let C be a completion of T in $\mathcal{M}$. Then, j as defined as $j(a) = S^a 0^\sim$ is an embedding of $\mathcal{M}$ onto an initial segment of $\text{ACT}(\mathcal{M}; C)$.*

Proof The function j is an injection since $\mathcal{M} \models \text{Pr}_T(\neg S^a 0 = S^b 0)$, for $a, b \in \mathcal{M}$ such that $a \neq b$. Similarly, j can be shown to be an embedding by provability in T of all true in $\mathcal{M}$ atomic sentences of the form $S^a 0 + S^b 0 = S^c 0$. Here also provability is understood in $\mathcal{M}$. Then, j is on a initial segment of $\text{ACT}(\mathcal{M}; C)$ by the provability in T the formulas $\forall x \leq S^a 0 (x = 0 \vee x = S 0 \vee \dots \vee x = S^a 0)$. More exactly, we use the fact, that for all $a \in \mathcal{M}$, $\mathcal{M} \models \text{Pr}_T(\neg \forall x \leq S^a 0 (x = 0 \vee x = S 0 \vee \dots \vee x = S^a 0))$. Thus, $0^\sim, S 0^\sim, \dots, S^a 0^\sim$ are the only elements in $\text{ACT}(\mathcal{M}; C)$ that are below $S^a 0^\sim$. This shows that j is onto an initial segment of $\text{ACT}(\mathcal{M}; C)$.

Later, when it is clear from the context, we shall write “ T proves φ ” without precising whether we mean provability in $\mathcal{M}$ or provability in our sense. Moreover:

Lemma 3.1.4 *If $A(v_0, \dots, v_{r-1})$ is a Δ_0 formula and $b_0, \dots, b_{r-1} \in \mathcal{M}$ then*

$$\mathcal{M} \models A(b_0, \dots, b_{r-1}) \text{ iff } \text{ACT}(\mathcal{M}; C) \models A(j(b_0), \dots, j(b_{r-1})).$$

Proof The thesis follows from the fact that $\mathcal{M} \subseteq_{\text{end}} \text{ACT}(\mathcal{M}; C)$. □

We shall prove later that no ACT-extension is elementary (in contrast with “ordinary” end-extensions) and that each such extension is proper, this will follow from the proof of Tarski’s theorem, i.e., Theorem 3.2.7.

It should be noticed that iterations of the construction of a model by means of the arithmetized completeness theorem give nothing essentially new. The model of the form $\text{ACT}(\text{ACT}(\mathcal{M}; C); D)$ has an elementary submodel of the form $\text{ACT}(\mathcal{M}; E)$. Let us be more precise here.

Proposition 3.1.5 *Let T be a primitive recursive theory in L_{PA} containing $I\Sigma_2$. Let $\mathcal{M} \models T$, let C be a completion of T in $\mathcal{M}$. Let $\mathcal{K} = \text{ACT}(\mathcal{M}; C)$ and let D be a completion in $\mathcal{K}$. Then there exists a completion E in $\mathcal{M}$ such that $\text{ACT}(\mathcal{M}; E) < \text{ACT}(\mathcal{K}; D)$.*

Proof We just give a hint. Simply, there is a natural candidate for E , it is

$$\{\varphi \in \mathcal{M} : \mathcal{M} \models \text{Tr}_2(C; \text{Tr}_2(D; \varphi))\}. \quad \square$$

We shall use the idea exposed above later, e.g. in Sects. 3.8, 3.9 and 3.10.

3.2 The Original Approach

In this section we present the usual approach to the incompleteness theorems. All the ideas below are due to Gödel [5]. We assume that we are given a fixed primitive recursive consistent theory T in the language of PA and proving all the axioms of $I\Delta_0 + \text{Exp}$. The material of this section is exposed from other points of view in many papers and textbooks, see, e.g. [3, 5–9].

Lemma 3.2.1 (Diagonal lemma) *Let $\psi(x)$ be a formula in one free variable x as indicated. Then there exists a sentence φ such that $T \vdash \varphi \equiv \psi(\ulcorner \varphi \urcorner)$.*

Proof Let ψ be given. Let F be a primitive recursive function associating with every formula $\delta(x)$ with exactly one free variable x the (Gödel number of the) sentence $\delta(\ulcorner \delta \urcorner)$, i.e. the result of substitution of the numeral corresponding to $\ulcorner \delta \urcorner$ to δ . By the representability theorem, there exists a formula $\alpha(x, v)$ such that for each δ , $T \vdash \alpha(\ulcorner \delta \urcorner, v) \equiv v = F(\delta)$. Let $\chi(x)$ be the formula $\exists v (\alpha(x, v) \& \psi(v))$ and let φ be $\chi(\ulcorner \chi \urcorner)$, so the Gödel number of φ is $F(\ulcorner \chi \urcorner)$. Then T proves

$$\begin{aligned} \varphi &\equiv \exists v (\alpha(\ulcorner \chi \urcorner, v) \& \psi(v)) \\ &\equiv \exists v (v = F(\ulcorner \chi \urcorner) \& \psi(v)) \\ &\equiv \exists v (v = \ulcorner \varphi \urcorner \& \psi(v)) \\ &\equiv \psi(\ulcorner \varphi \urcorner), \end{aligned}$$

i.e. φ has the desired property. $\square$

Observe that for $n > 0$ if $\psi(x)$ is Σ_n , then φ is Σ_n . Moreover, by changing $\chi(x)$ in the above proof to $\forall v [\alpha(x, v) \Rightarrow \psi(v)]$ we see that if ψ is Π_n then we may require φ to be Π_n .

There is a parametric version of the diagonal lemma. It is as follows.

Lemma 3.2.2 *Let $\psi(x, y_0, \dots, y_{r-1})$ be a formula with variables as indicated. Then there exists a formula $\varphi(y_0, \dots, y_{r-1})$ such that T proves*

$$\varphi(y_0, \dots, y_{r-1}) \equiv \psi(\ulcorner \varphi(y_0, \dots, y_{r-1}) \urcorner, y_0, \dots, y_{r-1}).$$

Proof The proof of this lemma is a simple modification of the proof of Lemma 3.2.1 and is left to the reader. $\square$

Let us rewrite the proof of the diagonal lemma in a version in which we work with primitive recursive functions rather than with formulas representing them. Let $\psi(x)$ be given. At first we used the function F such that for every formula $\delta(x)$, $F(\ulcorner \delta(x) \urcorner) = \ulcorner \delta(\ulcorner \delta(x) \urcorner) \urcorner$. Then we wrote the formula $\chi(x)$: $\psi(F(x))$. Finally we let φ be the sentence whose Gödel number is $F(\ulcorner \chi(x) \urcorner)$, i.e. $\chi(\ulcorner \chi(x) \urcorner)$. Then, working in T we have

$$\varphi \equiv \chi(\ulcorner \chi(x) \urcorner) \equiv \psi(\ulcorner F(\ulcorner \chi(x) \urcorner) \urcorner) \equiv \psi(\ulcorner \varphi \urcorner)$$

as required.

Gödel used the so-called *liar paradox* (“The statement I am just expressing is false”—is this sentence true or false?). Using the diagonal lemma (i.e., Lemma 3.2.1) he was able to express this paradox in arithmetic. As a matter of fact, he changed the statement of the liar paradox to “The statement I am just expressing admits no proof in T ”. The original statement of the liar paradox was used by A. Tarski in his proof of his theorem (i.e., 3.2.7).

Gödel needed some assumption on T for his first incompleteness theorem. He stated this assumption syntactically. He required T to be ω -consistent in the sense of the following definition. Say that T is ω -consistent if the following holds: for every formula $\xi(x)$, if all sentences $\xi(S^n 0)$ are provable in T , then the sentence $\exists x \neg \xi(x)$ is not provable in T . Observe that every ω -consistent theory is consistent (take $x = x$ as $\xi(x)$).

Theorem 3.2.3 (Gödel’s first incompleteness theorem) *Let φ be such that T proves $\varphi \equiv \neg \text{Pr}_T(\ulcorner \varphi \urcorner)$. Then*

- 1 If T is consistent then $T \not\vdash \varphi$.
- 2 If T is ω -consistent then $T \not\vdash \neg \varphi$.

Proof Assume that $T \vdash \varphi$. Then $T \vdash \text{Pr}_T(\ulcorner \varphi \urcorner)$ by the demonstrable Σ_1 -completeness. By the property of φ , $\text{Pr}_T(\ulcorner \varphi \urcorner)$ is equivalent in T to $\neg \varphi$, hence, T is inconsistent.

Assume that $T \vdash \neg\varphi$. Then, by the property of φ , $T \vdash \text{Pr}_T(\ulcorner\varphi\urcorner)$. Hence $T \vdash \exists z \text{Prov}_T(z, \ulcorner\varphi\urcorner)$, hence, by ω -consistency of T , exists $n \in \mathbb{N}$ such that $T \not\vdash \neg \text{Prov}_T(S^n 0, \ulcorner\varphi\urcorner)$. It follows that for this n we have $T \vdash \text{Prov}_T(S^n 0, \ulcorner\varphi\urcorner)$, for T is complete with respect to Δ_0 sentences. This n is the Gödel number of a proof of φ in T . Thus we have $T \vdash \varphi$ and T is inconsistent. $\square$

Observe that we applied the assumption of ω -consistency only to the Σ_1 formula $\exists z \text{Prov}_T(z, \ulcorner\varphi\urcorner)$. Also it has become customary in recent years to express the assumption in another way. One simply assumes that Σ_1 consequences of T are true (in the standard model). But what is really needed is that theorems of T of the form $\text{Pr}_T(\ulcorner\xi\urcorner)$, where ξ is a sentence, are true in $\mathbb{N}$.

Theorem 3.2.3 implies that every ω -consistent theory is incomplete. The assumption of ω -consistency was eliminated by Rosser [10]. The essence of his proof is to change (the formal counterpart of) the notion of provability. Say that a formula φ is *provable in Rosser's sense* if it is provable and below its smallest proof there is no proof of its negation. More exactly:

Theorem 3.2.4 (Rosser's theorem) *Let $\text{Prov}_T^R(x, \varphi)$ denote the formula*

$$\text{Prov}_T(x, \varphi) \ \& \ \forall y < x \ \neg \text{Prov}_T(y, \neg\varphi).$$

Let $\text{Pr}_T^R(\varphi)$ be $\exists x \text{Prov}_T^R(x, \varphi)$. Let φ be such that T proves $\varphi \equiv \neg \text{Pr}_T^R(\ulcorner\varphi\urcorner)$. Then if T is consistent then $T \not\vdash \varphi$ and $T \not\vdash \neg\varphi$.

Proof Assume firstly that $T \vdash \varphi$. Then $T \vdash \text{Pr}_T(\ulcorner\varphi\urcorner)$ by D1, moreover there exists $n \in \mathbb{N}$ such that $T \vdash \text{Prov}_T(S^n 0, \ulcorner\varphi\urcorner)$. Simultaneously $T \vdash \neg \text{Pr}_T^R(\ulcorner\varphi\urcorner)$ by the property of φ . It follows that there exists a proof of $\neg\varphi$ which is smaller than n , hence it is standard and T is inconsistent.

Assume now that $T \vdash \neg\varphi$. Then there exists a standard n such that $T \vdash \text{Prov}_T(S^n 0, \ulcorner\neg\varphi\urcorner)$. Simultaneously $T \vdash \text{Pr}_T^R(\ulcorner\varphi\urcorner)$, hence there exists $m \leq n$ such that $T \vdash \text{Prov}_T(S^m 0, \ulcorner\varphi\urcorner)$. Moreover $m \leq n$, so m is standard as well. Thus we have standard proofs of φ and $\neg\varphi$, so T is inconsistent. $\square$

Let Con_T be the *consistency statement*, i.e. Con_T is the following formula $\neg \text{Pr}_T(\ulcorner 0 = S0 \urcorner)$.

Theorem 3.2.5 (Gödel's second incompleteness theorem) *If T is consistent then $T \not\vdash \text{Con}_T$.*

Proof By Theorem 3.2.3 it suffices to show that $T \vdash \text{Con}_T \equiv \varphi$, where φ satisfies $T \vdash \varphi \equiv \neg \text{Pr}_T(\ulcorner\varphi\urcorner)$.

The implication $\varphi \Rightarrow \text{Con}_T$ is intuitively obvious, for φ being equivalent to unprovability of something, must imply consistency. The exact argument is as follows. By basic arithmetic we have $T \vdash 0 = S0 \Rightarrow \varphi$, so by D1 $T \vdash \text{Pr}_T(\ulcorner 0 = S0 \Rightarrow \varphi \urcorner)$. Now, working in T , we see that if $\neg \text{Con}_T$, i.e. $\text{Pr}_T(\ulcorner 0 = S0 \urcorner)$, then $\text{Pr}_T(\ulcorner\varphi\urcorner)$ (by D3), so $\neg\varphi$.

For the converse assume $\neg\varphi$, i.e. $\text{Pr}_T(\ulcorner\varphi\urcorner)$. Then we have $\text{Pr}_T(\ulcorner\text{Pr}_T(\ulcorner\varphi\urcorner)\urcorner)$ by D2. On the other hand, by the property of φ , we have $T \vdash \text{Pr}_T(\ulcorner\varphi\urcorner) \Rightarrow \neg\varphi$ and, by D2, $T \vdash \text{Pr}_T(\ulcorner\text{Pr}_T(\ulcorner\varphi\urcorner) \Rightarrow \neg\varphi\urcorner)$. Hence, by D3, we have $\text{Pr}_T(\ulcorner\neg\varphi\urcorner)$. Summing up we obtained $T \vdash \neg\varphi \Rightarrow [\text{Pr}_T(\ulcorner\varphi\urcorner) \& \text{Pr}_T(\ulcorner\neg\varphi\urcorner)]$, hence $T \vdash \neg\varphi \Rightarrow \text{Pr}_T(\ulcorner\varphi \& \neg\varphi\urcorner)$. But by logic we have $T \vdash (\varphi \& \neg\varphi) \Rightarrow 0 = S0$, so by D1 $T \vdash \text{Pr}_T(\ulcorner\varphi \& \neg\varphi \Rightarrow 0 = S0\urcorner)$. It follows that $T \vdash \neg\varphi \Rightarrow \text{Pr}_T(\ulcorner 0 = S0\urcorner)$ by D3. $\square$

We remark that this result does not hold for provability in Rosser's sense (the reason is that derivability conditions D2 and D3 fail for this notion). More exactly:

Remark 3.2.6 T proves Con_T^R , where Con_T^R is $\neg\text{Pr}_T^R(\ulcorner 0 = S0\urcorner)$.

Proof Assume, towards a contradiction, that $T + \neg\text{Con}_T^R$ is consistent and take $\mathcal{M} \models T + \neg\text{Con}_T^R$. Then, take a code d of a standard proof in T of $\neg 0 = S0$. Since $\mathcal{M} \models \exists z(\text{Prov}_T(z, 0 = S0) \& \forall y < z \neg\text{Prov}_T(y, \neg 0 = S0))$, any witness d' for this sentence is smaller than d . Hence it has to be a standard proof of $0 = S0$ and T is consistent. $\square$

The following result is known as Tarski's theorem on undefinability of truth. We do not require T to be primitive recursive, in Theorem 3.2.7 we assume merely that T is consistent, in the language of PA and contains $I\Delta_0 + \text{Exp}$.

Theorem 3.2.7 (Tarski) *There is no formula $\text{Tr}(x)$ such that for every sentence φ , T proves $\varphi \equiv \text{Tr}(\ulcorner\varphi\urcorner)$.*

Proof Assume that such a formula Tr exists. Pick a sentence φ such that T proves the equivalence $\varphi \equiv \neg\text{Tr}(\ulcorner\varphi\urcorner)$. For this φ the equivalence $\neg\text{Tr}(\ulcorner\varphi\urcorner) \equiv \varphi \equiv \text{Tr}(\ulcorner\varphi\urcorner)$ is provable, hence T is inconsistent. $\square$

We shall give several other proofs of Tarski's theorem (see Sects. 3.5 and 3.9). In fact, the argument in Sect. 3.5 is completely free of diagonalization. This is of some interest as this shows that there are other reasons for Tarski's theorem to hold and separates this theorem from other results of this section.

The following result is known as the arithmetical hierarchy theorem, it is due to S. C. Kleene and A. Mostowski. The assumptions on T are as usual, including T being primitive recursive.

Theorem 3.2.8 (The arithmetical hierarchy theorem) *The universal formula Tr_{Σ_n} is not equivalent (in T) to any Π_n formula. Similarly, the universal formula Tr_{Π_n} is not equivalent to any Σ_n formula.*

Proof We prove this first part, the second one follows from the first one by taking negations. Assume that $T \vdash \forall \xi [A(\xi) \equiv \text{Tr}_{\Sigma_n}(\xi)]$, where $A(x)$ is Π_n . Then $\neg A$ is Σ_n and by the diagonal lemma (i.e. 3.2.1) and the remark following it there exists a Σ_n sentence δ such that $T \vdash \delta \equiv \neg A(\ulcorner\delta\urcorner)$. Now we have $T \vdash \delta \equiv \neg A(\ulcorner\delta\urcorner) \equiv \neg\text{Tr}_{\Sigma_n}(\ulcorner\delta\urcorner) \equiv \neg\delta$, hence T is inconsistent. $\square$

The next result involving the diagonal lemma is due to A. Church. We remark that a slightly more involved argument yields a generalization in which the assumption that Σ_1 -consequences of T are true in the standard model is not needed, see Shoenfield [8].

Theorem 3.2.9 (Undecidability theorem) *Let T be a primitive recursive theory containing $I\Delta_0$ such that all its Σ_1 theorems are true in $\mathbb{N}$. Then, the set of Gödel numbers of theorems of T is Σ_1 but not Δ_1 .*

Proof The set of all theorems of T is Σ_1 , for it is described by the formula $\exists x \text{Prov}_T(x, \varphi)$. Since T contains $I\Delta_0$ it proves all true Σ_1 sentences. Then, by our assumptions, for $\xi \in \Sigma_1$, $\mathbb{N} \models \xi$ iff $T \vdash \xi$. The result follows from Theorem 3.2.8. $\square$

3.3 Formalized Incompleteness Theorems

All results in Sect. 3.2 were proved externally (“in mathematics”, say in ZFC set theory). It turns out that both Gödel’s incompleteness theorems and Rosser’s theorem are theorems of T itself. The arguments are just formalizations of arguments from Sect. 3.2, so are essentially the same, but slightly more delicate.

Theorem 3.3.1 (The formalized first incompleteness theorem) *Let φ be such that $T \vdash \varphi \equiv \neg \text{Pr}_T(\ulcorner \varphi \urcorner)$. Then T proves the following implications:*

$$\text{if } \text{Con}_T \text{ then } \neg \text{Pr}_T(\ulcorner \varphi \urcorner), \quad (3.3)$$

$$\begin{aligned} \text{if } \forall \theta, u \{ [\theta \in \Delta_0 \ \& \ \forall w \text{Pr}_T(\neg \theta(S^u 0, S^w 0))] \Rightarrow \neg \text{Pr}_T(\exists x \theta(S^u 0, x)) \} \\ \text{then } \neg \text{Pr}_T(\ulcorner \neg \varphi \urcorner). \end{aligned} \quad (3.4)$$

Proof Let us point out that Con_T is implied by the assumption of (3.4). If we substitute the formula $0 = S0$ as θ in the assumption of (3.4) we see that $\neg \text{Pr}_T(\exists x (0 = S0))$ holds and Con_T follows trivially.

Now, we work in T and the assumption of (3.3). Towards a contradiction assume that $\text{Pr}_T(\ulcorner \varphi \urcorner)$. Then by D2 we infer $\text{Pr}_T(\ulcorner \text{Pr}_T(\ulcorner \varphi \urcorner) \urcorner)$. Simultaneously we have $\text{Pr}_T(\ulcorner \varphi \urcorner) \Rightarrow \neg \varphi$ by the property of φ , hence $\text{Pr}_T(\ulcorner \text{Pr}_T(\ulcorner \varphi \urcorner) \Rightarrow \neg \varphi \urcorner)$ holds by D1. By D3 we infer $\text{Pr}_T(\ulcorner \neg \varphi \urcorner)$, hence $\text{Pr}_T(\ulcorner 0 = S0 \urcorner)$, so $\neg \text{Con}_T$, contrary to the assumption. Thus we proved (3.3).

For the second claim pick any $\mathcal{M} \models T$ and the assumption of (3.4) and assume that $\text{Pr}_T(\ulcorner \neg \varphi \urcorner)$ holds in $\mathcal{M}$. By the property of φ we get $\neg \varphi \Rightarrow \text{Pr}_T(\ulcorner \varphi \urcorner)$, so by D1, $\text{Pr}_T(\ulcorner \neg \varphi \Rightarrow \text{Pr}_T(\ulcorner \varphi \urcorner) \urcorner)$ holds in $\mathcal{M}$, hence by D3, $\text{Pr}_T(\ulcorner \text{Pr}_T(\ulcorner \varphi \urcorner) \urcorner)$ holds. In other words, $\text{Pr}_T(\ulcorner \exists z \text{Prov}_T(z, \ulcorner \varphi \urcorner) \urcorner)$ holds in $\mathcal{M}$, so by the assumption, the statement $\exists z \neg \text{Pr}_T(\ulcorner \neg \text{Prov}_T(S^z 0, \ulcorner \varphi \urcorner) \urcorner)$ holds in $\mathcal{M}$. Fix such z . Since $\neg \text{Prov}_T(S^z 0, \ulcorner \varphi \urcorner)$ is not provable in $\mathcal{M}$, then the statement $\text{Prov}_T(S^z 0, \ulcorner \varphi \urcorner)$ holds in $\text{ACT}(\mathcal{M}; C)$ for

some completion $C \in \mathcal{M}$. But this statement is Δ_0 , hence, by absoluteness of Δ_0 formulas, it is true in $\mathcal{M}$ itself. It follows that $\mathcal{M}$ satisfies $\text{Pr}_T(\ulcorner \varphi \urcorner)$ and $\text{Pr}_T(\ulcorner \neg \varphi \urcorner)$, contrary to $\mathcal{M} \models \text{Con}_T$ by the first sentence of the proof. $\square$

Theorem 3.3.2 (The formalized Rosser's theorem) *Let φ be such that T proves $\varphi \equiv \neg \text{Pr}_T^R(\ulcorner \varphi \urcorner)$. Then T proves $\text{Con}_T \Rightarrow [\neg \text{Pr}_T(\ulcorner \varphi \urcorner) \ \& \ \neg \text{Pr}_T(\ulcorner \neg \varphi \urcorner)]$.*

Proof Assume $\text{Pr}_T(\ulcorner \neg \varphi \urcorner)$. Then, $\exists n \text{Prov}_T(n, \ulcorner \neg \varphi \urcorner)$. Let us fix such an n . Moreover, by the properties of φ , we have $\text{Pr}_T(\ulcorner \neg \text{Pr}_T^R(\ulcorner \varphi \urcorner) \urcorner)$. If $\exists m < n \text{Prov}_T(m, \ulcorner \neg \varphi \urcorner)$, then we have $\neg \text{Con}_T$. On the other hand, if $\forall m < n \neg \text{Prov}_T(m, \ulcorner \neg \varphi \urcorner)$ then, by Σ_1 -completeness, we have $\text{Pr}_T(\ulcorner \forall m < n \neg \text{Prov}_T(m, \ulcorner \neg \varphi \urcorner) \urcorner)$. It follows that $\text{Pr}_T(\ulcorner \text{Pr}_T^R(\ulcorner \varphi \urcorner) \urcorner)$ and, in consequence, $\neg \text{Con}_T$.

Assume now that $\text{Pr}_T(\ulcorner \neg \varphi \urcorner)$. Then, $\text{Pr}_T(\ulcorner \text{Pr}_T^R(\ulcorner \varphi \urcorner) \urcorner)$ and $\exists n \text{Prov}_T(n, \ulcorner \neg \varphi \urcorner)$. If $\exists m < n \text{Prov}_T(m, \ulcorner \neg \varphi \urcorner)$ then $\neg \text{Con}_T$ so assume $\forall m < n \neg \text{Prov}_T(m, \ulcorner \neg \varphi \urcorner)$. Then, $\text{Pr}_T(\ulcorner \text{Prov}_T(S^n 0, \ulcorner \neg \varphi \urcorner) \urcorner)$ and $\text{Pr}_T(\ulcorner \forall m < S^n 0 \neg \text{Prov}_T(m, \ulcorner \neg \varphi \urcorner) \urcorner)$. So $\text{Pr}_T(\ulcorner \forall m (\text{Prov}_T(m, \ulcorner \neg \varphi \urcorner)) \Rightarrow \exists n < m \text{Prov}_T(n, \ulcorner \neg \varphi \urcorner) \urcorner)$ what is just $\text{Pr}_T(\ulcorner \neg \text{Pr}_T^R(\ulcorner \varphi \urcorner) \urcorner)$. Therefore $\neg \text{Con}_T$. $\square$

Theorem 3.3.3 (The formalized second incompleteness theorem) *T proves the implication $\text{Con}_T \Rightarrow \neg \text{Pr}_T(\ulcorner \text{Con}_T \urcorner)$.*

Proof By the proof of Theorem 3.2.5 $T \vdash \text{Con}_T \equiv \neg \text{Pr}_T(\ulcorner \varphi \urcorner)$ and $T \vdash \text{Con}_T \equiv \varphi$, where φ is such that $\varphi \equiv \neg \text{Pr}_T(\ulcorner \varphi \urcorner)$. Thus we have $T \vdash \text{Con}_T \Rightarrow \varphi$. By D1, $T \vdash \text{Pr}_T(\ulcorner \text{Con}_T \Rightarrow \varphi \urcorner)$. Then, by D3, $T \vdash \text{Pr}_T(\ulcorner \text{Con}_T \urcorner) \Rightarrow \text{Pr}_T(\ulcorner \varphi \urcorner)$. Reverting the implication we get $T \vdash \neg \text{Pr}_T(\ulcorner \varphi \urcorner) \Rightarrow \neg \text{Pr}_T(\ulcorner \text{Con}_T \urcorner)$. Since $T \vdash \text{Con}_T \equiv \neg \text{Pr}_T(\ulcorner \varphi \urcorner)$, we finally get $T \vdash \text{Con}_T \Rightarrow \neg \text{Pr}_T(\ulcorner \text{Con}_T \urcorner)$. $\square$

In the next few sections we give other proofs of the incompleteness theorems, in these new proofs the construction of a sentence by means of the diagonal lemma is not needed (though some version of diagonalization occurs in each of them).

3.4 Satisfaction Classes

In this section we define the notion of a satisfaction class. We shall devote the whole chapter (Chap. 5) to the study of this notion. But we define it just now because we shall need it as a technical tool earlier, in Sect. 3.5. Moreover the notion of a satisfaction class is closely related to the notion of a completion from Sect. 3.1.

Let $\mathcal{M} \models T$. By a *full satisfaction class for $\mathcal{M}$* we mean a subset D of $\text{Sent}^{\mathcal{M}}$ which satisfies the usual conditions on truth, i.e.,

1. for every $\varphi \in \text{Sent}^{\mathcal{M}}$, if φ is of the form $S^a 0 + S^b 0 = S^c 0$ then $\varphi \in D$ iff $a + b = c$,
2. the same for other atomic formulas,
3. for every $\varphi, \psi \in \text{Sent}^{\mathcal{M}}$, the conjunction $\varphi \ \& \ \psi$ is in D iff both, φ, ψ are in D ,
4. for every $\varphi \in \text{Sent}^{\mathcal{M}}$, $\neg \varphi$ is in D iff $\varphi \notin D$,

5. for every formula $\psi(\cdot)$ in the sense of $\mathcal{M}$, the sentence $\exists x \psi(x)$ is in D iff $\psi(S^a 0) \in D$ for some $a \in \mathcal{M}$.

Think of this as follows: D is just a notion of truth for all sentences in the sense of $\mathcal{M}$ (including nonstandard ones). Observe that a completion (as defined in Sect. 3.1) is also a notion of truth, but for another model.

The definition of a satisfaction class may be refined to allow *partial* satisfaction classes. One simply considers any class Γ of formulas, which is closed under subformulas and requires the same but only for formulas in Γ . We shall work merely with one such refinement, given by the quantifier rank of a formula. It is defined as follows. For a class of formulas Γ let $\exists\Gamma$ be the closure of Γ under the existential quantification. We let $Q_0 = \Delta_0$ and let Q_{n+1} be the closure of $Q_n \cup \exists Q_n$ under disjunction, negation and bounded quantification. We write $\text{qr}(\varphi) \leq n$, (the *quantifier rank of φ is $\leq n$*) if $\varphi \in Q_n$. We shall also say that a formula is a Q_n -formula. By induction on n one can show that for each $n \geq 1$ each formula φ in Q_n is in Δ_{n+1} , i.e., φ is equivalent to a Σ_{n+1} formula φ_0 and a Π_{n+1} formula φ_1 this equivalence is provable in PA and formulas φ_0 and φ_1 can be constructed effectively from φ . If we do not care about effectivity we may simply point out that, for each n , $\exists\Delta_n$ is included in Δ_{n+1} and that each Δ_n is closed under disjunction, negation and bounded quantification. We point out that the universal formula Tr_{Q_n} for Q_n -formulas itself is of quantifier rank $n + 1$. Indeed, we may simply modify a universal formula for Σ_{n+1} class, $\text{Tr}_{\Sigma_{n+1}}$.

Let $\mathcal{M} \models T$ and let $e \in \mathcal{M}$. We define the notion of a Q_e -satisfaction class as a subset D of $\text{Sent}^{\mathcal{M}} \cap Q_e^{\mathcal{M}}$ satisfying the following conditions:

1. for every $\varphi \in \text{Sent}^{\mathcal{M}}$, if φ is of the form $S^a 0 + S^b 0 = S^c 0$ then $\varphi \in D$ iff $a + b = c$,
2. the same for other atomic formulas,
3. for every $\varphi, \psi \in Q_e$, the conjunction $\varphi \& \psi$ is in D iff both φ, ψ are in D ,
4. for every $\varphi \in Q_e$, $\varphi \in D$ iff $\neg\varphi \notin D$,
5. for every formula $\psi(\cdot)$ such that the sentence $\exists x \psi(x)$ is in Q_e , this sentence is in D iff $\psi(S^a 0)$ is in D for some $a \in \mathcal{M}$.

Observe that these notions make sense for models for any theory T which is strong enough to arithmetize the language. Observe also that the definition of a satisfaction class is a single sentence in the language $L_T \cup \{D\}$, the definition of a Q_e -satisfaction class is a formula in the variable e in this language.

Robinson [11] was the first to realize that nonabsoluteness of the finiteness may occur at the very definitions of the language; the notions of a full and partial satisfaction classes were defined explicitly by Krajewski [12]. We shall prove later that if D is a Q_e satisfaction class for $\mathcal{M}$, where e is nonstandard, then D is not definable with parameters in $\mathcal{M}$ and the same holds for full satisfaction classes (this observation is immediate from Tarski's theorem, i.e., Theorem 3.2.7).

As pointed out above, we shall study satisfaction classes in Chap. 5, at the moment we merely prove the following result, Theorem 3.4.2 (I don't remember whether I know it from Krajewski or from R. Murawski).

Say that a satisfaction class D for $\mathcal{M}$ is *inductive* iff the structure $(\mathcal{M}, D)$ satisfies induction in its language. The definition of a full inductive satisfaction is not a single sentence, but one may consider the theory $T + “D$ is a full inductive satisfaction class” in the language $L_T \cup \{D\}$.

The following easy lemma is basic for many constructions in the theory of models of arithmetic, it is due to Robinson [11] and Friedman [13].

Lemma 3.4.1 (Overspill lemma) *Let T be any theory in any language containing L_{PA} such that T proves induction for formulas in its language. Let $\mathcal{M} \models T$ be non-standard, let $\varphi(x, y)$ be formula and let $a \in \mathcal{M}$. Assume that for every $n \in \mathbb{N}$ $\mathcal{M} \models \varphi(a, n)$. Then there exists a nonstandard $b \in \mathcal{M}$ such that $\mathcal{M} \models \forall n < b \varphi(a, n)$.*

Proof Let us consider $\psi(a, x) = \forall z \leq x \varphi(a, z)$. Clearly we have $\mathcal{M} \models \psi(a, 0)$. Moreover, if the conclusion fails $\psi(a, x)$ is true exactly about standard numbers. Thus we also have $\mathcal{M} \models \forall x [\psi(a, x) \Rightarrow \psi(a, x + 1)]$. So, by induction, $\mathcal{M} \models \forall x \psi(a, x)$ contrary to $\mathcal{M}$ being nonstandard. $\square$

Theorem 3.4.2 *Let $\mathcal{M} \models T$ be nonstandard and countable. Then $\mathcal{M}$ has an inductive Q_e -satisfaction class for some nonstandard $e \in \mathcal{M}$ iff $\mathcal{M}$ is recursively saturated.*

Proof For one direction let a type $p(v, b)$ with parameter $b \in \mathcal{M}$ be given. Observe that as e is nonstandard, every $\varphi \in p$, being standard, is in the class Q_e . We have: for every $n \in \mathbb{N}$ $\mathcal{M} \models \exists v \wedge_{\varphi < n} D(\varphi(S^v 0, S^b 0))$. It follows that this statement must hold also for some nonstandard $n_0 \in \mathcal{M}$. Pick $b \in \mathcal{M}$ such that $\mathcal{M} \models \wedge_{\varphi < n_0} D(\varphi(S^v 0, S^b 0))$. Such b realizes the whole type p because its elements are standard. This proves that if $\mathcal{M}$ has an inductive Q_e -satisfaction class then it is recursively saturated.

For the other direction let $\mathcal{M}$ be recursively saturated. We use the theorem of Barwise and, independently, Ressayre stating that any countable recursively saturated model is resplendent (cf. Theorem 2.2.2). It states that for any recursive theory T in an extended language if elementary diagram of $\mathcal{M}$ is consistent with T then there is an expansion $\mathcal{M}'$ of $\mathcal{M}$ by interpretations for new predicates and functions such that $\mathcal{M}'$ satisfies T . Our theory T is a theory stating that S is a Q_n -satisfaction class, for each $n \in \mathbb{N}$, and that full induction holds in an extended language. The theory T is consistent with the elementary diagram of $\mathcal{M}$ since each of its finite fragments is satisfied in $\mathcal{M}$ by the truth definition Tr_{Q_n} for Q_n formulas. Now, let D be an expansion of $\mathcal{M}$ satisfying T . Thus, for each $n \in \mathbb{N}$, D satisfies the formula stating that D is a satisfaction class for Q_n . Then, by overspill, D is a satisfaction class for Q_e , for some nonstandard e .

So, both implications are proved. $\square$

3.5 Tarski's Theorem

The following version of Tarski's theorem is convenient for model-theoretic applications (see, e.g. the proof of Lemma 5.2.10).

Theorem 3.5.1 *Let $\mathcal{M} \models \text{PA}$ and let $X \subseteq \mathcal{M}$. Assume that there exist $E \subseteq \mathcal{M}$ which is definable with parameters in $(\mathcal{M}, X)$ and codes truth in $\mathcal{M}$ in the following sense: for every formula φ and every $u \in \mathcal{M}$ we have $\varphi(S^u 0) \in E$ iff $\mathcal{M} \models \varphi(u)$. Then X is not definable in $\mathcal{M}$ with parameters.*

Proof Let E be definable in $(\mathcal{M}, X)$ by a formula ψ with a parameter w . Assume that X is definable in $\mathcal{M}$ by a formula ϱ with a parameter z . Then the Skolem closure of $\{w, z\}$ must be either standard or recursively saturated by the same argument as in Theorem 3.4.2, what is impossible. In order to ensure that this model is nonstandard we change the argument slightly, namely we pick also any nonstandard $i \in \mathcal{M}$ and do the same with the Skolem closure of $\{w, z, i\}$. Thus, if $\mathcal{M}$ is nonstandard, we are done. If $\mathcal{M}$ itself is standard, then we go to any of its elementary extension $\mathcal{K}$ and do the same with $\mathcal{K}$. $\square$

It is worth to know that $\mathcal{M}$ -finite parts of truth may be coded by some $\mathcal{M}$ -finite (hence, definable with parameter) set. Precisely:

Proposition 3.5.2 *If $\mathcal{M}$ is a countable recursively saturated model of PA and $b \in \mathcal{M}$, then there exists $E \in \mathcal{M}$ such that for every formula φ and every $u < b$ $\mathcal{M} \models \varphi(u)$ iff $\varphi(S^u 0) \in E$. By induction $E \in \mathcal{M}$.*

Proof By Theorem 3.4.2 it has an inductive satisfaction class D for Q_e -formulas for some nonstandard $e \in \mathcal{M}$. We let $E = \{\varphi(S^u 0) : \varphi < e \text{ \& } u < b \text{ \& } D(\varphi(S^u 0))\}$. $\square$

It should be noticed that the countability assumption in Proposition 3.5.2 is inessential, but we give no proof of this fact as we work in this book only with countable models.

We shall give one more argument for Tarski's theorem at the end of Sect. 3.9, that argument will be based on Berry's antinomy.

3.6 Scott and Kreisel's Proofs

In this section we present proofs of the incompleteness theorems due to D. Scott and G. Kreisel. We follow the exposition in Smoryński [14].

Scott's proof of the first incompleteness theorem: This time we shall give a proof of the existence of an independent sentence without writing it explicitly. The assumptions on T are as follows: T is a primitive recursive consistent theory in L_{PA} containing $I\Sigma_2$ and true in the standard model. So assume that T is complete, then, since T is true,

$T \vdash \text{Con}_T$. In particular, T proves the existence of a completion of T (in the sense of Sect. 3.1). Take such a completion C . Choose φ such that $T \vdash \varphi \equiv \neg C(\ulcorner \varphi \urcorner)$, using the diagonal lemma, i.e., 3.2.1. We claim that $T \not\vdash \varphi$ and $T \not\vdash \neg\varphi$. For if T proves φ , then T proves also $C(\ulcorner \varphi \urcorner)$ by the definition of a completion, so it proves $\neg\varphi$ by the choice of φ . Similarly, if T proves $\neg\varphi$ then it proves $C(\ulcorner \neg\varphi \urcorner)$ by the definition of a completion, so it proves φ by the choice of φ . $\square$

G. Kreisel used some sort of iterations of the arithmetized completeness theorem to give another proof of the second incompleteness theorem:

Kreisel's proof of the second incompleteness theorem: Let $\varphi_n : n \in \mathbb{N}$ be a recursive enumeration of all sentences in L_{PA} . Consider the tree consisting of all finite sequences of the form $\langle \varphi_0^{p_0}, \dots, \varphi_n^{p_n} \rangle$, where each p_i is either 0 or 1 and φ^p is either φ if $p = 0$ or $\neg\varphi$ if $p = 1$. Of course, this family is ordered by the relation of being an initial segment. Let U be the subtree of the previous one consisting of all consistent sequences, i.e., such sequences as above but satisfying $T \not\vdash \neg \bigwedge_{i=0}^n \varphi_i^{p_i}$.

Observe that there exists a linear ordering on each level of the tree U . We compare two sequences as above by picking the first place where they differ (i.e., have different “exponents”) and consider the one which has exponent 0 to be to the left to the one which has the exponent equal 1.

Suppose we are given a model $\mathcal{M} \models T + \text{Con}_T$. Consider the tree U in the sense of $\mathcal{M}$. Then there exists a canonical completion $C \in \mathcal{M}$. It is given by the leftmost branch of the tree $U^{\mathcal{M}}$. We let $\text{CCompl}_T(C)$ be an abbreviation for “ C is the canonical completion of T ”.

Assume that $T \vdash \text{Con}_T$. Begin with any $\mathcal{M}_0 \models T$. Then $\mathcal{M}_0 \models \text{Con}_T$, so there exists the canonical completion $C_0 \in \mathcal{M}_0$. Let $\mathcal{M}_1 = \text{ACT}(\mathcal{M}_0; C_0)$. This model still satisfies Con_T , so there exists the canonical completion C_1 in it, let $\mathcal{M}_2 = \text{ACT}(\mathcal{M}_1; C_1)$. Continuing in the same fashion we obtain a sequence $\mathcal{M}_0, \mathcal{M}_1, \dots$ of models obtained by means of the arithmetized completeness theorem using canonical completions in each step. Using the diagonal lemma (i.e., 3.2.1 above) we choose φ such that T proves

$$\varphi \equiv \forall C [\text{CCompl}(C) \Rightarrow \neg \text{Tr}_{\Sigma_2}(\ulcorner C(\varphi) \urcorner)].$$

Let $\varphi = \varphi_{n_0}$ in the above enumeration of all sentences. We let U_n be the tree U in the sense of $\mathcal{M}_n$, but restricted up to the n_0 -th level. We observe that $U_{n+1} \subseteq U_n$ for all $n \in \mathbb{N}$. The reason is that $\mathcal{M}_{n+1}$ is an end extension of $\mathcal{M}_n$, so if there exists a proof of inconsistency of some path in $\mathcal{M}_n$, then the same proof is the proof of inconsistency of the same path in $\mathcal{M}_{n+1}$. It follows that the leftmost path in U_{n+1} cannot be to the left from the leftmost path in U_n . Moreover, the leftmost path in U_n always changes with changing n . Let us see how the choice of φ and n_0 ensures this. In $\mathcal{M}_0$ the leftmost path is $\langle \varphi_0^{p_0}, \dots, \varphi_{n_0-1}^{p_{n_0-1}}, \varphi^p \rangle$, where φ is φ_{n_0} . By the property of φ , in $\mathcal{M}_1$ we must have $1 - p$ at level n_0 (i.e., 0 if $p = 1$ and 1 otherwise). It follows that the above path is inconsistent in $\mathcal{M}_1$, so the leftmost path in $\mathcal{M}_1$ lies properly to the right from the above path (which was the leftmost consistent path in $\mathcal{M}_0$). Thus, in each step when passing from $\mathcal{M}_n$ to $\mathcal{M}_{n+1}$ the leftmost consistent

path changes to the right. This is a contradiction as together we obtained infinitely many different paths in the finite binary tree U_0 . $\square$

This proof shows that if we iterate the arithmetized completeness theorem and take the canonical completion (given by the leftmost consistent path in the tree of possible partial completions), then we can do it only finitely many times, for this procedure has to stop after at most 2^{n_0} steps.

3.7 Jech's Argument

Jech [15] worked with models of set theory (where the notion of a model is expressible) and then used some conservation argument to show that his proof works for PA as well. We give a version of his proof which works for PA directly. To be more specific we assume that T is a primitive recursive consistent theory in the language of PA and containing $I\Sigma_2$. The idea is very natural, namely we use completions (in the sense described in Sect. 3.1) as an arithmetical substitute of a model.

Jech's proof of the second incompleteness theorem: Let $\varphi_n : n$ be a primitive recursive enumeration of formulas in one free variable. Consider the formula $\gamma(x)$:

$$\exists C [\text{Compl}(C) \ \& \ \text{Tr}_{\Sigma_2}(C; \neg\varphi_x(S^x 0))]. \quad (3.5)$$

This formula is φ_k for some k . Let A be the sentence $\gamma(S^k 0)$. We check that

$$T \vdash A \equiv \exists C [\text{Compl}(C) \ \& \ \text{Tr}_{\Sigma_2}(C, \neg A)]. \quad (3.6)$$

Working in T , we see that

$$\begin{aligned} A &\equiv \gamma(S^k 0) \\ &\equiv \exists C [\text{Compl}(C) \ \& \ \text{Tr}_{\Sigma_2}(C; \neg\varphi_k(S^k 0))] \\ &\equiv \exists C [\text{Compl}(C) \ \& \ \text{Tr}_{\Sigma_2}(C; \neg\gamma(S^k 0))] \\ &\equiv \exists C [\text{Compl}(C) \ \& \ \text{Tr}_{\Sigma_2}(C; \neg A)]. \end{aligned}$$

Call a completion C *positive* if $\text{Tr}_{\Sigma_2}(C; A)$ and *negative* otherwise. Then, by (3.6),

if a completion C is negative then

$$\text{Tr}_{\Sigma_2}(C; \forall D [\text{Compl}(D) \Rightarrow \text{Tr}_{\Sigma_2}(D; A)]). \quad (3.7)$$

Now assume, towards a contradiction, that $T \vdash \text{Con}_T$ and pick any model $\mathcal{M}$ of T . Then $\mathcal{M} \models \text{Con}_T$, so, by the arithmetized completeness theorem, $\mathcal{M} \models \exists C \text{ Compl}(C)$. We claim that

$$\mathcal{M} \models \exists C [\text{Compl}(C) \ \& \ \text{“}C \text{ is negative”}]. \quad (3.8)$$

Pick any completion $C \in \mathcal{M}$. If it is negative then we are done. Otherwise C is a positive completion in $\mathcal{M}$. Now, take $\mathcal{K} = \text{ACT}(\mathcal{M}; C)$. Since C is positive $\mathcal{K} \models \exists D [\text{Compl}(D) \ \& \ \text{Tr}_{\Sigma_2}(D, \neg A)]$. Now take a negative completion D in $\mathcal{K}$, and define $D' = D \cap \mathcal{M}$ (cf. Sect. 3.1 for iterations of the arithmetized completeness theorem). Then D' is a negative completion in $\mathcal{M}$.

We established that in any model $\mathcal{M}$ of $\mathbf{T}$ there is a negative completion. But now, let us take a negative completion C in $\mathcal{M}$ and construct $\mathcal{K} = \text{ACT}(\mathcal{M}; C)$. By 3.7, there are only positive completions in $\mathcal{K}$, a contradiction. $\square$

Of course, this translation of Jech's proof gives a version which is almost the same as the original Gödel's one, see Sect. 3.2. For (3.6) is just Gödel's property $\varphi \equiv \neg \text{Pr}_{\mathbf{T}}(\ulcorner \varphi \urcorner)$, by the arithmetized completeness theorem.

3.8 Nonstandard Models and Incompleteness

In this section we present proofs of the incompleteness theorems by means of non-standard models (but this nonstandardness must be expressed in a very strong way). Richard Kaye told the author that these arguments are based on the so-called *busy beaver problem*, cf. [16] for more in this direction.

The proof is based on some work with the following formula $\phi(x, y)$:

$$\forall \varphi, u \leq x \{ [\varphi \in \Delta_0 \ \& \ \exists w \text{Tr}_0(\varphi(S^u 0, S^w 0))] \Rightarrow \exists w < y \text{Tr}_0(\varphi(S^u 0, S^w 0)) \}. \quad (3.9)$$

There are some useful variants of this formula which will be used later in the book, e.g. in Sect. 4.3. These formulas, say $\phi_k(x, y)$, are as follows.

$$\forall \varphi, u \leq x \{ [\varphi \in \Pi_k \ \& \ \exists w \text{Tr}_{\Pi_k}(\varphi(S^u 0, S^w 0))] \Rightarrow \exists w < y \text{Tr}_{\Pi_k}(\varphi(S^u 0, S^w 0)) \}. \quad (3.10)$$

It is easy to see that $\mathbf{T} \vdash \forall x \exists y \phi(x, y)$. The reason is as follows. Work in $\mathbf{T}$. Fix x . Enumerate as $\{\varphi_i(S^{u_i} 0, z) : i < r\}$ all substitutions of the form $\varphi(S^u 0, z)$, where $\varphi \in \Delta_0$, $\varphi, u \leq x$. By induction on j it is easy to see that

$$\exists y \forall i \leq j \{ [\exists w \text{Tr}_0(\varphi_i(S^{u_i} 0, S^w 0))] \rightarrow \exists w < y \text{Tr}_0(\varphi_i(S^{u_i} 0, S^w 0)) \}.$$

We apply this observation to $j = r - 1$.

It follows that the function $F(x) = \min y : \phi(x, y)$ is well defined in $\mathbf{T}$. Clearly, this function dominates every Δ_0 function, hence it dominates each Σ_1 function.

We remark that every function F which dominates each Δ_0 function, dominates every Σ_1 function as well. For if $H(x) = y$ is a Σ_1 function, we write $H(x) = y \equiv \exists t K(x, y, t)$ with $K \in \Delta_0$ and we see that F dominates the function $x \mapsto \min\langle y, t \rangle : K(x, y, t)$ which dominates H .

The main property of $\phi(x, y)$ is:

$$\mathbf{T} \vdash \{ \text{Con}_{\mathbf{T}} \Rightarrow \forall x, y [\text{Pr}_{\mathbf{T}}(\phi(S^x 0, S^y 0)) \Rightarrow \phi(x, y)] \}. \quad (3.11)$$

While working in $\mathbf{T}$ we see that if $\neg\phi(x, y)$, then $\text{Pr}_{\mathbf{T}}(\neg\phi(S^x 0, S^y 0))$ —because $\phi(x, y)$ is Π_1 , its negation is Σ_1 and we may apply the demonstrable Σ_1 –completeness. So, we have the implication $\neg\text{Pr}_{\mathbf{T}}(\neg\phi(S^x 0, S^y 0)) \Rightarrow \phi(x, y)$. Now, by $\text{Con}_{\mathbf{T}}$, if $\text{Pr}_{\mathbf{T}}(\phi(S^x 0, S^y 0))$ then $\neg\text{Pr}_{\mathbf{T}}(\neg\phi(S^x 0, S^y 0))$. Hence (3.11) follows.

Granted (3.11) we claim that there exists $a = a(\mathbf{T})$ such that

$$\mathbf{T} \vdash [\text{Con}_{\mathbf{T}} \Rightarrow \forall b \neg\text{Pr}_{\mathbf{T}}(\phi(S^a 0, S^b 0))]. \quad (3.12)$$

Let us work in $\mathbf{T}$. Fix a . An exact bound on a value of a is to be specified later. Towards a contradiction, pick the smallest b for which $\text{Pr}_{\mathbf{T}}(\phi(S^a 0, S^b 0))$. By (3.11), $\phi(a, b)$ holds as well. We substitute in $\phi(x, y)$ the formula $\text{Pr}_{\mathbf{T}}(\phi(S^x 0, S^y 0))$ for φ and the parameter a for u . In order to obtain a contradiction, we change $\text{Pr}_{\mathbf{T}}(\phi(S^x 0, S^y 0))$ slightly. Firstly, we write this formula explicitly in its Σ_1 form, that is we write $\text{Pr}_{\mathbf{T}}(\phi(S^x 0, S^{w_0} 0))$ as $\exists w_1 H'(x, w_0, w_1)$, where $H' \in \Delta_0$. Then we join two variables w_0, w_1 to one—simply use a pair $w = \langle w_0, w_1 \rangle$ and write $(w)_i$ for w_i , for $i \in \{0, 1\}$. Let $H(x, w)$ be a formula $H'(x, (w)_0, (w)_1)$. Finally, we substitute in $\phi(a, b)$ the formula $H(x, w)$ for φ and we substitute a for a parameter u . Thus, we should choose $a \geq \ulcorner H(x, w) \urcorner$. Let us observe that the value of $H(x, w)$ is easily computable from the recursive index of $\mathbf{T}$.

Now, let us write our assumption $\text{Pr}_{\mathbf{T}}(\phi(S^a 0, S^b 0))$ as $\exists d \text{Prov}_{\mathbf{T}}(d, \phi(S^a 0, S^b 0))$. Let us fix such a d . Thus, by construction of the formula $H(x, w)$, we have $\text{Tr}_0(\ulcorner H(S^a 0, S^{(b,d)} 0) \urcorner)$, so also $\exists w \text{Tr}_0(\ulcorner H(S^a 0, S^w 0) \urcorner)$. Now, by $\phi(a, b)$, it holds that $\exists w < b \text{Tr}_0(\ulcorner H(S^a 0, S^w 0) \urcorner)$ and $\exists w < b H(a, w)$. However, w is a code for a pair, say $\langle b', d' \rangle$. Our usual coding of pairs, that is $\langle x, y \rangle = \frac{(x+y)(x+y+1)}{2} + x$ (cf. first section of this chapter), has the property that both elements are less or equal to the code of a pair. Thus, $b' < b$ and we obtain that $\exists d \text{Prov}_{\mathbf{T}}(d, \phi(S^a 0, S^{b'} 0))$ and $\text{Pr}_{\mathbf{T}}(\phi(S^a 0, S^{b'} 0))$. This contradicts the minimality of b and proves (3.12).

Observe that we have just obtained a very weak version of the first incompleteness theorem:

Let a be as above. Then the statement $\exists b \phi(S^a 0, S^b 0)$ is provable in $\mathbf{T}$, hence if $\mathbb{N} \models \mathbf{T}$ then for some $b \in \mathbb{N}$ we have $\mathbb{N} \models \phi(a, b)$. But, for this b , the statement $\phi(S^a 0, S^b 0)$ is unprovable.

Let us give the proofs of stronger versions.

Proof of the first incompleteness theorem [4] We prove firstly the formalized version stated as Theorem 3.3.1. Pick a model $\mathcal{M}$ of $\mathbf{T}$ such that $\mathcal{M} \models \text{Con}_{\mathbf{T}}$. Let us recall that $\text{Con}_{\mathbf{T}}$ is a consequence of predecessors of both implications of Theorem 3.3.1, i.e., (3.3) and (3.4) so we may use it freely in proving both implications. Pick $\phi(x, y)$ and $a \in \mathbb{N}$ as above, that is $\mathcal{M} \models \forall b \neg\text{Pr}_{\mathbf{T}}(\phi(S^a 0, S^b 0))$. Let us observe, that the whole theory $\{\neg\phi(S^a 0, S^b 0) : b \in \mathcal{M}\}$ is consistent with $\mathbf{T}$ in $\mathcal{M}$. Otherwise, by compactness in $\mathcal{M}$, $\mathbf{T}$ would prove $\bigvee_{b \leq c} \phi(S^a 0, S^b 0)$, for some $c \in \mathcal{M}$. But this is impossible as such a formula is simply equivalent to $\phi(S^a 0, S^c 0)$ and we have $\neg\text{Pr}_{\mathbf{T}}(\phi(S^a 0, S^c 0))$. Thus, by the arithmetized completeness theorem (see Sect. 3.1),

there exists a completion $C \in \mathcal{M}$ and a model $\mathcal{K} = \text{ACT}(\mathcal{M}; C)$ such that for every $b \in \mathcal{M}$ $\mathcal{K} \models \neg\phi(a, b)$. But $\text{T} \vdash \forall x \exists y \phi(x, y)$. It follows that $b_0 = \min\{b : \phi(a, b)\}$ (as defined in $\mathcal{K}$) is greater than all elements of $\mathcal{M}$ (think of this as follows: b_0 is not $\mathcal{M}$ -standard in $\mathcal{K}$). Thus, there exist $\varphi, u \leq a$ with that $\varphi \in \Delta_0$ such that for every $y \in \mathcal{M}$, $\mathcal{K} \models \neg\varphi(u, y)$ but $\mathcal{K} \models \exists y \varphi(u, y)$. Thus, $\mathcal{M} \models \neg\text{Pr}_T(\neg\exists y \varphi(S^u 0, y))$ because the statement $\exists y \varphi(S^u 0, y)$ is true in $\mathcal{K} = \text{ACT}(\mathcal{M}; C)$, therefore its negation cannot be provable in $\mathcal{M}$. On the other hand, $\mathcal{M} \models \forall y \neg\text{Pr}_T(\varphi(S^u 0, S^y 0))$. By the assumption (3.4) of 3.3.1, that is

$$\forall \theta, u \{ [\theta \in \Delta_0 \ \& \ \forall w \text{Pr}_T(\neg\theta(S^u 0, S^w 0))] \Rightarrow \neg\text{Pr}_T(\exists x \theta(S^u 0, x)) \},$$

we infer $\mathcal{M} \models \neg\text{Pr}_T(\exists y \varphi(S^u 0, y))$. This shows the second part of Theorem 3.3.1. The same argument establishes the nonformalized version, i.e. Theorem 3.2.3. $\square$

Proof of the second incompleteness theorem, i.e. 3.2.5 [4] It is quite easy to see that for every $a \in \mathbb{N}$ there exist models of T in which $\exists b \text{Pr}_T(\phi(S^a 0, S^b 0))$ holds. The quickest way to construct such models is as follows. Begin with any $\mathcal{M} \models \text{T}$. Enumerate as $\varphi_i(S^{u_i} 0, x) : i < r$ all substitutions of the form $\varphi(S^u 0, x)$ with $\varphi, u \leq a$ and $\varphi \in \Delta_0$. Construct a sequence $\mathcal{M}_i : i < r$ of models of T in the following way. If $\mathcal{M} \models \text{Pr}_T(\forall x \neg\varphi_0(S^{u_0} 0, x))$ we let $\mathcal{M}_0 = \mathcal{M}$. Otherwise by the arithmetized completeness theorem there exists $\mathcal{M}_0 \supseteq_{\text{end}} \mathcal{M}$ which is a model of T and in which the statement $\exists w \varphi_0(S^{u_0} 0, S^w 0)$ holds. This statement, being true in $\mathcal{M}_0$, is provable in this model. We iterate this construction, i.e. we apply it to $\mathcal{M}_0$ and $\varphi_1(S^{u_1} 0, x)$ to obtain $\mathcal{M}_1$, and so on. There are two possibilities: either at some point we get a model of $\neg\text{Con}_T$ and we are done or the final model $\mathcal{M}_{r-1}$ satisfies

$$\forall i < r (\text{Pr}_T(\forall x \neg\varphi_i(S^{u_i} 0, x)) \vee \exists w \text{Pr}_T(\varphi(S^{u_i} 0, S^w 0))).$$

This is so, because while passing to further extensions we do not loose the information we got as this information is Σ_1 . Working in $\mathcal{M}_{r-1}$ for each $i < r$ such that the second case holds we choose the smallest appropriate w and we let b be any element greater than the maximum of all these w . Such an element b is, provably in $\mathcal{M}_{r-1}$, a good witness for $\exists y \phi(a, y)$. Thus, if a is chosen as above $\mathcal{M}_{r-1} \models \neg\text{Con}_T$. $\square$

This argument gave the second incompleteness theorem (for each primitive recursive theory in L_{PA} containing $I\Sigma_2$). We remark that much weaker assumption suffices and the use of the arithmetized completeness theorem may be eliminated, cf. [17].

One more remark should be added here. In the proof of the first incompleteness theorem we used a model $\mathcal{K} = \text{ACT}(\mathcal{M}; C)$ in which the element $b_0 = \min\{b \in \mathcal{K} : \phi(a, b)\}$ is not $\mathcal{M}$ -standard. This gave a Σ_1 independent sentence. But this sentence need not be $\neg\text{Con}_T$, e.g., it could be $\neg\text{Con}_U$ for some theory U which is stronger than T (e.g. if $\text{T} = \text{PA}$, U could be ZFC). But iterations of the construction given by the arithmetized completeness theorem gave a model for $\neg\text{Con}_T$.

Let us give a proof of the formalized second incompleteness theorem (i.e. Theorem 3.3.3) along these lines.

Proof of Theorem 3.3.3 It suffices to see that every model $\mathcal{M}_j$ from the above proof is either equal $\mathcal{M}$ or can be chosen of the form $\text{ACT}(\mathcal{M}; E)$ for some completion $E \in \mathcal{M}$. This is so by Proposition 3.1.5. Hence this applies also to the final model $\mathcal{M}_{r-1}$. Granted this we see that $\neg \text{Con}_T$ belongs to some completion $E \in \mathcal{M}$. Moreover, this holds in every $\mathcal{M}$ which is a model of $T + \text{Con}_T$. Thus, $\text{Con}_T \Rightarrow \neg \text{Pr}_T(\text{Con}_T)$ is provable in this theory. $\square$

We stress that the heart of the matter in the argument presented in this section is the use of models in which $\exists b \text{Pr}_T(\phi(S^a 0, S^b 0))$ holds. Such models must be nonstandard in a strong sense of the word. The essential reason for the existence of nonstandard models (in this strong sense) is that otherwise the function $G(a) = \min\{b : \text{Pr}_T(\phi(S^a 0, S^b 0))\}$ would be well defined and would dominate each Σ_1 function (cf. (3.12)). But then G would have to dominate itself, by (3.11), what is impossible.

3.9 Incompleteness Theorems via Berry's Paradox

Before giving Kikuchi's proof from [18] we give some historical remarks. The proof is based on the so-called *Berry's paradox* (the least integer admitting no name involving less than two hundred words in English—this integer was just defined using less than two hundred words). As far as I know, Vopěnka [19] was the first to be able to transform this paradox to a proof of the second incompleteness theorem. His paper was unnoticed, however. Also in Boolos–Jeffrey [16] there is a footnote in which the authors say “Saul Kripke tells us that in the early 1960s he noticed a proof somewhat similar to the one given here” ([16], p. 291). Kikuchi's argument is an extension of the work of Boolos [20].

We begin an exposition of a version of Kikuchi's arguments from [18]. We work with the size of a formula (as the Gödel number) rather than counting the number of symbols in it. Moreover (this is a professional deviation of a model-theorist), we work with formulas with parameters. Our assumptions on T are as usual, i.e., T is a primitive recursive consistent theory in the language of Peano Arithmetic and containing PA.

Say that a formula φ with parameter $u \in \mathbb{N}$ *names* $d \in \mathbb{N}$ in T if

$$T \vdash \varphi(S^u 0, S^d 0) \ \& \ \exists! x \varphi(S^u 0, x).$$

Observe that (since T is consistent, by the assumption), every pair φ, u names at most one number d . It follows that for every natural number a there are at most $(a + 1)^2$ numbers named by $\varphi, u \leq a$. The formal counterpart of this notion, that is

$$\text{Pr}_T(\varphi(S^u 0, S^d 0) \ \& \ \exists! x \varphi(S^u 0, x))$$

will be denoted $\text{Nam}(\varphi, u, d)$. Let also $\text{Ber}(x, y)$ be the following formula

$$\forall \varphi, u \leq x \neg \text{Nam}(\varphi, u, y) \ \& \ \forall z < y \exists \varphi, u \leq x \text{Nam}(\varphi, u, z).$$

Of course, $\text{Ber}(x, y)$ expresses: “ y admits no name $\varphi(u, \cdot)$ with $\varphi, u \leq x$ and is the smallest with this property”. The notation Ber comes from Berry's paradox. Let us observe, that T recognizes that y is unique with respect to a given x , that is

$$T \vdash \forall x, y, y' (\text{Ber}(x, y) \ \& \ \text{Ber}(x, y') \Rightarrow y = y'). \quad (3.13)$$

Now, let us turn to the proof of the second incompleteness theorem.

Proof of the second incompleteness theorem, [18] Let Nam and Ber be as above, let $a = \ulcorner \text{Ber}(x, y) \urcorner$, and let $e = (a + 1)^2$ for convenience. Then

$$T \vdash \text{Con}_T \equiv \exists b \leq S^e 0 \text{Ber}(S^a 0, b). \quad (3.14)$$

The implication $\leftarrow$ is immediate (if such b exists then T does not prove something). For the converse it suffices to observe that Con_T implies that every pair φ, u names at most one element.

Now, let us work in $T + \text{Con}_T$. To complete the proof we need to show $\neg \text{Pr}_T(\text{Con}_T)$.

By (3.14), $\exists b \leq S^e 0 \text{Ber}(S^a 0, b)$. Fix such b . Assume that $\text{Pr}_T(\text{Ber}(S^a 0, S^b 0))$. Then $\text{Pr}_T(\exists x \text{Ber}(S^a 0, x))$. It follows then by (3.13) and derivability conditions, that $\text{Pr}_T(\exists! x \text{Ber}(S^a 0, x))$. Thus we get $\text{Nam}(\text{Ber}, S^a 0, S^b 0)$, contrary to the choice of b . Summing up, in $T + \text{Con}_T$ we proved that there exists b such that

$$\text{Ber}(S^a 0, S^b 0) \text{ and } \neg \text{Pr}_T(\text{Ber}(S^a 0, S^b 0)).$$

Moreover, for this b , we have

$$\forall c < b \exists \varphi, u \leq a \text{Nam}(\varphi, u, c).$$

Now, exactly as in Sect. 3.8 we iterate the Arithmetized Completeness Theorem. That is, we choose any $\mathcal{M} \models T + \text{Con}_T$. We choose $b \in \mathcal{M}$ as above, i.e., b is the element of $\mathcal{M}$ such that $\text{Ber}(a, b)$ holds in $\mathcal{M}$. For this b we have $\neg \text{Pr}_T(\text{Ber}(S^a 0, S^b 0))$. We take any completion $C \in \mathcal{M}$ such that $\mathcal{K} = \text{ACT}(\mathcal{M}; C)$ satisfies $\neg \text{Ber}(a, b)$. But $\mathcal{K}$ satisfies also $\forall c < b \exists \varphi, u \leq a \text{Nam}(\varphi, u, c)$ (this is Σ_1 true statement, so is provable in T , so is in C) so either $\mathcal{K} \models \neg \text{Con}_T$ or the smallest b' such that $\mathcal{K} \models \neg \text{Ber}(a, b')$ is strictly greater than b . If $\mathcal{K} \models \neg \text{Con}_T$ then we are done as this shows that $\mathcal{M} \models \neg \text{Pr}_T(\text{Con}_T)$. Otherwise we repeat this construction. Observe that $\neg \text{Ber}(a, b)$ and $\neg \text{Ber}(a, b')$ hold in each further model, for the statement $\forall c < y \exists \varphi, u \leq a \text{Nam}(\varphi, u, c)$ is Σ_1 and is preserved in each further model. After e steps we must obtain a model of $\neg \text{Con}_T$, by (3.14). By Proposition 3.1.5 this iterated extension is (essentially) obtained by a single application of the Arithmetized Completeness Theorem. It follows that $\neg \text{Pr}_T(\text{Con}_T)$ holds in each model of $T + \text{Con}_T$, so is provable in this theory.

Together we obtained both versions of Gödel's second incompleteness theorem, i.e., Theorems 3.2.5 and 3.3.3. Theorem 3.2.5 follows from the fact that one of the models obtained by means of the Arithmetized Completeness Theorem has to be a model of $\neg\text{Con}_T$. $\square$

Proof of the formalized first incompleteness theorem [18] We worked with the statement $\exists b \leq S^e 0 \text{ Ber}(S^a 0, b)$ of which we proved in T that it is equivalent to Con_T , see (3.14). This statement is not Π_1 . But

$$T \vdash \exists b \leq S^e 0 \text{ Ber}(S^a 0, b) \equiv \exists y \leq S^e 0 \forall \varphi, u \leq S^a 0 \neg \text{Nam}(\varphi, u, y), \quad (3.15)$$

and the right-hand-side is Π_1 . Hence we shall work with the right-hand-side of (3.15). We abbreviate it as γ .

We have just seen that $T + \text{Con}_T \vdash \neg \text{Pr}_T(\text{Con}_T)$, hence $T + \text{Con}_T \vdash \neg \text{Pr}_T(\gamma)$ by (3.14), (3.15) and derivability conditions. This gives one half of the Formalized First Incompleteness Theorem. For the second half we state at first the formal version of $\omega\text{-Con}_T$ for Δ_0 formulas:

$$\forall \varphi, u \{ [\varphi \in \Delta_0 \ \& \ \forall d \text{ Pr}_T(\varphi(S^u 0, S^d 0))] \rightarrow \neg \text{Pr}_T(\exists x \neg \varphi(S^u 0, x)) \}. \quad (3.16)$$

Let us observe that this is just the assumption of the formalized first incompleteness theorem, cf. Theorem 3.3.1. Clearly, $\neg\gamma$ is Σ_1 , so let A be a Δ_0 formula such that $T \vdash (\neg\gamma) \equiv \exists x A(x)$. Working in $T + \omega\text{-Con}_T$, assume $\text{Pr}_T(\neg\gamma)$, i.e., $\text{Pr}_T(\exists x A(x))$. By (3.16) there exists x such that $\neg \text{Pr}_T(\neg A(S^x 0))$. By the completeness of T with respect to Δ_0 -sentences, for this x we have $\text{Pr}_T(A(S^x 0))$. But, obviously $T \vdash \omega\text{-Con}_T \rightarrow \text{Con}_T$ (just apply the assumption to the formula $x = x$). It follows from Con_T that $\text{Pr}_T(A(S^x 0))$ implies $A(x)$ and $\exists x A(x)$. Thus, we proved in $T + \omega\text{-Con}_T$ that if $\text{Pr}_T(\neg\gamma)$ then $\neg\gamma$, i.e., $\neg\text{Con}_T$. Simultaneously we have also Con_T and therefore $T + \omega\text{-Con}_T$ is inconsistent. Thus, we showed $T + \omega\text{-Con}_T \vdash \neg \text{Pr}_T(\neg\gamma)$. But, by derivability conditions, $\neg \text{Pr}_T(\neg\gamma) \equiv \neg \text{Pr}_T(\neg\text{Con}_T)$. Summing up, we obtained the second half of the Formalized First Incompleteness Theorem: $T + \omega\text{-Con}_T \vdash \neg \text{Pr}_T(\neg\text{Con}_T)$. $\square$

It should be noticed that non-formalized version of the First Incompleteness Theorem may be proved along the same lines.

We show that Kikuchi's argument may be adapted to give a proof of the strong version of Tarski's theorem (Kikuchi [18] writes that his work was inspired by the work of Boolos [20] in which some weak version of Tarski's theorem is proved).

Proof of Tarski's Theorem 3.2.7 Assume that a formula $A(\cdot)$ is such that

$$T \vdash \varphi \equiv A(\ulcorner \varphi \urcorner) \text{ for every sentence } \varphi \quad (3.17)$$

but T is consistent. We define (inside T): the pair $\langle \psi, u \rangle$ *names* w if

$$A(\ulcorner \psi(S^u 0, S^w 0) \ \& \ \forall x < S^w 0 \neg \psi(S^u 0, x) \urcorner) \quad (3.18)$$

Observe that for natural numbers u, w we are talking about the sentence in L_T . Moreover, if T is consistent then every pair $\langle \psi, u \rangle$ in which ψ is a formula and u is standard, names at most one object w . It follows that for every $k, r \in \mathbb{N}$

$$T \vdash \forall \varphi, u \leq S^k 0 [(\exists w < S^r 0 \langle \varphi, u \rangle \text{ names } w) \Rightarrow \exists! w < S^r 0 \langle \varphi, u \rangle \text{ names } w]. \quad (3.19)$$

Working in T we see that if some $\langle \varphi, u \rangle$ names w , then $\varphi(u, w) \& \forall x < w \neg \varphi(u, x)$ holds by (3.17), hence w is uniquely determined by $\langle \varphi, u \rangle$. We show that for every $k \in \mathbb{N}$

$$T \vdash \exists b \leq (k+1)^2 \forall \varphi, u \leq k \langle \varphi, u \rangle \text{ does not name } b. \quad (3.20)$$

For there are at most $(k+1)^2$ pairs $\langle \varphi, u \rangle$ with $\varphi, u \leq k$ and each such pair names at most one object. Let $B(x, y)$ be

$$\forall \varphi, u \leq x \{[\exists w \langle \varphi, u \rangle \text{ names } w] \rightarrow \exists w < y \langle \varphi, u \rangle \text{ names } w\}, \quad (3.21)$$

let $C(x, y)$ be $B(x, y) \& \forall z < w \neg B(x, z)$. We let $k = \ulcorner C(x, y) \urcorner$ and apply (3.20) to obtain $b \leq (k+1)^2$ which is the smallest object which admits no name $\langle \varphi, u \rangle$ with $\varphi, u \leq k$. This is a contradiction because the formula $C(x, y)$ with parameter k names it. $\square$

3.10 Incompleteness via Grelling's Antinomy

Say that a predicate is *heterological* iff it is not true of itself. The question is whether being heterological is heterological itself. This clearly leads to a paradox, called *Grelling's antinomy* in the literature. Cezary Cieřliński [21] was able to transform this antinomy to a proof of Gödel's incompleteness theorem. We give an exposition of his argument. Let T be a primitive recursive theory in L_{PA} containing $I\Sigma_2$. Let φ_i be a primitive recursive enumeration of formulas in one free variable v and let $\text{het}_T(x)$ be $\exists C [\text{Compl}_T(C) \& \text{Tr}_2(C(\neg \varphi_x(S^x 0)))]$ (we use notation from Sect. 3.1). Thus, $\text{het}_T(x)$ expresses “ $S^x 0$ does not have the property φ_x in some ACT-extension”. Let also HET_T be the sentence $\text{het}_T(\ulcorner \text{het}_T(x) \urcorner)$. Obviously, T proves the implication $\text{HET}_T \Rightarrow \text{Con}_T$. Moreover, we have, in T , $\text{het}_T(x) \equiv \exists C \text{ Compl}_{T+\neg \varphi_x(S^x 0)}$, hence $\text{het}_T(x) \equiv \text{Con}_{T+\neg \varphi_x(S^x 0)}$. It follows that $\text{het}_T(x) \equiv \neg \text{Pr}_T(\ulcorner \varphi_x(S^x 0) \urcorner)$. Substituting this to the definition of HET_T we infer $\text{HET}_T \equiv \neg \text{Pr}_T(\ulcorner \text{het}_T(\text{het}_T(x)) \urcorner)$, hence $\text{HET}_T \equiv \neg \text{Pr}_T(\ulcorner \text{HET}_T \urcorner)$. In other words, the sentence $\xi = \text{HET}_T$ has the usual Gödel's property $\xi \equiv \neg \text{Pr}_T(\ulcorner \xi \urcorner)$, so the further part of the argument may be carried out exactly like in the usual proofs, see Sect. 3.2.

3.11 Σ_1 -Closed Models

We have seen in Sect. 3.8 that if we require a model $\mathcal{M}$ of T to be nonstandard in a strong sense of the word then it must satisfy $\neg\text{Con}_T$. This idea was developed further by Adamowicz and Bigorajska [22]. This is of interest because their argument shows a link between the idea $\exists_1$ -closed model (known in model theory, see [23]) and the second incompleteness theorem. Say that a model $\mathcal{M}$ of a theory T is $\exists_1$ -closed if for every quantifier free formula $\varphi(v_0, \dots, v_r)$ and every $a_1, \dots, a_r \in \mathcal{M}$, if some extension $\mathcal{K} \models T$ of $\mathcal{M}$ satisfies $\exists v_0 \varphi(v_0, a_1, \dots, a_r)$, then $\mathcal{M}$ satisfies this formula. When working with models of PA and related theories it is more natural to work with Σ_1 formulas rather than $\exists_1$ ones. This leads to the following definition. Let T be a theory in L_{PA} and containing $I\Delta_0 + \text{Exp}$. Say that a model $\mathcal{M}$ for T is Σ_1 -closed if for every Δ_0 formula $\varphi(v_0, \dots, v_r)$ and every $a_1, \dots, a_r$, if the sentence $\exists v_0 \varphi(v_0, a_1, \dots, a_r)$ is true in some Δ_0 -elementary extension $\mathcal{K}$ of $\mathcal{M}$ such that $\mathcal{K} \models T$, then this sentence holds in $\mathcal{M}$. Observe that in the definition we may replace the class of Δ_0 formulas by the class of Σ_1 formulas without changing the notion.

The difficulty is that PA has no Σ_1 -closed model. But the theory $\{\varphi : \varphi \text{ is a sentence of class } \Pi_2 \text{ such that } \text{PA} \vdash \varphi\}$ has such models (and this is the first trick).

The reason why PA has no Σ_1 -closed model is presented in [24] (cf. Proposition 2.4). We sketch briefly their argument.

A standard model $\mathbb{N}$ is a Σ_1 -closed model for a theory T if and only if $\mathbb{N} \models T$ and T implies Π_1 theory of $\mathbb{N}$. Thus, $\mathbb{N}$ cannot be a Σ_1 -closed model for PA.

In order to rule out the possibility of an existence of a nonstandard Σ_1 -closed model for PA we state some useful properties. For $T \supseteq I\Delta_0$,

- (i) if $\mathcal{M}$ is a Σ_1 -closed model for T then $\mathcal{M}$ is a Σ_1 -closed model for $\Pi_1(T)$,
- (ii) if $\mathcal{M}$ is a Σ_1 -closed model and $\mathcal{N} <_{\Sigma_1} \mathcal{M}$, then $\mathcal{N}$ is a Σ_1 -closed model for $\Pi_2(T)$,
- (iii) if $\mathcal{M}$ and $\mathcal{N}$ are Σ_1 -closed models for T and $\mathcal{M} <_{\Delta_0} \mathcal{N}$, then $\mathcal{M} <_{\Sigma_2} \mathcal{N}$.

Now, assume that a nonstandard model $\mathcal{M}$ is a Σ_1 -closed model for T . Then, take a nonstandard element $a \in \mathcal{M}$ and consider a model $\mathcal{K} = K_{\Sigma_1}(\mathcal{M}, a)$, a submodel of $\mathcal{M}$ on elements Σ_1 definable from a . It is known that $\mathcal{K} <_{\Sigma_1} \mathcal{M}$ and $\mathcal{K} \not\models B\Sigma_1 + \text{Exp}$. Then, by (i) and (ii), $\mathcal{K}$ and $\mathcal{M}$ are Σ_1 -closed models for $\Pi_1(T)$. Thus, by (iii), $\mathcal{K} <_{\Sigma_2} \mathcal{M}$. Then, since $B\Sigma_1$ is Π_3 axiomatized it would be preserved in $\mathcal{K}$ if it is satisfied in $\mathcal{M}$. Therefore $\mathcal{M} \not\models B\Sigma_1 + \text{Exp}$. This rules out that PA has a nonstandard Σ_1 -closed model.

The main existence theorem for Σ_1 -closed models (Lemma 3.11.3) is a consequence of the following observation.

Lemma 3.11.1 *Let $\{\mathcal{M}_i : i \in I\}$ be a Δ_0 -elementary chain of structures for L_{PA} . Then the union $\mathcal{M}$ of this chain is a Δ_0 -elementary extension of each $\mathcal{M}_i$.*

Proof Pick $i_0 \in I$. As pointed out in the proof of Lemma 2.1.1, the value of each term t on each valuation $\bar{a} = a_0, \dots, a_{r-1}$ in $\mathcal{M}_{i_0}$ is the same when evaluated in $\mathcal{M}$

and in $\mathcal{M}_{i_0}$. Thus it suffices to verify that for every Δ_0 -formula φ and every valuation $\bar{a} = a_0, \dots, a_{r-1}$ in $\mathcal{M}_{i_0}$, $\mathcal{M} \models \varphi(\bar{a})$ iff $\mathcal{M}_{i_0} \models \varphi(\bar{a})$. This is proved by induction on φ . For atomic φ this is immediate and the steps corresponding to conjunction and negation are immediate. The case of bounded quantifier is also easy, we leave it to the reader. $\square$

Lemma 3.11.2 *Let D be a theory in L_{PA} containing $I\Delta_0 + \text{Exp}$ such that D has a Π_2 axiomatization. Then whenever $\{\mathcal{M}_i : i \in I\}$ is a Δ_0 -elementary chain of models of D , the union $\bigcup_{i \in I} \mathcal{M}_i$ is a model of D .*

Proof Let $\mathcal{M} = \bigcup_{i \in I} \mathcal{M}_i$. Let γ be an axiom of D , we may assume that γ is of the form $\forall x \exists y \varphi(x, y)$, where $\varphi \in \Delta_0$. Pick $a \in \mathcal{M}$. Pick $i \in I$ such that $a \in \mathcal{M}_i$. Then $\mathcal{M}_i \models \exists b \varphi(a, b)$, for $\mathcal{M}_i$ is a model of D . Then for this b , $\mathcal{M} \models \varphi(a, b)$ because $\varphi \in \Delta_0$ (see Lemma 3.11.1). $\square$

The following lemma is proved in [22].

Lemma 3.11.3 *Let T be a consistent theory having a Π_2 axiomatics. Then T has a Σ_1 -closed model.*

The above lemma may be proved by constructing a Δ_0 -elementary chain of countable models of D . Whenever, there is a formula $\exists x \varphi(x, a_1, \dots, a_r)$, which holds in some Δ_0 elementary extension $\mathcal{K}$ but not in the sum of models constructed so far, we put $\mathcal{K}$ in the Δ_0 -elementary chain. The sum of this chain has to be Σ_1 -closed.

Let T be a recursive, consistent theory in $\mathcal{L}_{PA}$ containing $I\Delta_0 + \text{Exp}$. Let U be the set of Π_2 -consequences of T , i.e.,

$$U = \{\varphi : \varphi \text{ is a } \Pi_2 \text{ sentence and } T \vdash \varphi\}.$$

By Lemma 3.11.3, U has a Σ_1 -closed model, pick such a model $\mathcal{M}$. Consider the following formula $ZA(\varphi, x)$:

$$\neg \text{Pr}_T(\neg \varphi(S^x 0)).$$

This formula expresses that $\varphi(S^x 0)$ is consistent with T .

Lemma 3.11.4 *Let $\mathcal{M}$ be a Σ_1 -closed model of U . Then if $\mathcal{M} \models \text{Con}_T$ then ZA is universal for Σ_1 formulas, i.e., $\mathcal{M}$ satisfies*

$$\forall \varphi \in \Sigma_1 \forall x [\text{Tr}_{\Sigma_1}(\varphi(S^x 0)) \equiv ZA(\varphi, x)].$$

Proof Work in $\mathcal{M}$. Pick $\varphi, a \in \mathcal{M}$ and assume $\text{Tr}_{\Sigma_1}(\varphi(S^a 0))$. It follows from derivable Σ_1 -completeness (cf. (3.1)) that $\text{Tr}_{\Sigma_1}(\varphi(S^a 0))$ is provable inside $\mathcal{M}$ that is $\text{Pr}_T(\text{Tr}_{\Sigma_1}(\varphi(S^a 0)))$. But $\text{Tr}_{\Sigma_1}(\varphi(S^a 0))$ is in $\mathcal{M}$ provably equivalent to $\varphi(S^a 0)$ so, by derivability conditions, $\text{Pr}_T(\varphi(S^a 0))$. Now, by Con_T , $\neg \text{Pr}_T(\neg \varphi(S^a 0))$.

For the converse, assume $ZA(\varphi, a)$, that is $\neg \text{Pr}_T(\neg \varphi(S^a 0))$. Since $\text{Tr}_{\Sigma_1}(\varphi(S^a 0))$ is provably in $I\Delta_0 + \text{Exp}$ equivalent to $\varphi(S^a 0)$, we have $\neg \text{Pr}_T(\neg \text{Tr}_{\Sigma_1}(\varphi(S^a 0)))$. By

derivable Σ_1 -completeness, we have $\text{Pr}_T(\gamma(\bar{u}))$, for every true in $\mathcal{M}$ Σ_1 formula $\gamma(\bar{z})$ with parameters $\bar{u} \in \mathcal{M}$. Thus, we have that $\neg \text{Pr}_{T+\text{Diag}_{\Delta_0}}(\neg \text{Tr}_{\Sigma_1}(\varphi(S^a 0)))$. We may infer that the theory $T + \text{Diag}_{\Delta_0}(\mathcal{M}) + \text{Tr}_{\Sigma_1}(\varphi(S^a 0))$ is consistent in the sense of $\mathcal{M}$. It follows that it is also consistent in the external sense of the word; there are more proofs in $\mathcal{M}$ than in our world. Then, we take a model $\mathcal{K} \models T + \text{Diag}_{\Delta_0}(\mathcal{M})$ in which $\text{Tr}_{\Sigma_1}(\exists z \psi(a, z))$ holds. (Observe that we cannot use the arithmetized completeness theorem here for the existence of a completion is Σ_4 , hence far beyond possibilities of U .) Thus, such $\mathcal{K}$ is a Δ_0 -elementary extension of $\mathcal{M}$ and, since $\mathcal{M}$ is Σ_1 -closed, $\mathcal{M} \models \text{Tr}_{\Sigma_1}(\exists z \psi(a, z))$. $\square$

Observe that Lemma 3.11.4 does the job. Since ZA is Π_1 , we can repeat the proof of the arithmetical hierarchy theorem (Theorem 3.2.8 above) to show that $\mathcal{M} \models \neg \text{Con}_T$. It follows that T cannot prove Con_T because if T proved it, then Con_T being Π_1 would have to hold in $\mathcal{M}$.

3.12 Some Extensions of PA

The theory PA is incomplete, so it is natural to work out some stronger theories. Many such extensions of PA were studied in the literature, we say a few words only about two.

The first possibility of extending PA is to add a new rule of inference, the so-called ω -rule. This is as follows

$$\frac{\theta \vee \varphi(S^a 0), \quad \text{for all } a \in \mathbb{N}}{\theta \vee \forall x \varphi(x)}. \quad (3.22)$$

The side formula θ need not occur in the statement of the rule. Of course, each mathematician working in any area but logic accepts (3.22) as a correct rule of inference (“if we proved that all a have the property $\varphi(x)$ then we proved $\forall a \varphi(a)$ ”). But the difficulty here is that a proof involving this rule is not a finite object and, hence, there is no chance for a computer to check whether a given proof is correct.¹ This seems to be a sensible limitation for the notion of an acceptable proof, that is why we do not accept this as rule of proofs in the study of logical foundations of arithmetic. Anyway, it will serve as a technical tool in Chap. 5.

There is another remark which we would like to put here. Set theory is usually considered as a foundation of the whole mathematics. One may write down the ω -rule for set theory as well. But this rule is able to yield some information about natural numbers and its addition gives no information about sets that are higher in von Neumann’s hierarchy, thus we get a theory which is incomplete (because of forcing extensions).

Proposition 3.12.1 *Peano arithmetic with the ω -rule is complete.*

¹The reader knowing some complexity theory will realize that the question whether a sequence of formulas is a proof should be decidable in polynomial time.

Proof By induction on $\varphi(x_{i_0}, \dots, x_{i_r})$ we prove

for every sequence $\langle n_{i_0}, \dots, n_{i_r} \rangle$ the statement $\varphi(S^{n_{i_0}}0, \dots, S^{n_{i_r}}0)$ is provable with the ω -rule iff it is true in $\mathbb{N}$.

We leave the details to the reader. The ω -rule is used to handle the quantifier step of induction. $\square$

We would like to point out that the set of sentences which are derivable with k uses of the ω -rule is definable, but these definitions are more and more complicated when k grows and we have no single formula describing the notion of provability with the ω -rule (this follows from Tarski's theorem, i.e. Theorem 3.2.7 and Proposition 3.12.1). Namely, the formula describing the family of sentences provable with k uses of the ω -rule has some universality property. We shall work this out in Sect. 5.2.

We would like to say a few words about another possibility of extending PA. Let T be as usual. Consider the following scheme

$$\forall x [\text{Pr}_T(\ulcorner \varphi(S^x 0) \urcorner) \Rightarrow \varphi(x)]. \quad (3.23)$$

This scheme is known as the (*uniform*) *reflection principle* and is denoted by RFN_T . One may extend PA by adding the scheme (3.23) to its axioms. Observe that this scheme is not derivable in PA (take $0 = S(0)$ as φ for example). The reason for this underivability is that in nonstandard model there may exist a proof of $\varphi(S^a 0)$, but such a proof need not be a standard object. This is just an element of our model of which this model thinks that it is the Gödel number of a proof. But if this object is nonstandard then we cannot transform it to a real proof in the external sense.

The result below describes provable instances of some weaker version of the reflection principle (the so-called *local reflection principle*). The local reflection principle for a *sentence* φ is a formula

$$\text{Pr}_T(\ulcorner \varphi \urcorner) \Rightarrow \varphi. \quad (3.24)$$

In (3.24), we restrict reflection to hold only for sentences but again taking φ as $0 = S(0)$ shows that they are unprovable in a consistent theory T. Moreover, Löb's theorem shows that if an instance of a local reflection principle $\text{Pr}_T(\ulcorner \varphi \urcorner) \Rightarrow \varphi$ is provable then φ is also provable. Let us remark, that the converse implication is trivially true. Thus, the provable instances of (3.24) are provable because of a simple reason, provability of a consequent of an implication.

Proposition 3.12.2 (Löb's theorem) *Let φ be a sentence such that $\text{PA} \vdash \text{Pr}_{\text{PA}}(\ulcorner \varphi \urcorner) \Rightarrow \varphi$. Then $\text{PA} \vdash \varphi$.*

Proof Assume $\text{PA} \vdash \text{Pr}_T \Rightarrow \varphi$ and, towards a contradiction, that $\text{PA} \not\vdash \varphi$. Then $\text{PA} + \neg\varphi$ is consistent and by the second incompleteness theorem $\text{PA} + \neg\varphi \not\vdash \text{Con}_{\text{PA} + \neg\varphi}$. Some transformations yield $\text{PA} + \neg\varphi \not\vdash \neg\text{Pr}_{\text{PA}}(\ulcorner \varphi \urcorner)$ and, by the deduction theorem, $\text{PA} \not\vdash \neg\varphi \Rightarrow \neg\text{Pr}_{\text{PA}}(\ulcorner \varphi \urcorner)$. This contradicts the provability of the reflection principle for φ . $\square$

Löb's theorem answers the question raised by Leon Henkin, whether the sentence asserting its own provability is provable. Simply, for any such a sentence its local reflection principle is provable.

Corollary 3.12.3 *Let φ be such that $\text{PA} \vdash \varphi \equiv \text{Pr}_{\text{PA}}(\ulcorner \varphi \urcorner)$. Then $\text{PA} \vdash \varphi$.*

We merely remark that the reflection principle (3.23) is equivalent to its variant in which we may quantify over formulas, provided they are of given complexity:

$$\forall \varphi, x \{ [\varphi \in \Sigma_m \ \& \ \text{Pr}_T(\varphi(S^x 0))] \Rightarrow \text{Tr}_{\Sigma_m}(\varphi(S^x 0)) \} : m \in \mathbb{N}. \quad (3.25)$$

Indeed, fix m . Let φ, x be given, where $\varphi \in \Sigma_m$. Assume that $\text{Pr}_T(\varphi(S^x 0))$. Then also $\text{Pr}_T(\text{Tr}_{\Sigma_m}(\varphi(S^x 0)))$ by the properties of universal formulas, and we may apply (3.23) to the formula Tr_{Σ_m} with parameter $\varphi(S^x 0)$.

References

1. Adamowicz, Z., & Zbierski, P. (1997). *Logic of mathematics: a modern course of classical logic*. A Wiley-Interscience publication. New York: Wiley. ISBN 978-0471060260.
2. Kaye, R. (1991). *Models of Peano arithmetic*. Oxford: Oxford University Press. ISBN: 019853213X.
3. Hájek, P., & Pudlák, P. (1993). *Metamathematics of first-order arithmetic*. Perspectives in mathematical logic. Berlin: Springer.
4. Kotlarski, H. (1994). On the incompleteness theorems. *Journal of Symbolic Logic*, 59(4), 1414–1419. ISSN: 0022-4812. <https://doi.org/10.2307/2275714>.
5. Gödel, K. (1931). Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38, 173–198.
6. Feferman, S. (1960). Arithmetization of metamathematics in a general setting. eng. *Fundamenta Mathematicae*, 49(1), 35–92. <http://eudml.org/doc/213578>.
7. Lindström, P. (2016). *Aspects of incompleteness*. Cambridge: Cambridge University Press. ISBN 9781316716854. <https://doi.org/10.1017/9781316716854>.
8. Shoenfield, J. R. (1967). *Mathematical logic*. Boston: Addison-Wesley.
9. Takeuti, G. (1987). *Proof theory*. North-Holland.
10. Rosser, B. (1936). Extensions of some theorems of Gödel and Church. *Journal of Symbolic Logic*, 1(3), 87–91. ISSN: 0022-4812. <https://doi.org/10.2307/2269028>.
11. Robinson, A. (1963). On languages which are based on non-standard arithmetic. *Nagoya Mathematical Journal*, 22, 83–117. ISSN: 0027-7630. <https://doi.org/10.1017/S0027763000011065>.
12. Krajewski, S. (1976). Non-standard satisfaction classes. In: W. Marek, M. Srebrny & A. Zarach (Eds.), *Set theory and hierarchy theory a memorial tribute to Andrzej Mostowski*, Lecture notes in mathematics (Vol. 537). Berlin: Springer.
13. Friedman, H. (1973). Countable models of set theories. In: A. R. D. Mathias & H. Rogers (Eds.), *Cambridge summer school in mathematical logic*. Lecture Notes in Mathematics (Vol. 337, pp. 539–573).
14. Smoryński, C. (1977). The incompleteness theorems. *Handbook of mathematical logic*. Studies in logic and the foundations of mathematics (Vol. 90, pp. 821–865). Amsterdam: Elsevier. ISBN: 9780444863881. [https://doi.org/10.1016/S0049-237X\(08\)71123-6](https://doi.org/10.1016/S0049-237X(08)71123-6).
15. Jech, T. (1994). On Gödel's second incompleteness theorem. *Proceedings of the American Mathematical Society*, 121, 311–313.

16. Boolos, G. S., & Jeffrey, R. C. (1998). *Computability and logic* (3rd ed.). Cambridge: Cambridge University Press.
17. Kotlarski, H. (1998). Other proofs of old results. *Mathematical Logic Quarterly*, 44(4), 474–480. ISSN: 09425616. <https://doi.org/10.1002/malq.19980440406>.
18. Kikuchi, M. (1994). A note on Boolos' proof of the incompleteness theorem. *Mathematical Logic Quarterly*, 40(4), 528–532. ISSN: 09425616. <https://doi.org/10.1002/malq.19940400409>.
19. Vopěnka, P. (1966). A new proof of the Gödel's result of non-provability of consistency. *Bull. Acad. Polon. Sci.*, 14, 111–116.
20. Boolos, G. S. (1989). A new proof of the Gödel incompleteness theorem. *Notices Am. Math. Soc.*, 36, 388–390.
21. Cieśliński, C. (1999). Gödel's incompleteness theorems and semantical antinomies. PhD thesis. Warsaw University.
22. Adamowicz, Z., & Bigorajska, T. (2001). Existentially closed structures and Gödel's second incompleteness theorem. *The Journal of Symbolic Logic*, 66(1), 349–356. ISSN: 00224812. <http://www.jstor.org/stable/2694926>.
23. Hodges, W. (1993). *Model theory*. Cambridge: Cambridge University Press. ISBN 9780511551574. <https://doi.org/10.1017/CBO9780511551574>.
24. Adamowicz, Z., Cordon-Franco, A., & F. Félix Lara-Martín. (2016). Existentially closed models in the framework of arithmetic. *The Journal of Symbolic Logic*, 81(2), 774–788. ISSN: 0022-4812. <https://doi.org/10.1017/jsl.2015.36>.

Chapter 4

Transfinite Induction



In this chapter we present an approach to the classical results due to Gentzen. In 1936, Gentzen proved consistency of arithmetic using transfinite induction up to ε_0 . His proof is given in many textbooks, see, e.g. [1]. Gentzen's argument used the so-called *cut elimination* procedure. Much later another approach to this sort of problems via *Herbrand's theorem* was developed. This approach is given in details in Hájek-Pudlák [2]. In late 1980s still another approach, essentially model-theoretic in character, was developed by Zygmunt Ratajczyk [3] and, independently, Rick Sommer [4], see also Sommer [5]. They used essentially some ideas developed by Ketonen and Solovay [6]. We give an exposition of this approach.

4.1 Indicators

Let T be a recursive theory in the language of PA containing $I\Delta_0 + \text{Exp}$. A recursive function $Y(x, y) = z$ is an *indicator* (for models of T) if for every countable $M \models T$ and any $a, b \in M$ we have: there exists an initial segment I of M such that $a \in I < b$ and $I \models T$ iff the value $Y(a, b)$, calculated in M , is nonstandard. A healthy way of thinking of this is the following. The value $Y(a, b)$ is a sort of distance from a to b (though not in the sense of metric spaces, it is not symmetric), hence the main property of indicators is: there exists an I with the properties stated in the definition iff the distance from a to b is nonstandard.

An indicator $Y(x, y)$ for models of T is *well behaved* if T proves its monotonicity in y , that is

$$\forall x, y, y' (y \leq y' \Rightarrow Y(x, y) \leq Y(x, y')) \quad (4.1)$$

and if, for each $n \in \mathbb{N}$, T proves that the function $x \mapsto \min y : Y(x, y) \geq n$ is total. In what follows we construct only well behaved indicators.

Now, we characterize indicators.

Proposition 4.1.1 (J. Paris) *Let T satisfy the above assumptions. Let $Y(x, y)$ be a well behaved indicator for models of T . Then*

1. *If $\mathbb{N} \models T$ then $\mathbb{N} \models \forall x, z \exists y Y(x, y) \geq z$,*
2. *if T is consistent then there exists a model of T satisfying $\neg \forall x, z \exists y Y(x, y) \geq z$.*
3. *If F is a recursive function such that for some formula representing F , T proves “ F is total” then there exists $z \in \mathbb{N}$ such that F is dominated by the function $x \mapsto \min y : Y(x, y) \geq z$.*

It follows that if $\mathbb{N} \models T$, then the statement $\forall x, z \exists y Y(x, y) \geq z$ is independent from T .

Proof For the first part pick $x, z \in \mathbb{N}$. It suffices to show that every countable non-standard model $\mathcal{M}$ of T models also $\exists y Y(x, y) \geq z$. So let $\mathcal{M}$ be given and pick a nonstandard $y \in \mathcal{M}$. Then the value of $Y(x, y)$ is nonstandard (because some model of T , namely $\mathbb{N}$, is between x and y), so this value is greater than z .

For the second part we pick any countable nonstandard $\mathcal{M} \models T$ and any non-standard $a \in \mathcal{M}$. Assume that $\mathcal{M} \models \forall x, z \exists y Y(x, y) \geq z$, for otherwise $\mathcal{M}$ has the desired property. We let $b = \min\{y : Y(a, y) \geq a\}$. Then there exists $I \models T$ between a, b as a was chosen to be nonstandard. If $I \models \forall x, z \exists y Y(x, y) \geq z$, then we pick $b' \in I$ such that $I \models Y(a, b') \geq a$. Then b' satisfies the same formula in $\mathcal{M}$ (because Y is recursive, i.e., Σ_1). This is a contradiction with minimality of b , indeed, $b' < I < b$.

The third part follows from the fact that in the proof of the second part a was arbitrarily small in the nonstandard part of $\mathcal{M}$. If the third part is false then, by compactness and monotonicity of $Y(x, y)$ (cf. Eq. (4.1)) , we could find a model $\mathcal{M}$ with a nonstandard element a such that $Y(a, F(a))$ is nonstandard. But then we would have an initial segment I satisfying T with $a \in I < F(a)$. This contradicts the fact that F is a total recursive function in T (I does not satisfy totality of F). $\square$

Observe that Proposition 4.1.1 gives an independent statement (for a given T satisfying the assumptions) in a way drastically different than the proof of Gödel’s theorem. Moreover, this statement is Π_2 , whilst Gödel’s consistency statement is Π_1 . We shall see in later sections that a refinement of this construction gives a deep information about PA.

4.2 Transfinite Induction in PA

Observe that there is no problem with formalizing the definition of an ordinal (see Sect. 1.4) in PA. The reason is that as a matter of fact we obtained there a primitive recursive function which associates with every n the (Gödel number of a) Σ_1 formula Pol_n (and the auxiliary definition of $<_n$), hence one can use the formula $\exists n \text{Tr}_{\Sigma_1}(\text{Pol}_n(\alpha))$ as the definition of an ordinal below ε_0 . We shall write that α is an ordinal in the sense of the model under consideration, or that $\alpha < \varepsilon_0$.

Our goal is to prove transfinite induction principle below ε_0 in PA. The exact formulation is in Theorem 4.2.1 and Corollary 4.2.2.

For (a notation for) an ordinal ϱ and a natural number m we define $L(\varrho, \Sigma_m)$ to be the scheme of minimum for Σ_m formulas over ϱ , i.e. the scheme

$$[\exists \alpha < \varrho A(\alpha)] \rightarrow \exists \alpha [A(\alpha) \ \& \ \forall \beta < \alpha \neg A(\beta)],$$

where A is a Σ_m formula, possibly with other free variables. We also denote by $\omega_m(\alpha)$ the sequence of ordinals defined by induction in the following manner: $\omega_0(\alpha) = \alpha$, $\omega_{m+1}(\alpha) = \omega^{\omega_m(\alpha)}$. Thus, $\omega_0(\omega) = \omega$, more generally $\omega_m(\omega) = \omega_{m+1}$ and $\omega_m(1) = \omega_m$ in this notation. The sequence $\omega_m(\alpha)$ is defined, in particular, for finite α .

The following result is a variant of Gentzen's proof of transfinite induction in PA. This variant is due to Mints [7], see also [2, 4, 5, 8].

Theorem 4.2.1 *For all $l \geq 1$ and all k, m*

$$I\Sigma_{m+l} \vdash L(\omega_{m+1}(k), \Sigma_l).$$

Let $\text{TI}(\varrho, \Pi_m)$ denote the scheme of transfinite induction for Π_m formulas over ϱ , that is

$$[\psi(0) \ \& \ \forall \beta < \varrho (\forall \gamma < \beta \psi(\gamma) \Rightarrow \psi(\beta))] \Rightarrow \forall \beta < \varrho \psi(\beta),$$

where $\psi \in \Pi_m$. Observe that the scheme $\text{TI}(\varrho, \Pi_m)$ is equivalent to $L(\varrho, \Sigma_m)$.

Corollary 4.2.2 *For every $k, m \in \mathbb{N}$ PA proves $\text{TI}(\omega_m, \Sigma_k)$.* □

Lemma 4.2.3 *For each k, l $I\Sigma_l \vdash L(\omega^k, \Sigma_l)$.*

Proof Let $A(\cdot)$ be a Σ_l formula. Work in $I\Sigma_l$. Assume $\exists \alpha < \omega^k A(\alpha)$; we construct the smallest such α . An α given by the assumption must be of the form $\alpha = \omega^{k-1} \cdot m_{k-1} + \dots + \omega^0 \cdot m_0$. We shall simply choose the required sequence $m_{k-1}, \dots, m_0$ by applying the scheme of minimum to a sequence $C_{k-1}, \dots, C_0$ of formulas. We let $C_{k-1}(m_{k-1})$ be

$$\exists \langle m_{k-2}, \dots, m_0 \rangle \exists \alpha [\alpha = \omega^{k-1} \cdot m_{k-1} + \dots + \omega^0 \cdot m_0 \ \& \ A(\alpha)].$$

By the assumption $\exists m_{k-1} C_{k-1}(m_{k-1})$ and by the scheme of minimum for Σ_l formulas (which is equivalent to Σ_l induction) there exists a smallest m_{k-1} such that $C_{k-1}(m_{k-1})$. This m_{k-1} is a parameter in further formulas. We find m_{k-2} in the same manner. That is we write $C_{k-2}(m_{k-2})$:

$$\exists \langle m_{k-3}, \dots, m_0 \rangle \exists \alpha [\alpha = \omega^{k-1} \cdot m_{k-1} + \dots + \omega^0 \cdot m_0 \ \& \ A(\alpha)]$$

and get the smallest m_{k-2} . Iterate this procedure k times. This gives the sequence $m_{k-1}, \dots, m_0$, so gives an ordinal $\alpha = \omega^{k-1} \cdot m_{k-1} + \dots + \omega^0 \cdot m_0$; clearly this is the smallest witness for $A(\cdot)$. □

Observe that the argument gives a proof which heavily depends on k , indeed, the greater k is the longer procedure is applied in the process of choosing the sequence of coefficients $m_{k-1}, \dots, m_0$, therefore of α .

Lemma 4.2.4 *Let $\nu \geq \omega$ be an ordinal. Then, for each l , $I\Delta_0 + L(\nu, \Sigma_{l+1}) \vdash L(\omega^\nu, \Sigma_l)$.*

Observe that Theorem 4.2.1 is an immediate consequence of Lemmas 4.2.3 and 4.2.4.

The proof of Lemma 4.2.4 is similar to that of Lemma 4.2.3, but slightly more delicate. The reason is that we cannot work so freely; indeed, the α to be found may have an expansion of nonstandard length (we are working in a fixed model for $I\Delta_0 + L(\nu, \Sigma_{l+1})$).

Let $A(\cdot)$ be a Σ_l formula; we shall obtain a proof by reducing the problem to Σ_{l+1} formulas and minimum principle over ν .

We define recursively two sequences $B_s(\cdot)$ and $C_s(\cdot)$ of Σ_{l+1} formulas. $B_0(\alpha_0)$ is

$$\begin{aligned} & [\exists m_0 \neq 0 \exists r \exists \langle m_1, \dots, m_{r-1} \rangle \exists \langle \alpha_1, \dots, \alpha_{r-1} \rangle \exists \alpha \\ & \nu > \alpha_0 > \dots > \alpha_{r-1} \ \& \ \alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_{r-1}} \cdot m_{r-1} \ \& \ A(\alpha)] \ \& \\ & [\forall \gamma < \alpha \ \forall r \ \forall \langle m_0, \dots, m_{r-1} \rangle \ \forall \langle \alpha_1, \dots, \alpha_{r-1} \rangle \ \forall \alpha \\ & (\nu > \gamma > \alpha_1 > \dots > \alpha_{r-1} \ \& \ \alpha = \omega^\gamma \cdot m_0 + \omega^{\alpha_1} \cdot m_1 + \dots + \omega^{\alpha_{r-1}} \cdot m_{r-1}) \rightarrow \neg A(\alpha)]. \end{aligned}$$

Thus, if we assume $\exists \alpha < \omega^\nu A(\alpha)$, then $B_0(\alpha_0)$ is just a description of the smallest possible exponent in the expansion $\alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_{r-1}} \cdot m_{r-1}$. Observe that B_0 is Σ_l & Π_l , so is Σ_{l+1} . The formula $C_0(m_0)$ is defined similarly; this will be the description of the smallest possible coefficient m_0 . $C_0(m_0)$ is

$$\begin{aligned} & \exists \alpha_0 [B_0(\alpha_0) \ \& \ [\exists r \exists \langle m_1, \dots, m_{r-1} \rangle \exists \langle \alpha_1, \dots, \alpha_{r-1} \rangle \exists \alpha \\ & \nu > \alpha_0 > \dots > \alpha_{r-1} \ \& \ \alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_{r-1}} \cdot m_{r-1} \ \& \ A(\alpha)] \ \& \\ & [\forall k < m_0 \ \forall r \ \forall \langle m_1, \dots, m_{r-1} \rangle \ \forall \langle \alpha_1, \dots, \alpha_{r-1} \rangle \ \forall \alpha \\ & (\nu > \alpha_0 > \dots > \alpha_{r-1} \ \& \ \alpha = \omega^{\alpha_0} \cdot k + \omega^{\alpha_1} \cdot m_1 + \dots + \omega^{\alpha_{r-1}} \cdot m_{r-1}) \rightarrow \neg A(\alpha)]]. \end{aligned}$$

Once again, $C_0(\cdot)$ is Σ_{l+1} . Observe moreover that in $I\Delta_0 + L(\nu, \Sigma_{l+1})$ we have

$$[\exists \alpha < \omega^\nu A(\alpha)] \rightarrow \exists \alpha_0, m_0 [B_0(\alpha_0) \ \& \ C_0(m_0)]. \quad (*)$$

Now assume that $B_0, \dots, B_{s-1}$ and $C_0, \dots, C_{s-1}$ are constructed, we write down B_s and C_s . $B_s(\alpha_s)$ is

$$\begin{aligned} & \exists \langle \alpha_0, \dots, \alpha_{s-1} \rangle \exists \langle m_0, \dots, m_{s-1} \rangle \{ [\forall i < s \ \text{Tr}_{\Sigma_{l+1}}(B_i(\alpha_i) \ \& \ C_i(m_i))] \ \& \\ & [\exists r \exists m_s \exists \langle \alpha_{s+1}, \dots, \alpha_r \rangle \exists \langle m_{s+1}, \dots, m_r \rangle \exists \alpha \\ & \alpha_s > \alpha_{s+1} > \dots > \alpha_r \ \& \ \alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_r} \cdot m_r \ \& \ A(\alpha)] \ \& \\ & [\forall \gamma < \alpha_s \ \forall r \ \forall m_s \ \forall \langle \alpha_{s+1}, \dots, \alpha_r \rangle \ \forall \langle m_{s+1}, \dots, m_r \rangle \ \forall \alpha \\ & (\alpha_{s+1} > \dots > \alpha_r \ \& \ \alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^\gamma \cdot m_s + \dots + \omega^{\alpha_r} \cdot m_r) \rightarrow \neg A(\alpha)] \}. \end{aligned}$$

Observe that this is still a Σ_{l+1} formula, because by the collection principle the quantifier $\forall i < s \ \text{Tr}_{\Sigma_{l+1}}$ does not extend the class of this formula in $I\Sigma_{l+1}$.

$C_s(m_s)$ is constructed in a similar manner, namely it is

$$\begin{aligned}
& \exists \langle \alpha_0, \dots, \alpha_{s-1} \rangle \exists \langle m_0, \dots, m_{s-1} \rangle \{ [\forall i < s \text{Tr}_{\Sigma_{l+1}}(B_i(\alpha_i) \& C_i(m_i))] \& \\
& \quad \exists \alpha_s B_s(\alpha_s) \& [\exists r \exists \langle \alpha_{s+1}, \dots, \alpha_r \rangle \exists \langle m_{s+1}, \dots, m_r \rangle \exists \alpha \\
& \quad \alpha_s > \alpha_{s+1} > \dots > \alpha_r \& \alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_r} \cdot m_r \& A(\alpha)] \& \\
& [\forall k < m_s \forall r \forall \langle \alpha_{s+1}, \dots, \alpha_r \rangle \forall \langle m_{s+1}, \dots, m_r \rangle \forall \alpha (\alpha_s > \alpha_{s+1} > \dots > \alpha_r \& \\
& \quad \alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_s} \cdot k + \dots + \omega^{\alpha_r} \cdot m_r) \rightarrow \neg A(\alpha)] \}.
\end{aligned}$$

Once again, this is a Σ_{l+1} formula. It follows that the formula $D(\eta): \exists i \text{Tr}_{\Sigma_{l+1}}(B_i(\eta))$ is still Σ_{l+1} .

Granted this we see that there exists η such that $D(\eta)$, hence there exists the smallest η so that $D(\eta)$. For this η there exists i such that $\text{Tr}_{\Sigma_{l+1}}(B_i(\eta))$. Pick sequences $\langle \alpha_0, \dots, \alpha_i \rangle$ and $\langle m_0, \dots, m_i \rangle$ such that for each $j \leq i$ $\text{Tr}_{\Sigma_{l+1}}(B_j(\alpha_j) \& C_j(m_j))$. Obviously, the ordinal $\alpha = \omega^{\alpha_0} \cdot m_0 + \dots + \omega^{\alpha_i} \cdot m_i$ is the smallest such that $A(\alpha)$ and the proof of Lemma 4.2.4 and, hence, of Theorem 4.2.1, is finished. $\square$

4.3 Totality of Functions in Hardy Hierarchy

In this section we show how to work with Hardy hierarchy in $I\Sigma_n$ and related theories.

The first difficulty is that the definition of fundamental sequences involved transfinite induction: $\{\omega^\gamma\}(b) = \omega^{\{\gamma\}(b)}$. This is avoided as follows. Let λ be a limit ordinal. Using its short normal form (1.7) we write it as $\lambda = \beta_1 + \omega^{\delta_1}$, where $\beta_1 \gg \omega^{\delta_1}$. Now we write δ_1 in its short Cantor normal form: $\delta_1 = \beta_2 + \omega^{\delta_2}$, where $\beta_2 \gg \omega^{\delta_2}$. We continue in the same fashion till we get δ_r of the form $\delta_r = m$, where $0 < m < \omega$. Then we let $\{\delta_{r-1}\}(b) = \omega^{m-1} \cdot b$. Then we come back to the original ordinal λ . Granted this, we see that $I\Sigma_{m+1}$ defines correctly fundamental sequences up to each $\omega_{m+1}(k)$. The argument above gives the existence conditions and Theorem 4.2.1 gives uniqueness of fundamental sequences.

Our next goal is to indicate how to formalize in PA the definition of Hardy hierarchy. Let h be a function. Let $h_0(x) = x$. Now, we write the formula describing the process sketched in Lemma 1.5.7. That is we write $h_\alpha(u) = w$ iff there exist two sequences $\alpha_0, \dots, \alpha_{r-1}$ and $u_0, \dots, u_{r-1}$ such that $\alpha_0 = \alpha$, $u_0 = u$, for each $j < r - 2$ if α_j is limit then $\alpha_{j+1} = \{\alpha_j\}(u_j)$ and $u_{j+1} = u_j$ and if α_j is non limit then $\alpha_{j+1} = \alpha_j - 1$ and $u_{j+1} = h(u_j)$, $\alpha_{r-2} > 0$, $\alpha_{r-1} = 0$ and $u_{r-1} = w$. Clearly this can be written in PA.

Before going further we give some more details in order to show that if h is Σ_1 then we obtain also a Π_1 definition of h_α . Let $B(\alpha, u, \gamma)$ be the following formula:

$$\begin{aligned}
& \exists r, \langle \alpha_0, \dots, \alpha_{r-1} \rangle, \langle u_0, \dots, u_{r-1} \rangle \{ \alpha = \alpha_0 \& \gamma = \alpha_{r-1} \& u = u_0 \& \\
& \quad \forall j < r - 2 \{ \alpha_j \neq 0 \& [\text{Lim}(\alpha_j) \Rightarrow \alpha_{j+1} = \{\alpha_j\}(u_j)] \& \\
& \quad [\text{Succ}(\alpha_j) \Rightarrow \alpha_{j+1} = \alpha_j - 1 \& u_{j+1} = h(u_j)] \}.
\end{aligned}$$

We assert that for each $k \in \mathbb{N}$ $I\Sigma_{m+1}$ proves “for all $\alpha < \omega_{m+1}(k)$ and all u there exists the smallest γ such that $B(\alpha, u, \gamma)$ ”. Once again, this follows from Theorem 4.2.1 because transfinite induction for Π_s formulas is equivalent to the scheme of minimum

for Σ_s formulas. Obviously, we must have $\gamma = 0$ (otherwise the conditions determine the next items of the sequences of α 's and u 's). Then we see that if r is such that $\alpha_{r-1} = \gamma = 0$ then $w = u_{r-1}$ is just $h_\alpha(u)$. In other words, for each $k \in \mathbb{N}$, $I\Sigma_{m+1}$ proves that $h_{\omega_{m+1}(k)}$ is total. It follows that the formula $h_{\omega_{m+1}(k)}(u) = w$ is Π_1 as well, i.e., we have two definitions: one of the class Σ_1 and another one of the class Π_1 (their equivalence is provable in $I\Sigma_{m+1}$).

As usual, we say that h is a Δ_m -function in T if there exists Σ_m formula and a Π_m formula, whose equivalence is provable in T and both define h . Our goal is to show

Theorem 4.3.1 *Let h be a Δ_1 function. Let $m \in \mathbb{N} \setminus \{0\}$ and let $I\Sigma_m$ prove that h is total. Let $k \in \mathbb{N}$. Then $I\Sigma_m$ proves $\forall \alpha < \omega_m(k) \forall a h_\alpha(a) \downarrow$.*

Of course, the most interesting case is when h is the usual successor function, $h(x) = S(x) = x + 1$. Observe that the statements to be proved are Π_2 .

Proof of Theorem 4.3.1 At first we prove the following version lemma on compositions (cf. Lemma 1.5.5):

$$\forall \alpha < \omega_m(k) \forall \beta < \omega_m(k) \forall b, c, d \{ [\beta \gg \alpha \ \& \ h_\alpha(b) = c \ \& \ h_\beta(c) = d] \Rightarrow h_{\beta+\alpha}(b) = d \}. \quad (4.2)$$

For every $k \in \mathbb{N}$ this is shown exactly like in the proof of Lemma 1.5.5, using Π_1 transfinite induction $TI(\omega_m(k), \Pi_1)$, which is provable by Theorem 4.2.1 and the remark following it. In order to see that this is, indeed, a Π_1 formula we use the remark before the statement of the theorem. In the predecessor of the implication we use the Σ_1 form of the definition of the Hardy hierarchy and in the successor we use its Π_1 form.

Now let $m = 1$. By induction on k we prove

$$\text{for every } k \in \mathbb{N}, I\Sigma_1 \vdash \forall j, a h_{\omega^k \cdot j}(a) \downarrow. \quad (4.3)$$

Fix any model $\mathcal{M}$ of $I\Sigma_1$, we prove that the above statements hold in $\mathcal{M}$. For $k = 0$ we must prove that $\mathcal{M} \models \forall j, a h_j(a) \downarrow$ and this is immediate by induction on j (with parameter a). So assume the conclusion for k , i.e., $\mathcal{M} \models \forall j, a h_{\omega^k \cdot j}(a) \downarrow$. Then we apply induction on j to see that $\mathcal{M}$ satisfies $\forall j h_{\omega^{k+1} \cdot j}(a) \downarrow$, for any parameter $a \in \mathcal{M}$. For $j = 0$ there is nothing to check. For $j = 1$ this is just the conclusion for k and $j = a$. In order to obtain the conclusion for $j + 1$ we write $h_{\omega^{k+1} \cdot (j+1)}(a)$ as $h_{\omega^{k+1} + \omega^{k+1} \cdot j}(a)$ and we apply the induction assumption to obtain $b = h_{\omega^{k+1} \cdot j}(a)$. Then, we apply the conclusion for k to get $h_{\omega^{k+1}}(b) = h_{\omega^k \cdot b}(b)$. Observe that formal induction on j was applied to the formula $h_{\omega^k \cdot j}(a) \downarrow$ with parameter a .

Now let $m > 1$. This time we shall show that for every $k \in \mathbb{N}$, $I\Sigma_m$ proves the statement

$$\forall \alpha < \omega_{m-1}(k) h_{\omega^\alpha}(a) \downarrow. \quad (4.4)$$

Observe that by Theorem 4.2.1, $I\Sigma_m$ proves transfinite induction for Π_2 formulas over $\omega_{m-1}(k)$. Equation (4.4) is obvious for $\alpha = 0$ and for limit α . Also, the limit step is evident by the same argument as for (4.3), but with α instead of k . $\square$

We assumed in Theorem 4.3.1 that $m > 0$. For $m = 0$ the situation is different because of the following result.

Theorem 4.3.2 (R. Parikh) *Let $H(x, y)$ be a Δ_0 formula such that $I\Delta_0$ proves $\forall x \exists y H(x, y)$. Then there exists $m \in \mathbb{N}$ such that $I\Delta_0$ proves $\forall x \exists y < x^m H(x, y)$.*

Proof Assume the contrary, for every $m \in \mathbb{N}$ the theory $I\Delta_0 + \exists x \forall y < x^m \neg H(x, y)$ is consistent. By compactness, let $(\mathcal{M}, a)$ be a model of $I\Delta_0 + \{\forall y < x^m \neg H(c, y)\}$, where c is a new constant and a is its interpretation. Let I be the initial segment of $\mathcal{M}$ defined as $I = \sup\{a^k : k \in \mathbb{N}\}$. Then $I \models I\Delta_0$, indeed, it is a cut in a model of $I\Delta_0$. In order to derive a contradiction it suffices to show that I cannot satisfy $\exists y H(a, y)$. This follows from the fact that Δ_0 formulas are absolute between a model of $I\Delta_0$ and its initial segments. $\square$

Our following goal is to give some information about iterations of quickly growing Skolem functions F_k . They are defined as follows.

$$F_k(x) = \min y : \phi_k(x, y), \quad (4.5)$$

where the formula $\phi_k(x, y)$ is given by (3.10) in Sect. 3.8. We give some information about Hardy iterations of the function F_k , for a fixed $k \in \mathbb{N}$.

It will be convenient to work directly with the relation ϕ_k rather than with the function F_k determined by it (because the use of minimum puts the formula to a higher class of complexity). Say that Z is ϕ_k -scattered if for all $a \in Z \setminus \{\max Z\}$, $\phi_k(a, h^Z(a))$ holds. Let $D_k(\alpha)$ be an abbreviation for

$$\forall a \exists Z [\min Z \geq a \ \& \ Z \text{ is } \alpha\text{-large} \ \& \ \phi_k\text{-scattered}]. \quad (4.6)$$

Here is a generalization of Theorem 4.3.1.

Theorem 4.3.3 *For every $m > 0$ and every $t, r \in \mathbb{N}$, $I\Sigma_{m+t+1}$ proves $\forall \alpha < \omega_m(r)$ $D_t(\alpha)$.*

Once again, if one wants to use Theorem 4.2.1, then one obtains a weaker result (because the formula to be proved by transfinite induction is Π_{t+3}), hence we have perform some additional work.

Proof of Theorem 4.3.3 Exactly as in the proof of Theorem 4.3.1 our starting point is a reformulation of the lemma on compositions. It is as follows:

$$\text{Let } A, B \text{ be sets, such that } \min(B) = \max(A) \text{ and } A \text{ is } \alpha\text{-large and } B \text{ is } \beta\text{-large, where } \beta \gg \alpha. \text{ Then } A \cup B \text{ is } \beta + \alpha\text{-large.} \quad (4.7)$$

This has nothing to do with being ϕ_t -scattered. The restriction of (4.7) to $\alpha, \beta < \omega_m(r)$ is provable in $I\Sigma_m$. Of course, this holds if A, B are ϕ_t -scattered.

Observe that the proof of $\forall a \exists b \phi(a, b)$ requires some induction. This fact is proved as follows. Pick a . Enumerate as $\varphi_i(S^{u_i}0, v) : i < s$ all substitutions of the form $\varphi(S^u0, v)$ with $\varphi, u \leq a$ and $\varphi \in \Pi_t$. Then we apply induction on j in

$$j \leq s \Rightarrow \{\exists z \forall i \leq j [(\exists w \text{Tr}_{\Pi_i}(\varphi_i(S^{u_i}0, S^w0))) \Rightarrow \exists w \leq z \text{Tr}_{\Pi_i}(\varphi_i(S^{u_i}0, S^w0))]\}.$$

This formula is Σ_{t+2} . Thus, we have

$$I\Sigma_{t+2} \vdash \forall a \exists z \forall \varphi, u \leq a [\exists w \text{Tr}_{\Pi_t}(\varphi(S^u0, S^w0)) \Rightarrow \exists w \leq z \text{Tr}_{\Pi_t}(\varphi(S^u0, S^w0))]. \quad (4.8)$$

Now we assert that

$$\forall r \in \mathbb{N} \quad I\Sigma_{t+2} \vdash \forall \alpha < \omega^r D_t(\alpha). \quad (4.9)$$

This is proved exactly as (4.3). Thus, we proceed by induction on r . For $r = 1$ we apply (4.8), in the inductive step we apply, again (4.8) and the analogue of lemma on compositions in the form stated as (4.7). We leave the details to the reader. $\square$

Our following goal is to indicate the possibility of formalizing the proof of the approximation lemma (i.e., Theorem 1.6.2) in $I\Sigma_m$. In the proof we used the auxiliary notion of a pair of sets approximating given function, see the proof of Lemma 1.6.4. The proof was by transfinite induction. Observe that the quantifier “there exist two sets B_1, B_2 ” may be bounded, indeed, the coding of sets by means of binary expansions has the property that $B \subseteq A \Rightarrow B \leq A$, hence we may bound this quantifier to A . Thus, the statement of Lemma 1.6.4 is of the class Π_1 .

In the proof of Lemma 1.6.4 we used the following version of the pigeonhole principle:

$$[\forall i < a_0 - 2 \exists z < a_0 - 1 \xi(i, z)] \Rightarrow [\exists s < a_0 - 1 \forall i < a_0 - 2 \exists z < a_0 - 1 [\xi(i, z) \ \& \ z \neq s]],$$

where ξ is a Δ_0 formula. This principle is easily proved by Σ_1 -induction. Summing up, we obtain:

Lemma 4.3.4 *Let $m > 0$. Then for every $k \in \mathbb{N}$, $I\Sigma_m$ proves “if A is $\omega_m(k)$ -large set, then for every finite function g there exists an $\omega_{m-1}(k)$ -large subset B of A which is an approximation for g ”.* $\square$

4.4 Hardy Largeness and Indicators

In this section we present the main part of the construction which yields initial segments satisfying PA and related theories. As pointed out earlier, this is Ratajczyk’s [3] approach to the Pudlák’s principle (hence, to the Paris–Harrington’s result, cf. Theorem 4.4.4); Sommer’s [4, 5] approach to proof-theoretic problems via models is slightly different. We begin with case $m = 1$ for motivational purpose.

Let $\mathcal{M}$ be a nonstandard model of $I\Sigma_1$. By Theorem 4.3.1, for every standard $k \in \mathcal{M}$,

$$\mathcal{M} \models \forall a \ h_{\omega^k}(a) \downarrow,$$

hence by the argument of overspill lemma (Lemma 3.4.1), the same statement holds for some nonstandard $k \in \mathcal{M}$. Fix such $k \in \mathcal{M}$. Pick also nonstandard $a \in \mathcal{M}$. Then the set $A = [a, h_{\omega^k}(a)]$ is ω^k -large. Thus, there exists an ω^k -large set in $\mathcal{M}$, pick such one and denote it by A . From now on we denote by h^A the successor function in the sense of A . By Lemma 4.3.4, given a function $g \in \mathcal{M}$, A has a k -large subset B that is an approximation for g . Consider the function

$$g(x) = \begin{cases} \text{the } w \text{ such that } \text{Tr}_0(\varphi(S^u 0, S^w 0)) & \text{if } x \text{ is of the form } \varphi(S^u 0, v) \text{ with} \\ & \varphi \in \Delta_0 \text{ and } \exists! w \text{ Tr}_0(\varphi(S^u 0, S^w 0)) \\ 0 & \text{otherwise} \end{cases} \quad (4.10)$$

and restrict its domain to $(< \max A)$ to make it $\mathcal{M}$ -finite. Let B be a k -large subset of A approximating g . Let us show the reason of working with sets approximating such a function. We have: for every $b \in B \setminus \{\max B\}$ for every substitution $\varphi(S^u 0, v) < b - 2$ with $\varphi \in \Delta_0$, if $\exists! w \varphi(S^u 0, S^w 0)$, then the appropriate w is either strictly below $h^B(b)$ or is above $\max B$. Pick any initial segment I of $\mathcal{M}$ closed under successor and bounded in B (as k is nonstandard, many such cuts exist, one of them is just $\mathbb{N}$). We let $J = \sup\{b_i : i \in I\}$, where, as usual $B = \{b_0, \dots, b_k\}$ in increasing order. This is a natural candidate for a model for $I\Sigma_1$ (the property of B will ensure this), but there are two minor difficulties here. We must make sure that J is a substructure of $\mathcal{M}$, that is, J is closed under multiplication. Moreover, we must ensure that (at least for standard) φ and $u \in J$, the substitution $\varphi(S^u 0, v)$ is in J .

These difficulties are overcome as follows. We change the construction slightly. We let $r = \max r : 2r \leq k$ and $C = \{h_{\omega^r, i}^A(a_0) : i \text{ is such that } h_{\omega^r, i}^A(a_0) \downarrow\}$. Then by Lemma 1.8.1, the set C is ω^r -large and $h_{\omega^r}^A$ -scattered. Well, we did not require A to be of the form $[a, b]$, but clearly, $h(a) \geq a + 1$ for all a , and hence, $h_{\omega^r}^A(a) \geq S_{\omega^r}(a) \geq S_{\omega^2}(a) \geq 2^a \cdot a$ for all a , hence C is scattered with respect to exponentiation. It follows that each cut J as defined above (but with C rather than A) is closed under exponentiation, and hence it is closed under addition and multiplication. Thus we apply the approximation lemma to C and obtain an r -large set $B \subseteq C$.

This trick allows us to overcome the second difficulty as well. The reason is that the function $\varphi, u \mapsto \varphi(S^u 0, v)$, i.e., the substitution function, being primitive recursive, is dominated by h_{ω^r} (because r is nonstandard), hence if $\varphi, u \leq c$, where $c \in C \setminus \{\max C\}$, then $\varphi(S^u 0, v) \leq h^C(c)$ and we may apply the approximation lemma.

Let us show how the property of B ensures that each cut J as above satisfies $I\Sigma_1$. For $b \in B \setminus \{\max B\}$ we have in $\mathcal{M}$: for $\varphi, u \leq b - 2$ if $\exists! w \text{ Tr}_0(\varphi(S^u 0, S^w 0))$ then $\exists w < h_2^B(b) \text{ Tr}_0(\varphi(S^u 0, S^w 0))$ or $\forall w < \max B \text{ Tr}_0(\neg \varphi(S^u 0, S^w 0))$. It follows that for each cut J as above

$$J \models \exists w \varphi(u, w) \quad \text{iff} \quad \mathcal{M} \models \exists w < \max B [\varphi(u, w) \ \& \ \forall z < w \neg \varphi(u, z)]$$

for all $\varphi \in \Delta_0$ and $u \in J$.

Granted this reduction of Σ_1 -truth in J to truth in $\mathcal{M}$ we argue as follows. Assume that

$$J \models [\exists x \varphi(0, x, u)] \ \& \ \forall t [(\exists x \varphi(t, x, u)) \Rightarrow \exists x \varphi(t + 1, x, u)].$$

It follows that $\mathcal{M} \models \exists x < \max B \varphi(0, x, u)$ and for each $t \in J$,

$$\mathcal{M} \models (\exists x < \max B \varphi(t, x, u)) \Rightarrow \exists x < \max B \varphi(t+1, x, u).$$

By overspill lemma, this holds for each $t < t_0$ for some $t_0 \in \mathcal{M} \setminus J$. In particular, $J \models \forall t \exists x \varphi(t, x, u)$ as required.

Let us sum up. Below we call an indicator for T a function which has the decisive property: $Y(x, y)$ is nonstandard iff there exists an initial segment of the model under consideration satisfying T such that $x \in I < y$. We formulate the results for models of $I\Sigma_m$ with an extra subset (named by a new predicate) because we shall use this stronger version in Chap. 5.¹

- Theorem 4.4.1** 1. *If $\mathcal{M}$ is a nonstandard model of $I\Sigma_1$ then there exist a set $C \in \mathcal{M}$ which is scattered with respect to exponentiation and ω^s -large for some nonstandard $s \in \mathcal{M}$.*
2. *If $C \in \mathcal{M}$ satisfies the properties stated in part 1 then the function $Y(x, y) = \max z : h_{\omega^z}^C(x) \leq y$ is an indicator for initial segments J satisfying $I\Sigma_1$ such that J is closed under the successor in the sense of C . Moreover, if $Y(x, y)$ is nonstandard then the appropriate I may be chosen in such a way that $C \cap I$ is unbounded in I and the structure $(I, C \cap I)$ satisfies Σ_1 induction in its language.*
3. *If F is a recursive function such that (for some formula representing F) $I\Sigma_1$ proves “ F is total”, then there exists $s \in \mathbb{N}$ such that F is dominated by S_{ω^s} .*

Proof The first claim and the main part of claim 2 were proved above. For the moreover clause of claim 2 we simply remark that we could apply the same construction but for the enumeration $\varphi(S^u 0, v) : r$ of formulas with parameter A and do the same. For the third part one should only remark that in particular we obtained an ω^k -large set $A \in \mathcal{M}$ being of the form $[a, S_{\omega^k}(a)]$ with nonstandard k and were able to find initial segments J of $\mathcal{M}$ such that J is not closed under S_{ω^r} , provided $r \in \mathcal{M}$ is nonstandard. Now, if $F(a)$ is not bounded by any S_{ω^s} , for some $s \in \mathbb{N}$ then, by overspill, it is not bounded by S_{ω^r} , for some nonstandard $r \in \mathcal{M}$. The contradiction follows by the existence of a cut I such that $a \in I < S_{\omega^r}(a)$ such that $I \models I\Sigma_1$ and such that F is not defined for a in I . $\square$

Before going to the case $m > 1$ let us observe that the heart of the matter in the construction presented above was the following: if $\varphi, u \leq b$, where $b \in B$, then either all statements of the form $\exists w < z \varphi(u, w)$ for $z \in B, z \geq h_2^B(b)$ are true in $\mathcal{M}$ or all these statements are false. That is we may bound quantifier to any (large enough) element of B without changing truth of the statement obtained in this manner. It is convenient to introduce two sets: E , the set of these sentences as above that are true and A , the set of negations of elements of E . These sets play the role of Σ_1 and Π_1 truth (in initial segments of $\mathcal{M}$ determined as above). It is convenient to introduce one more set $D = \{b_{2i} : i \text{ is such that } b_{2i} \text{ exists}\}$, where $B = \{b_i : i\}$ in increasing order. Then D is $k/2$ -large (or $(k-1)/2$ -large if $\text{Card} B$ is odd) and has the decisive

¹Unfortunately, these parts of the book presenting the work from [9] were not finished.

property: for all $d \in D \setminus \{\max D\}$ for all $\varphi, u \leq d$ if $\varphi \in \Delta_0$ then for each $t, s \in D$ strictly greater than d , the equivalence

$$\exists w < t \operatorname{Tr}_0(\varphi(S^w 0, S^u)) \equiv \exists w < s \operatorname{Tr}_0(\varphi(S^w 0, S^u))$$

holds in $\mathcal{M}$.

We shall use analogues of these sets when working with case $m > 1$ in order to state inductive conditions for the construction. Let us go to the case when $m > 1$. The result will be as follows.

Theorem 4.4.2 *Let $m > 1$ be fixed.*

1. *For every $k \in \mathbb{N}$, $I\Sigma_m$ proves $\forall a \exists b [a, b]$ is $\omega_m(k)$ -large.*
2. *Let $\mathcal{M} \models I\Sigma_m$ and let $A \in \mathcal{M}$ be $\omega_m(k)$ -large for all standard $k \in \mathbb{N}$. Then there exists an initial segment I of $\mathcal{M}$ such that $A \cap I$ is unbounded in I and the structure $(I, A \cap I)$ satisfies $I\Sigma_m$ in its language.*
3. *The function $Y(x, y) = \max k : S_{\omega_m(k)}(x) \leq y$ is an indicator for models of $I\Sigma_m$.*
4. *If f is a recursive function such that (for some Σ_1 definition of f) $I\Sigma_m$ proves totality of f , then there exists $k \in \mathbb{N}$ such that $f(x)$ is dominated by $S_{\omega_m(k)}$.*

Similar result holds for full PA:

Theorem 4.4.3 1. *For every $k \in \mathbb{N}$ PA proves $\forall a \exists b [a, b]$ is ω_k -large.*

2. *Let $\mathcal{M} \models \text{PA}$ and let $A \in \mathcal{M}$ be ω_k -large for all standard $k \in \mathbb{N}$. Then there exists an initial segment I of $\mathcal{M}$ such that $A \cap I$ is unbounded in I and the structure $(I, A \cap I)$ satisfies PA in its language.*
3. *The function $Y(x, y) = \max k : S_{\omega_k}(x) \leq y$ is an indicator for models of PA.*
4. *If f is a recursive function such that for some Σ_1 definition of f , PA proves that f is total, then f is dominated by S_{ω_k} for some $k \in \mathbb{N}$.*

Proof of Theorem 4.4.2 Fix $m \in \mathbb{N}$, $m > 1$. The first part of the theorem is just a restatement of Theorem 4.3.1. Let us go to the main construction. Fix $\mathcal{M} \models I\Sigma_m$. Let c be a nonstandard element of $\mathcal{M}$. Then, for every standard k , $\mathcal{M}$ satisfies “there exists an $\omega_m(k+1)$ -large set U with $c \leq \min U$ ”, so by overspill lemma (i.e., Lemma 3.4.1) the same holds in $\mathcal{M}$ for some nonstandard $k \in \mathbb{N}$. Pick such a nonstandard k and an appropriate set $U \in \mathcal{M}$. By Lemma 1.8.3 there exist a subset C of U which is $\omega_m(k)$ -large and $\omega_m(k)$ -scattered. In particular, C is ω^ω -scattered. We construct a sequence $D_1, \dots, D_m$ of sets in the sense of $\mathcal{M}$ such that

1. $C \supseteq D_1 \supseteq \dots \supseteq D_m$,
2. D_j is $\omega_{m-j}(k-j)$ -large for $j < m$ and D_m is $\frac{k-m}{2}$ -large if $k-m$ is even and is $\frac{k-m-1}{2}$ -large if $k-m$ is odd,
3. for every $j \leq m$ and every $d \in D_j$ we have: for every $\varphi, u_j, \dots, u_m \leq d$ if $\varphi \in \Delta_0$ and $Q_{j-1}, \dots, Q_0$ is a sequence of quantifiers, then for every $\langle f_{j-1}, \dots, f_0 \rangle, \langle g_{j-1}, \dots, g_0 \rangle \in [D_j]^j$ with $d < f_{j-1} < \dots < f_0$ and $d < g_{j-1} < \dots < g_0$, we have

$$\begin{aligned}
& Q_{j-1}w_{j-1} < f_{j-1} Q_{j-2}w_{j-2} < f_{j-2} \cdots Q_0w_0 < f_0 \\
& \quad \text{Tr}_0(\varphi(S^{w_0}0, \dots, S^{w_{j-1}}0, S^{u_j}0, \dots, S^{u_m}0)) \\
& \text{iff} \\
& Q_{j-1}w_{j-1} < g_{j-1} Q_{j-2}w_{j-2} < g_{j-2} \cdots Q_0w_0 < g_0 \\
& \quad \text{Tr}_0(\varphi(S^{w_0}0, \dots, S^{w_{j-1}}0, S^{u_j}0, \dots, S^{u_m}0))
\end{aligned} \tag{4.11}$$

holds in $\mathcal{M}$.

The construction is as follows. We begin with the function which differs only notationally from (4.10). That is we let

$$g_1(x) = \begin{cases} \min w_0 : \text{Tr}_0(\varphi(S^{w_0}0, S^{u_1}0, \dots, S^{u_m}0)) & \text{if } x \text{ is of the form} \\ & \varphi(v_0, S^{u_1}0, \dots, S^{u_m}0), \\ & \text{with } \varphi \in \Delta_0 \text{ and} \\ & \exists w_0 \text{Tr}_0(\varphi(S^{w_0}0, S^{u_1}0, \dots, S^{u_m}0)) \\ 0 & \text{in other cases} \end{cases} \tag{4.12}$$

We let B_1 be an $\omega_{m-1}(k)$ -large subset of C which is an approximation of g_1 . Writing $B_1 = \{b_0, \dots, b_{\text{Card } B_1 - 1}\}$ we take as D_1 the set consisting of these $b \in B_1$ which have even indices. By Lemma 1.8.3, this set is $\omega_{m-1}(k-1)$ -large. Part 3 of (4.11) is checked exactly as above.

We let

$$A_1 = \{\forall v_0 \varphi(v_0, S^{u_1}0, \dots, S^{u_m}0) : \varphi \in \Delta_0, \varphi, u_1, \dots, u_m \leq \max D_1, \\ \forall w_0 \text{Tr}_0(\varphi(S^{w_0}0, S^{u_1}0, \dots, S^{u_m}0))\}$$

and

$$E_1 = \{\exists v_0 \varphi(v_0, S^{u_1}0, \dots, S^{u_m}0) : \varphi \in \Delta_0, \varphi, u_1, \dots, u_m \leq \max D_1, \\ \exists w_0 \text{Tr}_0(\varphi(S^{w_0}0, S^{u_1}0, \dots, S^{u_m}0))\}.$$

We use these sets, or, more exactly, A_1 , to define the function g_2 . We let

$$g_2(x) = \begin{cases} \min w_1 : \ulcorner \forall v_0 \varphi(v_0, S^{w_1}0, S^{u_2}0, \dots, S^{u_m}0) \urcorner \in A_1 & \text{if } x \text{ is } \forall v_0 \varphi(v_0, v_1, S^{u_2}0, \dots, S^{u_m}0), \\ & \varphi \in \Delta_0 \text{ and there exists } w_1 \text{ as above} \\ 0 & \text{in other cases.} \end{cases} \tag{4.13}$$

Let B_2 be an $\omega_{m-2}(k-1)$ -large subset of D_1 that is an approximation of g_2 and let D_2 be the set of these $b \in B_2$ which have even numbers in the increasing enumeration of B_2 . Then D_2 is $\omega_{m-2}(k-2)$ -large.

Pick $b \in B_2$ such that $h_2^{B_2}(b) \downarrow$. Then for $\varphi, u_2, \dots, u_m \leq b$, the substitution $\forall v_0 \varphi(v_0, v_1, S^{u_2}0, \dots, S^{u_m}0)$ is smaller than $h^{B_2}(b) - 2$, hence the appropriate w_1 is either below $h_2^{B_2}(b)$ or is greater or equal $\max B_2$. Thus, for $d \in D_2$ and $\varphi, u_2, \dots, u_m \leq d$ and every $t_1, s_1 \in D_2$, if $d < t_1, s_1$ and $\varphi \in \Delta_0$, then

$$\begin{aligned}
& \exists w_1 < t_1 \ulcorner \forall v_0 \varphi(v_0, S^{w_1}0, S^{u_2}0, \dots, S^{u_m}0) \urcorner \in A_1 \\
& \text{iff} \\
& \exists w_1 < s_1 \ulcorner \forall v_0 \varphi(v_0, S^{w_1}0, S^{u_2}0, \dots, S^{u_m}0) \urcorner \in A_1.
\end{aligned} \tag{4.14}$$

It follows that for each $d \in D_2$ and each $\varphi, u_2, \dots, u_m \leq d$ and each $t_1 < t_0$ and $s_1 < s_0$ in D_2 and strictly greater than d , if $\varphi \in \Delta_0$, then

$$\begin{aligned}
& \exists w_1 < t_1 \forall w_0 < t_0 \text{Tr}_0(\varphi(S^{w_0}0, S^{w_1}0, S^{u_2}0, \dots, S^{u_m}0)) \\
& \text{iff} \\
& \exists w_1 < s_1 \forall w_0 < s_0 \text{Tr}_0(\varphi(S^{w_0}0, S^{w_1}0, S^{u_2}0, \dots, S^{u_m}0)).
\end{aligned} \tag{4.15}$$

A similar reasoning with $\neg\varphi$ establishes part 3 of the required properties of D_2 , i.e., (4.11).

We iterate this construction. So let D_j be given and have the properties listed above. We let A_j be the set of all sentences $\forall v_{j-1} \exists v_{j-2} \dots Q v_0 \varphi(v_0, \dots, v_{j-1}, S^{u_j}0, \dots, S^{u_m}0)$ (where Q is the appropriate quantifier) such that $\varphi \in \Delta_0$, and there exists a sequence $d < t_{j-1} < \dots < t_0$ of elements of D_j such that $\varphi, u_j, \dots, u_m \leq d$ and $\forall w_{j-1} < t_{j-1} \exists w_{j-2} < t_{j-2} \dots Q w_0 < t_0 \text{Tr}_0(\varphi(S^{w_0}0, \dots, S^{w_{j-1}}0, S^{u_j}0, \dots, S^{u_m}0))$. We let

$$g_j(x) = \begin{cases} \min w_j : \ulcorner \forall v_{j-1} \exists v_{j-2} \dots Q v_0 \varphi(v_0, \dots, v_{j-1}, S^{w_j}0, S^{u_{j+1}}0, \dots, S^{u_m}0) \urcorner \in A_j \\ \quad \text{if } x \text{ is } \forall v_{j-1} \exists v_{j-2} \dots Q v_0 \varphi(v_0, \dots, v_{j-1}, v_j, S^{u_{j+1}}0, \dots, S^{u_m}0) \\ \quad \text{with } \varphi \in \Delta_0 \text{ and } w_j \text{ exists} \\ 0 \quad \text{otherwise.} \end{cases} \tag{4.16}$$

Let B_{j+1} be an $\omega_{m-j-1}(k-j)$ -large subset of D_j which is an approximation for g_{j+1} and let D_{j+1} be the set of these $b \in B_{j+1}$ which have even numbers in the increasing enumeration of B_{j+1} . Then conditions 1 and 2 are obvious, so let us indicate why 3 holds. Pick $d \in D_{j+1}$ and $\varphi, u_{j+1}, \dots, u_m \leq d$ be given. Then the substitution $\forall v_{j-1} \dots Q v_0 \varphi(v_0, \dots, v_j, S^{u_{j+1}}0, \dots, S^{u_m}0)$ is below the next element of B_{j+1} , hence for $t_j, s_j \in D_{j+1}$ strictly greater than d we have: $\exists w_j < t_j \ulcorner \forall v_{j-1} \exists v_{j-2} \dots Q v_0 \varphi(v_0, \dots, v_{j-1}, S^{w_j}0, S^{u_{j+1}}0, \dots, S^{u_m}0) \urcorner \in A_j$ iff $\exists w_j < s_j \ulcorner \forall v_{j-1} \exists v_{j-2} s_j \dots Q v_0 \varphi(v_0, \dots, v_{j-1}, S^{w_j}0, S^{u_{j+1}}0, \dots, S^{u_m}0) \urcorner \in A_j$, hence we may apply the inductive assumption.

The last of these sets, D_m , is of nonstandard finite cardinality. Let I be any initial segment of $\mathcal{M}$, closed under successor and bounded in D_m . Let $J_I = \{e \in \mathcal{M} : e < d_i \text{ for some } i \in I\}$, where $D_m = \{d_0, \dots, d_{\text{Card}(D_m)-1}\}$ in increasing order. In order to show that $J_I \models I\Sigma_m$ we show a reduction of Σ_m -truth in J_I to truth in $\mathcal{M}$.

We claim that for each $j \leq m$ we have:

$$\begin{aligned}
& \text{for each } \Delta_0 \text{ formula } \varphi \text{ and each } u_j, \dots, u_m \in J_I \\
& J_I \models \forall w_{j-1} \exists w_{j-2} \dots Q w_0 \varphi(w_0, \dots, w_{j-1}, u_j, \dots, u_m)
\end{aligned} \tag{4.17}$$

iff

there exists $d \in D_j$ greater than all parameters $u_j, \dots, u_m$ and a sequence $t_{j-1} < \dots < t_0$ of elements of D_m strictly greater than d such that

$$\mathcal{M} \models \forall w_{j-1} < t_{j-1} \exists w_{j-2} < t_{j-2} \dots Q w_0 < t_0 \varphi(w_0, \dots, w_{j-1}, u_j, \dots, u_m). \quad (4.18)$$

This fact is proved by induction on j , case $j = 1$ being immediate. Assume the fact for j , we show it for $j + 1$. Assume

$$J_I \models \forall w_j \exists w_{j-1} \dots Q w_0 \varphi(w_0, \dots, w_j, u_{j+1}, \dots, u_m).$$

Pick $d \in D_j \cap J_I$ greater than all parameters $u_{j+1}, \dots, u_m$. Let $t_j \in D_j$ be the next element of D_j . Consider the formula obtained from the above by bounding the first quantifier to t_j . This formula is true in J_I , so by the inductive assumption there exists a sequence $t_{j-1}, \dots, t_0$ as desired. For the converse assume that d and a sequence $t_j, \dots, t_0$ like on the right hand side exists. Then each sequence $s_j, \dots, s_0$ of elements of D_{j+1} has the same property. Pick $w_j \in J_I$. Find $d' \in D_{j+1}$ which is greater than d and w_j . Pick a sequence $s_j < s_{j-1} < \dots < s_0$ of elements of D_{j+1} with $s_j > d'$. Then

$$\mathcal{M} \models \exists w_{j-1} < s_{j-1} \dots Q w_0 s_0 \varphi(w_0, \dots, w_{j-1}, w_j, u_{j+1}, \dots, u_m)$$

by (4.11). By the inductive assumption

$$J_I \models \exists w_{j-1} \dots Q w_0 \varphi(w_0, \dots, w_{j-1}, w_j, u_{j+1}, \dots, u_m).$$

Let us show how to use the above reduction to show that $J_I \models I\Sigma_m$. Let $\varrho(u, v)$ be a Π_m formula and assume that $J_I \models \exists v \varrho(u, v)$. Write ϱ in the form $\forall w_{m-1} \exists w_{m-2} \dots \varphi(w_0, \dots, w_{m-1}, u, v)$ with $\varphi \in \Delta_0$. Pick an appropriate $v \in J_I$ and let $u_m = \langle u, v \rangle$. Pick a sequence $t_{m-1} < \dots < t_0$ of elements of D_m such that $u_m \leq d < t_{m-1}$ for some $d \in D_j$. By the above reduction, $\mathcal{M} \models \forall w_{m-1} < t_{m-1} \exists w_{m-2} < t_{m-2} \dots Q w_0 \varphi(w_0, \dots, w_{m-1}, u, v)$. But the smallest such v exists in $\mathcal{M}$ (because this is a model for $I\Delta_0$), and this element is the smallest which satisfies ϱ in J_I .

Granted this construction we see that part 1 of Theorem 4.4.2 is just a reformulation of Theorem 4.3.1. Part 2 was proved above. Let us merely point out that in order to obtain initial segments satisfying $I\Sigma_m$ in the language with the additional predicate letter for A one applies the same construction, but allows formulas with parameter A in the definition of functions g_j . Part 3 is immediate by parts 1 and 2, and part 4 is a direct consequence of part 2. $\square$

Proof of Theorem 4.4.3. This is the same argument as that for Theorem 4.4.2. The only difference is that we begin with a set $A \in \mathcal{M}$ which is ω_m -large for some nonstandard $m \in \mathcal{M}$ and perform the above construction $m - 1$ times. We leave the details to the reader. $\square$

Here is a version of the main result of J. Paris and L. Harrington [10].

Theorem 4.4.4 *Let $k \geq 2$. Then $I\Sigma_{k-1}$ does not prove the following statement: “for every a there exists b such that for every partition $L : [a, b]^{k+1} \rightarrow 3^{k-1}$ there exists*

a set $D \subseteq [a, b]$ which is homogeneous for L and satisfies $\text{Card}(D) > \min(D) + \frac{k(k-1)}{2} + 1$ ".

Proof By formalizing the proof of Theorem 1.10.9 we see that otherwise $I\Sigma_{k-1}$ would prove the statement "for every a there exists b such that the interval $[a, b]$ is ω_k -large". But this contradicts the fourth point of Theorem 4.4.2. $\square$

Let us formulate a reverse problem. Namely, let a set A have the following property: for every partition $R: [A]^m \rightarrow s$ there exists a homogeneous ξ -large set $H \subseteq A$. We merely remark that in [11], we give some lower bounds on how large A must be.

References

1. Takeuti, G. (1987). *Proof theory*.
2. Hájek, P., & Pudlák, P. (1993). *Metamathematics of first-order arithmetic*. Perspectives in mathematical logic. Berlin: Springer.
3. Ratajczyk, Z. (1988). A combinatorial analysis of functions provably recursive in Σ_n . *Fundamenta Mathematicae* 3(130), 191–213. issn: 0016-2736. <https://doi.org/10.4064/fm-130-3-191-213>.
4. Sommer, R. (1990.) *Transfinite induction and hierarchies of functions generated by transfinite recursion within Peano arithmetic*. Ph.D thesis, University of California, Berkeley.
5. Sommer, R. (1995). Transfinite induction within Peano arithmetic. *Annals of Pure and Applied Logic* 76(3), 231–289. issn: 01680072. [https://doi.org/10.1016/0168-0072\(95\)00029-G](https://doi.org/10.1016/0168-0072(95)00029-G).
6. Ketonen, J., & Solovay, R. (1981). Rapidly growing Ramsey functions. *The Annals of Mathematics* 113(2), 267. <https://doi.org/10.2307/2006985>.
7. Mints, G. E. (1971). Exact estimation of the provability of transfinite induction in initial parts of arithmetic. *Zapiski Nauchnykh Seminarov; Leningrad. Otdel. Mat. Inst. Steklov. (LOMI)* 20, 134–144; 286.
8. Kotlarski, H., & Ratajczyk, Z. (1990). Inductive full satisfaction classes. *Annals of Pure and Applied Logic* 47(3), 199–223. issn: 01680072. [https://doi.org/10.1016/0168-0072\(90\)90035-Z](https://doi.org/10.1016/0168-0072(90)90035-Z).
9. Kotlarski, H., & Ratajczyk, Z. (1990). More on induction in the language with a satisfaction class. *Mathematical Logic Quarterly* 36(5), 441–454. issn: 09425616. <https://doi.org/10.1002/malq.19900360509>.
10. Paris, J., & Harrington, L. (1977). A Mathematical Incompleteness in Peano Arithmetic. *Handbook of mathematical logic*. Studies in logic and the foundations of mathematics (Vol. 90, pp. 1133–1142). Amsterdam: Elsevier. isbn: 9780444863881. [https://doi.org/10.1016/S0049-237X\(08\)71130-3](https://doi.org/10.1016/S0049-237X(08)71130-3).
11. Bigorajska, T., & Kotlarski, H. (2006). Partitioning alpha-large sets: Some lower bounds. *Transactions of the American Mathematical Society* 358(11), 4981–5002. issn: 00029947. <https://doi.org/10.1090/S0002-9947-06-03883-9>.

Chapter 5

Satisfaction Classes



We introduced the notions of a full satisfaction class and of a Q_e -satisfaction class in Sect. 3.4 as a technical tool for the proof of Tarski's theorem (cf. Sect. 3.5). In this chapter we shall study these notions in their own right.

As pointed out in Sect. 3.4 Robinson [1] was the first to realize that one can treat seriously nonabsoluteness of the finite in the very definitions of the language. Robinson defined the so-called *external satisfaction* (defined by means of Skolem functions) and showed it does not apply to all formulas in the sense of the model under consideration. Moreover, even when it applies, it need not be equivalent to something more familiar, the so-called *internal satisfaction*, defined as membership in D , where D is obtained in the following way. Consider the model $\langle \mathbb{N}, \models \rangle$ (for the language of $\text{PA} +$ one unary predicate letter for $\{\varphi : \mathbb{N} \models \varphi\}$) and extend this model elementarily. Let D be the interpretation of the above in such model.

The subject of satisfaction classes was rejuvenated by Krajewski [2] (under the inspiration of the late Professor A. Mostowski). He gave the appropriate definitions and proved that satisfaction classes are never unique (except standard model).

5.1 Satisfaction Classes: Generalities

Before proving Krajewski's result we observe that all satisfaction classes for a given $\mathcal{M}$ coincide on standard formulas. Precisely

Theorem 5.1.1 *Let $\mathcal{M}$ be a model of PA and let D be a satisfaction class for $\mathcal{M}$ (either full or Q_e for some $e \in \mathcal{M}$). Then for every formula φ of L_{PA} (if $\varphi \in Q_e$ in the latter case), and every $a \in \mathcal{M}$, $\mathcal{M} \models \varphi(a)$ iff $\varphi(S^a 0) \in D$.*

Proof Immediate, by induction on formulas. □

Krajewski's result mentioned above was for special models (and involved no continuum hypothesis), nowadays it is much easier to work with countable recursively saturated models. Precisely

Theorem 5.1.2 *Let $\mathcal{M}$ be a countable model of PA and let D be a full satisfaction class for $\mathcal{M}$. Assume that $\langle \mathcal{M}, D \rangle$ is recursively saturated. Then $\mathcal{M}$ has continuum many full satisfaction classes, indeed, D has continuum many automorphic images, i.e. $\{E \subseteq \mathcal{M} : \exists g \in \text{Aut}(\mathcal{M}) E = g * D\}$ is of power $2^{\aleph_0}$.*

We shall prove later (see Theorem 5.2.3) that each nonstandard model admitting a full satisfaction class is recursively saturated, moreover it admits a full satisfaction class D such that $\langle \mathcal{M}, D \rangle$ is recursively saturated. We also point out that the same results hold for Q_e -satisfaction classes, provided e is a nonstandard element of the model considered.

Proof of Theorem 5.1.2: By Chang–Makkai theorem (i.e., Theorem 2.3.1) it suffices to show that D is not definable in $\mathcal{M}$ with parameters. So assume that

$$\langle \mathcal{M}, D \rangle \models \forall z \forall \varphi [D(\varphi(S^z 0)) \equiv A(\varphi(S^z 0), p)],$$

where $A(\cdot, \cdot)$ is an L_{PA} formula and $p \in \mathcal{M}$. This is impossible by the proof of Tarski's theorem (i.e. Theorem 3.2.7). We use the parametric version of the diagonal lemma (i.e. Lemma 3.2.2) and get a formula φ such that $\text{PA} \vdash [\varphi(x) \equiv \neg A(\varphi(S^x 0), x)]$. Then in $\mathcal{M}$ we have

$$D(\varphi(S^x 0)) \equiv \varphi(x) \equiv \neg A(\varphi(S^x 0), x) \equiv \neg D(\varphi(S^x 0)).$$

Contradiction and D cannot be definable. □

The reader familiar with the idea from Sect. 3.5 will observe that one could equally well check that in the situation described in the assumption of Theorem 5.1.2 the Skolem closure of $\{p\}$ in $\mathcal{M}$ would be recursively saturated, what is impossible.

We proved earlier (see Theorem 3.4.2) that a countable model of PA is recursively saturated iff it has an inductive satisfaction class for Q_e formulas for some nonstandard $e \in \mathcal{M}$. This cannot be generalized to the existence of full inductive satisfaction classes (but may be generalized to noninductive full satisfaction classes, see Sect. 5.2). The reason is that the existence of a full inductive satisfaction class leads to a theory which is much stronger than PA. One could, e.g., consider the theories $\text{PA} + “D \text{ is a full inductive satisfaction class}”$ and $\text{PA} + “D \text{ is a full satisfaction class}” + \text{induction for } \Sigma_n \text{ formulas in the language } L_{\text{PA}} + D$. In later sections we shall give an information on conditions on $\text{Th}(\mathcal{M})$ which are equivalent (for a countable recursively saturated model $\mathcal{M}$) to the existence of D such that $\langle \mathcal{M}, D \rangle$ satisfies the appropriate theory like the ones above.

5.2 Noninductive Satisfaction Classes

The goal of this section is to show that a countable model for PA is recursively saturated iff it has a full satisfaction class. Moreover, this satisfaction class may be chosen to be very pathological. In order to state what we have in mind, let us define a sequence of sentences which are disjunctions of copies of the statement $0 = 0$. But (as we shall see below) it is important to show the distribution of parentheses in these disjunctions. So define A_0 to be the statement $0 = 0$ and let A_{i+1} be $(A_i \vee A_i)$. Thus each disjunct of the form $0 = 0$ is hidden as deeply inside parentheses as possible.

Theorem 5.2.1 (Kotlarski, Krajewski, Lachlan [3]) *Let $\mathcal{M}$ be a countable recursively saturated model for PA. Then $\mathcal{M}$ has a full satisfaction class, indeed, given a nonstandard $i \in \mathcal{M}$ there exists a full satisfaction class D for $\mathcal{M}$ making A_i false.*

Observe that one could prove D -truth of each A_i by induction on i , so the conclusion is that induction in $\langle \mathcal{M}, D \rangle$ fails for the satisfaction class D given by the theorem.

Corollary 5.2.2 *The theory $\text{PA} + “D \text{ is a full satisfaction class}”$ is conservative over PA, that is for every sentence φ in the language of PA, if $\text{PA} + “D \text{ is a full satisfaction class}”$ proves φ , then already PA proves φ .*

Let us remark that this corollary fails if we require D to satisfy some induction. The theory of a satisfaction class with Σ_n induction, for $n \geq 1$, is characterized in [4] and [5]. On the other hand, in the paper [6] a serious gap was spotted by Albert Visser. Namely, it is not obvious, that Δ_0 induction suffices to show that all nonstandard logical axioms are in D . A discussion of this problem can be found in an appendix of [7]. In [7], Wcisło and Łełyk showed that PA with Δ_0 induction for the language with a satisfaction class is not conservative over PA. Later, Łełyk in [8] reestablished the result from [6] showing how to prove that all logical axioms are true in D using just Δ_0 induction for the language with D . In fact, Theorem 141 from [8] proves the, so called, Global Reflection Principle for this theory.

The second main result of this section is the following converse to Theorem 5.2.1.

Theorem 5.2.3 (Lachlan [9]) *If $\mathcal{M} \models \text{PA}$ is nonstandard and admits a full satisfaction class (or a satisfaction class for Q_e formulas for some nonstandard $e \in \mathcal{M}$) then $\mathcal{M}$ is recursively saturated.*

Observe that in Theorem 3.4.2 we obtained the same conclusion from the assumption of the existence of an *inductive* satisfaction class (in fact, that argument used Σ_1 induction).

Both results stated above seem to be very non intuitive, so let us develop some ideas which lead them. Let $\mathcal{M} \models \text{PA}$. Consider the following game (suggested by one of the constructions of indicators, due to J. Paris). There are two players: **Stupid** and **Dishonest**.¹ **Stupid** asks questions of the form “does $\xi \in D$ ”, where ξ is a sentence in the sense of $\mathcal{M}$. **Dishonest** he answers “yes” or “no”. If his answer is positive and ξ begins with an existential quantifier, say ξ is $\exists z \varphi(z)$, then he is also obliged to show $z \in \mathcal{M}$ such that he fixes the answer to “does $\varphi(S^z 0) \in D$ ”. The definition of “**Dishonest** wins” is natural: his answers do not contradict the definition of a satisfaction class. Well, how is **Dishonest** obliged to play in order to win the game? The idea is that he keeps the information given up to the place he is and answers “yes” or “no” if the next question is in a sense near to the previous information and answers anything he wishes otherwise. Simply he is supposed to take care that **Stupid** *has no chance to verify* that he does not know what he is talking about... Thus, what is needed, is some notion of sentences being “near” or “far apart”. J. Paris suggested to use ω -logic for this purpose. Thus, the appropriate notion would be “ φ is k -near to $\psi_0, \dots, \psi_{r-1}$ ” if one of the implications $\bigwedge_{i < r} \psi_i \Rightarrow \varphi$, $\bigwedge_{i < r} \psi_i \Rightarrow \neg \varphi$ is provable in ω -logic by a proof of height at most k . We said a few words about the ω -rule in Sect. 3.12. Let us make the idea presented above precise.

Let PA be formulated in a finite language L without function symbols in which only symbols $\neg$, $\vee$ and $\exists$ are allowed (other symbols are treated as abbreviations in the standard manner). Let $\mathcal{M} \models \text{PA}$ be given. Consider the language $L(\mathcal{M})$ obtained from L by adjoining constants naming elements of $\mathcal{M}$. We shall also use the language $^*L(\mathcal{M})$, which is defined as above, but *in the sense of* $\mathcal{M}$ (thus nonstandard sentences are allowed).

The system of ω -logic that we have in mind, $\mathcal{L}(\mathcal{M})$, has all sentences of $^*L(\mathcal{M})$ as the set of all formulas. Its axioms are: all true atomic sentences, negated false atomic sentences, and all sentences of the form $\neg \varphi \vee \varphi$. Let Ax denote the set of axioms. The rules of inference are:

Expansion	$\frac{\psi}{\varphi \vee \psi}$	
Commutativity	$\frac{\varphi \vee \psi}{\psi \vee \varphi}$	
Associativity	$\frac{\varphi \vee (\psi \vee \chi)}{(\varphi \vee \psi) \vee \chi}$	$\frac{(\varphi \vee \psi) \vee \chi}{\varphi \vee (\psi \vee \chi)}$
Contraction	$\frac{\varphi \vee \varphi}{\varphi}$	
Cut	$\frac{\chi \vee \varphi \quad \neg \chi \vee \psi}{\varphi \vee \psi}$	
$\exists$ -introduction	$\frac{\varphi(a)}{\exists x \varphi(x)}$	$(a \text{ in } \mathcal{M})$
$\mathcal{M}$ -rule	$\frac{\neg \varphi(a) \text{ for all } a \in \mathcal{M}}{\neg \exists x \varphi(x)}$	

¹I believe that in English the names **Naive** and **Unfair** would give a better description of the intention of this game.

We also repeat each rule with the additional formula θ as a parameter. For example the repetition of the $\mathcal{M}$ -rule is:

$$\frac{\theta \vee \neg\varphi(a) \text{ for all } a \text{ in } \mathcal{M}}{\theta \vee \neg\exists x \varphi(x)}.$$

We define the notion of provability in $\mathcal{L}(\mathcal{M})$. For any set $\Gamma \subseteq {}^*L(\mathcal{M})$ we let Γ' be

$\Gamma \cup \{\theta \in {}^*L(\mathcal{M}) : \theta \text{ is an immediate consequence of a sentence or sentences in } \Gamma \cup \text{Ax by one of the rules}\}.$

We define Γ^α by induction on α :

$$\Gamma^\alpha = \Gamma \cup \left(\bigcup_{\beta < \alpha} \Gamma^\beta \right)'.$$

We stress that here ordinals are in the sense of external world, not in the sense of $\mathcal{M}$. We shall also write $\Gamma \vdash_\alpha \varphi$ for $\varphi \in \Gamma^\alpha$ and read “ φ is provable from Γ by means of a proof of height at most α ”. If Γ is omitted then we mean that $\Gamma = \emptyset$ (i.e. only axioms are allowed in the derivation). Because of the $\mathcal{M}$ -rule a proof of a sentence θ may be infinite and may involve an infinite external subset of Γ .

Thus we have the notion of φ being k -near to $\psi_0, \dots, \psi_{r-1}$: this is $\vdash_k [\wedge_{i < r} \psi_i \Rightarrow \varphi]$ or the same for $\neg\varphi$.

Observe that for $k \in \mathbb{N}$ the notion $\vdash_k \varphi$ is a formula in L_{PA} in variable φ . We point out that this is not a single formula in which k would be a variable (the reader is asked to write these formulas down to see that there is no possibility to treat uniformly consecutive applications of the $\mathcal{M}$ -rule). Anyway, for each k the notion $\Gamma \vdash_k \varphi$ is expressible in $\mathcal{M}$, provided Γ is definable in $\mathcal{M}$. Moreover, it is easy to check that this $\Gamma \vdash_k \varphi$ is definable by a recursive in k sequence of formulas. This remark is the heart of the matter in the proof of the following lemma, which shows that if $\mathcal{M}$ is recursively saturated, then φ is provable in $\mathcal{M}$ -logic implies that φ is provable by means a proof of finite height.

Lemma 5.2.4 *Under the assumptions and notation above, if $\mathcal{M}$ is recursively saturated, Γ is definable in $\mathcal{M}$, $\varphi \in \mathcal{M}$ and $\Gamma \vdash_\alpha \varphi$ for some α , then there exists $k \in \mathbb{N}$ such that $\Gamma \vdash_k \varphi$ holds.*

Proof Assume the contrary. Let α be the smallest infinite ordinal such that there exists $\varphi \in \mathcal{M}$ such that $\Gamma \vdash_\alpha \varphi$ and for no $\beta < \alpha$, $\Gamma \vdash_\beta \varphi$. We assert that $\alpha = \omega$. The reason is that otherwise φ would have to be inferred from some formulas either by the $\mathcal{M}$ -rule or by another one, but in each case at least one premise would have to be in $\Gamma^\beta \setminus \bigcup_{\delta < \beta} \Gamma^\delta$, for some infinite $\beta < \alpha$. This contradicts α being the smallest ordinal with the property stated above. Knowing that $\alpha = \omega$ we see that φ had to be obtained by the $\mathcal{M}$ -rule, either with or without parameter (other rules are finite). We consider the case of the $\mathcal{M}$ -rule with parameter. Thus, φ is of the form $\theta \vee \neg\exists x \psi(x)$. Moreover, for every $a \in \mathcal{M}$, the statement $\theta \vee \neg\psi(a)$ is in Γ^k for

some finite k . In other words, the type $p(a) = \{\Gamma \not\vdash_k \theta \vee \neg\psi(a) : k\}$ is omitted. By recursive saturation of $\mathcal{M}$, this type cannot be finitely realized. Thus, for some k_0 , we have, in $\mathcal{M}$: $\forall a \Gamma \vdash_{k_0} \theta \vee \neg\psi(a)$. But then $\alpha \leq k_0 + 1 < \omega$. Contradiction. $\square$

Now, we proceed to a construction of a satisfaction class. Observe that we need no game in order to have an exact criterion when one may put a sentence ξ to the class which is supposed to be constructed (but this was a good intuition). The construction will be more or less a repetition of the proof of the completeness theorem. In order to repeat this proof, we need, as usual, a version of the deduction theorem.

Lemma 5.2.5 *Let $\Gamma \subseteq {}^*L(\mathcal{M})$ be a set of sentences and let $\varphi, \psi \in {}^*L(\mathcal{M})$ be sentences. Then*

$$\Gamma \cup \{\varphi\} \vdash \psi \text{ implies } \Gamma \vdash \neg\varphi \vee \psi$$

and

$$\Gamma \cup \{\varphi\} \vdash \psi, \neg\psi \text{ implies } \Gamma \vdash \neg\varphi.$$

Proof The second part follows from the first using the cut rule, so we prove the first part. The idea is to go through given proof of ψ from Γ, φ and to replace each occurrence of any formula ϱ in this proof by the formula $\neg\varphi \vee \varrho$. More precisely, by induction on α we prove that if $\Gamma, \varphi \vdash_\alpha \psi$, then $\Gamma \vdash \neg\varphi \vee \psi$. If $\alpha = 0$, then ψ is either an axiom or an element of Γ or $\psi = \varphi$. If $\psi = \varphi$ then $\neg\psi \vee \varphi$ is an axiom. If ψ is an axiom, then we apply the expansion rule. If $\psi \in \Gamma$ then the result is obvious.

Assume the fact for all $\alpha' < \alpha$. Let ψ be such that $\Gamma, \varphi \vdash_\alpha \psi$. Then ψ was derived from some sentences proved by proofs of height $< \alpha$. We show how to handle just one rule (commutativity with parameter), leaving all other cases to the reader. So let ψ be of the form $\theta \vee (\tau \vee \xi)$ and was obtained from $\theta \vee (\xi \vee \tau)$. Thus we have $\Gamma \vdash_{\alpha'} \theta \vee (\xi \vee \tau)$ for some $\alpha' < \alpha$. By the inductive assumption we have $\Gamma \vdash \neg\varphi \vee (\theta \vee (\xi \vee \tau))$. We use the associativity rule to infer $(\neg\varphi \vee \theta) \vee (\xi \vee \tau)$, then we use the commutativity rule with parameter $\neg\varphi \vee \theta$ to infer $(\neg\varphi \vee \theta) \vee (\tau \vee \xi)$ and we use the associativity rule again. We leave other cases to the reader. $\square$

As usual, a set of sentences Γ is *consistent* if no contradiction is derivable from it.

Lemma 5.2.6 *Let $\Gamma \subseteq {}^*L(\mathcal{M})$ be a set of sentences, where $\mathcal{M}$ is countable. Then Γ can be extended to a full satisfaction class iff Γ is consistent.*

Proof One direction is obvious. Indeed, it suffices to check that if D is a full satisfaction class, then it is closed under all rules of inference, hence by transfinite induction on α we see that $D \not\vdash_\alpha 0 = 1$ and the same must be true for each $\Gamma \subseteq D$. The converse is proved much like the completeness theorem. We enumerate all sentences in the sense of $\mathcal{M}$ as $\varphi_i : i \in \mathbb{N}$ using the countability assumption. By induction on i we construct a sequence D_i of finite sets of sentences such that for each i , $\Gamma \not\vdash \neg \bigwedge (D_i)$ and either $\varphi_i \in D_{i+1}$ or $\neg\varphi_i \in D_{i+1}$. Moreover, if φ_i is of the form $\exists x \psi(x)$ and belongs to D_{i+1} , then $\psi(a) \in D_{i+1}$ for some $a \in \mathcal{M}$ as well. We let D_0 be empty. Consider the first sentence φ_0 . If $\Gamma \not\vdash \neg\varphi_0$ we let $D'_0 = \{\varphi_0\}$,

otherwise we let $D'_0 = \{\neg\varphi_0\}$. If φ_0 does not begin with the existential quantifier or $\Gamma \vdash \neg\varphi_0$, then we let $D_0 = D'_0$. Otherwise we write φ_0 as $\exists x \psi(x)$ and we assert that there exists $a \in \mathcal{M}$ such that $\Gamma \not\vdash \neg(\varphi_0 \& \psi(a))$. Otherwise for all $a \in \mathcal{M}$ we have $\Gamma \vdash \neg(\varphi_0 \& \psi(a))$ and, by the $\mathcal{M}$ -rule, $\Gamma \vdash \forall x [\varphi_0 \Rightarrow \neg\psi(x)]$. Now, some transformations yield $\Gamma \vdash \varphi_0 \Rightarrow \forall x \neg\psi(x)$, hence $\Gamma \vdash \neg\varphi_0$, a contradiction. Thus, we pick such $a \in \mathcal{M}$ and let $D_0 = \{\varphi_0, \psi(a)\}$. Further steps are the same, we leave the details to the reader. $\square$

Of course, now we are obliged to show that $\emptyset$ is consistent in $\mathcal{M}$ -logic. The difficulty is that nonstandard sentences have no natural interpretation. Fortunately, for recursively saturated $\mathcal{M}$ we need to consider only proofs of finite height by Lemma 5.2.4.

Lemma 5.2.7 *Let $\mathcal{M}$ be nonstandard. Let k be a nonstandard element of $\mathcal{M}$ and let A_k be the disjunction of copies of $0 = 0$ as before the statement of Theorem 5.2.1. Then for every $m \in \mathbb{N}$ we have $\not\vdash_m A_k$. In particular, if $\mathcal{M}$ is recursively saturated, then $\mathcal{M}$ -logic is consistent.*

Of course, Theorem 5.2.1 follows from these lemmas. The second claim of Lemma 5.2.7 follows immediately from the first one and Lemma 5.2.4. For the first part we must develop some machinery of the so-called *approximations of formulas* (they have nothing to do with the approximations of functions as in Sect. 1.6, but we follow the terminology from the literature, clearly there is no possibility of confusing these two notions of approximations).

The idea is as follows. Let φ be formula. We define a tree of its subformulas in the natural manner. Thus we put $U_0 = \{\varphi\}$. Assume that levels $U_0, \dots, U_{r-1}$ are constructed. Pick a formula $\psi \in U_{r-1}$. If ψ is of the form $\neg\varrho$ then we put ϱ to U_r (and make it $\prec$ -greater than $\neg\varrho$). If ψ is of the form $\xi \vee \zeta$ then we put both ξ, ζ to U_r and make them greater than ψ . If ψ is of the form $\exists x \varrho(x)$ then we put ϱ to U_r and make it greater than ψ . If ψ is atomic we do nothing (recall that we are working without function symbols, so terms are either variables or constants). Let $E(\varphi, m)$ denote this tree up to the m th level. We replace each formula $\gamma(\bar{x})$ from the m th level by a new predicate letter $P_\gamma(\bar{x})$ (with the same free variables as those of γ) if it is not atomic and by itself if it is atomic and come back with formulas obtained in this manner using the structure given by our tree. The resulting formula $\text{Appr}(m, \varphi)$ has many new predicate letters, but has the same structure as φ up to depth m . We do the same with proofs. This sort of construction reflects the structure of formulas to the extent that from a proof we get a proof in an additional logic in which new predicates letters are allowed (countably many for each finite arity), but the interpretations of these new letters are completely arbitrary. In particular, the consecutive approximations of A_k are $p, p \vee p, (p \vee p) \vee (p \vee p), \dots$, where p is a propositional constant (i.e. 0-ary predicate letter). None of them is a theorem of this logic, so A_k cannot admit a proof of finite height. Well, where are the problems which occur here?

Unfortunately, if we replace in a proof all formulas with their m -th approximations we do not get a valid proof. Let us consider some examples. Firstly, if we take $\text{Appr}(m, \neg\varphi \vee \varphi)$ then this formula is no longer our axiom. Therefore, we replace only subformulas of formulas from a given proof, e.g., axioms become $\neg\text{Appr}(m, \varphi) \vee \text{Appr}(m, \varphi)$.

Secondly, if we have two formulas in the proof, $\varphi(a, b)$ and $\varphi(a, c)$ then, after applying $\exists$ -introduction rule, we may get the same formula $\exists x \varphi(a, x)$. Thus, our approximations of $\varphi(a, b)$ and $\varphi(a, c)$ should take this into account. We should exchange $\varphi(a, b)$ and $\varphi(a, c)$ with the same predicate, say P_φ , and different arguments. Therefore, for all formulas which differ only in used variables and constants we should use the same predicate symbol.

Now, we should show that if we have a proof p of height $k \in \mathbb{N}$ then we can replace subformulas of the original formulas used in the proof by their m approximations, for arbitrary $m \in \mathbb{N}$, so that the proof remains valid. We should remember that that we should left intact all logical operators which are used in rules of inferences in a proof p . Since p has a standard height all such operators are also at standard height bounded by some function of k . We merely remark that if p is just an axiom we may take $\text{Appr}(n, \theta) \vee (\neg\text{Appr}(m, \varphi) \vee \text{Appr}(m, \varphi))$. We refer an interested reader to Kaye's book [10] for a detailed proof. Kaye's proof takes into account a language with function symbols which adds some extra complications.

We remark that the machinery developed above works for the usual logic (i.e. the one without the ω -rule) as well. The following is an immediate corollary.

Corollary 5.2.8 *Let $\mathcal{M} \models \text{PA} + \neg\text{Con}_{\text{PA}}$ and let $d \in \mathcal{M}$ be such that $\mathcal{M} \models \text{Prov}_{\text{PA}}(d, 0 = S0)$. Then d has nonstandard rank (as a proof tree).*

Proof Indeed, otherwise we could take an approximation d' of such a proof d . If d has a standard rank then d' would be a standard, valid proof possibly with some new predicates. These formulas may have nonstandardly many variables but only finitely many of them are used in the proof rules. Since these predicates do not occur in the axioms of PA we can replace them with formulas of PA over relevant variables only. In this way, we obtain a proof of $0 = S0$ from PA. A contradiction. □

We go to the proof of Lachlan's result, i.e. to Theorem 5.2.3. Observe that every model of $I\Sigma_1$ has well behaved universal Σ_n formulas for every $n \in \mathbb{N}$, hence by the argument of Theorem 3.4.2 it is recursively saturated with respect recursive types contained in Σ_n . But it need not be saturated with respect to types containing formulas higher in the hierarchy. It is surprising that recursive saturation may be proved in an essentially other way than in Theorem 3.4.2.

Proof of Theorem 5.2.3 Let $\mathcal{M} \models \text{PA}$ have a full satisfaction class D . Let $p = \{\varphi_0(v, b), \varphi_1(v, b), \dots\}$ be a recursive type in variable v and with parameter $b \in \mathcal{M}$.

We may assume that for all $i \in \mathbb{N}$ $\mathcal{M} \models \forall v [\varphi_{i+1}(v, b) \Rightarrow \varphi_i(v, b)]$, otherwise we pass to the type $\{\varphi(v, b_0), \varphi_0(v, b) \& \varphi_1(v, b), \dots\}$. In the formulas below we omit the parameter b to simplify notation.

We let $\delta_0(v)$ be $\neg\varphi_0(v)$ and $\delta_{i+1}(v)$ be $\varphi_i(v) \& \neg\varphi_{i+1}(v)$. It is easy to see that if $u \in \mathcal{M}$ satisfies all $\neg\delta_i$, $i \in \mathbb{N}$, then u realizes $p(v)$.

From now on assume towards a contradiction that $\mathcal{M}$ omits $p(v)$. It follows that we have

$$\text{for every } u \in \mathcal{M} \text{ there exists } j \in \mathbb{N} \text{ such that } \mathcal{M} \models \delta_j(u). \quad (5.1)$$

Let us remark that since $\forall v(\varphi_{i+1}(v) \Rightarrow \varphi_i(v))$ the above j is unique.

The key of Lachlan's proof is a construction of a sequence $\gamma_i(v) : i \in \mathbb{N}$ of formulas in the sense of $\mathcal{M}$. Fix a nonstandard $m \in \mathbb{N}$. We let $\gamma_0(v)$ be the trivial formula $v = v$. Assume that we know the formula $\gamma_i(v)$. We let $\gamma_{i+1}(v)$ be

$$(\neg\exists x \gamma_i(x) \& v = v) \vee (\exists x \gamma_i(x) \& \xi_0(v)).$$

Now, we describe a procedure for writing the formulas $\xi_0(v), \xi_1(v), \dots, \xi_m(v)$. One can prove by backward induction from m that this sequence is well defined. The formula $\xi_0(v)$ is

$$[\exists x (\gamma_i(x) \& \delta_0(x) \& \varphi_0(v)) \vee \xi_1(v)],$$

where $\xi_1(v)$ is

$$(\neg\exists x (\gamma_i(x) \& \delta_0(x))) \& [\exists x (\gamma_i(x) \& \delta_1(x) \& \varphi_1(v)) \vee \xi_2(v)].$$

Now, if $\xi_j(v)$ is of the form

$$\varrho \& [\exists x (\gamma_i(x) \& \delta_j(x) \& \varphi_j(v)) \vee \xi_{j+1}(v)],$$

then $\xi_{j+1}(v)$ is

$$\varrho \& \neg\exists x (\gamma_i(x) \& \delta_j(x)) \& [\exists x (\gamma_i(x) \& \delta_{j+1}(x) \& \varphi_{j+1}(v)) \vee \xi_{j+2}(v)].$$

As $\xi_m(v)$ we put the contradictory formula $v \neq v$. Let us observe that in writing ξ_{j+1} we use only that part of the formula $\xi_j(v)$ which is already well defined. One can verify that the formula $\xi_j(v)$, for $j < m$, has the form

$$\left(\bigwedge_{k < j} \neg\exists x (\gamma_i(x) \& \delta_k(x)) \right) \& [\exists x (\gamma_i(x) \& \delta_j(x) \& \varphi_j(v)) \vee \xi_{j+1}(v)].$$

This completes the construction of the formula γ_{i+1} , and, hence, of the whole sequence. Let us point out, that for each $\gamma_i(v)$ we construct separately the sequence $\xi i_j(v)$.

Observe that this is a procedure inside $\mathcal{M}$, indeed, as a matter of fact from the formula representing p we get the formula defining the sequence δ_i and then we get the formula representing the sequence γ_i . But the formulas γ_i are nonstandard in an essential way, so when we speak about their truth, we must use a satisfaction class. In the formulas below we write x rather than $S^x 0$ for $x \in \mathcal{M}$.

We assert that

$$\text{if } D(\exists x \gamma_i(x)) \text{ then there exists } n \in \mathbb{N} \text{ such that } D(\exists x (\gamma_i(x) \& \delta_n(x))). \quad (5.2)$$

Indeed, otherwise every $x \in \mathcal{M}$ such that $D(\gamma_i(x))$ would satisfy $\neg\delta_n$ for all $n \in \mathbb{N}$ what is impossible by claim (5.1).

Now we claim that

$$\text{for every } i \leq m, \quad D(\exists x \gamma_i(x)). \quad (5.3)$$

Assume that $D(\forall x \neg\gamma_i(x))$ holds and, in particular, that $i > 0$. Thus $\gamma_i(v)$ is not equivalent (in the sense of D -truth) to $v = v$. Hence, from the definition of γ_i we have $D(\exists x \gamma_{i-1}(x))$. By (5.2), there exists $n \in \mathbb{N}$ such that $D(\exists x (\gamma_{i-1}(x) \& \delta_n(x)))$ holds. Let n be the smallest with this property. Then

$$D(\neg\exists x (\gamma_{i-1}(x) \& \delta_j(x))) \text{ holds for all } j < n.$$

Now, let us show that

$$D(\forall v [\gamma_i(v) \equiv \varphi_n(v)]).$$

For, we may prove within D by a finite proof that

$$D(\forall v (\gamma_i(v) \equiv [\exists x (\gamma_{i-1}(x) \& \delta_n(x) \& \varphi_n(v)) \vee \xi_{n+1}])).$$

Then, we observe that one of conjuncts in ξ_{n+1} is a false sentence $\neg\exists x (\gamma_{i-1}(x) \& \delta_n(x))$ and it occurs at a standard depth in ξ_{n+1} . Therefore we have,

$$D(\forall v (\gamma_i(v) \equiv \exists x (\gamma_{i-1}(x) \& \delta_n(x) \& \varphi_n(v)))).$$

and, consequently,

$$D(\forall v [\gamma_i(v) \equiv \varphi_n(v)]).$$

Let us observe that checking this requires D to satisfy finite Tarski's conditions, induction in $(\mathcal{M}, D)$ is not needed. Since p is a type we have $\mathcal{M} \models \exists x \varphi_n(x)$ and we obtain a contradiction with the assumption that $D(\forall x \neg\gamma_i(x))$. Thus, we showed (5.3).

In the last paragraph we proved also that

$$\begin{aligned} \text{if } n \text{ is the smallest such that } D(\exists x [\gamma_i(x) \& \delta_n(x)]), \\ \text{then } D(\forall x [\gamma_{i+1}(x) \equiv \varphi_n(x)]). \end{aligned} \quad (5.4)$$

We let $n(i)$ be the smallest n such that $D(\exists x [\gamma_i(x) \& \delta_n(x)])$. Thus $n(i) \in \mathbb{N}$ for all $i \in \mathcal{M}$.

We claim that

$$n(i) < n(i + 1) \text{ for all } i \in \mathcal{M}. \quad (5.5)$$

Indeed, by the definition of $n(i)$ we have $D(\exists x [\gamma_i(x) \& \delta_{n(i)}(x)])$ and $D(\exists x [\gamma_{i+1}(x) \& \delta_{n(i+1)}(x)])$. Hence, by claim (5.4), $D(\forall x [\gamma_{i+1}(x) \equiv \varphi_{n(i)}(x)])$ and $D(\exists x [\psi_{n(i)}(x) \& \delta_{n(i+1)}(x)])$ holds. By the definition of δ_j this means that $D(\exists x [\varphi_{n(i)}(x) \& \neg \varphi_{n(i+1)}(x)])$ and so the same is true in $\mathcal{M}$. Since $\varphi_{n_i}(x) \Rightarrow \varphi_k(x)$ for all $k \leq n_i$, we have that $n(i) < n(i + 1)$.

Now we pick any nonstandard $i \in \mathcal{M}$ and, by (5.5), we get a strictly decreasing sequence $n(i), n(i - 1), n(i - 2), \dots$ of natural numbers. This is impossible and so p cannot be omitted in $\mathcal{M}$.

Observe that by working with small enough nonstandard m, i we obtain Q_e formulas, so this argument work in the case of satisfaction classes for Q_e formulas as well. $\square$

Smith [11] (cf. also [12, 13]) proved the following result.

Theorem 5.2.9 *Let $\mathcal{M}$ be a nonstandard model of PA. Then there exists a formula $\psi(x)$ in the language of PA in the sense of $\mathcal{M}$ such that for each full satisfaction class D for $\mathcal{M}$, the set $\{m \in \mathcal{M} : D(\psi(S^m 0))\}$ is not definable in $\mathcal{M}$ with parameters.*

We begin with a construction of a formula $\psi(x)$ from a given sequence $\langle \varphi_1(x), \varphi_2(x), \dots \rangle$ which we will specify later. The formula $\psi(x)$ will be a combination of conjunctions and disjunctions of these formulas with some special distribution of parentheses. We let

$$\begin{aligned} \psi_0(x) &= \varphi_0(x), \\ \psi_1(x) &= \varphi_0(x) \& \varphi_1(x), \\ \psi_2(x) &= \varphi_0(x) \& (\varphi_1(x) \vee \varphi_2(x)), \\ \psi_3(x) &= \varphi_0(x) \& (\varphi_1(x) \vee (\varphi_2(x) \& \varphi_3(x))), \\ &\dots \end{aligned}$$

Now, let a nonstandard model $\mathcal{M}$ of PA be given. Fix a nonstandard $m \in \mathcal{M}$. We apply the above construction m times. The resulting formula $\psi_m(x)$ will be our $\psi(x)$.

It will be convenient for us to present $\psi(x)$ with the help of additional formulas $\xi_i(x)$, for $i = m, \dots, 1$. To simplify the definition of $\xi_i(x)$, we assume that m is even. Now,

$$\begin{aligned} \xi_m(x) &= \varphi_m(x), \\ \xi_{m-1}(x) &= \varphi_{m-1}(x) \vee \varphi_m(x), \\ \xi_{m-2}(x) &= \varphi_{m-2}(x) \& (\varphi_{m-1}(x) \vee \varphi_m(x)), \\ \xi_{m-3}(x) &= \varphi_{m-3}(x) \vee (\varphi_{m-2}(x) \& (\varphi_{m-1}(x) \vee \varphi_m(x))), \\ &\dots \end{aligned}$$

Simply, $\xi_{m-1}(x) = \varphi_m(x) \circ \xi_m(x)$, where $\circ$ is $\vee$ or $\&$ depending whether m is even or odd, respectively. Now, we can present $\psi(x)$ as

$$\begin{aligned}
\psi(x) &= \varphi_0(x) \ \& \ \xi_1(x), \\
\psi(x) &= \varphi_0(x) \ \& \ (\varphi_1(x) \vee \xi_2(x)), \\
\psi(x) &= \varphi_0(x) \ \& \ (\varphi_1(x) \vee (\varphi_2(x) \ \& \ \xi_3(x))), \\
\psi(x) &= \varphi_0(x) \ \& \ (\varphi_1(x) \vee (\varphi_2(x) \ \& \ (\varphi_3(x) \vee \xi_4(x)))), \\
&\dots
\end{aligned}$$

The formula $\psi(x)$ can be seen as $\psi_i(x)$ with $\xi_{i+1}(x)$ added as the most nested conjunct or disjunct.

The set defined by $\psi(x)$ is largely determined by this formula rather than by a satisfaction class under consideration. Let us observe that, for $i \in \mathbb{N}$, φ_i is placed at a standard depth in the syntactic tree of ψ . More exactly, we have: for every standard k and each full satisfaction class D for $\mathcal{M}$

$$D(\forall x [\psi(x) \Rightarrow \psi_{2k}(x)]) \quad \text{and} \quad D(\forall x [\psi_{2k+1}(x) \Rightarrow \psi(x)]). \quad (5.6)$$

Now, to prove (5.6) it suffices to observe that, for a given $k \in \mathbb{N}$, $\psi(x)$ is just $\psi_{2k}(x)$ with $\varphi_{2k}(x)$ replaced by a conjunct $\varphi_{2k}(x) \ \& \ \xi_{2k+1}(x)$. Similarly, $\psi(x)$ is just $\psi_{2k+1}(x)$ with $\varphi_{2k+1}(x)$ replaced by a disjunction $\varphi_{2k+1}(x) \vee \xi_{2k+2}(x)$. Equation (5.6) follows by the closure of D on proofs of finite length. Once again, we stress that to check (5.6) no induction in $(\mathcal{M}, D)$ is needed.

Now, we find a sequence $\varphi_i(x)$ which will work for us. Let $\theta_i(y)$ be a recursive enumeration of formulas (without parameters) in the variable y . Let p_i be the natural enumeration of primes, thus, $p_0 = 2, p_1 = 3, \dots$. We denote by $|$ the divisibility relation, i.e., $u|w$ is an abbreviation for $\exists z \ u \cdot z = w$.

For $k \in \mathbb{N}$ let $\varphi_{2k}(x)$ be

$$[\neg \exists y \ \theta_k(y) \ \& \ p_k | x \ \& \ p_k^2 \nmid x] \vee [\exists y \ \theta_k(y) \ \& \ \exists z \ (p_k^{2+z} | x \ \& \ p_k^{3+z} \nmid x \ \& \ \theta_k(z))].$$

Let also $\varphi_{2k+1}(x)$ be

$$\forall q > p_k \text{ if } q \text{ is prime then } q \nmid x.$$

Let ψ_m be as above for this sequence φ_i . Then we may recover the set defined by $\theta_i(y)$ from $\psi(x)$. More precisely

Lemma 5.2.10 *If D is a full satisfaction class for $\mathcal{M}$, then*

$$\{b : \theta_i(b)\} = \{b : D(\exists x [\psi(x) \ \& \ p_i^{b+2} | x \ \& \ p_i^{b+3} \nmid x])\}.$$

Derivation of Theorem 5.2.9 from Lemma 5.2.10: If $\psi(x)$ is $\psi_m(x)$ for some nonstandard $m \in \mathcal{M}$ and D is a full satisfaction class for $\mathcal{M}$, then

$$\mathcal{M} \models \theta_i(b) \quad \text{iff} \quad D(\exists x [\psi(x) \ \& \ p_i^{b+2} | x \ \& \ p_i^{b+3} \nmid x])$$

by Lemma 5.2.10. It follows that $\{x : D(\psi(x))\}$ cannot be definable in $\mathcal{M}$ with parameters by Tarski's theorem (i.e. Theorem 3.2.7). As a matter of fact we use here the version stated as 3.5.1. $\square$

Proof of Lemma 5.2.10: Assume $\mathcal{M} \models \theta_i(b)$. We find a such that

$$D(\psi(a) \ \& \ p_i^{b+2} | a \ \& \ p_i^{b+3} \not| a).$$

For $j < i$ we put $c_j = 0$ if $\neg \exists y \theta_j(y)$ and any c_j such that $\theta_j(c_j - 1)$ if $\exists y \theta_j(y)$. We let a be $2^{1+c_0} \cdot 3^{1+c_1} \cdots p_{i-1}^{1+c_{i-1}} \cdot p_i^{b+2}$ and we assert that $\psi(a)$ is D -true. By (5.6) it suffices to check that $D(\psi_{2i+1}(a))$ holds. But $\varphi_{2j}(a)$, for $j \leq i$ are true in $\mathcal{M}$, hence they are D -true. Moreover, by construction of a , $\varphi_{2i+1}(a)$ is true and, consequently, D -true. It follows that $\psi_{2i+1}(a)$ is D -true. Therefore, $\exists x [\psi(x) \ \& \ p_i^{b+2} | x \ \& \ p_i^{b+3} \not| x]$ is in D , hence b is in the set on the right hand side. This proves inclusion $\subseteq$.

For the converse assume that $\exists x [\psi(x) \ \& \ p_i^{b+2} | x \ \& \ p_i^{b+3} \not| x]$ is in D . Then, there exists $a \in \mathcal{M}$ such that $\psi(a) \in D$ and $\mathcal{M} \models p_i^{b+2} | a \ \& \ p_i^{b+3} \not| a$. Then, for all $j < i$, $\varphi_{2j+1}(a)$ are false. It follows that all $\varphi_{2j}(a)$, for $j \leq i$, are true. In particular $\varphi_{2i}(a)$ holds. Thus, $\exists z [p_i^{2+z} | a \ \& \ p_i^{3+z} \not| a \ \& \ \theta_i(z)]$. But such z is uniquely determined by a , hence $z = b$, so $\theta_i(b)$ holds. $\square$

S. Smith found the above results to prove that the countability assumption is needed in Theorem 5.2.1. To be more specific, let us fix an arithmetical formula $x \in y$ expressing the fact x belongs to a set coded by y . As $x \in y$ one can simply take a formula $p_x | y$, where p_x is the x -th prime in a model. Now, let us recall that a set X is coded in a model $\mathcal{M}$ if there exists a in $\mathcal{M}$ such that for all b from $\mathcal{M}$, $b \in X$ if and only if $\mathcal{M} \models b \in a$. Kaufmann [14] proved using an extra set-theoretic assumption (the so-called $\diamond$, cf. Kunen [15], p. 80 for more in this direction) that there exist recursively saturated models $\mathcal{M}$ (of any given complete extension of PA) such that

$$\text{for every } X \subseteq \mathcal{M} \text{ if } X \cap (< a) \text{ is coded in } \mathcal{M}, \text{ for all } a \in \mathcal{M}, \quad (5.7) \\ \text{then } X \text{ is definable with parameters in } \mathcal{M}.$$

Kaufmann's model is of cardinality $\aleph_1$ and is ω_1 -like, that is for every $a \in \mathcal{M}$, the set $\{b \in \mathcal{M} : b < a\}$ is countable. Since his paper, such models are called *rather classless* in the literature. Shelah [16] proved an absoluteness result, from which it follows that the existence of such models is provable in ZFC alone, but no one knows the actual proof. The simplest construction (using $\diamond$) seems to be the one due to Kossak, cf. [17]. Schmerl [18] studied such models in greater depth. We omit the proof of the existence of such models because we restrict ourselves to countable ones in this book. We merely remark that a recursively saturated model having the property (5.7) must be uncountable. Otherwise, by Theorem 3.4.2, it has an inductive satisfaction class D for Q_e -formulas for some nonstandard $e \in \mathcal{M}$. But then D is not definable by Tarski's theorem (see Theorem 3.5.1) but its intersections with $(< a)$ are $\mathcal{M}$ -finite because $(\mathcal{M}, D)$ satisfies induction in its language.

Smith's result is

Theorem 5.2.11 *If $\mathcal{M}$ is a recursively saturated model of PA which has the property (5.7), then it admits no full satisfaction class.*

Proof It is an immediate corollary to Theorem 5.2.9 that if $\mathcal{M}$ satisfies the assumption (5.7), then $\mathcal{M}$ admits no full satisfaction class D such that $(\mathcal{M}, D)$ makes all axioms of PA true (that is, also all instances of induction for nonstandard formulas are supposed to be D -true). The reason is that if $\mathcal{M}$ admitted such a D , then the set D -defined by ψ , i.e. the set $E = \{u : D(\psi(u))\}$, would have the property of being piecewise coded. Indeed, we could prove the following within D ,

$$D(\forall z \exists w \forall y \leq z (y \in w \equiv \psi(y))).$$

Thus, the following would hold in $(\mathcal{M}, D)$

$$\forall a \exists b \forall y \leq a (y \in b \equiv D(\psi(y)))$$

that is $E \cap (< a)$ would be $\mathcal{M}$ -finite for all $a \in \mathcal{M}$. But E is undefinable contrary to the property (5.7).

In order to prove that $\mathcal{M}$ does not admit a D which need not make all axioms of PA true Smith applied the above and some tricks taken from the proof of Lachlan's theorem, i.e. Theorem 5.2.3.

We shall use the notation θ_i , φ_i and ψ_i as constructed above for the proof of Theorem 5.2.9. Observe at first that, by Lemma 5.2.10, the formula ψ_{2k+1} codes all information about $\theta_0, \dots, \theta_k$. More exactly, for each $k \in \mathbb{N}$ we have:

$$\theta_i(y) \equiv \exists x [\psi_{2k+1}(x) \ \& \ p_i^{2+y} | x \ \& \ p_i^{3+y} \nmid x] \text{ for all } i \leq k.$$

Let us introduce two abbreviations. If ϱ is a formula then by Cl_ϱ we denote the formula expressing that the set described by ϱ is piecewise coded, i.e.

$$\forall z \exists w \forall y \leq z [y \in w \equiv \varrho(y)].$$

We shall express this also as “ ϱ determines (or codes) a class”.

Let also $\varrho \sim \theta_i$ be an abbreviation for “ ϱ codes θ_i ”, i.e.

$$\forall y \{\theta_i(y) \equiv \exists x [\varrho(x) \ \& \ p_i^{2+x} | y \ \& \ p_i^{3+i} \nmid y]\}. \quad (5.8)$$

Now we repeat the tricks from the proof of Theorem 5.2.3. We define a sequence $\gamma_i(x)$ for $i \in \mathcal{M}$ by induction on i . We let $\gamma_0(x)$ be $x = x$ and $\gamma_{i+1}(x)$ be

$$[\neg \text{Cl}_{\gamma_i} \ \& \ x = x] \vee [\text{Cl}_{\gamma_i} \ \& \ \Delta(x)].$$

The formula $\Delta(x)$ is defined inductively with the help of additional formulas $\Delta_k(x)$, for $k \leq m$. Intuitively, $\Delta(x)$ will pick the smallest k such that $\gamma_i(x)$ does not code $\theta_k(x)$ and, in this case, $\Delta(x)$ will be equivalent to ψ_{2k+1} . Therefore, the formulas

$\Delta(x)$ and, in consequence, γ_{i+1} will code all the formulas $\theta_0, \dots, \theta_{k+1}$. Now, let us define the sequence $\Delta_k(x)$ by backward induction from some fixed nonstandard m . Firstly, for $j \leq m$, let δ_j be a formula

$$\bigwedge_{r < j} \gamma_i \sim \theta_r.$$

For standard j the bracketing of this formula may be arbitrary and, as it will occur later, for nonstandard j the exact bracketing will be of no importance as our reasoning will never reach these nonstandard formulas. We let $\Delta_m(x)$ to be $\neg x = x$. Now, we define $\Delta_{k-1}(x)$, for $k > 0$, as

$$(\delta_{k-1}) \ \& \ [(\neg(\gamma_i \sim \theta_{k-1}) \ \& \ \psi_{2(k-1)+1}(x)) \vee \Delta_k(x)].$$

We let $\Delta(x) = \Delta_0(x)$. It is instructive to look what occurs in $\Delta(x)$ at standard depth. So, we can present $\Delta(x)$ as

$$\begin{aligned} \Delta(x) &= \Delta_0(x) \\ \Delta(x) &= \delta_0 \ \& \ \{(\neg(\gamma_i \sim \theta_0) \ \& \ \psi_1(x)) \vee \Delta_1(x)\} \\ \Delta(x) &= \delta_0 \ \& \ \{(\neg(\gamma_i \sim \theta_0) \ \& \ \psi_1(x)) \vee ((\delta_1) \ \& \ [(\neg(\gamma_i \sim \theta_1) \ \& \ \psi_3(x)) \vee \Delta_2(x)])\} \\ &\dots \end{aligned}$$

As usual, this construction is done inside $\mathcal{M}$, but speaking about truth of these formulas makes sense only if a satisfaction class D for $\mathcal{M}$ is given. So let us fix a satisfaction class D .

Assume firstly that $\gamma_i(x)$ determines a class (in the sense of D). Hence $\gamma_i(x)$ is equivalent to some standard formula with a parameter) because $\mathcal{M}$ is rather classless. It follows that γ_i cannot code in the sense of (5.8) the information about all θ_k . Thus there exists the smallest $n_i \in \mathbb{N}$ such that γ_i does not code θ_{n_i} . We have shown

$$\begin{aligned} &\text{if } \gamma_i \text{ codes a class, then} \\ &\text{there exists the smallest } n_i \in \mathbb{N} \text{ such that } \gamma_i \text{ does not code } \theta_{n_i}. \end{aligned} \quad (5.9)$$

In this situation $\gamma_{i+1}(x)$ is equivalent, under D , to ψ_{2n_i+1} . For we write γ_{i+1} as

$$(\neg \text{Cl}_{\gamma_i} \ \& \ x = x) \vee [\text{Cl}_{\gamma_i} \ \& \ \Delta(x)].$$

But $\neg \text{Cl}_{\gamma_i}$ is D -false by the assumption. So, γ_{i+1} is D -equivalent to

$$\delta_0 \ \& \ \{(\neg(\gamma_i \sim \theta_0) \ \& \ \psi_1(x)) \vee \Delta_1(x)\}$$

If $n_i = 0$, then δ_1 is a false conjunct of $\Delta_1(x)$, thus $\gamma_{i+1}(x) \equiv \psi_1(x)$ is D -true. If $n_i = 1$ then δ_0, δ_1 and $\neg(\gamma_i \sim \theta_1)$ are true and δ_2 is false. So, γ_{i+1} is D -equivalent to ψ_3 .

In general, all $\delta_0, \dots, \delta_{n_i}$ and $\neg(\gamma_i \sim \theta_{n_i})$ are true while δ_{n_i+1} is false. Thus, by a finite D -proof, one can show that $\gamma_{i+1} \equiv \psi_{2n_i+1}$. Summing up, we have

$$\text{if } \gamma_i \text{ codes a class, then } \gamma_{i+1} \equiv \psi_{2n_i+1} \text{ is } D\text{-true.} \quad (5.10)$$

Observe that in this situation ψ_{2n_i+1} codes all the information about $\theta_0, \dots, \theta_{n_i}$ and is standard, hence it codes a class. It follows that n_{i+1} is defined and $n_{i+1} > n_i$ because γ_i does not code θ_{n_i} . Thus we have

$$\text{if } \gamma_i \text{ codes a class then } \gamma_{i+1} \text{ codes a class and } n_{i+1} > n_i. \quad (5.11)$$

Now assume that γ_i does not code a class. Thus $i > 0$ (because the formula $x = x$ codes a class), so let us look at γ_{i-1} . We have that $\gamma_i(x)$ is D -equivalent to

$$(\neg \text{Cl}_{\gamma_{i-1}} \ \& \ x = x) \vee (\text{Cl}_{\gamma_{i-1}} \ \& \ \xi) \text{ for some } \xi.$$

But $x = x$ is not D -equivalent to γ_i , so $\neg \text{Cl}_{\gamma_{i-1}}$ cannot hold, so γ_{i-1} codes a class. By (5.11), γ_i also codes a class. It follows that all γ_i determine classes.

Now let $i \in \mathcal{M}$ be nonstandard. We must have $n_i > n_{i-1} > n_{i-2} > \dots$ by (5.11). This is impossible as all n_i 's are standard by (5.9) and D cannot exist. $\square$

We remark also that the results of this section show a curious phenomenon: if $\mathcal{M}$ is countable, then $\mathcal{M}$ -logic is consistent iff $\mathcal{M}$ is recursively saturated. Indeed, both conditions are equivalent to the existence of a full satisfaction class for $\mathcal{M}$. Smith also transferred the proof of Theorem 5.2.3 to a more direct proof of inconsistency of $\mathcal{M}$ -logic for a non recursively saturated $\mathcal{M}$.

5.3 Inductive Full Satisfaction Classes

In this section we shall give some information on theories $\Sigma_m\text{-PA}(D)$ with induction for Σ_m formulas in $L_{\text{PA}} \cup \{D\}$ and similarly for full $\text{PA}(D)$. We give no proofs of these results. They require some work with higher ordinals and extension of the methods developed above, see [5]. We would like to point out that model theoretic methods in proof theory were developed further, see Avigad and Sommer [19].

We define transfinite iterations of a version of ω -logic. Let T be, as usual, a consistent, primitive recursive theory in L_{PA} containing PA . We define formulas T^α and Γ_n^α for $\alpha < \varepsilon_0$ by the following induction. We let $T^0 = \text{PA}$, and $T^\lambda = \bigcup_{\alpha < \lambda} T^\alpha$ for limit λ . If T^α is defined then Γ_n^α are defined as follows. We let $\Gamma_0^\alpha(\varphi)$ be $\text{Pr}_{T^\alpha}(\varphi)$, $\Gamma_{n+\frac{1}{2}}^\alpha(\varphi)$ be “ φ is of the form $\eta \vee \forall z \psi(z)$ and $\forall z \Gamma_n^\alpha(\eta \vee \psi(S^z 0))$ ”, $\Gamma_{n+1}^\alpha(\varphi)$ be $\exists \langle \xi_0, \dots, \xi_{r-1} \rangle [\forall i < r \ \Gamma_{n+\frac{1}{2}}^\alpha(\xi_i) \ \& \ \text{Pr}_{T^0}((\bigwedge_{i < r} \xi_i) \Rightarrow \varphi)]$, and, finally, $T^{\alpha+1}$ is

$T^\alpha \cup \{\neg \Gamma_n^\alpha(0 = 1) : n\}$. This definition makes sense in any $\mathcal{M} \models T$ (then n ranges over $\mathcal{M}$). One obtains a single formula $\text{Pr}_{T^\alpha}(\varphi)$ in two variables α, φ and a primitive recursive sequence, over n , of formulas $\Gamma_n^\alpha(\varphi)$ in variables α, φ .

We have the following description of the existence of full inductive satisfaction classes in terms of consistency of this system of ω -logic.

Theorem 5.3.1 *Let $\mathcal{M}$ be a countable recursively saturated model of PA and let $m \in \mathbb{N}, m > 0$. Then $\mathcal{M}$ has a full Σ_m -inductive satisfaction class iff for every $k \in \mathbb{N}$, $\mathcal{M} \models \neg \Gamma_k^{\omega_m(k)}(0 = 1)$.*

We describe now the arithmetical part of the theory $\text{PA}(D)$ in terms of transfinite induction. We give no proofs, because in this book we restricted ourselves to ordinals below ε_0 . Now we have to go to higher ordinals, up to $\varepsilon_{\varepsilon_0}$. We do not develop this stuff, this section is written for the readers familiar with this. Let $\omega_m(\alpha)$ be as defined previously. For $\tau < \varepsilon_0$ we denote by ε_τ the τ th epsilon ordinal. Let $\text{TI}(\alpha)$ denote the scheme of transfinite induction over α (for all formulas).

Theorem 5.3.2 *Fix $m \in \mathbb{N}$. Then the set of L_{PA} consequences of $I\Sigma_m(D)$ is axiomatized by PA together with $\text{TI}(\varepsilon_{\omega_m(k)}) : k \in \mathbb{N}$.*

Corollary 5.3.3 *The set of L_{PA} consequences of full $\text{PA}(D)$ is axiomatized by PA together with $\text{TI}(\varepsilon_{\omega_k}) : k \in \mathbb{N}$.*

References

1. Robinson, A. (1963). On languages which are based on non-standard arithmetic. *Nagoya Mathematical Journal*, 22, 83–117. ISSN: 0027-7630. <https://doi.org/10.1017/S0027763000011065>.
2. Krajewski, S. (1976). Non-standard satisfaction classes. In W. Marek, M. Srebrny & A. Zarach (Eds.), *Set Theory and Hierarchy Theory A Memorial Tribute to Andrzej Mostowski*. Lecture notes in mathematics (Vol. 537). Berlin: Springer.
3. Kotlarski, H., Krajewski, S., & Lachlan, A. H. (1981). Construction of satisfaction classes for nonstandard models. *Canadian Mathematical Bulletin*, 24(3), 283–293. ISSN: 0008-4395. <https://doi.org/10.4153/CMB-1981-045-3>.
4. Kotlarski, H., & Ratajczyk, Z. (1990). Inductive full satisfaction classes. *Annals of Pure and Applied Logic*, 47(3), 199–223. ISSN: 01680072. [https://doi.org/10.1016/0168-0072\(90\)90035-Z](https://doi.org/10.1016/0168-0072(90)90035-Z).
5. Kotlarski, H., & Ratajczyk, Z. (1990). More on induction in the language with a satisfaction class. *Mathematical Logic Quarterly*, 36(5), 441–454. ISSN: 09425616. <https://doi.org/10.1002/malq.19900360509>.
6. Kotlarski, H. (1986). Bounded induction and satisfaction classes. *Mathematical Logic Quarterly*, 32(31–34), 531–544. ISSN: 09425616. <https://doi.org/10.1002/malq.19860323107>.
7. Weislo, B., & Łętyk, M. (2017). Notes on bounded induction for the compositional truth predicate. *The Review of Symbolic Logic*, 10(3), 455–480. ISSN: 1755-0203. <https://doi.org/10.1017/S1755020316000368>.
8. Łętyk, M. (2017). *Axiomatic theories of truth, bounded induction and reflection principles*. Ph.D. thesis. Warsaw University. <https://depotuw.ceon.pl/handle/item/2266>.

9. Lachlan, A. H. (1981). Full satisfaction classes and recursive saturation. *Canadian Mathematical Bulletin*, 24(3), 295–297. ISSN: 0008-4395. <https://doi.org/10.4153/CMB-1981-046-0>.
10. Kaye, R. (1991). *Models of Peano arithmetic*. Oxford: Oxford University Press. ISBN: 019853213X.
11. Smith, S. T. (1984). *Nonstandard syntax and semantics and full satisfaction classes for models of arithmetic*. Ph.D. thesis. Yale University.
12. Smith, S. T. (1987). Nonstandard characterizations of recursive saturation and resplendency. *Journal of Symbolic Logic*, 52(3), 842. ISSN: 0022-4812. <https://doi.org/10.2307/2274370>.
13. Smith, S. T. (1989). Nonstandard definability. *Annals of Pure and Applied Logic*, 42(1), 21–43. ISSN: 01680072. [https://doi.org/10.1016/0168-0072\(89\)90064-X](https://doi.org/10.1016/0168-0072(89)90064-X).
14. Kaufmann, M. (1977). A rather classless model. *Proceedings of the American Mathematical Society*, 62(2), 330. ISSN: 00029939. <https://doi.org/10.2307/2041038>.
15. Kunen, K. (1992). *Set theory, an introduction to independence proofs*. Studies in logic and foundations of mathematics (Vol. 102). Amsterdam: North-Holland. ISBN: 0444868399.
16. Shelah, S. (1978). Models with second order properties II. Trees with no undefined branches. *Annals of Mathematical Logic*, 14(1), 73–87. ISSN: 00034843. [https://doi.org/10.1016/0003-4843\(78\)90009-8](https://doi.org/10.1016/0003-4843(78)90009-8).
17. Kossak, R., & Kotlarski, H. (1988). Results on automorphisms of recursively saturated models of PA. *Fundamenta Mathematicae*, 129(1), 9–15. ISSN: 0016-2736. <https://doi.org/10.4064/fm-129-1-9-15>.
18. Schmerl, J. (1981). Recursively saturated, rather classless models of Peano arithmetic. In M. Lerman, J. H. Schmerl, & R. I. Soare (Eds.), *Logic Year 1979-80*. Lecture notes in mathematics (Vol. 859). Berlin: Springer.
19. Avigad, J., & Sommer, R. (1997). A model-theoretic approach to ordinal analysis. *The Bulletin of Symbolic Logic*, 3(1), 17–52. <https://projecteuclid.org:443/euclid.bsl/1182353487>.

Index

Symbols

$L(\mathcal{M})$, 92
 $L(\mathcal{Q}, \Sigma_m)$, 75
 $\mathcal{Q}_e$ -satisfaction class, 54
 $\mathcal{Q}_n$ -formula, 54
 $[A]^m$, 1
 Con_T , 50
 Con_T^R , 51
 Δ_0 formula, 44
 $\text{Diag}(\mathcal{M})$, 41
 $\text{ElemDiag}(\mathcal{M})$, 41
 $\text{KS}(\mu; a)$, 17
 $\text{LD}(\alpha, \beta)$, 32
 $\text{LM}(\alpha)$, 8
 $\mathcal{L}(\mathcal{M})$, 92
 $\mathcal{M}$ -logic, 92
 $\text{Nit}(h; \alpha, a)$, 17
 Π_n -completion, 45
 $\text{Prov}_T(y, x)$, 45
 RKS , 17
 $\text{RM}(\alpha)$, 8
 Σ_n -completion, 45
 Σ_1 -closed model, 66
 $P_m(A)$, 1
 $\text{TI}(\alpha)$, 105
 $\text{TI}(\mathcal{Q}, \Pi_m)$, 75
 Tr_0 , 44
 Tr_n , 44
 Tr_{Δ_0} , 44
 Tr_{Π_n} , 44
 Tr_{Σ_n} , 44
 α -large partition, 20
 α -large set, 14
 α -large set in Grzegorczyk–Wainer sense, 17
 α -large set in the sense of slow growing hierarchy, 18

α -large set, original definition, 16
 α -small set, 14
 $\alpha \gg \beta$, 9
 $\alpha \ggg \beta$, 9
 $\alpha \rightarrow (\beta)_\gamma^m$, 2
 $\beta \Rightarrow_b \alpha$, 9
 $\mathbb{I}_i(x)$, 14
 $\exists_1$ -closed model, 66
 $\langle x, y \rangle$, 44
 ω -consistency, 49, 64
 ω -logic, 92
 ω -rule, 68
 ω_m , 19
 $\omega_m(\alpha)$, 75
 $\omega_{(n)}(\alpha, c)$, 27
 ϕ_k -scattered set, 79
 $\text{Pr}_T(\ulcorner \varphi \urcorner)$, 45
 $\text{psn}(\alpha)$, 12
 $\text{tow}_k(n)$, 6
 ε_0 , 8
 f -scattered sets, 18
 $f(x) \downarrow$, 15
 $v_\delta(\alpha)$, 32
 RFN_T , 69

A

ACT-extension, 47
 Addition of ordinals, 8
 Approximation of a finite function, 18, 19
 Approximately large sets, 18
 Arithmetical hierarchy theorem, 51
 Arithmetized completeness theorem, 46
 Arrow notation: Hardy, 29
 Arrow notation in Ramsey theory, 2
 At most α -large set, 14

B

Berry's paradox, 62
 Bounded formula, 44
 Bounded quantifiers, 44
 Branch in a tree, 2
 Busy beaver problem, 59

C

Cantor normal form, 8
 Chain of structures, 40
 Church's theorem, 52
 Coefficient in Cantor normal form, 8
 Coloring, 1
 Completion of T , 45
 Consistency statement, 50
 Cut elimination, 73

D

Derivability conditions, 45
 Derivable Σ_1 -completeness, 45
 Diagonal lemma, nonparametric version, 48
 Diagonal lemma, the parametric version, 49
 Diagram of a model, 41

E

Elementary diagram of a model, 41
 Elementary equivalence, 39
 Elementary extension, 40
 Elementary substructure, 40
 Exactly α -large set, 14
 Exponents in Cantor normal form, 8
 Extension of a model, 39

F

Fast growing hierarchy, 17
 Finite Ramsey theorem, 4
 Formalized first incompleteness theorem, 52
 Formalized Rosser's theorem, 53
 Formalized second incompleteness theorem, 53
 Full Cantor normal form of an ordinal, 8
 Full satisfaction class, 53
 Fundamental sequences, 9

G

Gödel's first incompleteness theorem, 49
 Gödel's second incompleteness theorem, 50
 Grelling's antinomy, 65
 Grzegorzcyk–Wainer hierarchy, 17

H

Hardy hierarchy, 14
 Herbrand's theorem, 73
 Homogeneous set, 2

I

Indicator for models of T , 73
 Indicator, well behaved, 73
 Inductive satisfaction class, 55
 Infinite Ramsey theorem, 2
 Iterated exponentiation, 14

K

König's infinity lemma, 2
 $KS(\mu; a)$, 17

L

Leftmost exponent of an ordinal, 8
 Levels in the tree, 2
 Liar paradox, 49
 Limit ordinal, 8
 $LM(\alpha)$, 8
 Local reflection principle, 69

M

Monochromatic set, 2
 Multiplication of ordinals, 9

N

Notations for ordinals, 8
 Number of iterations, 17

O

Open formulas, 44
 Ordinal ε_0 , 8
 Overspill lemma, 55

P

Pairing function, 44
 Partial satisfaction class, 54
 Partition, 1
 Provability in Rosser sense, 50
 Provability of transfinite induction, 75
 Pseudonorm of an ordinal, 12
 Pudlák's principle, 18

Q

Quantifier-free formulas, 44

Quantifier rank of a formula, 54

R

Ramsey theorem, finite form, 4

Ramsey theorem, infinite form, 2

Rather classless models, 101

Recursive saturation, 40

Reflection principle, 69

Relation $\beta \Rightarrow_b \alpha$, 9

Resplendency, 41

RFN_T, 69

Rightmost exponent of an ordinal, 8

RM(α), 8Robinson's theory Q , 44

Root of a tree, 2

Rosser consistency, 51

Rosser's theorem, 50

S

Saturation, 40

Scattered, 18

Schwichtenberg–Wainer hierarchy, 17

Set theoretic tree, 2

Short Cantor normal form, 9

Slow growing hierarchy based on h , 18

Submodel, 39

Substructure of a model, 39

Successor ordinal, 8

T

Tarski's theorem, 51

The relation $\alpha \gg \beta$, 9The relation $\alpha \ggg \beta$, 9

The relation RKS, 17

The scheme of Π_m transfinite induction over ϱ , 75The scheme of Σ_m minimum over ϱ , 75The sequence ω_m , 19The sequence $\omega_m(\alpha)$, 75The sequence $\omega_{(n)}(\alpha, c)$, 27The ω -rule, 68TI(α), 105TI(ϱ, Π_m), 75

Tree, 2

Tree (in the number-theoretic sense), 27

Type Σ_n , 40Type over $\mathcal{M}$, 40*type recursive*, 40**U**

Undecidability theorem, 52

Uniform reflection principle, 69

Union of a chain of structures, 40

Universal formulas, 44

W

Well behaved indicator, 73